

Answers to

Theory of Computation

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Cover: Flauros, the sixty-fourth demon and the Duke of Hell, who gives true answers of all things past, present, and future. Although, you should verify what he says for yourself. (Unless you can get him inside a triangle and then you're good, although whether a triangle of roads, or rivers, or planets, is most unclear.) From *Dictionnaire Infernal*, Jacques Albin Simon Collin de Plancy, 1863.

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Chapter I: Mechanical Computation

I.1.10 It is like a program in that it is single-purpose, that is, just as we may have a program that doubles its input and does nothing else, so also we may have such a Turing machine. It is unlike a program in that it is hardware.

It is like the computers we have on our desks today in that it mechanically converts the input into the output. A Turing machine is unlike such a computer in that it is purpose-built, for a single job.

I.1.11 The definition describes the machine. Further description of the machine's action, which involves the definitions of configurations and the transitions, follows Definition 1.4. In some ways it is an extension of the definition.

Restated, as A Bauer says in a comment to this question on Stack Exchange, why is the road not part of the car?

I.1.12 It is not a question of determining whether a particular mathematical statement is true. Obviously, if $x = 2$ then we can determine whether $x > 3$. Similarly, we can determine the truth or falsity of $\forall x \in \mathbb{N} [x^2 > 3]$. Rather, the *Entscheidungsproblem* asks for an algorithm that inputs any mathematical statement at all, and outputs 'T' or 'F'.

I.1.13

(A) Here is the sequence of tapes traced out in the course of the computation of the predecessor of 2.

Step	Configuration	Step	Configuration	Step	Configuration
0		3		6	
1		4		7	
2		5			

(B) This is the tape diagram sequence for the sum of 2 and 2.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8			
4		9			

(c) This is the predecessor computation.

$$\begin{aligned}
 &\langle q_0, 1, \varepsilon, 1 \rangle \vdash \langle q_0, 1, 1, \varepsilon \rangle \vdash \langle q_0, B, 11, \varepsilon \rangle \vdash \langle q_1, 1, 1, \varepsilon \rangle \vdash \langle q_1, B, 1, \varepsilon \rangle \vdash \langle q_2, 1, \varepsilon, \varepsilon \rangle \\
 &\vdash \langle q_2, B, \varepsilon, 1 \rangle \vdash \langle q_3, 1, \varepsilon, \varepsilon \rangle
 \end{aligned}$$

There is no instruction for state q_3 so $\langle q_3, 1, \varepsilon, \varepsilon \rangle$ is a halting configuration.

The sum of 2 and 2 is a little longer.

$$\begin{aligned} \langle q_0, 1, \varepsilon, 1B11 \rangle &\vdash \langle q_0, 1, 1, B11 \rangle \vdash \langle q_0, B, 11, 11 \rangle \vdash \langle q_1, B, 11, 11 \rangle \vdash \langle q_1, 1, 11, 11 \rangle \\ &\vdash \langle q_2, 1, 11, 11 \rangle \vdash \langle q_2, 1, 1, 111 \rangle \vdash \langle q_2, 1, \varepsilon, 1111 \rangle \vdash \langle q_2, B, \varepsilon, 11111 \rangle \\ &\vdash \langle q_3, B, \varepsilon, 11111 \rangle \vdash \langle q_3, 1, \varepsilon, 1111 \rangle \vdash \langle q_4, B, \varepsilon, 1111 \rangle \vdash \langle q_5, 1, \varepsilon, 111 \rangle \end{aligned}$$

I.1.14

- (A) To show that some operation on negative numbers (or any mathematical object, for that matter) can be computed, name a representation for them with strings over a finite alphabet and then show how to use a Turing machine to do mechanical computations on those strings. For instance, to show that multiplication of two integers is mechanical, we could use the alphabet $\Sigma = \{B, 0, \dots, 9, +, -\}$ and represent each integer as a pair $\langle \text{sign}, \text{natural number} \rangle$. Making a Turing machine that has four cases, for the four possible pairs of signs, is tedious but perfectly possible.
- (B) If by “problem” we mean that the machine fails to halt, this is not the source of the problem. Rather, this machine is a straightforward infinite loop.
- (C) We could give a machine states that are not sequential. The only limitation is that we use the convention that the initial state is q_0 . For instance, this machine fits the definition $\mathcal{P} = \{q_0 B B q_{50}, q_0 11 q_{50}, q_{50} 11 q_1, q_{50} 11 q_1\}$, and on any input reaches state q_{50} after one step.

I.1.15 The halting state is q_4 and the working states are q_0, q_1, q_2 and q_3 .

I.1.16 The trace is not too interesting.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6		10	
3		7			

I.1.17

- (A) The result of running the mystery machine on input 11 is this.

Step	Configuration	Step	Configuration
0		2	
1		3	

- (B) On input 1011 it gives this.

Step	Configuration	Step	Configuration
0		2	
1		3	

- (C) The input 110 gives this.

Step	Configuration	Step	Configuration
0		3	
1		4	
2			

(D) With the input 1101 the machine gives this sequence of tapes.

Step	Configuration	Step	Configuration
0		4	
1		5	
2			
3			

(E) This is what happens when the initial tape is empty.

Step	Configuration
0	
1	

I.1.18 This is the transition table.

Δ	B	$\emptyset$	1
q_0	Bq_4	Rq_0	Rq_1
q_1	Bq_4	Rq_2	Rq_0
q_2	Bq_4	Rq_0	Rq_3
q_3	—	—	—

I.1.19 The machine $\mathcal{P}_{\text{blankones}} = \{q_0BBq_2, q_01Bq_1, q_1BRq_0, q_111q_0\}$ does this on a tape with two 1's.

Step	Configuration	Step	Configuration
0		3	
1		4	
2		5	

I.1.20

(A) This machine does the job; note that the head never moves.

$$\mathcal{P}_{\text{singlebitnot}} = \{q_0BBq_1, q_0\emptyset 1q_1, q_01\emptyset q_1\}$$

This shows input $\sigma = \emptyset$.

Step	Configuration
0	0 q_0
1	1 q_1

- (B) This machine does ‘and’. Note that the states are not numbered sequentially; sometimes making a gap is convenient when you are writing the machine. For instance, here state q_{100} is used as the halting state, since it was clear there would not be a hundred states and so this one would be available. Similarly, q_{10} is for what the machine does if the first bit is 0 while q_{20} is for the case where the first bit is 1.

$$\mathcal{P}_{\text{singlebitand}} = \{q_0BBq_{100}, q_0\theta Bq_{10}, q_01Bq_{20}, q_{10}BRq_{11}, q_{10}\theta\theta q_{100}, q_{10}11q_{100}, \\ q_{11}BBq_{100}, q_{11}\theta\theta q_{100}, q_{11}1\theta q_{100}, q_{20}BRq_{21}, q_{20}\theta\theta q_{100}, q_{20}1\theta q_{100}, \\ q_{21}BBq_{100}, q_{21}\theta\theta q_{100}, q_{21}11q_{100}\}$$

This shows what the machine does on input $\sigma = 01$.

Step	Configuration	Step	Configuration
0	0 1 q_0	2	1 q_{11}
1	1 q_{10}	3	0 q_{100}

- (c) This is the ‘or’ machine. As in the prior item, the states are not numbered sequentially. Note also that this machine is basically the same as the ‘or’ machine.

$$\mathcal{P}_{\text{singlebitor}} = \{q_0BBq_{100}, q_0\theta Bq_{10}, q_01Bq_{20}, q_{10}BRq_{11}, q_{10}\theta\theta q_{100}, q_{10}11q_{100}, \\ q_{11}BBq_{100}, q_{11}\theta\theta q_{100}, q_{11}11q_{100}, q_{20}BRq_{21}, q_{20}\theta\theta q_{100}, q_{20}1\theta q_{100}, \\ q_{21}BBq_{100}, q_{21}\theta 1q_{100}, q_{21}11q_{100}\}$$

Here is what it does on input $\sigma = 01$.

Step	Configuration	Step	Configuration
0	0 1 q_0	2	1 q_{11}
1	1 q_{10}	3	1 q_{100}

I.1.21 This machine uses q_0 to slide to the right end, appends the 01, and uses q_2 to slide back to the left.

$$\mathcal{P}_{\text{append01}} = \{q_0B\theta q_1, q_0\theta Rq_0, q_01Rq_0, q_1B1q_2, q_1\theta Rq_1, q_11Lq_2, q_2BRq_3, q_2\theta Lq_2, q_21Lq_2\}$$

This example shows what the machine does, on input $\sigma = 11$.

Step	Configuration	Step	Configuration	Step	Configuration
0	1 1 q_0	4	1 1 0 q_1	8	1 1 0 1 q_2
1	1 1 q_0	5	1 1 0 1 q_2	9	1 1 0 1 q_2
2	1 1 q_0	6	1 1 0 1 q_2	10	1 1 0 1 q_3
3	1 1 0 q_1	7	1 1 0 1 q_2		

I.1.22 This machine works.

$$\mathcal{P}_{\text{constantthree}} = \{q_0BBq_2, q_01Bq_1, q_1BRq_0, q_111q_0, q_2B1q_2, q_21Rq_3, q_3B1q_3, q_31Rq_4, \\ q_4B1q_4, q_411q_5, q_5BLq_6, q_51Lq_6, q_6BBq_7, q_61Lq_7\}$$

Here is the machine acting on an input of 2.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9			

I.1.23 This machine computes successor.

$$\mathcal{P}_{\text{successor}} = \{q_0B1q_1, q_01Lq_0\}$$

An example is this tape sequence for input 3.

Step	Configuration
0	
1	
2	

I.1.24

- (A) We begin with the head at the start of a sequence of 1's. Erase that first 1, then slide to the end of the sequence, and past a blank. Put two 1's, and then slide back to the start of the initial sequence. Iterate that. The machine has nine instructions and alphabet $\Sigma = \{1, B\}$.

$$\mathcal{P}_{\text{doubler}} = \{q_0BBq_9, q_01Bq_1, q_1BRq_2, q_11Rq_2, q_2BRq_3, q_21Rq_2, q_3B1q_4, q_31Rq_3, q_4BBq_4, q_41Rq_5, \\ q_5B1q_6, q_511q_6, q_6BLq_7, q_61Lq_6, q_7BRq_8, q_71Lq_7, q_8BRq_9, q_811q_0\}$$

Here is the sequence of tapes for the input 2. It shows the blanks because there can be two of them, which can be confusing. It is in halves just to allow a page break in the middle.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9		14	

This is the second half.

Step	Configuration	Step	Configuration	Step	Configuration
15		20		25	
16		21		26	
17		22		27	
18		23		28	
19		24			

(B) To double a number represented in binary just append a 0 to the right side. The machine has three instructions and $\Sigma = \{0, 1, B\}$.

$$\mathcal{P}_{\text{doublerbinary}} = \{q_0 B B q_3, q_0 0 0 q_3, q_0 1 R q_1, q_1 B 0 q_2, q_1 0 R q_1, q_1 1 R q_1, q_2 B R q_3, q_2 0 L q_2, q_2 1 L q_2\}$$

On input 6 it gives this tape sequence.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6			
3		7			

I.1.25 This Turing machine works.

$$\mathcal{P}_{\text{odd}} = \{q_0 B B q_{100}, q_0 1 R q_1, q_1 B L q_{100}, q_1 1 B q_2, q_2 B L q_3, q_2 1 L q_3, q_3 B B q_4, q_3 1 B q_4, q_4 B R q_5, q_4 1 1 q_5, q_5 B R q_0, q_5 1 1 q_0\}$$

For instance, on input 3 it does this

Step	Configuration	Step	Configuration	Step	Configuration
0		3		6	
1		4		7	
2		5		8	

and on 4 it does this.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9			

I.1.26 While this Turing machine is moving right, if it hits a comma then it replaces it with a 1 and closes up to the left. That is, it scans left to the start of the string, erases the first 1, and then starts moving right again.

$$\mathcal{P}_{\text{addany}} = \{q_0 \text{BL} q_{10}, q_0 1 \text{R} q_0, q_0, 1 q_1, q_1 \text{BR} q_2, q_1 1 \text{L} q_1, q_1, \text{L} q_1, q_2 \text{BR} q_0, q_2 1 \text{B} q_2, \\ q_{10} \text{BR} q_{100}, q_{10} 1 \text{L} q_{10}, q_{10}, \text{L} q_{10}\}$$

For instance, on input 11, 1, 11 it does this.

Step	Configuration	Step	Configuration	Step	Configuration
0		6		12	
1		7		13	
2		8		14	
3		9		15	
4		10		16	
5		11		17	

(It is split into two sets of tables to give the formatter a place to put a page break.)

Step	Configuration	Step	Configuration	Step	Configuration
18		23		28	
19		24		29	
20		25		30	
21		26		31	
22		27		32	

I.1.27 No. Given a configuration $C = \langle q, s, \tau_L, \tau_R \rangle$, one preceding configuration is $\hat{C} = C$ with $\mathcal{I} = q s s q$.

I.1.28

- (A) The machine can flip the first bit, slide right and flip the second bit, etc. The fact that we know the number of bits in advance means that we can just have four groups of states: first flip and slide, second flip and slide, etc. (We could do an arbitrary number of bits but it would be a little harder.) At the end we slide left to position the head at the left end of the non-blank characters.

Although the question says that you need not produce a machine, here is one. In this machine, state q_0 flips the first bit and then q_1 slides right. The next flip and slide combination is q_2 and q_3 , etc., until the machine gets to the fourth bit. Then it moves the head back to the start with q_7 .

$$\mathcal{P}_{\text{bitwiseNOT}} = \{q_0 B B q_1, q_0 0 1 q_1, q_0 1 0 q_1, q_1 B R q_2, q_1 0 R q_2, q_1 1 R q_2, q_2 B B q_3, q_2 0 1 q_3, q_2 1 0 q_3, q_3 B R q_4, q_3 0 R q_4, q_3 1 R q_4, q_4 B B q_5, q_4 0 1 q_5, q_4 1 0 q_5, q_5 B R q_6, q_5 0 R q_6, q_5 1 R q_6, q_6 B B q_7, q_6 0 1 q_7, q_6 1 0 q_7, q_7 B R q_8, q_7 0 L q_7, q_7 1 L q_7\}$$

For instance, on input 0110 it does this.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		9	
1		6		10	
2		7		11	
3		8		12	
4					

- (B) The machine has two cases, depending on what it finds for b_3 . If $b_3 = 0$ then it moves right, copying the next bit into the current one, and then at the end it appends a 0. Likewise, if $b_3 = 1$ then it moves right, copying, and then at the end it appends a 1. As with the prior item, because we know in advance that there are four bits we can just have three groups of states.
- (C) One implementation is to expect two four-bit strings, $a_3 a_2 a_1 a_0$ and $b_3 b_2 b_1 b_0$, separated by a blank. The machine reads a_3 and goes into one of two states, r_0 or r_1 , depending on whether $a_3 = 0$ or $a_3 = 1$. From state r_0 it slides past the blank separator, then reads b_3 , and changes it to the logical 'and' of b_3 and 0. Similarly, from state r_1 it slides past the blank separator, reads b_3 , and changes it to the logical 'and' of b_3 and 1. Repeat that for a total of four iterations.

I.1.29

- (A) Under the convention that the machine starts with its head under the first non-blank character if there are any, $\mathcal{P} = \{q_0BBq_1, q_011q_0\}$ will halt only if the input is blank.
- (B) The machine $\mathcal{P} = \{q_0BBq_0, q_011q_1\}$ will fail to halt only if the input is blank.
- (C) The machine $\mathcal{P} = \{q_0BRq_1, q_01Rq_1, q_1BBq_1, q_111q_2\}$ will halt if and only if the second character is 1. (Trying the same idea for the first character won't work because of our convention that the first character is blank only if all characters are blank, so there is only one input string with a blank first character.)

I.1.30 The strategy here is to scan right. If the machine sees a b then it goes into a state, q_{20} that signifies it has seen one b and if the next character is also a b then it will halt and go to the accepting state. If it sees an a then it goes into state q_{10} , signifying that the number of b's in the consecutive count is zero.

$$\mathcal{P}_{\text{twobs}} = \{q_0BBq_4, q_0aRq_{10}, q_0bRq_{20}, q_{10}BBq_4, q_{10}aRq_{10}, q_{10}bRq_{20}, q_{20}BBq_4, q_{20}aRq_0, q_{20}bbq_3\}$$

As an example, on input abba it does this.

Step	Configuration	Step	Configuration
0		2	
1		3	

Note that it ends in the accept state. In contrast, on input aba the machine ends in the reject state.

Step	Configuration	Step	Configuration
0		3	
1		4	
2			

I.1.32 For this machine, if the first character is 0 then it puts a 1 in its place, marking that the string is in the language. Otherwise, it puts a 0. Then it blanks the remaining characters, and returns to the first character.

$$\mathcal{P}_{\text{starts with zero}} = \{q_001q_1, q_010q_1, q_0B0q_{10}, q_10Rq_2, q_11Rq_2, q_20Bq_3, q_21Bq_3, q_2BBq_4, q_3BRq_2, q_4BLq_4, q_409q_{10}, q_411q_{10}\}$$

This is the action of the machine on 01101.

Step	Configuration	Step	Configuration	Step	Configuration
0		6		12	
1		7		13	
2		8		14	
3		9		15	
4		10		16	
5		11		17	

I.1.33 For this machine, if the first character is not a 0 then it puts a 0 in its place, marking that the string as not in the language. If the first character is 0, then it checks whether the second character is 1. If so, it leaves it alone, otherwise it puts 0, marking that the input is not in the language. Then it blanks the remaining characters, and returns to the single character marking whether the input is a member.

$$\mathcal{P}_{\text{starts with } 01} = \{q_0 0Bq_1, q_0 10q_3, q_0 B0q_{10}, q_1 BRq_2, q_2 0Rq_4, q_2 1Rq_4, q_2 B0q_{10}, q_3 0Rq_4, q_4 0Bq_5, q_4 1Bq_5, q_4 BBq_6, q_5 BRq_4, q_6 00q_{10}, q_6 11q_{10}, q_6 BLq_6\}$$

This is the machine on 01101.

Step	Configuration	Step	Configuration	Step	Configuration
0	$\{01101\}$ q_0	6	$\{B1BB1\}$ q_5	11	$\{B1BBBB\}$ q_6
1	$\{B1101\}$ q_1	7	$\{B1BB1\}$ q_4	12	$\{B1BBBB\}$ q_6
2	$\{B1101\}$ q_2	8	$\{B1BBBB\}$ q_5	13	$\{B1BBBB\}$ q_6
3	$\{B1101\}$ q_4	9	$\{B1BBBBB\}$ q_4	14	$\{B1BBBBB\}$ q_6
4	$\{B1B01\}$ q_5	10	$\{B1BBBBB\}$ q_6	15	$\{B1BBBBB\}$ q_{10}
5	$\{B1B01\}$ q_4				

I.1.34 The main thing with this machine is that there could be zero-many 0's. That is, the machine should accept a string that starts with 1. On the other hand, if the input string is 000, without a 1, then it is not in the language.

With that, the machine reads past initial 0's, if any, blanking them out. If it find a 1 then it leaves it alone, marking that the input is in the language. Then it blanks the remaining characters, and returns to the single character marking whether the input is a member. Otherwise it puts 0, marking that the input is not in the language.

$$\mathcal{P}_{\text{contains } 1} = \{q_0 0Bq_1, q_0 1Rq_2, q_0 B0q_{10}, q_1 BRq_0, q_2 0Bq_3, q_2 1Bq_3, q_2 BLq_4, q_3 BRq_2, q_4 00q_{10}, q_4 11q_{10}, q_4 BLq_4\}$$

This is the machine with input 00110.

Step	Configuration	Step	Configuration	Step	Configuration
0	$\{00110\}$ q_0	5	$\{BB110\}$ q_2	10	$\{BB1BBB\}$ q_4
1	$\{B0110\}$ q_1	6	$\{BB1B0\}$ q_3	11	$\{BB1BBB\}$ q_4
2	$\{B0110\}$ q_0	7	$\{BB1B0\}$ q_2	12	$\{BB1BBB\}$ q_4
3	$\{BB110\}$ q_1	8	$\{BB1BB\}$ q_3	13	$\{BB1BBB\}$ q_{10}
4	$\{BB110\}$ q_0	9	$\{BB1BBB\}$ q_2		

I.1.35 The idea of this machine is that it starts with two groups of 1's. The head blanks the first 1 in the first group, then moves over and blanks the last 1 in the second group. At that point the relation of less-than-or-equal exists between the original two groups of 1's if and only if it exists between what remains of the two groups. Iterating the machine ends with one of two cases: the first string is empty (in which case we must blank out what's left of the second string, if any, and print a 1), and the second string is empty but the first is not (in this case we blank out what's left of the first string). The first case is handled by instructions starting with q_{50} and the second with instructions starting with q_{75} .

$$\begin{aligned} \mathcal{P}_{\text{leq}} = \{ & q_0 \text{BR} q_{50}, q_0 \text{1B} q_1, q_1 \text{BR} q_2, q_1 \text{11} q_2, q_2 \text{BR} q_3, q_2 \text{1R} q_2, q_3 \text{BL} q_{75}, q_3 \text{1R} q_4, q_4 \text{BL} q_5, q_4 \text{1R} q_4, \\ & q_5 \text{BB} q_6, q_5 \text{1B} q_6, q_6 \text{BL} q_7, q_6 \text{11} q_7, q_7 \text{BL} q_8, q_7 \text{1L} q_7, q_8 \text{BR} q_0, q_8 \text{1L} q_8, \\ & q_{50} \text{BR} q_{51}, q_{50} \text{11} q_{51}, q_{51} \text{B1} q_{100}, q_{51} \text{1B} q_{52}, q_{52} \text{BR} q_{51}, q_{52} \text{11} q_{51}, \\ & q_{75} \text{BL} q_{76}, q_{75} \text{11} q_{76}, q_{76} \text{BL} q_{100}, q_{76} \text{1B} q_{77}, q_{77} \text{BL} q_{76}, q_{77} \text{1L} q_{76} \} \end{aligned}$$

For instance, on inputs 2 and 3 it does this.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9		14	

(This tape sequence is split in two to leave a place for a line break.)

Step	Configuration	Step	Configuration	Step	Configuration
15		21		27	
16		22		28	
17		23		29	
18		24		30	
19		25		31	
20		26			

I.1.36 The strategy is to blank out the first character, slide to the end, and then blank out the final character (if it matches the first), and then repeat. When the string is empty, the machine puts a 1 on the tape, showing success.

The machine below remembers that the first character was an a by moving to state q_{20} . It remembers a b by moving to state q_{30} . Starting in q_{40} it moves back to the start of the remaining input string. State q_{60} is for the cases where the machine detects that the input was not a palindrome, while state q_{80} is for when the

machine finds it is one. Finally, states $q_1, \dots, q_4$ handle the case that the input string (either the initial string or what is left of the initial string after some end-trimming) has one character.

$$\begin{aligned} \mathcal{P}_{\text{pal}} = \{ & q_0 \text{BB} q_{80}, q_0 \text{aB} q_1, q_0 \text{bB} q_3, q_1 \text{BR} q_2, q_1 \text{aR} q_2, q_1 \text{bR} q_2, q_2 \text{BL} q_{80}, q_2 \text{aa} q_{20}, q_2 \text{bb} q_{20}, \\ & q_3 \text{BR} q_4, q_3 \text{aR} q_4, q_3 \text{bR} q_4, q_4 \text{BL} q_{80}, q_4 \text{aa} q_{30}, q_4 \text{bb} q_{30}, \\ & q_{20} \text{BL} q_{21}, q_{20} \text{aR} q_{20}, q_{20} \text{bR} q_{20}, q_{21} \text{BB} q_{60}, q_{21} \text{aB} q_{40}, q_{21} \text{bb} q_{60}, \\ & q_{30} \text{BL} q_{31}, q_{30} \text{aR} q_{30}, q_{30} \text{bR} q_{30}, q_{31} \text{BB} q_{60}, q_{31} \text{aa} q_{60}, q_{31} \text{bB} q_{40}, \\ & q_{40} \text{BL} q_{41}, q_{40} \text{aL} q_{41}, q_{40} \text{bL} q_{41}, q_{41} \text{BR} q_0, q_{41} \text{aL} q_{41}, q_{41} \text{bL} q_{41}, \\ & q_{60} \text{BB} q_{100}, q_{60} \text{aB} q_{61}, q_{60} \text{bB} q_{61}, q_{61} \text{BL} q_{60}, q_{61} \text{aa} q_{60}, q_{61} \text{bb} q_{60}, \\ & q_{80} \text{B1} q_{100}, q_{80} \text{a1} q_{100}, q_{80} \text{b1} q_{100} \} \end{aligned}$$

As an example, this is the action of the machine on abba.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6		10	
3		7		11	

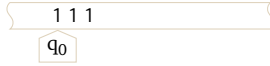
(The tape sequence is in two parts to leave room for a page break.)

Step	Configuration	Step	Configuration	Step	Configuration
12		16		20	
13		17		21	
14		18		22	
15		19			

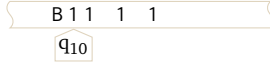
A second example is the action of the machine on aba.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9		14	

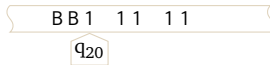
I.1.37 The strategy is to make two copies of the starting block of 1's, then move the head to the start of the first of the two copies. This is an example initial configuration.



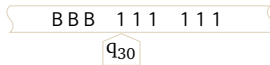
The machine strips the leading 1, moves right, puts in the 1 that will start the first copy block, and then puts in the 1 to start the second block. With that, the head moves back to the start.



Now comes the complication. The machine strips the leading 1, then moves right and puts in a 1 at the end of the first copy block. But the place where it must put this 1 is the blank separating the two copy blocks. So it must then move right and put in a blank before moving to the end of the second copy block. At that end it puts two 1's, the first to mark the stripped leading 1 and the second to mark the inserted blank. With that, the head moves back to the start.



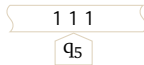
The machine iterates the prior sequence of steps until it finds that the leading sequence of 1's is now blank. To finish, it moves the head to the start of the first copy block.



I.1.38 Let the machine have the same configuration at two different steps $\mathcal{C}(t) = \mathcal{C}(\hat{t})$ with $t < \hat{t}$. By determinism, $\mathcal{C}(t+1) = \mathcal{C}(\hat{t}+1)$. Likewise, $\mathcal{C}(t+i) = \mathcal{C}(\hat{t}+i)$ for $i = 2, \dots$. This means that the machine cycles every $\hat{t} - t$ steps, because $i = \hat{t} - t$ gives that $\mathcal{C}(t + (\hat{t} - t)) = \mathcal{C}(\hat{t} + (\hat{t} - t))$ but $\mathcal{C}(t + (\hat{t} - t)) = \mathcal{C}(\hat{t})$.

This sufficient condition is not necessary, since we can make a machine that never repeats its configuration but also never halts. One way is to make a machine that writes a 1 to the tape, then moves right, and then repeats.

I.1.39 Consider this configuration, with the machine in state q_5 .



It could have arrived at that configuration after this configuration and instruction.



or this



or for that matter, this.



I.2.2 Church's Thesis is an assertion that 'mechanically computable' (by a discrete and deterministic machine) is equivalent to 'computable by a Turing machine'. It is not a theorem because we cannot derive it from the traditional axioms for mathematics.

In some ways it is more like a definition than a theorem. But it is not a usual definition, where as a convenience we take a word so that we don't have to keep saying a phrase over and over, as where we define 'bachelor' as 'unmarried man' or 'linearly dependent' as ' $x_0 = a_1x_1 + \dots + a_nx_n$ '. Rather, it is more like Calculus's definition of the derivative in that it expresses the introduction of a fundamental concept.

I.2.3

- (A) As with a program, a Turing Machine can implement an algorithm. But an algorithm may be implemented by more than one Turing Machine. For instance, we can do a shell sort of a list of five numbers with two different Turing Machines, if only by renaming some of the states. In the words of M Davis (Davis 2016), “[Church’s Thesis] says nothing about what algorithms ARE. It is about what algorithms can and can not DO.” *Comment:* there is no widely agreed-upon definition of algorithm, which is odd as algorithms are a central study in Computer Science.
 - (B) A Turing Machine is a single-purpose device. For instance, we may write a Turing Machine to multiply two of its inputs. But today we use “computer” to mean a general-purpose, that is, programmable, device. *Comment:* we will later see that there are general-purpose Turing Machines, so the distinction is blurry, one of connotation rather than of precise definition.
- I.2.4** Anything that can be done on the modern computers that we have in our pockets can be done, in principle, on an old-timey-style mechanical device. For more detail, see ?? B; the low level construction details covered there can be done with gears and levers. So the phrase ‘mechanically computable’ is not about the exact implementation, it covers any way of computing that is discrete and deterministic, and physically realizable.
- I.2.5**
- (A) The tape, as given in the definition, is not infinite. It is unbounded.
 For an analogy, consider the dictionary definition of a book. In that definition there is no upper limit on the number of pages. But no book, no actual bound collection of pages, has infinitely many. So also, no halting Turing machine computation uses infinitely much tape.
 While it is true that you can write a Turing machine that does not halt and that just keeps using more and more tape, at no step of the calculation is the amount of tape infinite. It never runs out — it is unbounded — but it is never infinite.
 - (B) It is a model. A model works by discarding some aspects of the situation that it models. See (Wikipedia contributors 2016). (Besides, whether the universe is finite — either just currently finite, or finite into any unbounded future — is not clear. In any event it is not a mathematical question.)
- I.2.6** Church’s Thesis is the third.
- The first is mistaken because Church’s Thesis speaks only to discrete and deterministic mechanisms. The second is mistaken because Church’s Thesis does not speak to the capabilities, or limitations, of human computers. There are people who posit that humans can do more than mechanisms can do, and people who speculate that there may be machines that that can do more than people can, even in principle, but Church’s Thesis speaks to an equivalence between what can be done by *any* discrete and deterministic mechanism and what can be done by this one kind, Turing machines. The last misses the mark for much the same reason as the second.
- I.2.7** (From (Andreas Blass 2015).) The first benefit that we get from this thesis is that it lets us connect formal mathematical theorems to real-world issues of computability. For example, the theorem that the word problem for groups is Turing-undecidable has the real-world interpretation that no algorithm can solve all instance of the word problem.
- The second benefit is in mathematics itself, specifically in computability theory. Published proofs that something is Turing-computable almost never proceed by exhibiting a Turing-machine program, or indeed a program in any actual computing language. Sometimes, if the matter is simple enough, they provide some sort of pseudo-code. Most often, though, they merely give an informal description of an algorithm. It is left to the reader to see that this actually does give an algorithm (in the intuitive sense) and therefore, by the Church-Turing thesis, could be simulated by a Turing machine. The usual situation is that, although experts in Turing-machine programming would (if they existed) be able to routinely convert the intuitive algorithm into a Turing machine program, the program would be too large and complicated to be worth writing down.
- I.2.8** Lance Fortnow says it perfectly, “Computation is about process, about the transitions made from one state of the machine to another. Computation is not about the input and the output, point A and point B, but the journey. . . . You can feed a Turing machine an infinite digits of a real number (Siegelmann), have computers interact with each other (Wegner-Goldin), or have a computer that perform an infinite series of tasks (Denning) but in all these cases the process remains the same, each step following Turing’s model.” (Fortnow 2010).

I.2.9

- (A) Yes, this is clear. To compute $f \circ g(x)$ we compute $y = g(x)$ and then compute $f(y)$.
 (B) The composition $f \circ g$ may be computable, or may be not computable. On the one hand, the constant function $g(x) = 0$ is computable, and the composition $f \circ g(x)$ equals $f(0)$. Because $f(0)$ is fixed, not variable, this result is computable.

On the other hand, if $g(x) = x$ then the composition is $f \circ g(x) = f(x)$ is not computable.

I.2.10 We can simulate a ternary machine using an ordinary programming language, that is, on a binary machine. We can simulate a binary machine on a Turing Machine. Thus we can simulate a ternary machine on a Turing machine. So, anything that can be computed on a ternary machine can be computed on a Turing machine.

The converse is even easier: to simulate a binary machine on a ternary machine just pick a trit and avoid using it.

I.2.11 The role of Church's Thesis here is that it allows us to avoid exhibiting Turing machines as sets of four-tuples to do these computations. Instead we can describe how to do them inside a usual programming environment.

- (A) Let f_0 be computed by the Turing machine $\mathcal{P}_0$ and let f_1 be computed by TM_1 . For $x \in \mathbb{N}$ this will compute $h(x)$: run $\mathcal{P}_0$ on input x for one step, then $\mathcal{P}_1$ on x for a step, then $\mathcal{P}_0$ for another step, etc., until each halts. When that happens, if it ever does, output 1.
 (B) As with the prior item, for $x \in \mathbb{N}$ compute the function on input x by: run $\mathcal{P}_0$ on x for one step, then $\mathcal{P}_1$ on x for a step, then $\mathcal{P}_0$ for another step, etc. The difference here is that if either halts then output 1.
 (C) Suppose f is computed by $\mathcal{P}$. Fix $x \in \mathbb{N}$. Run $\mathcal{P}$ on input 0 for a step. Then run $\mathcal{P}$ on input 0 for an additional step and run $\mathcal{P}$ on 1 for a step. After that, run $\mathcal{P}$ on input 0 for another step, run $\mathcal{P}$ on 1 for another step, and start a run of $\mathcal{P}$ on input 2. In this way we cycle among the machine running on different inputs.

Whenever a computation of $\mathcal{P}$ on some input i halts, check whether what is left on its tape equals x . If so, output 1.

- (D) Let f_0 be computed by the Turing machine $\mathcal{P}_0$ and let f_1 be computed by TM_1 . For $x \in \mathbb{N}$, run $\mathcal{P}_0$ on input x until it halts. Suppose the output is y . Run $\mathcal{P}_1$ on y until it halts, and then the output is $h(x)$. Since each function f_0 and f_1 was given as total, that output will appear eventually, and so h is also total.
 (E) This is the same as the prior item except that the function h will be partial since there is some input for which $\mathcal{P}_0$ does not halt.

I.2.12 Here is a procedure that is intuitively mechanically computable, and so by Church's Thesis is computable. Let the input be n . For $i = 0$ compute $f(i)$ and see if it equals n . If so, output 0. If not we iterate: take $i = 1$ and compute $f(1)$. If so, output 0. Otherwise, repeat until the procedure outputs some value; if it never does then we are in the second case of h 's specification.

I.2.13 This procedure is intuitively mechanically computable and so by Church's Thesis is computable. Let the input be n . Run the Turing machine computing f on input n until it halts, if it ever does. If so then run the Turing machine computing $g(n)$ until it halts, if ever. After that, if after that ever comes, output 1. Otherwise we are in the second case of h 's specification.

I.2.14 If there is a root then this procedure will find it eventually. If there is no root then this procedure will just run forever, in an unbounded search.

I.2.15 Perhaps predictably, you end with nothing since there is no bill still in your possession.

I.2.17 The strategy is to move right on tape 0, writing the complementary bit onto tape 1 until the tape 0 head hits a blank. Then the machine moves left on tape 1. (Below, rather than write the pair of characters as $\langle x, y \rangle$ we just have xy .)

Δ	00	01	0B	10	11	1B	B0	B1	BB
q_0	00, q_1	01, q_1	01, q_1	10, q_1	11, q_1	10, q_1	B0, q_1	B1, q_1	LL, q_2
q_1	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	LL, q_2
q_2	0L, q_2	0L, q_2	0R, q_3	1L, q_2	1L, q_2	1R, q_3	BL, q_2	BL, q_2	BR, q_3

(Many of the input pairs cannot happen. For example, if the machine is in state q_0 then the input 00 cannot

happen.)

I.2.18 The strategy is to move right on the two tapes, writing the logical and onto tape 1 until on both tapes the heads hit blanks. Then the machine moves left on tape 1. (Below, instead of writing the character pair as $\langle x, y \rangle$, we just write xy .)

Δ	00	01	0B	10	11	1B	B0	B1	BB
q_0	00, q_1	00, q_1	0B, q_1	10, q_1	11, q_1	1B, q_1	BB, q_1	BB, q_1	LL, q_2
q_1	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	RR, q_0	LL, q_2
q_2	LL, q_2	LL, q_2	0R, q_3	LL, q_2	LL, q_2	1R, q_3	BL, q_2	BL, q_2	BR, q_3

(Many of these input pairs cannot happen. For example, if the machine is in state q_1 then the input B0 cannot happen.)

V I.3.10 ‘Primitive recursive’ is a collection of functions, while ‘primitive recursion’ is a way to combine functions.

V I.3.11 There the definition gives 1 for all y , including $y = 0$.

V I.3.12

(A) $f(0) = 0$, $f(1) = f(2 \cdot 1 - 2) = f(0) = 0$

(B) Expanding the definition gives $f(2) = f(2 \cdot 2 - 2) = f(2)$, leaving the value undefined.

V I.3.13

(A) Following the definition gives this; for instance, $F(1) = F(1 - 1) = F(0) = 42$.

n	0	1	2	3	4	5	6	7	8	9	10
$t(n)$	42	42	42	42	42	42	42	42	42	42	42

(B) This

$$F(y) = \begin{cases} g() & \text{-- if } y = 0 \\ h(F(z), z) & \text{-- if } y = S(z) \end{cases}$$

works where g is the no-input constant function $g() = 42$ and $h(a, b) = a$.

By the way, since F is constant we can construct it as a primitive recursive function without using the schema of primitive recursion. Use composition: $F(y) = S(S(\dots Z(y))) = S^{42} \circ Z(y)$.

V I.3.14 One way is to note that it is the composition $\text{plus_three} = S \circ (S \circ S)(x)$.

V I.3.15 Starting with this

$$\text{is_zero}(y) = \begin{cases} 1 & \text{-- if } y = 0 \\ 0 & \text{-- if } y = S(z) \end{cases}$$

we take g to be the zero input function composition $g() = S(Z())$ and take h to be the two-input zero function $h(a, b) = Z(a, b) = 0$.

V I.3.16

(A) These are the first eleven values; obviously this is the sequence of squares.

n	0	1	2	3	4	5	6	7	8	9	10
$t(n)$	0	1	4	9	16	25	36	49	64	81	100

(B) Because s is a function of one input, the bookkeeping says that we need to describe it in this way.

$$s(y) = \begin{cases} g() & \text{-- if } y = 0 \\ h(s(z), z) & \text{-- if } y = S(z) \end{cases}$$

So g is a function of no arguments and is therefore constant, $g() = Z() = 0$. For h we want $h(s(z), z) = s(z) + (2 \cdot (z + 1) - 1) = s(z) + (2z + 1)$. So we take $h(a, b) = \text{plus}(a, S(\text{product}(S(S(Z()))), b))$.

V I.3.17

(A) Here are the first eleven triangular numbers.

n	0	1	2	3	4	5	6	7	8	9	10
$t(n)$	0	1	3	6	10	15	21	28	36	45	55

(B) Because t is a function of one input, the bookkeeping is that we need to describe it as here.

$$t(y) = \begin{cases} 0 & \text{-- if } y = 0 \\ h(t(z), z) & \text{-- if } y = S(z) \end{cases}$$

So g is a function of no arguments and therefore is a constant function, $g() = Z() = 0$. As to h , observe that for $n > 0$ we have $t(n) = t(n-1) + n$, because there is one dot in row 1, two dots in row 2, etc. So we want $h(t(z), z) = t(z) + (z+1)$ and putting that into the format required gives $h(a, b) = \text{plus}(a, S(b))$.

V I.3.18

(A) These are the first six values, suggesting that $d(n) = n^3$.

n	0	1	2	3	4	5
$t(n)$	0	1	8	27	64	125

(B) The given d is a function of one input so the arity bookkeeping has g be a function of no inputs and h be a function of two.

$$d(y) = \begin{cases} g() & \text{-- if } y = 0 \\ h(d(z), z) & \text{-- if } y = S(z) \end{cases}$$

Take g to be the constant $g() = Z() = 0$. To get $h(d(z), z) = d(z) + 3(z+1)^2 + 3(z+1) + 1$ only requires some (perhaps a little tedious) translation into low-level functions. Write $d(z) + 3(z+1)^2 + 3(z+1) + 1$ as $s + t + u + v$. Then $h(a, b) = \text{plus}(s, \text{plus}(t, \text{plus}(u, v)))$ where $s(a, b) = a$, and $u(a, b) = \text{product}(S(S(S(Z()))))$, $\text{plus}(b, S(Z()))$, and $v(a, b) = S(Z())$, and $t(a, b)$ is this.

$$\text{product}(S(S(S(Z()))), \text{product}(\text{plus}(b, S(Z())), \text{plus}(b, S(Z()))))$$

V I.3.19

(A) Here are the ten values.

n	1	2	3	4	5	6	7	8	9	10
$h(n)$	1	3	7	15	31	63	127	255	511	1023

(B) This is a function of one input so the arity bookkeeping gives that we want this.

$$H(y) = \begin{cases} g() & \text{-- if } y = 0 \\ h(H(z), z) & \text{-- if } y = S(z) \end{cases}$$

Take g to be the constant function $g() = 1 = S(Z())$. To get $h(H(z), z) = 2 \cdot H(z) + 1$ take $h(a, b) = S(\text{product}(a, S(S(Z()))))$.

V I.3.20 The derivation for factorial is this.

$$\text{fact}(y) = \begin{cases} g() & \text{if } y = 0 \\ h(\text{fact}(z), S(z)) & \text{if } y = S(z) \end{cases}$$

Here g is a function of no inputs $g() = 1 = S(Z())$ and h is a function of two variables $h(a, b) = \text{product}(a, S(b))$.

V I.3.21

- (A) Assume both are greater than zero. We know what to do if $a > b > 0$. For the $a < b$ case we can switch the order of the two, $\gcd(a, b) = \gcd(b, a)$, because for both sides of the equation we just find all of the common divisors and take the maximum. And for $a = b$, the greatest common divisor is a .

$$\gcd(a, b) = \begin{cases} 0 & \text{-- if } a = b = 0 \\ a & \text{-- if } b = 0 \text{ and } a \neq 0 \\ b & \text{-- if } a = 0 \text{ and } b \neq 0 \\ a & \text{-- if } a = b \\ \gcd(b, a) & \text{-- if } a < b \\ \gcd(a - b, b) & \text{-- if } a > b \end{cases}$$

For fun we can write a program (it collapses some of the above cases).

```
(define (gcd-first a b)
  (cond
    [(zero? a) b]
    [(zero? b) a]
    [(> b a) (gcd-first b a)]
    [else (gcd-first (- a b) b)]))
```

- (B) These are easy to compute by hand but here they came from the program

```
> (gcd-first 28 12)
4
> (gcd-first 104 20)
4
> (gcd-first 300009 25)
1
```

- (C) Again, for fun we did this as Racket script.

```
(define (gcd-second a b)
  (if (zero? b)
      a
      (gcd-second b (remainder a b))))
```

Here is the result.

```
> (gcd-second 28 12)
4
> (gcd-second 104 20)
4
> (gcd-second 300009 25)
1
```

V I.3.23 These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$.

- (A) The constant function that always returns zero, $\mathcal{C}_0(\vec{x})$, is included in the definition of primitive recursion, Definition 3.8, as $\mathcal{Z}(\vec{x}) = 0$. The function that always returns 1 is then $\mathcal{C}_1(\vec{x}) = \mathcal{S}(\mathcal{Z}(\vec{x}))$, the function that returns 2 is $\mathcal{C}_2(\vec{x}) = \mathcal{S}(\mathcal{S}(\mathcal{Z}(\vec{x})))$, etc. Because successor, zero, and composition are all permitted in the construction of a primitive recursive function, each $\mathcal{C}_k$ is primitive recursive.

- (B) Use proper subtraction to define $\max(\{x, y\}) = y + (x \dot{\div} y)$. If $x \geq y$ then it reduces to $\max(\{x, y\}) = y + (x - y) = x$, while if $x < y$ then it is $\max(\{x, y\}) = y + 0 = y$. The parts of the right-hand side of $\max(\{x, y\}) = y + (x \dot{\div} y)$ are all primitive recursive, so the $\max$ function is also primitive recursive. Similarly $\min(\{x, y\}) = x + (y \dot{\div} \max(\{x, y\}))$.

- (C) Let $\text{absdiff}(x, y) = (x \dot{\div} y) + (y \dot{\div} x)$. If $x \geq y$ then only the second term is zero and $\text{absdiff}(x, y) = x - y$, as desired. The $y > x$ case is similar. Since this expression is made from primitive recursive parts put together using primitive recursive combinators, $\text{absdiff}(x, y)$ is therefore primitive recursive.

V I.3.24 These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators, and therefore the function is primitive recursive.

- (A) This primitive recursion defines sign .

$$\text{sgn}(y) = \begin{cases} 0 & \text{-- if } y = 0 \\ 1 & \text{-- if } y = \mathcal{S}(z) \end{cases}$$

Note that in the first clause we write 0 as an abbreviation for $Z(y)$, and we write 1 to abbreviate $\mathcal{S}(Z(y))$.
 (B) A primitive recursion will do.

$$\text{negsign}(y) = \begin{cases} 1 & \text{if } y = 0 \\ 0 & \text{if } y = \mathcal{S}(z) \end{cases}$$

But also note that $\text{negsign}(y) = 1 - \text{sgn}(y)$, that is, $\text{negsign}(y) = 1 \dot{-} \text{sgn}(y)$.

(c) The first is $\text{lessthan}(x, y) = \text{sgn}(y \dot{-} x)$ while the other is $\text{greaterthan}(x, y) = \text{sgn}(x \dot{-} y)$.

V I.3.25 These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators and therefore the function is primitive recursive.

(A) We have $\text{not}(x) = 1 \dot{-} x$. For the two-input ones, $\text{and}(x, y) = \text{sgn}(x) \cdot \text{sgn}(y) = \text{product}(\text{sgn}(x), \text{sgn}(y))$ and $\text{or}(x, y) = \text{sgn}(x + y)$ (an alternative is $\text{or}(x, y) = 1 \dot{-} ((1 \dot{-} x) \cdot (1 \dot{-} y))$).

(B) $\text{equal}(x, y) = \text{negsign}(\text{lessthan}(x, y) + \text{greaterthan}(x, y))$

V I.3.26 These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators, and therefore the function is primitive recursive.

(A) $\text{notequal}(x, y) = \text{sign}(\text{lessthan}(x, y) + \text{greaterthan}(x, y))$

(B) The first is $m(x) = 7 \cdot \text{equal}(x, 1) + 9 \cdot \text{equal}(x, 5) + 2 \cdot (\text{and}(\text{notequal}(x, 1)), \text{notequal}(x, 5))$. The second is similar, $n(x, y) = 7 \cdot \text{equal}(x, 1) \cdot \text{equal}(y, 2) + 9 \cdot \text{equal}(x, 5) \cdot \text{equal}(y, 5)$.

V I.3.27 These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators and therefore the function is primitive recursive.

(A) Since g is given as primitive recursive, the function $h(a, b) = a + g(b)$ is also primitive recursive. With that, this primitive recursion will do the bounded sum.

$$f(y) = \begin{cases} 0 & \text{if } y = 0 \\ h(f(z), z) & \text{if } y = \mathcal{S}(z) \end{cases}$$

An example is called for. We will expand $f(4)$.

$$\begin{aligned} f(4) &= h(f(3), 3) = f(3) + g(3) \\ &= h(f(2), 2) + g(3) = f(2) + g(2) + g(3) \\ &= h(f(1), 1) + g(2) + g(3) = f(1) + g(1) + g(2) + g(3) \\ &= h(f(0), 0) + g(1) + g(2) + g(3) = f(0) + g(0) + g(1) + g(2) + g(3) = g(0) + \cdots + g(4) \end{aligned}$$

(B) This is similar to bounded sum.

(c) Consider the bounded product function.

$$P_p(\vec{x}, m) = p(\vec{x}, 0) \cdot p(\vec{x}, 1) \cdots p(\vec{x}, m-1) = \prod_{0 \leq i < m} p(\vec{x}, i)$$

There are two things to note here. First, if $P_p(\vec{x}, m) = 1$ then $P_p(\vec{x}, i) = 1$ for all $i < m$. Second, if there is an m where $P_p(\vec{x}, m) = 0$ then $P_p(\vec{x}, n) = 0$ for all $m \leq n$. Thus the sequence $P_p(\vec{x}, 0), P_p(\vec{x}, 1), P_p(\vec{x}, 2), \dots$ consists of a number of 1's followed by all 0's. The initial 1's go until the first n for which the predicate $p(\vec{x}, n)$ gives 0.

Next consider the bounded sum function.

$$S_{P_p}(\vec{x}, m) = \sum_{0 \leq i < m} P_p(\vec{x}, i)$$

The prior paragraph gives that $S_{P_p}(\vec{x}, m)$ is the least number n with $n < m$ such that $p(\vec{x}, n) = 0$.

Now define $M(\vec{x}, m)$ by cases.

$$M(\vec{x}, m) = \begin{cases} m & \text{-- if } S_{P_p}(\vec{x}, m) = 0 \\ S_{P_p}(\vec{x}, m) & \text{-- otherwise} \end{cases}$$

- V **I.3.28** These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators and therefore the function is primitive recursive.

(A) We can define U with a primitive recursion.

$$U(\vec{x}, m) = \begin{cases} 1 & \text{-- if } m = 0 \\ p(\vec{x}, z) \cdot U(\vec{x}, z) & \text{-- if } m = S(z) \end{cases}$$

For instance, here is the computation of $U(\vec{x}, 3)$.

$$\begin{aligned} U(\vec{x}, 3) &= p(\vec{x}, 2) \cdot U(\vec{x}, 2) \\ &= p(\vec{x}, 2) \cdot p(\vec{x}, 1) \cdot U(\vec{x}, 1) \\ &= p(\vec{x}, 2) \cdot p(\vec{x}, 1) \cdot p(\vec{x}, 0) \cdot U(\vec{x}, 0) \\ &= p(\vec{x}, 2) \cdot p(\vec{x}, 1) \cdot p(\vec{x}, 0) \cdot 1 \end{aligned}$$

(B) First an example with $m = 3$. Consider this.

$$\hat{E}(\vec{x}, 3) = 1 \div [(1 \div p(\vec{x}, 2)) \cdot (1 \div p(\vec{x}, 1)) \cdot (1 \div p(\vec{x}, 0))]$$

If any $p(\vec{x}, i)$ is 1 then the entire expression in square brackets is 0, and so $\hat{E}$ is 1. If all of the $p(\vec{x}, i)$ are 0 then the expression in square brackets is 1 and so $\hat{E}$ is 0.

Consequently, take $E(\vec{x}, m) = 1 \div \hat{E}(\vec{x}, m)$ where the latter function is below.

$$\hat{E}(\vec{x}, m) = \begin{cases} 0 & \text{-- if } m = 0 \\ (1 \div p(\vec{x}, z)) \cdot \hat{E}(\vec{x}, z) & \text{-- if } m = S(z) \end{cases}$$

- (C) This follows straight from the definition on noting that one suitable bound on the quotient k is the dividend y , that is, $\text{divides}(x, y) = \exists d \leq y [y = x \cdot d]$.
 (D) As in the prior item, we need only find a suitable bound. Here, $\text{prime}(y) = 1$ if and only if both $1 < y$ and there is no $x < y$ with $x > 1$ and $\text{divides}(x, y)$. We have verified that all of those components are primitive recursive.

- V **I.3.29** These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators and therefore the function is primitive recursive.

(A) The table is straightforward.

a	0	1	2	3	4	5	6	7
$\text{rem}(a, 3)$	0	1	2	0	1	2	0	1

(B) The relationship is not true when $\text{rem}(a, 3) = 2$.

(C) This follows from the prior item: item (1) is 0, item (2) is also 0, and item (3) is $\text{rem}(z, 3) + 1$.

$$\text{rem}(a, 3) = \begin{cases} 0 & \text{-- if } a = 0 \\ 0 & \text{-- if } a = S(z) \text{ and } \text{rem}(z, 3) + 1 = 3 \\ \text{rem}(z, 3) + 1 & \text{-- if } a = S(z) \text{ and } \text{rem}(z, 3) + 1 \neq 3 \end{cases}$$

(D) Here is the explicit definition given in the form for primitive recursion.

$$\text{rem}(a, 3) = \begin{cases} 0 & \text{-- if } a = 0 \\ \text{product}(S(\text{rem}(z, 3)), \text{notequal}(S(\text{rem}(z, 3)), 3)) & \text{-- if } a = S(z) \end{cases}$$

(E) Change the 3's to b 's.

$$\text{rem}(a, b) = \begin{cases} 0 & \text{-- if } a = 0 \\ \text{product}(S(\text{rem}(z, b)), \text{notequal}(S(\text{rem}(z, b)), b)) & \text{-- if } a = S(z) \end{cases}$$

V **I.3.30** These answers use familiar notation. For instance, we write $x + y$ instead of $\text{plus}(x, y)$. We will also omit explicit mention that the expressions are made from primitive recursive parts put together using primitive recursive combinators and therefore the function is primitive recursive.

(A) The division is easy.

a	0	1	2	3	4	5	6	7	8	9	10
$\text{div}(a, 3)$	0	0	0	1	1	1	2	2	2	3	3

(B) One way to say it is that $\text{div}(a + 1, 3) = \text{div}(a, 3)$ except when $\text{rem}(a + 1, 3) = 0$.

(C) Based on the prior item, this is the explicit definition given in the form for primitive recursion (again, the order of the two inputs is not important).

$$\text{div}(a, 3) = \begin{cases} 0 & \text{-- if } a = 0 \\ \text{plus}(\text{div}(z, 3), \text{equal}(\text{rem}(S(z), 3), 0)) & \text{-- if } a = S(z) \end{cases}$$

(D) Replace the 3's with b 's.

$$\text{div}(a, b) = \begin{cases} 0 & \text{-- if } a = 0 \\ \text{plus}(\text{div}(z, b), \text{equal}(\text{rem}(S(z), b), 0)) & \text{-- if } a = S(z) \end{cases}$$

V **I.3.31** This starts with bounded minimization.

$$f(x, y) = \lfloor x/y \rfloor = \min_{t \leq x} [(t + 1) \cdot y > x]$$

For instance, $f(9, 2) = \lfloor 9/2 \rfloor = \min_{t \leq 9} [(t + 1) \cdot 2 > 9]$ gives $t + 1 = 5$, so $t = 4$. Another example is that $f(8, 2) = \lfloor 8/2 \rfloor = \min_{t \leq 8} [(t + 1) \cdot 2 > 9]$ also gives $t + 1 =$ and so $t = 4$. Putting it in terms of previously defined functions, for instance changing the multiplication dot to a call to the product function, is routine.

V **I.3.32**

(A) $G(\langle 3, 1 \rangle) = 2^4 \cdot 3^2 = 144$, $G(\langle 2, 2, 2 \rangle) = 2^3 \cdot 3^3 \cdot 5^3 = 27\,000$.

(B) The first two are $A_0 = \langle 3, 2, 1 \rangle$ and $A_1 = \langle 1, 0 \rangle$. For the third, the prime factorization is $343 = 7^3$. Since there is no factor of 2 (or 3 or 5), there is no such sequence A_2 .

(C) If it did not use successor then $A_0 \langle 1, 0 \rangle$ and $A_1 = \langle 1 \rangle$ would have the same encoding.

(D) The first is easy, $\bar{F}(0) = G(\langle \rangle) = 1$. Because $F(0) = 1$ we have $\bar{F}(1) = G(\langle \rangle F(0)) = 2^2 = 4$. Then $F(1) = 1$ so $\bar{F}(2) = G(\langle \rangle F(0), F(1)) = 2^2 \cdot 3^2 = 36$. Finally, $F(2) = 2$ so $\bar{F}(3) = G(\langle \rangle F(0), F(1), F(2)) = 2^2 \cdot 3^2 \cdot 5^3 = 4500$.

V **I.3.33** This is an implementation.

```
(define (M x)
  (if (<= x 100)
      (M (+ x 11)))
      (- x 10)))
```

(A) The output values are that $M(x) = 91$ if $x \in \{0, \dots, 101\}$, and otherwise $M(x) = x - 10$.

(B) That this function is primitive recursive follows from ?? 3.26 and ?? 3.23.

V **I.3.34** Every initial function in Definition 3.8 is total. Every primitive recursive function is derived from the initial functions using a finite number of applications of function composition and primitive recursion. Both of these combinators yield total function outputs from total function inputs. (Technically, the last two sentences use induction.)

V **I.4.10** A total recursive function is any computable function that converges on all inputs, that is, it is computed by a machine that halts on all inputs. A primitive recursive function is one that can be derived with the schema in Definition 3.8.

Every primitive recursive function is a total recursive function. The converse does not hold — there are total recursive functions that are not primitive recursive, such as Ackermann's.

- V **I.4.11** The value $\mathcal{H}_4(2, 0) = 1$ is immediate from the definition. Next, $\mathcal{H}_4(2, 1) = \mathcal{H}_3(2, \mathcal{H}_4(2, 0))$, giving $\mathcal{H}_2(2, \mathcal{H}_3(2, 0)) = \mathcal{H}_1(2, \mathcal{H}_2(2, 0)) = 2$.

The last three are similar but tedious so a script is the best way to proceed. This is a copy of the straightforward transcription of the definition of $\mathcal{H}$ given in the section body.

```
(define (H n x y)
  (cond
    [(= n 0) (+ y 1)]
    [(and (= n 1) (= y 0)) x]
    [(and (= n 2) (= y 0)) 0]
    [(and (> n 2) (= y 0)) 1]
    [else (H (- n 1) x (H n x (- y 1)))]))
```

These calls give the answers.

```
> (H 4 2 0)
1
> (H 4 2 1)
2
> (H 4 2 2)
4
> (H 4 2 3)
16
> (H 4 2 4)
65536
```

The last one takes some time; below, the first number is milliseconds of CPU time.

```
cpu time: 155726 real time: 157987 gc time: 53317
65536
```

So it takes 156 seconds, about three minute (on a standard business laptop). Here is the output value summary.

y	0	1	2	3	4
$\mathcal{H}(4, 2, y)$	1	2	4	16	65 536

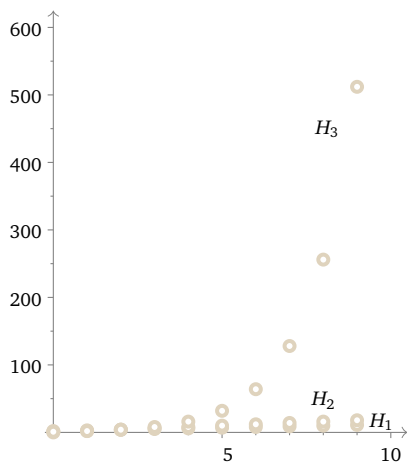
- V **I.4.12** A script yields that $\mathcal{H}_4(3, 3) = 7\,625\,597\,484\,987$. Dividing by $60 \cdot 60 \cdot 24 \cdot 365.25$ gives the number of years as about a quarter million, 241 640.
- V **I.4.13** We have that $\mathcal{H}_3(3, 3) = 27$ and $\mathcal{H}_2(2, 2) = 4$ so the ratio is 6.75.
- V **I.4.14** We can get the values with Lemma 4.2, or the Racket procedure given in the section body.

```
> (for ([y (in-range 10)])
  (printf "~a,~a,\n" y (H 1 2 y)))
(0,2),
(1,3),
(2,4),
(3,5),
(4,6),
(5,7),
(6,8),
(7,9),
(8,10),
(9,11),
```

This summarizes the output.

y	0	1	2	3	4	5	6	7	8	9
$\mathcal{H}_1(2, y)$	2	3	4	5	6	7	8	9	10	11
$\mathcal{H}_2(2, y)$	0	2	4	6	8	10	12	14	16	18
$\mathcal{H}_3(2, y)$	1	2	4	8	16	32	64	128	256	512

Here is a graph.



V I.4.15 Writing a small script makes the job of computing easier. Here is the table of values.

	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$y = 5$
$k = 0$	1	2	3	4	5	6
$k = 1$	2	3	4	5	6	7
$k = 2$	3	5	7	9	11	13
$k = 3$	5	13	29	61	125	253

V I.4.16 This Racket program illustrates the definition.

```
(define (g x y)
  (if (= 100 (+ x y))
      0
      1))

(define (f x)
  (define (f-helper y) ; value of x inherited from enclosing defn of f
    (if (= 0 (g x y))
        y
        (f-helper (+ 1 y))))
  (f-helper 0))
```

Racket will compute the values.

```
> (f 0)
100
> (f 1)
99
> (f 50)
50
> (f 100)
0
```

The computation of $(f\ 101)$ has to be terminated as it runs forever.

- (A) $f(0) = 100$
- (B) $f(1) = 99$
- (C) $f(50) = 50$
- (D) $f(100) = 0$
- (E) This is not defined.

This is another description of f .

$$f(x) = \begin{cases} 100 - x & \text{if } x \leq 100 \\ \uparrow & \text{otherwise} \end{cases}$$

V I.4.17

This Racket program illustrates the definition.

```
(define (g-mult x y)
  (if (= 100 (* x y))
      0
      1))

(define (f-mult x)
  (define (f-helper y) ; value of x inherited from enclosing defn of f-mult
    (if (= 0 (g-mult x y))
        y
        (f-helper (+ 1 y))))
  (f-helper 0))
```

Racket will compute some of the values.

```
> (f 1)
100
> (f 50)
2
> (f 100)
1
```

- (A) The value of $f(0)$ is undefined because there is no y where $0 \cdot y$ equals 100.
- (B) $f(1) = 100$
- (C) $f(50) = 2$
- (D) $f(100) = 1$
- (E) This is not defined because there is no y where $101 \cdot y$ equals 100.

V I.4.18 As stated, the largest known prime Fermat number is $F_4 = 65\,537$.

- (A) $f(0) = 5$
- (B) $f(1) = 17$
- (C) We don't know if this is defined as we don't know of any Fermat primes larger than F_{50} .
- (D) We don't know if this is defined because we don't know of any Fermat primes larger than F_{F_4} .

V I.4.19 For each input x to the function f , search for the first y where $y^3 \geq x^2$.

- (A) $f(0) = 0$
- (B) $f(1) = 1$
- (C) $f(50) = 14$ since $50^2 = 2500$ while $14^3 = 2744$.
- (D) $f(100) = 22$
- (E) $f(x) = \lceil x^{2/3} \rceil$ where the 'ceiling' notation denotes taking the first integer greater than or equal to $x^{2/3}$

V I.4.20 Recall that the number $d \in \mathbb{N}$ divides the number $n \in \mathbb{N}$ if there is a $k \in \mathbb{N}$ with $k \geq 1$ such that $d \cdot k = n$.

- (A) We have $\text{notrelprime}(0, 0) = 0$ as the number $d = 2$ divides them both. It divides $n = 0$ because there is a $k \in \mathbb{N}$ such that $2 \cdot k = 0$, namely $k = 0$.
- (B) The only number that divides 1 is 1 itself, so $f(1) \uparrow$.
- (C) $f(2) = 2$
- (D) $f(3) = 3$
- (E) $f(4) = 2$
- (F) $f(42) = 2$
- (G) If $x = 0$ then $f(x) = 0$. If $x = 1$ then $f(x)$ is undefined. Otherwise $f(x) = p$ where p is the smallest prime that divides x

V I.4.21 This Racket script helps with the calculations.

```
(define (g x y)
  (ceiling (- (/ (add1 x) (add1 y))
             1)))

(define (f x)
  (define (f-helper y)
    (if (= 0 (g x y))
        y
        (f-helper (add1 y))))
  (f-helper 0))
```

(A) Here is the output.

```
> (f 0)
0
> (f 1)
1
> (f 2)
2
> (f 3)
3
> (f 4)
4
> (f 5)
5
> (f 6)
6
```

The explanation of those values comes from this table of the relevant values of $g(x, y)$.

$g(x, y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$y = 5$	$f(x)$
$x = 0$	0						0
$x = 1$	1	0					1
$x = 2$	2	1	0				2
$x = 3$	3	1	1	0			3
$x = 4$	4	2	1	1	0		4
$x = 5$	5	2	1	1	1	0	5

(B) We will argue that $f(x) = x$. Fix x . To have $g(x, y) = 0$ we need that $(x + 1)/(y + 1)$ is less than or equal to 1. The first y giving that is x .

V I.4.22 No. All the functions in that collection would be total, they would all halt on all inputs. But some computable functions are not total—some effective procedures do not halt on some inputs.

V I.4.23 The proof of Lemma 4.2 shows that $\mathcal{H}_1(x, y) = x + y$. The definition of $\mathcal{H}$ gives this.

$$\mathcal{H}_2 = \begin{cases} 0 & \text{-- if } y = 0 \\ \mathcal{H}_1(x, \mathcal{H}_2(x, y - 1)) & \text{-- otherwise} \end{cases}$$

We will use induction to show that $\mathcal{H}_2(x, y) = x \cdot y$. The base step is $y = 0$, and in this case $x \cdot y = 0$, which matches the first clause. Now the inductive step. Suppose that $\mathcal{H}_2(x, y) = x \cdot y$ for $y = 0, y = 1, \dots, y = k$. For $y = k + 1$, the ‘otherwise’ clause applies (as y can’t be zero) so $\mathcal{H}_2(x, y) = \mathcal{H}_1(x, \mathcal{H}_2(x, y - 1)) = x + x \cdot k = x \cdot (1 + k) = x \cdot y$.

The other equality is similar. The definition of $\mathcal{H}$ gives this.

$$\mathcal{H}_3 = \begin{cases} 1 & \text{-- if } y = 0 \\ \mathcal{H}_2(x, \mathcal{H}_3(x, y - 1)) & \text{-- otherwise} \end{cases}$$

We will use induction to show that $\mathcal{H}_3(x, y) = x^y$. For the $y = 0$ base step we have $x^0 = 1$, which matches the first clause. For the inductive step assume the inductive hypothesis that $\mathcal{H}_3(x, y) = x^y$ for $y = 0, y = 1, \dots, y = k$. Consider $y = k + 1$. As y can’t be zero the ‘otherwise’ clause applies, giving $\mathcal{H}_3(x, y) = \mathcal{H}_2(x, \mathcal{H}_3(x, y - 1)) = x \cdot x^k = x^{1+k} = x^y$.

V I.4.24 In each application of the recursion, either the first argument decreases or else the first argument remains the same and the third argument decreases. Further, each time that the third argument reaches zero, the first argument decreases, so it eventually reaches zero as well. If the first argument is zero then the computation terminates.

V I.4.25 The function `remtwo` just toggles.

y	0	1	2	3	4	5	...
<code>remtwo(y)</code>	0	1	0	1	0	1	...

(A) The associated primitive recursion is also simple

$$\text{remtwo}(y) = \begin{cases} 0 & \text{-- if } y = 0 \\ 1 \div \text{remtwo}(z) & \text{-- if } y = S(z) \end{cases}$$

(As we often do, for clarity here we use abbreviations. For instance, instead of the function $Z()$ we write the number 0 and we similarly abbreviate $\mathcal{S}(Z())$ with 1.)

(B) Let $f(n) = \mu y [y + \text{remtwo}(n) = 0]$. If n is even then $\text{remtwo}(n) = 0$ and $y = 0$ suffices. If n is odd then the $\text{remtwo}(n) = 1$ and there is no natural number y such that $y + 1 = 0$.

V I.4.26 We can compute $f(x)$ by running the machine until it halts. This is the computation with $x = 0$.

Step	Configuration
0	
1	
2	
3	

Here is the computation with input $x = 1$.

Step	Configuration	Step	Configuration
0		3	
1		4	
2		5	

The same for $x = 2$.

Step	Configuration	Step	Configuration	Step	Configuration
0		3		6	
1		4		7	
2		5			

This is 3.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		7	
1		5		8	
2		6		9	
3					

This is 4.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6		10	
3		7		11	

Finally, this is the computation for $f(5)$.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9			

This table summarizes.

x	0	1	2	3	4	5
$f(x)$	3	5	7	9	11	13

V I.4.27 To compute $f(x)$ we can run the machine until it halts. This is the computation with $x = 0$,

Step	Configuration
0	
1	

and this is the computation with input $x = 1$.

Step	Configuration
0	
1	
2	

This is the same for $x = 2$.

Step	Configuration
0	1 1 q₀
1	1 1 q₁
2	1 1 1 q₂

The $x = 3$ computation does this.

Step	Configuration
0	1 1 1 q₀
1	1 1 1 q₁
2	1 1 1 1 q₂

This is $x = 4$.

Step	Configuration
0	1 1 1 1 q₀
1	1 1 1 1 q₁
2	1 1 1 1 1 q₂

Finally, this is the computation for $f(5)$.

Step	Configuration
0	1 1 1 1 1 q₀
1	1 1 1 1 1 q₁
2	1 1 1 1 1 1 q₂

The table below summarizes.

x	0	1	2	3	4	5
$f(x)$	2	3	3	3	3	3

For this machine the running time is constant. The machine in the prior exercise does the same job but its running time is a linear function of the input size.

- V I.4.28** One way to compute $f(x)$ for various values is to run the machine until it halts.
 (A) This is the computation with $x = 0$.

Step	Configuration	Step	Configuration
0		3	
1		4	
2			

(B) This is $x = 1$.

Step	Configuration	Step	Configuration
0		3	
1		4	
2			

(C) This is the same for $x = 2$.

Step	Configuration	Step	Configuration
0		3	
1		4	
2			

(D) The table summarizes.

x	0	1	2
$f(x)$	4	4	4

This machine ignores its input, moves right two squares and back left two, and then halts. So its running time is constant, $f(x) = 4$.

V I.4.29 These are not too hard to compute by hand but a small program is also handy.

```
;; 3n+1 function
(define (H n)
  (if (even? n)
      (/ n 2)
      (+ (* 3 n) 1)))

(define (C n)
  (define (C-helper n k)
    (if (= 1 n)
        k
        (C-helper (H n) (add1 k))))
  (C-helper n 0))
```

(A) The script gives this.

```
> (H 4)
2
> (H (H 4))
1
> (H (H (H 4)))
4
```

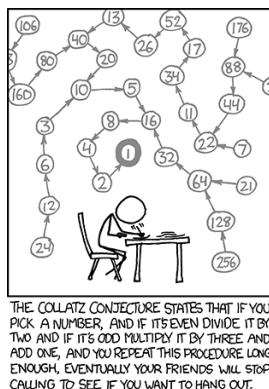


FIGURE 1, FOR QUESTION I.4.29: Courtesy xkcd.com

So we have this.

$$H^k(4) \begin{array}{c|cccc} k & 0 & 1 & 2 & 3 \\ \hline & 4 & 2 & 1 & 4 \end{array}$$

(B) The script says this.

```
> (C 4)
2
```

This matches the result of the prior item, that after the second application of H the result is 1.

(C) Here are the successive values.

```
> (H 5)
16
> (H (H 5))
8
> (H (H (H 5)))
4
> (H (H (H (H 5))))
2
> (H (H (H (H (H 5)))))
1
```

Thus $C(5) = 5$, which agrees with the program's result.

```
> (C 5)
5
```

(D) The successive values are 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, and 1, so $C(11) = 14$.

(E) Here are the values.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$C(n)$	0	1	7	2	5	8	16	3	19	6	14	9	9	17	15	4	12	20	20

Figure 1 on page 32 shows that XKCD has something to say about the Collatz conjecture.

V I.4.30 If D were primitive recursive then it would be some f_i . But then $D(i) = f_i(i) + 1 = D(i) + 1$, which is impossible.

I.A.1 This command line run starts with five 1's.

```
$ ./turing-machine.rkt -f machines/pred.tm -c "1" -r "11111"
step 0: q0: *1*1111
step 1: q0: 1*1*111
step 2: q0: 11*1*11
step 3: q0: 111*1*1
step 4: q0: 1111*1*
step 5: q0: 11111*B*
step 6: q1: 1111*1*B
step 7: q1: 1111*B*B
step 8: q2: 111*1*BB
```

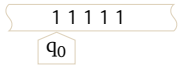
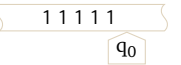
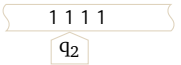
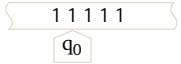
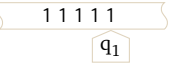
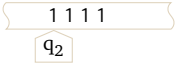
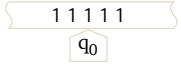
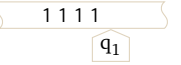
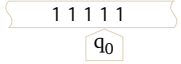
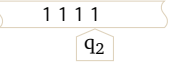
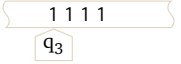
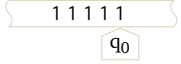
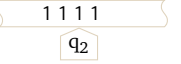


```

step 9: q2: 11*1*1BB
step 10: q2: 1*1*11BB
step 11: q2: *1*111BB
step 12: q2: *B*1111BB
step 13: q3: B*1*111BB
step 14: HALT

```

Here it is in the tape graphic format.

Step	Configuration	Step	Configuration	Step	Configuration
0		5		10	
1		6		11	
2		7		12	
3		8		13	
4		9			

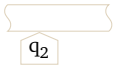
This is the command line invocation for an empty tape,

```

$ ./turing-machine.rkt -f machines/pred.tm -c " "
step 0: q0: *B*
step 1: q1: *B*B
step 2: q2: *B*BB
step 3: q3: B*B*B
step 4: HALT

```

and the matching sequence of tape diagrams.

Step	Configuration
0	
1	
2	
3	

I.A.2 This simulates $1 + 2$.

```

$ ./turing-machine.rkt -f machines/pred.tm -c "1" -r " 11"
step 0: q0: *1* 11
step 1: q0: 1*B*11
step 2: q1: 1*B*11
step 3: q1: 1*1*11
step 4: q2: 1*1*11
step 5: q2: *1*111
step 6: q2: *B*1111
step 7: q3: *B*1111
step 8: q3: B*1*111
step 9: q4: B*B*111
step 10: q5: BB*1*11
step 11: HALT

```

and this is the matching tape diagram.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6		10	
3		7			

The command line for $0 + 2$

```
$ ./turing-machine.rkt -f machines/pred.tm -c " " -r "11"
```

gives these tape diagrams.

Step	Configuration	Step	Configuration	Step	Configuration
0		3		6	
1		4		7	
2		5		8	

The command line for $0 + 0$

```
$ ./turing-machine.rkt -f machines/pred.tm -c " "
```

gives this sequence of tape diagrams.

Step	Configuration	Step	Configuration	Step	Configuration
0		3		6	
1		4		7	
2		5		8	

I.A.3 This ten-instruction machine

$$\mathcal{P}_{\text{addthree}} = \{q_0 1 R q_0, q_0 B B q_1, q_1 B 1 q_1, q_1 1 R q_2, q_2 B 1 q_2, q_2 1 R q_3, \\ q_3 B 1 q_3, q_3 1 1 q_4, q_4 1 L q_4, q_4 B R q_5\}$$

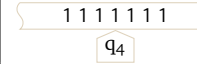
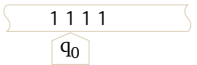
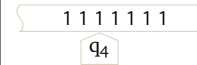
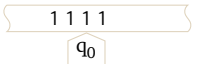
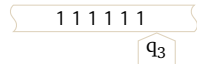
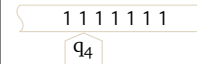
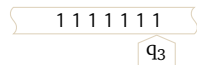
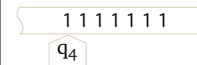
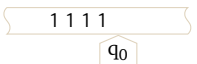
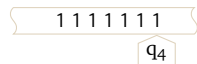
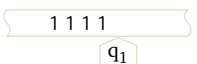
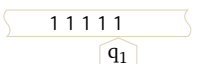
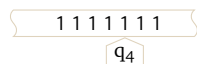
matches the source for the simulator.

```
0 1 R 0
0 B B 1
1 B 1 1
1 1 R 2
2 B 1 2
2 1 R 3
3 B 1 3
3 1 1 4
4 B R 5
4 1 L 4
```

This command line for input 4

```
$ ./turing-machine.rkt -f machines/addthree.tm -c "1" -r "111"
```

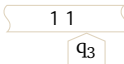
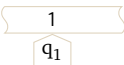
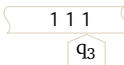
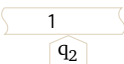
gives these tape diagrams.

Step	Configuration	Step	Configuration	Step	Configuration
0		7		14	
1		8		15	
2		9		16	
3		10		17	
4		11		18	
5		12		19	
6		13			

The command line for 0

```
$ ./turing-machine.rkt -f machines/addthree.tm -c " " "
```

gives this sequence of tape diagrams.

Step	Configuration	Step	Configuration	Step	Configuration
0		4		8	
1		5		9	
2		6		10	
3		7		11	

I.A.4

(A) Something like this will do.

$$\mathcal{P}_0 = \{q_{10}0Rq_{10}, q_{10}1Rq_{10}, q_{10}BBq_{100}\}$$

(B) This will back up, blanking out cells.

$$\mathcal{P}_1 = \{q_{20}0Bq_{21}, q_{20}1Bq_{21}, q_{20}BBq_{100}, q_{21}0Lq_{20}, q_{21}1Lq_{20}, q_{21}BLq_{20}\}$$

(C) Here, roughly q_0 is where none of the substring is matched, q_1 is where 0 has been seen and the machine is looking for the 1, and q_2 is where 01 has been seen and the machine is looking for the 0. Then the four states q_3, q_4, q_5 , and q_6 mark a failure (they end with q_6 writing 0 on a blank tape). The four states q_7, q_8 ,

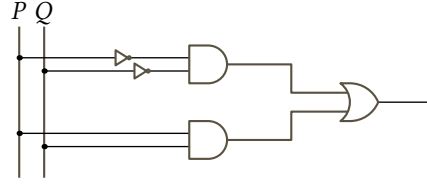
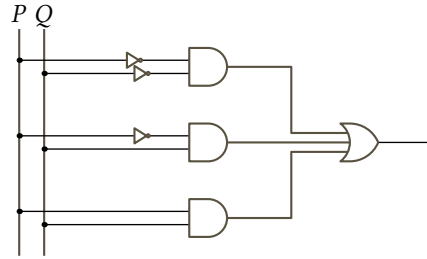
FIGURE 2, FOR QUESTION I.B.1: Circuit for $P \text{ XOR } Q$.

FIGURE 3, FOR QUESTION I.B.2: Circuit for the implication operation.

q_9 , and q_{10} mark a success.

$$\begin{aligned} \mathcal{P}_{\text{decide}_{010}} = \{ & q_0 0Rq_1, q_0 1Rq_0, q_0 BBq_3, q_1 0Rq_1, q_1 1Rq_2, q_1 BBq_3, q_2 0Rq_7, q_2 11q_0, q_2 BBq_3, \\ & q_3 0Rq_3, q_3 1Rq_3, q_3 BLq_4, q_4 0Bq_5, q_4 1Bq_5, q_4 BBq_6, \\ & q_5 0Lq_4, q_5 1Lq_4, q_5 BLq_4, q_6 00q_{100}, q_6 10q_{100}, q_6 B0q_{100}, \\ & q_7 0Rq_7, q_7 1Rq_7, q_7 BLq_8, q_8 0Bq_9, q_8 1Bq_9, q_8 BBq_{10}, \\ & q_9 0Lq_8, q_9 1Lq_8, q_9 BLq_8, q_{10} 01q_{100}, q_{10} 11q_{100}, q_{10} B1q_{100} \} \end{aligned}$$

V I.B.1

(A) Here is the table.

P	Q	$P \text{ XOR } Q$
0	0	0
0	1	1
1	0	1
1	1	0

The statement is $(\neg P \wedge \neg Q) \vee (P \wedge Q)$.

(B) Figure 2 on page 36 gives the circuit.

V I.B.2

(A) The definition gives this input/output behavior

P	Q	$P \rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

so a Boolean expression is $(\neg P \wedge \neg Q) \vee (\neg P \wedge Q) \vee (P \wedge Q)$.

(B) Figure 3 on page 36 gives the circuit.

V I.B.3 The statement is $(\neg P \wedge Q) \vee (P \wedge Q)$. Figure 4 on page 37 gives the circuit.

V I.B.4 The two circuits are in Figure 5 on page 37.

(A) In the expression, each clause matches a line of the table with a 1 output.

$$(\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge Q \wedge \neg R) \vee (P \wedge \neg Q \wedge R)$$

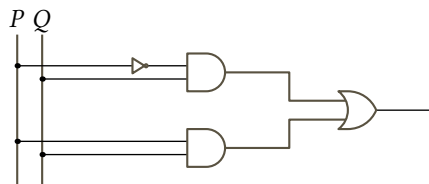
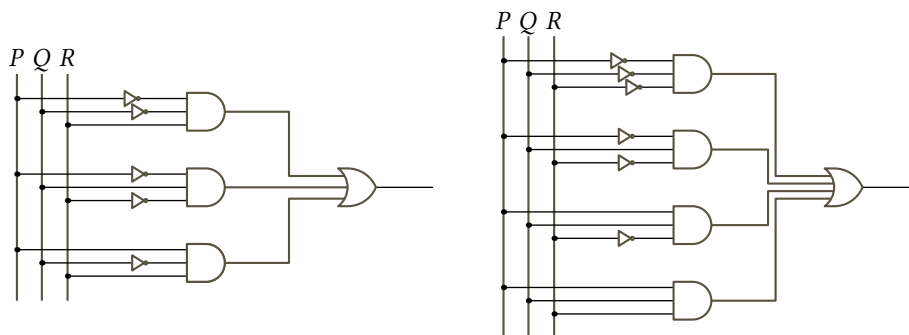
FIGURE 4, FOR QUESTION I.B.3: Circuit for the unnamed operation (we could call it Q).

FIGURE 5, FOR QUESTION I.B.4: Circuits for the two unnamed operation tables.

(B) The clauses match the 1 lines.

$$(\neg P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge R).$$

V I.B.5 Here is the input/output behavior.

P	Q	R	
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

That table immediately gives the Disjunctive Normal Form expression.

$$(\neg P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge \neg R) \vee (P \wedge \neg Q \wedge R) \vee (P \wedge Q \wedge R)$$

Figure 6 on page 38 gives the circuit.

V I.B.6 Here a line has a 1 in the output column if and only if two or three of the inputs are 1's.

P	Q	R	
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

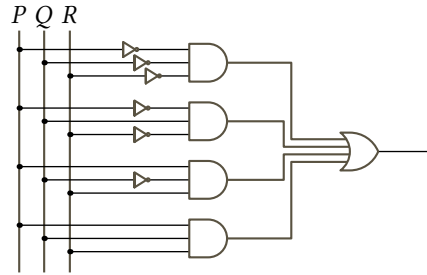
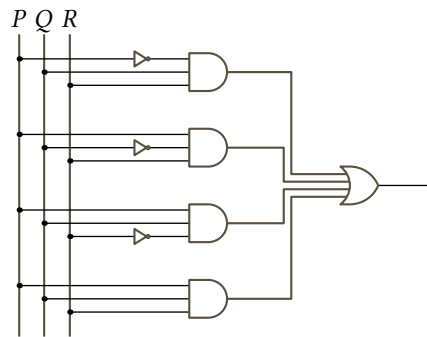
FIGURE 6, FOR QUESTION I.B.5: Circuit for $P = R$.

FIGURE 7, FOR QUESTION I.B.6: Circuit for the three-input majority operator.

This is a Disjunctive Normal Form expression.

$$(\neg P \wedge Q \wedge R) \vee (P \wedge \neg Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge R)$$

Figure 7 on page 38 gives the circuit.

V I.B.7

(A) Here a line has a 1 in the output column if and only if three or four of the inputs are 1's.

P	Q	R	S	
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

This DNF expression contains all the clauses where at least three of the variables are without negations.

$$(\neg P \wedge Q \wedge R \wedge S) \vee (P \wedge \neg Q \wedge R \wedge S) \vee (P \wedge Q \wedge \neg R \wedge S) \vee (P \wedge Q \wedge R \wedge \neg S) \vee (P \wedge Q \wedge R \wedge S)$$

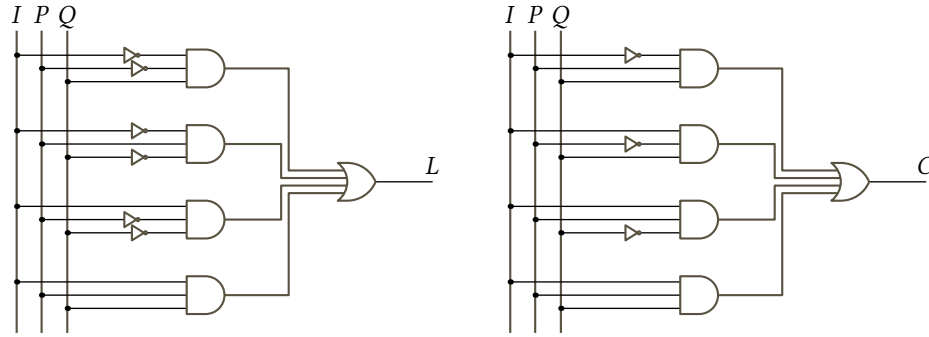


FIGURE 8, FOR QUESTION I.B.8: Partial sum and carry circuits for one addition column.

- (B) The DNF expression must consist of the clauses where at least three of the variables are not negated. There are $\binom{5}{3} = 10$ such clauses with exactly three non-negated variables, $\binom{5}{4} = 5$ clauses with one negated variable, and $\binom{5}{5} = 1$ with no negated variables.

$$\begin{aligned}
 &(\neg P \wedge \neg Q \wedge R \wedge S \wedge T) \vee (\neg P \wedge Q \wedge \neg R \wedge S \wedge T) \vee (\neg P \wedge Q \wedge R \wedge \neg S \wedge T) \vee (\neg P \wedge Q \wedge R \wedge S \wedge \neg T) \\
 &\vee (P \wedge \neg Q \wedge \neg R \wedge S \wedge T) \vee (P \wedge \neg Q \wedge R \wedge \neg S \wedge T) \vee (P \wedge \neg Q \wedge R \wedge S \wedge \neg T) \\
 &\vee (P \wedge Q \wedge \neg R \wedge \neg S \wedge T) \vee (P \wedge Q \wedge \neg R \wedge S \wedge \neg T) \vee (P \wedge Q \wedge R \wedge \neg S \wedge \neg T) \\
 &\vee (\neg P \wedge Q \wedge R \wedge S \wedge T) \vee (P \wedge \neg Q \wedge R \wedge S \wedge T) \vee (P \wedge Q \wedge \neg R \wedge S \wedge T) \\
 &\vee (P \wedge Q \wedge R \wedge \neg S \wedge T) \vee (P \wedge Q \wedge R \wedge S \wedge \neg T) \\
 &\vee (P \wedge Q \wedge R \wedge S \wedge T)
 \end{aligned}$$

V I.B.8

- (A) Here is the tabular version. The carries are in the second, third, and fifth columns from the right.

$$\begin{array}{r}
 \begin{array}{cccccc}
 & 1 & 1 & 10 & 11 & 1 \\
 + & 1 & 0 & 0 & 1 & \\
 \hline
 1 & 0 & 1 & 0 & 0 &
 \end{array}
 \end{array}$$

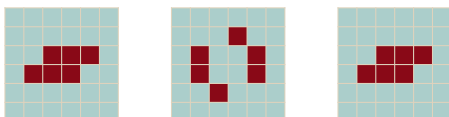
- (B) The definition gives these two input/output behaviors. What is called the Least significant bit here is often called a 'full adder'. (Also common is a 'half adder' which does not take a Carry in, so it only has two inputs.)

Carry in I	P	Q	Least significant bit L	Carry out C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

A Boolean expression for L is $(\neg I \wedge \neg P \wedge Q) \vee (\neg I \wedge P \wedge \neg Q) \vee (I \wedge \neg P \wedge \neg Q) \vee (I \wedge P \wedge Q)$. For C a DNF expression is $(\neg I \wedge P \wedge Q) \vee (I \wedge \neg P \wedge Q) \vee (I \wedge P \wedge \neg Q) \vee (I \wedge P \wedge Q)$.

- (c) Figure 8 on page 39 gives the circuits.

V I.C.4 The tub is a still life. The toad oscillates with a period of two.



- V I.C.5 No. For instance, the discussion of the simplest nontrivial pattern gives two unequal 3×3 gameboards where the next generation is entirely dead. So from a dead board you cannot tell what was the prior board.

Also, because there are pairs of boards that go next to the same board by counting there must also be some boards with no predecessor, with zero-many boards that come before. Such a board is a ‘Garden of Eden’. For these, were we to try running the clock backward, it is not a question of choosing which of several boards lead to the current one — there is no such board.

V I.C.6

(A) For 3×3 each cell is either alive or dead so there are 2^9 grids. For $n \times n$ it is 2^{n^2} .

(B) With $2^9 = 512$ many 3×3 gameboards, we use a program.

```
; Input a natural number less than 2^9, output a grid with associated 0's and 1's
; For instance, n=28 is 11100 in binary so this returns the vector of vectors [[0 0 0] [0 1 1] [1 0 0]]
(define (number->three-by-three-grid n)
  (if (>= n (expt 2 9))
      (error "input too large; must be less than 2^9")
      (let* ([bitstr (number->string n 2)]
             [pad-bitstr (make-string (- 9 (string-length bitstr)) #\0)]
             [length-9-bitstr (string-append pad-bitstr bitstr)]
             [input-style-bitstr (string-replace (string-replace length-9-bitstr "0" ".") "1" "*")]
             [list-of-lines (list (substring input-style-bitstr 0 3)
                                  (substring input-style-bitstr 3 6)
                                  (substring input-style-bitstr 6 9))])
        (parse-lines-to-grid list-of-lines))))

; Test if a grid is empty
(define (grid-empty? g)
  (let ([size (grid-size g)])
    (grid-equal? g (grid-create (first size) (second size)))))

; Count the number of 3x3 grids that are alive after the specified number of generations
(define (number-alive generations)
  (let ([counter 0])
    (for ([n (in-range (expt 2 9))])
      (displayln (~a "n=" n "\ngrid=" (grid->string (number->three-by-three-grid n))))
      (let* ([initial-u (universe (number->three-by-three-grid n) (list 0 0))]
             [list-of-universes (run initial-u generations)]
             [ending-grid (universe-grid (last list-of-universes))])
        (when (not (grid-empty? ending-grid))
          (set! counter (add1 counter))
          (displayln "not blank"))
        (displayln (~a "Number of 3x3 boards surviving is " counter)))))
```

Running the program gives this.

```
> (number-alive 1)
... lots of lines ...
n=510
grid=***
***
**.
not blank
n=511
grid=***
***
***
not blank
Number of 3x3 boards surviving is 461
```

(c) Again using the program gives this.

```
> (number-alive 10)
... lots of lines ...
n=510
grid=***
***
**.
not blank
n=511
grid=***
***
***
not blank
Number of 3x3 boards surviving is 249
```


V **I.D.8** Rewrite $2^{65\,536}$ as $(10^{\log_{10} 2})^{65\,536}$. That is about $(10^{0.3010299957})^{65\,536} \approx 10^{19728.30}$. So there are 19 729 digits.

V **I.D.9** This is about uniformity. Imagine that we have an enumeration of the primitive recursive functions f_0, f_1 , etc. Suppose that level zero is $\mathcal{A}_0 = f_9$, and level one is $\mathcal{A}_1 = f_{121}$, and $\mathcal{A}_2 = f_{800}$, etc. What we have is that the function associating the subscript on the $\mathcal{A}$ with the subscript on the f is not primitive recursive. That is, in this imagined scenario, $\mathcal{A}_i = f_{g(i)}$ but g is not primitive recursive.

It is a little like saying that all the layers of a cake are delicious, but the cake was made where the flavor of layer 1 was picked with no regard for the flavor of layer 0, perhaps coffee cake and hummingbird cake.

V **I.D.10**

(A) We can get this function as a composition $f(y) = S(S(y))$. The successor functions are level 1. Composition adds two to that so overall f is level 3.

(B) We have this.

$$\text{pred}(y) = \begin{cases} Z(x) = 0 & \text{if } x = 0 \\ \mathcal{I}_0(z) = z & \text{if } y = S(z) \end{cases}$$

The zero function is level zero, as is the projection function. Primitive recursion adds three so the predecessor function is level 3.

V **I.D.11** In the definition of $\mathcal{A}$, evaluation of the $k = 0$ clause clearly terminates. In each of the other two clauses, either k decreases or k does not decrease but y does. When y reaches zero then k decreases, so it eventually reaches zero as well, which we have already noted gives a terminating evaluation.

V **I.D.12**

(A) We will verify this statement by induction on k (the ‘in general’ follows immediately from it).

$$\mathcal{A}(k, y + 1) > \mathcal{A}(k, y) \tag{*}$$

In the $k = 0$ base step, $\mathcal{A}(0, y + 1) = y + 2$ while $\mathcal{A}(0, y) = y + 1$, so $(*)$ is clear.

For the inductive step assume that $(*)$ holds for $k = 0, 1, \dots, n$ and consider $k = n + 1$. Then the definition of $\mathcal{A}$ gives the equality in $\mathcal{A}(n + 1, y + 1) = \mathcal{A}(n, \mathcal{A}(n + 1, y)) > \mathcal{A}(n + 1, y)$, while the inequality follows from Lemma D.2.A.

(B) We will prove this statement by induction on y .

$$\mathcal{A}(k + 1, y) \geq \mathcal{A}(k, y + 1) \tag{*}$$

The second clause of $\mathcal{A}$ ’s definition is $\mathcal{A}(k + 1, 0) = \mathcal{A}(k, 1)$ so the $y = 0$ base step is immediate.

The inductive step starts with the inductive hypothesis that $(*)$ holds for $y = 0, 1, \dots, n$. Consider the $y = n + 1$ case, involving $\mathcal{A}(k + 1, n + 1)$. The third clause of $\mathcal{A}$ ’s definition is that $\mathcal{A}(k + 1, n + 1) = \mathcal{A}(k, \mathcal{A}(k + 1, n))$. Then $\mathcal{A}(k + 1, n) \geq \mathcal{A}(k, n + 1) \geq (n + 1) + 1$ holds first by the inductive hypothesis and second by Lemma D.2.A. With that, by the ‘in general’ clause of the prior exercise, Lemma D.2.B, $\mathcal{A}(k, \mathcal{A}(k + 1, n)) \geq \mathcal{A}(k, (n + 1) + 1)$, and therefore $\mathcal{A}(k + 1, n + 1) \geq \mathcal{A}(k, (n + 1) + 1)$, as desired.

(C) The statement here is that $\mathcal{A}(k, y) > k$. Repeatedly applying the prior item, Lemma D.2.c, shows this.

$$\mathcal{A}(k, y) = \mathcal{A}(k - 1, y + 1) = \mathcal{A}(k - 2, y + 2) = \dots = \mathcal{A}(0, y + (k + 1))$$

Then the first clause of the definition is that $\mathcal{A}(0, y + (k + 1)) = y + (k + 1)$, which is greater than k .

(D) For $\mathcal{A}(k + 1, y) > \mathcal{A}(k, y)$, start with Lemma D.2.c and then apply Lemma D.2.B.

$$\mathcal{A}(k + 1, y) \geq \mathcal{A}(k, y + 1) > \mathcal{A}(k, y)$$

The ‘in general’ follows on iterating.

(E) This is the statement.

$$\mathcal{A}(k + 2, y) > \mathcal{A}(k, 2y) \tag{*}$$

We will do induction on y . For the base step of $y = 0$ we use the ‘in general’ part of the prior item, Lemma D.2.E, to get $\mathcal{A}(k+2, y) = \mathcal{A}(k+2, 0) > \mathcal{A}(k, 0) = \mathcal{A}(k, 2y)$.

For the inductive step assume that (*) holds for $y = 0, \dots, n$ and consider the $y = n+1$ case. The third clause in the definition of $\mathcal{A}$ is that $\mathcal{A}(k+2, n+1) = \mathcal{A}(k+1, \mathcal{A}(k+2, n))$. The inductive hypothesis is that $\mathcal{A}(k+2, n) > \mathcal{A}(k, 2n)$ and so the ‘in general’ clause of Lemma D.2.B gives this.

$$\mathcal{A}(k+2, n+1) = \mathcal{A}(k+1, \mathcal{A}(k+2, n)) > \mathcal{A}(k+1, \mathcal{A}(k, 2n))$$

Apply Lemma D.2.c to that.

$$\mathcal{A}(k+2, n+1) > \mathcal{A}(k+1, \mathcal{A}(k, 2n)) \geq \mathcal{A}(k, \mathcal{A}(k, 2n) + 1)$$

Now, by Lemma D.2.A, $\mathcal{A}(k, 2n) \geq 2n+1$ and so $\mathcal{A}(k, 2n) + 1 \geq 2n+2 = 2(n+1)$. Thus $\mathcal{A}(k+2, n+1) > \mathcal{A}(k, 2(n+1))$, as required.

V I.E.2 Here is the code.

```
# swap-two.loop  Input x,y and output y,x
r2 = r0
r0 = r1
r1 = r2
```

And here is a sample invocation.

```
jim@millstone:src/scheme/prologue$ ./loop.rkt -f machines/swap-two.loop -p "1 2 3" -o 2
2 1
```

V I.E.3 We can get the effect of the first with two increments in a row, and we can get the effect of the second by first zeroing and then incrementing.

V I.E.4 The rotate left code is an adaptation of the rotate right code.

```
# rotate-shift-left.loop  input x,y,z, output y,z,x
r3 = r0
r0 = r1
r1 = r2
r2 = r3
```

Here is a sample invocation.

```
jim@millstone:src/scheme/prologue$ ./loop.rkt -f machines/rotate-shift-left.loop -p "1 2 3" -o 3
2 3 1
```

Putting the two together gives this.

```
# rotate-shift-right
r3 = r2
r2 = r1
r1 = r0
r0 = r3
# now rotate-shift-left
r3 = r0
r0 = r1
r1 = r2
r2 = r3
```

Here is a sample invocation.

```
jim@millstone:src/scheme/prologue$ ./loop.rkt -f machines/concatenate-rotations.loop -p "1 2 3" -o 3
1 2 3
```

V I.E.5 Here is a program.

```
# power.loop  Return r0 ^ r1
r2 = r1      # save the inputs in higher registers
r1 = r0      #
r0 = 0
r0 = r0 + 1  # cover the r1=0 case
r3 = 0      # Initialize
r3 = r3 + 1  # to 1
loop r2
  r0 = 0
loop r3
```

```
loop r1
  r0 = r0 + 1
end
end
r3 = r0
end
```

And here are two sample invocations.

```
jim@millstone:src/scheme/prologue$ ./loop.rkt -f machines/power.loop -p "3 2"
9
jim@millstone:src/scheme/prologue$ ./loop.rkt -f machines/power.loop -p "3 0"
1
```

I.E.6 The first does this; observe that it runs five times.

```
> (for ([i (in-range 5)])
      (set! i 1)
      (displayln (~a "i=" i)))
i=1
i=1
i=1
i=1
i=1
```

The second does this.

```
> (for ([i (in-range 5)])
      (displayln (~a "i=" i))
      (set! i 1))
i=0
i=1
i=2
i=3
i=4
```


Chapter II: Background

V II.1.20 To show that the map is one-to-one we will show that $g(n) = g(\hat{n})$ can only happen if $n = \hat{n}$. So let $g(n) = g(\hat{n})$, giving $3n = 3\hat{n}$. Divide by three to get $n = \hat{n}$.

To show that the map is onto we will show that every member of the codomain $y \in \{3k \mid k \in \mathbb{N}\}$ is the image of some member of the domain. Because y is a multiple of three, $y/3$ is a natural number. Since $y = g(y/3)$ we have that y is indeed the image of a member of the domain.

V II.1.21 This is a type error. It is not the sets that are one-to-one or onto; rather, it is a correspondence f between the sets. For $D = \{n^2 \mid n \in \mathbb{N}\}$ and $C = \{n^3 \mid n \in \mathbb{N}\}$ the natural such correspondence $f: D \rightarrow C$ is $f(n) = n^{3/2}$.

V II.1.22

- (A) The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = 2x$ is one-to-one but not onto. Verifying that it is one-to-one is routine: assume that $f(x) = f(\hat{x})$ for $x, \hat{x} \in \mathbb{Z}$, so that $2x = 2\hat{x}$, and then cancel the 2's to get that $x = \hat{x}$. To show that it is not onto we need only name one codomain element $y \in \mathbb{Z}$ that is not the image of any domain element. One such is $y = 1$, since the image of every domain element is even.
- (B) The function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $g(x) = 2x - 1$ is also one-to-one but not onto. The one-to-one argument works just as it did in the prior item: assume that $g(x) = g(\hat{x})$ for $x, \hat{x} \in \mathbb{Z}$ so that $2x - 1 = 2\hat{x} - 1$. Add 1 to both sides and cancel the 2's to get $x = \hat{x}$. This function is not onto because the codomain element $y = 2$ is not the image of any domain element, since it is even but the number $2x - 1$ is odd for any input integer x .
- (C) They are not inverse. For instance, $f \circ g(1) = f(g(1)) = f(1) = 2$.

V II.1.23

- (A) The function $f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(n) = n + 1$ is one-to-one but not onto. To check that it is one-to-one suppose that $f(n_0) = f(n_1)$ for $n_0, n_1 \in \mathbb{N}$, so that $n_0 + 1 = n_1 + 1$, and then subtract 1 from both sides to get $n_0 = n_1$. It is not onto because the codomain element $0 \in \mathbb{N}$ is not the image of any domain element.
- (B) The map $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) = n + 1$ is a correspondence—it is both one-to-one and onto. The verification that it is one-to-one goes: assume that $f(n_0) = f(n_1)$ so that $n_0 + 1 = n_1 + 1$, and subtract 1 to conclude that $n_0 = n_1$. The verification that it is onto goes: fix some codomain element $y \in \mathbb{Z}$ and note that $x = y - 1$ is a domain element which satisfies that $f(x) = y$.
- (C) The function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = 2n$ is one-to-one but not onto. For one-to-one, assume that $n_0, n_1 \in \mathbb{N}$ are such that $f(n_0) = f(n_1)$. Then $2n_0 = 2n_1$ and cancelling the 2's gives that $n_0 = n_1$. It is not onto because the codomain element $y = 1$ is not the image of any domain element, since all such images are even.
- (D) The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 2n$ is one-to-one but not onto. The one-to-one verification is as in the prior item: suppose that $f(n_0) = f(n_1)$ for some $n_0, n_1 \in \mathbb{Z}$, so that $2n_0 = 2n_1$, and cancel the 2's to conclude that $n_0 = n_1$. Also as in the prior item, this map is not onto because the codomain element $y = 1$ is not in the range of the function, as all of the range elements are even.
- (E) The function $f: \mathbb{Z} \rightarrow \mathbb{N}$ given by $f(n) = |n|$ is not one-to-one but it is onto. It is not one-to-one because the two domain elements $n_0 = -1$ and $n_1 = 1$ map to the same output, $f(n_0) = 2 = f(n_1)$. To show that f is an onto map consider a codomain element $y \in \mathbb{N}$, and observe that if we take the domain element $x \in \mathbb{Z}$ such that $x = y$ then $f(x) = y$.

V II.1.24

- (A) This function is a correspondence. To verify that it is one-to-one suppose that $f(q_0) = f(q_1)$ for $q_0, q_1 \in \mathbb{Q}$, so that $q_0 + 3 = q_1 + 3$, and subtract 3 from both sides to get that $q_0 = q_1$. For onto, consider a codomain element $y \in \mathbb{Q}$ and observe that the domain element $q \in \mathbb{Q}$ given by $q = y - 3$ satisfies that $f(q) = y$.
- (B) This function is not a correspondence. It is not onto because the codomain element $1/2 \in \mathbb{Q}$ is not the image of any domain element $z \in \mathbb{Z}$, since each such image $f(z) = z + 3$ is an integer. (This function is

one-to-one: suppose that $f(z_0) = f(z_1)$, so that $z_0 + 3 = z_1 + 3$, and subtract 3 from both sides to conclude that $z_0 = z_1$. But because it is not onto, it is not a correspondence.)

- (c) This is not a correspondence. It is not even a function, because it is not well-defined — the two $q_0 = 1/2$ and $q_1 = 2/4$ are equal, $q_0 = q_1$, but they are not associated with the same $|a \cdot b|$'s since $|1 \cdot 2| = 2$ and $|2 \cdot 4| = 8$.

V II.1.25

- (A) This set is finite. It has the same cardinality as $I = \{0, 1, 2\}$, as witnessed by the function $f: I \rightarrow \{1, 2, 3\}$ given by $f(x) = x + 1$, which is one-to-one and onto by inspection.
- (B) The set $S = \{0, 1, 4, 9, 16, \dots\}$ of perfect squares is infinite. There is no correspondence between some $I = \{0, 1, \dots, n-1\}$ and S since there is no onto function — for any $f: I \rightarrow S$ the set $\{f(0), f(1), \dots, f(n-1)\}$ cannot equal S because it has a largest element but S has no largest element.
- (C) The set of primes is infinite. The argument from the prior item applies.
- (D) By the Fundamental Theorem of Arithmetic, a fifth degree polynomial has at most five real number roots. Thus this set has cardinality at most five, and so is finite.

V II.1.26

- (A) The map $f: \{0, 1, 2\} \rightarrow \{3, 4, 5\}$ given by $f(0) = 3$, $f(1) = 4$, and $f(2) = 5$, that is, $f(x) = x + 3$, is a correspondence. By inspection each element of the codomain is associated with one and only one element of the domain.
- (B) The function $f(x) = x^3$ is a correspondence between these sets. For each perfect cube there is one and only one associated integer cube root.

V II.1.27

- (A) The function f given by $0 \mapsto \pi$, $1 \mapsto \pi + 1$, $3 \mapsto \pi + 2$, and $7 \mapsto \pi + 3$ is both one-to-one and onto, by inspection.
- (B) Let $E = \{2n \mid n \in \mathbb{N}\}$ and $S = \{n^2 \mid n \in \mathbb{N}\}$. Consider the map $f: E \rightarrow S$ given by $f(x) = (x/2)^2$. To show that f is one-to-one, suppose that $f(x_0) = f(x_1)$. Then $(x_0/2)^2 = (x_1/2)^2$ and because these are natural numbers their square roots are equal, $x_0/2 = x_1/2$. Thus $x_0 = x_1$ and the map is one-to-one.
For onto, consider $y \in S$. Because y is a perfect square there is a natural number $n \in \mathbb{N}$ so that $y = n^2$. Let $x = 2n$. Clearly x is even and $f(x) = y$.
- (c) The second interval is $2/3$ -rds the width of the first, so consider the function $f(x) = (2/3) \cdot x - (5/3)$. It is one-to-one because $f(x_0) = f(x_1)$ implies that $(2/3) \cdot x_0 - (5/3) = (2/3) \cdot x_1 - (5/3)$, and straightforward algebra gives that $x_0 = x_1$. To show that f is onto, fix $y \in (-1 \dots 1)$. Let $x = (y + (5/3)) / (2/3)$ (this came from solving $y = (2/3)x - (5/3)$ for x). When y is an element of the codomain $(-1 \dots 1)$ then this x is an element of the domain $(1 \dots 4)$, and more algebra shows that $f(x) = y$.

- V II.1.28 The problem statement is ambiguous as to which set is the domain and which is the codomain. We will prove that $f: (0 \dots 1) \rightarrow (1 \dots \infty)$ given by $f(x) = 1/x$ is one-to-one and onto.

For one-to-one assume that $f(x_0) = f(x_1)$, where $x_0, x_1 \in (0 \dots 1)$. Then $1/x_0 = 1/x_1$ and cross-multiplication gives $x_1 = x_0$. For onto suppose that $y \in (1 \dots \infty)$ and note that if $x = 1/y$ then $f(x) = y$ and also $x \in (0 \dots 1)$.

- V II.1.29 Call the sets $D = \{1, 2, 3, 4, \dots\}$ and $C = \{7, 10, 13, 16, \dots\}$. We will show that the function $f: D \rightarrow C$ described by the formula $f(x) = 3(x - 1) + 7$ gives a correspondence between the sets. This function is one-to-one since $f(x_0) = f(x_1)$ implies that $3(x_0 - 1) + 7 = 3(x_1 - 1) + 7$, and subtracting 7, dividing by 3, and adding 1 gives that $x_0 = x_1$. It is onto because if y is an element of the codomain C then it has the form $y = 3n + 4$ for $n \in \mathbb{N}^+$. That equation gives $y = 3(n - 1) + 7$. Some algebra gives $(y - 7)/3 = n - 1$ for $n \in \mathbb{N}^+$ and we have that $n = ((y - 7)/3) + 1$ is an element of $\mathbb{N}^+ = D$ with $f(n) = y$.

V II.1.30

- (A) The obvious map is this.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	48	49	50	51	52	53	54	55	56	57

It is one-to-one by inspection, meaning that looking over the association shows that never do two different domain elements map to the same codomain element. Similarly, it is also onto by inspection.

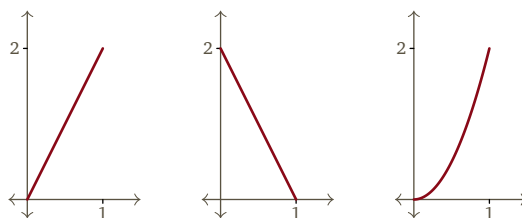


FIGURE 9, FOR QUESTION II.1.32: The correspondences $f_0, f_1, f_2: (0..1) \rightarrow (0..2)$ given by $f_0(x) = 2x$, $f_1(x) = 2-2x$, and $f_2(x) = 2x^2$.

- (B) The inverse map associates the same pairs of elements in the prior item's table, but the domain and codomain swap places so here A is the domain and C is the codomain.

y	48	49	50	51	52	53	54	55	56	57
x	0	1	2	3	4	5	6	7	8	9

As in the prior item, this map is one-to-one and onto by inspection.

- V II.1.31 For each pair of sets we could define a function whose domain is the first set, or a function whose domain is the second set. We choose whichever is convenient.

- (A) Write the even numbers as $E = \{2n \mid n \in \mathbb{N}\}$. Consider the function $f: \mathbb{N} \rightarrow E$ defined by $f(m) = 2m$. To verify that f is one-to-one, assume that two inputs yield the same output $f(m) = f(\hat{m})$, giving $2m = 2\hat{m}$, and divide by 2 to get that therefore the two inputs are the same, $m = \hat{m}$.

To verify that f is onto, consider a member of the codomain $y \in E$, so that $y = 2n$ for some $n \in \mathbb{N}$. Now, $n = y/2$ has the property that $f(n) = y$. Note that n is an element of the domain $\mathbb{N}$ because as a member of the codomain, y is even, and thus dividing by two yields a natural number.

- (B) Write the odd numbers as $M = \{2n + 1 \mid n \in \mathbb{N}\}$. Consider $g: \mathbb{N} \rightarrow M$ given by $n \mapsto 2n + 1$. To verify that g is one-to-one, assume that $g(m_0) = g(m_1)$, so that $2m_0 + 1 = 2m_1 + 1$, subtract 1, and divide by 2, to conclude that $m_0 = m_1$.

To verify that g is onto, start with an element of the codomain $y \in M$. Then it has the form $y = 2k + 1$ for some natural number k . That means there is a natural number k that maps under g to y , namely $k = (y-1)/2$.

- (C) One answer is to cite the two prior exercise parts and that the inverse of a correspondence is also a correspondence, to get that the composition $g \circ f^{-1}$ is a correspondence from the even numbers to the odds.

A direct answer is to consider the map $p: E \rightarrow M$ given by $p(n) = n + 1$. Verify that p is one-to-one and onto as in the prior two items.

- V II.1.32 The natural correspondence $f_0: (0..1) \rightarrow (0..2)$ is $f_0(x) = 2x$. For the other two there are many choices but two more correspondences $f_0, f_1, f_2: (0..1) \rightarrow (0..2)$ are $f_1(x) = 2 - 2x$ and $f_2(x) = 2x^2$. Figure 9 on page 47 shows the three graphs. They are different functions because they return different values on the input $x = 1/3$.

Each function is one-to-one and onto because the figure shows that every horizontal line at $y \in (0..2)$ intercepts its graph in one and only one point.

For an algebraic verification that they are correspondences we can use f_1 as a model. It is one-to-one because if $f(x_0) = f(x_1)$ then $2 - 2x_0 = 2 - 2x_1$, and then subtracting 2 and dividing by -2 yields that $x_0 = x_1$. It is onto because if $y \in (0..2)$ then the number $x = (y - 2)/(-2)$ is an element of $(0..1)$ and has the property that $f(x) = 2 - 2 \cdot ((y - 2)/(-2)) = y$.

- V II.1.33 The function

$$\hat{f}(n) = \begin{cases} 1 & \text{-- if } n = 0 \\ 0 & \text{-- if } n = 1 \\ n^2 & \text{-- if } n > 1 \end{cases}$$

is different than Example 1.8's function f because they give different outputs for the input 0. That $\hat{f}$ is onto is clear, as every perfect square is the output associated with some input. That $\hat{f}$ is a one-to-one function is also

clear, although it is a little messier to write down. Assume $\hat{f}(x_0) = \hat{f}(x_1)$ and then there are four cases: both x_0 and x_1 are members of the set $\{0, 1\}$, only the first one is a member, only the second is a member, or neither is a member. All four cases are easy.

- V II.1.34 To check that f is one-to-one suppose that $f(x_0) = f(x_1)$, so that $c^{x_0} = c^{x_1}$. Take the logarithm base c of both sides, giving $x_0 = x_1$.

For onto, consider a codomain element $y \in (0 .. \infty)$. The number $x = \log_c(y)$ is defined and is an element of $\mathbb{R}$, the domain of f . Of course, $f(x) = f(\log_c(y)) = c^{\log_c(y)} = y$.

- V II.1.35 The two $P_2 = \{2^k \mid k \in \mathbb{N}\} = \{1, 2, 4, 8, \dots\}$ and $P_3 = \{3^k \mid k \in \mathbb{N}\} = \{1, 3, 9, \dots\}$ are subsets of $\mathbb{N}$. The natural correspondence $g: P_2 \rightarrow P_3$ is to associate elements having the same exponent, $g(2^k) = 3^k$. This function is one-to-one because if $g(2^{k_0}) = g(2^{k_1})$ then $3^{k_0} = 3^{k_1}$, and taking the logarithm base 3 of both sides shows that the inputs are the same, $k_0 = k_1$. This function is onto because if $y \in P_3$ then it has the form $y = 3^k$, and is the image of $2^k \in P_2$.

The obvious generalization is that if two natural numbers $a, b \in \mathbb{N}$ are greater than 1 then $P_a = \{a^k \mid k \in \mathbb{N}\}$ and $P_b = \{b^k \mid k \in \mathbb{N}\}$ are sets of natural numbers that have the same cardinality. The argument is as in the prior paragraph.

- V II.1.36 Of course, for each item there are many possible answers.

- (A) One is $f_0: \mathbb{N} \rightarrow \mathbb{N}$ given by $f_0(n) = n + 1$. It is one-to-one because if $f_0(n) = f_0(\hat{n})$ then $n + 1 = \hat{n} + 1$ and so $n = \hat{n}$. It is not onto because no $n \in \mathbb{N}$ maps to 0.

A second one is $f_1: \mathbb{N} \rightarrow \mathbb{N}$ given by $f_1(n) = n^2$. It is one-to-one because if $f_1(n) = f_1(\hat{n})$ then $n^2 = \hat{n}^2$ and because these are natural numbers they are nonnegative, so $n = \hat{n}$. This function is not onto because no natural number maps to 2.

- (B) This function

$$f_0(x) = \begin{cases} x/2 & \text{-- if } x \text{ is even} \\ (x-1)/2 & \text{-- if } x \text{ is odd} \end{cases} \quad \begin{array}{c|cccccc} x & 0 & 1 & 2 & 3 & 4 & 5 & \dots \\ f_0(x) & 0 & 0 & 1 & 1 & 2 & 2 & \dots \end{array}$$

is onto because every $y \in \mathbb{N}$ is the image of $x = 2y$. It is not one-to-one because $f_0(1) = f_0(0)$.

A second one is $f_1(x) = \lfloor \sqrt{x} \rfloor$ (recall that the floor operation, $\lfloor n \rfloor$, gives the largest integer less than or equal to n).

$$\begin{array}{c|cccccccccccc} x & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & \dots \\ f_1(x) & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & \dots \end{array}$$

It is onto because every $y \in \mathbb{N}$ is the image of $x = y^2$. It is not one-to-one because $f_1(1) = f_1(2)$.

- (c) One such map is $f_0(x) = 0$. It is not one-to-one because $f_0(0) = f_0(1)$. It is not onto because no natural number input maps to 1.

A non-constant such function is the square of the prior item's second one, $f_1(x) = (\lfloor \sqrt{x} \rfloor)^2$.

$$\begin{array}{c|cccccccccccc} x & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & \dots \\ f_1(x) & 0 & 1 & 1 & 1 & 4 & 4 & 4 & 4 & 4 & 9 & 9 & \dots \end{array}$$

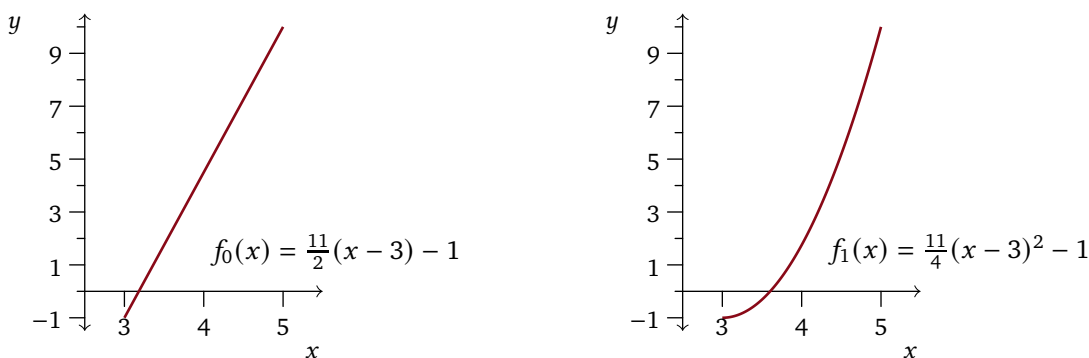
It is not one-to-one because $f_1(1) = f_1(2)$. It is not onto because no number maps to 2, as 2 is not a square.

- (D) The natural function that is both one-to-one and onto is the identity function, $f_0(x) = x$. It is one-to-one because if $f_0(n) = f_0(\hat{n})$ then $n = \hat{n}$, just straight from the definition of f_0 . It is onto because any codomain element $y \in \mathbb{N}$ is the image under f_0 of itself, $f_0(y) = y$.

Another correspondence swaps even and odd numbers.

$$f_1(x) = \begin{cases} x+1 & \text{-- if } x \text{ is even} \\ x-1 & \text{-- if } x \text{ is odd} \end{cases} \quad \begin{array}{c|cccccc} x & 0 & 1 & 2 & 3 & 4 & 5 & \dots \\ f_1(x) & 1 & 0 & 3 & 2 & 5 & 4 & \dots \end{array}$$

Observe first that $f_1: \mathbb{N} \rightarrow \mathbb{N}$; that is, if $x \in \mathbb{N}$ then $f_1(x) \in \mathbb{N}$. To check that f_1 is one-to-one suppose that $f_1(n) = f_1(\hat{n})$. There are two cases. If n is odd then $f_1(n) = n - 1$ is even, and so $f_1(\hat{n})$ is even also,

FIGURE 10, FOR QUESTION II.1.37: Two correspondences between $(3..5)$ and $(-1..11)$.

implying that $\hat{n}$ is odd, and therefore $n - 1 = \hat{n} - 1$, giving that $n = \hat{n}$. The even case is similar. To check that f_1 is onto consider a codomain point $y \in \mathbb{N}$. Here also there are two cases and we will only do the odd one, so suppose that y is odd. Then $y = f_1(x)$ where $x = y - 1$, and x is an element of the domain $\mathbb{N}$ because if y is odd then $y - 1 \geq 0$.

V II.1.37 Let $D = (3..5)$ and $C = (-1..10)$.

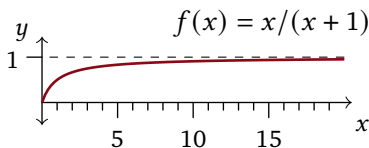
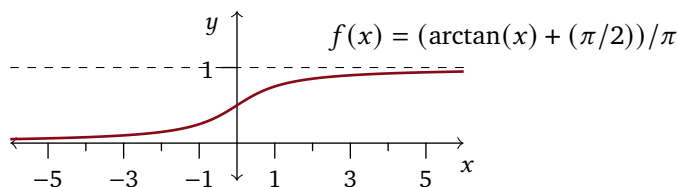
One correspondence is $f_0(x) = (11/2)(x - 3) - 1$. Whether $f_0: D \rightarrow C$ is not obvious so we start with that: if $3 < x < 5$ then subtracting 3, multiplying by $11/2$, and subtracting 1 from all terms gives $-1 < f_0(x) < 10$. Next we show that this map is onto. If $y \in C$ then $y = f_0(x)$ where $x = (2/11)(y + 1) + 3$. To verify that this input x is an element of the domain D , start with $-1 < y < 10$, add 1 to all terms, multiply by $2/11$, and add 3, ending with $3 < x < 5$. Finally, we show that the map f_0 is one-to-one. If $f_0(x) = f_0(\hat{x})$ then $(11/2)(x - 3) - 1 = (11/2)(\hat{x} - 3) - 1$ and adding 1 to both sides, multiplying by $2/11$, and finally adding 3, gives that $x = \hat{x}$.

Another correspondence is $f_1(x) = (11/4)(x - 3)^2 - 1$. As with f_0 we start by verifying that $f_1: D \rightarrow C$: if $3 < x < 5$ then subtracting 3, squaring, multiplying by $11/4$, and subtracting 1 from all terms gives $-1 < f_1(x) < 10$. To show that f_1 is onto assume that $y \in C$. Then $y = f_1(x)$ where $x = \sqrt{(4/11)(y + 1)} + 3$. To check that $x \in D$, start with $-1 < y < 10$ and add 1 to all terms, multiply by $4/11$, square (this doesn't change the direction of the $<$'s because all numbers are nonnegative), and add 3, ending with $3 < x < 5$. To show that this map is one-to-one, assume $f_1(x) = f_1(\hat{x})$ so that $(11/4)(x - 3)^2 - 1 = (11/4)(\hat{x} - 3)^2 - 1$. To both sides add 1, multiply by $4/11$, take the square root (which is unambiguous because the x 's are all greater than zero) and then add 3, giving that $x = \hat{x}$.

Figure 10 on page 49 shows that the two functions are unequal. It also gives an alternative demonstration that each is a correspondence because for each, every horizontal line at $y \in (-1..10)$ intersects the graph one and only one point.

V II.1.38 In each case we produce a function from one set to the other and verify that it is both one-to-one and onto. For the third item, note that just because a set appears first in the question does not mean that it must be the domain of the correspondence.

- (A) Where $S_4 = \{4k \mid k \in \mathbb{N}\}$ and $S_5 = \{5k \mid k \in \mathbb{N}\}$, let $f: S_4 \rightarrow S_5$ be $f(x) = (5/4)x$. It is one-to-one because $f(x_0) = f(x_1)$ implies that $(5/4)x_0 = (5/4)x_1$, so $x_0 = x_1$. It is onto because if we consider a codomain element $y = 5k$ for some $k \in \mathbb{N}$ then clearly $y = f(4k)$.
- (B) These two sets are the same set, that is, they contain the same numbers. They correspond via the identity function.
- (C) Write $T = \{0, 1, 3, 6, \dots\} = \{n(n+1)/2 \mid n \in \mathbb{N}\}$ (these are called the triangular numbers). Consider the function $f: \mathbb{N} \rightarrow T$ defined by $f(n) = n(n+1)/2$. This map is one-to-one because the continuous function $\hat{f}: \mathbb{R} \rightarrow \mathbb{R}$ given by $\hat{f}(x) = x(x+1)/2$ is a parabola with roots at 0 and -1 and so the part in the positive numbers, with $x \geq 0$, has that a horizontal line at y will intersect the graph in at most one point. The fact that $\hat{f}$ is one-to-one gives that its restriction f is also one-to-one. The function f is onto by definition, since T is defined as the range of that function.

FIGURE 11, FOR QUESTION II.1.39: A correspondence between $(0.. \infty)$ and $(0.. 1)$.FIGURE 12, FOR QUESTION II.1.42: A correspondence between $\mathbb{R}$ and $(0.. 1)$.**V II.1.39**

- (A) One is $f(x) = 2x$. This map is one-to-one because if $f(x) = f(\hat{x})$ then $2x = 2\hat{x}$ and so $x = \hat{x}$. This map is onto because where $0 < y < 2$, the number $x = y/2$ has the properties that $f(x) = y$ and $0 < x < 1$.
- (B) One is $f(x) = (b - a) \cdot x + a$. Note that $f: (0.. 1) \rightarrow (a.. b)$ since if $0 < x < 1$ then multiplying by $b - a$ gives $0 < x < (b - a)$ (the $<$'s maintain their direction because $a < b$), and adding a gives $a < x < b$. Further, this function is one-to-one: because $b - a \neq 0$ this line does not have slope 0 and so the function passes the horizontal line test (for an algebraic argument, start with $f(x) = f(\hat{x})$, subtract a and divide by $b - a$, and conclude that $x = \hat{x}$). Finally, to show that this function is onto, fix $y \in (a.. b)$. Note that $f(x) = y$ where $x = (y - a)/(b - a)$. What remains is to check that $x \in (0.. 1)$: starting from $a < y < b$, subtract a and divide by $b - a$ to get $0 < (y - a)/(b - a) < 1$ (as earlier, note that the $<$'s maintain their direction because $b - a > 0$).
- (C) One is $f(x) = x + a$. This function is one-to-one because $f(x) = f(\hat{x})$ implies that $x + a = \hat{x} + a$ and so $x = \hat{x}$. It is onto because if $y \in (a.. \infty)$ then $y = f(x)$ where $x = y - a$ (and $x \in (0.. \infty)$).
- (D) Similar triangles gives $x/(x + 1) = f(x)/1$. Figure 11 on page 50 shows that it is a correspondence because for each $y \in (0.. 1)$, the horizontal line at that height intercepts the function's graph in exactly one point.

V II.1.40 Consider the function $f: \mathbb{N} \rightarrow S$ given by $f(m) = {}^{m+2}\sqrt{2} = 2^{1/(m+2)}$. It is one-to-one because if $f(m) = f(\hat{m})$ then $2^{1/(m+2)} = 2^{1/(\hat{m}+2)}$. Take the logarithm of both sides $\lg(2^{1/(m+2)}) = \lg(2^{1/(\hat{m}+2)})$ to get $(1/(m+2)) \cdot \lg(2) = (1/(\hat{m}+2)) \cdot \lg(2)$, and so $1/(m+2) = 1/(\hat{m}+2)$. Thus $m = \hat{m}$.

This function is onto because if $y \in S$ then $y = 2^{1/n}$ for some $n \in \mathbb{N}$ with $n \geq 2$. Thus, $y = 2^{1/(x+2)} = f(x)$ for $x \in \mathbb{N}$.

V II.1.41 Recall that just as every natural number has a finite decimal representation, and that this representation is unique, so also every natural number has a unique binary representation.

- (A) View each string of bits σ starting with 1 as the binary representation of a natural number n . Then $f: B \rightarrow \mathbb{N}$ given by $\sigma \mapsto n - 1$ is clearly a correspondence.
- (B) Prefix each string $\tau \in B^*$ with 1, so $\sigma = 1 \frown \tau$. Now the prior argument applies.

V II.1.42 The tangent function is a correspondence from $(-\pi/2.. \pi/2)$ to $\mathbb{R}$ so its inverse, arctangent, is a correspondence in the other direction. Thus, $g(x) = (\arctan(x) + (\pi/2))/\pi$ gives a correspondence from $\mathbb{R}$ to $(0.. 1)$. See Figure 12 on page 50, which shows that g is one-to-one and onto because for each element y of the codomain $(0.. 1)$, the horizontal line at y intercepts the function's graph in exactly one point.**V II.1.43**

- (A) The map $g: I_0 \rightarrow I_1$ given by $g(x) = x \cdot (r_1/r_0)$ is onto because if $y \in I_1 = [0.. 2\pi r_1)$ then $y = g(x)$ where $x = y \cdot (r_0/r_1)$, and $0 \leq y < 2\pi r_1$ implies that $0 \leq y \cdot (r_0/r_1) < 2\pi r_0$. To show that this map is one-to-one suppose $g(x) = g(\hat{x})$. Then $x \cdot (r_1/r_0) = \hat{x} \cdot (r_1/r_0)$ and so $x = \hat{x}$.
- (B) The map $g(x) = a + x \cdot (b - a)$ is a correspondence. To verify that it is one-to-one, suppose that $g(x) = g(\hat{x})$

so that $a + x \cdot (b - a) = a + \hat{x} \cdot (b - a)$. Subtract a and divide by $b - a$ to conclude that $x = \hat{x}$. This map is also onto because $y \in [a..b)$ is the image of $x = (y - a)/(b - a)$, and $a \leq y < b$ implies that $0 \leq y - a < b - a$, and so $0 \leq (y - a)/(b - a) < 1$.

- (c) By the prior item there are correspondences $f: [0..1) \rightarrow [a..b)$ and $g: [0..1) \rightarrow [c..d)$. Then the composition $g \circ f^{-1}$ has domain $[a..b)$, codomain $[c..d)$, and is a correspondence.

(This is also easy to check by considering the map $f: [a..b) \rightarrow [c..d)$ given by $f(x) = c + (x - a) \cdot (d - c)/(b - a)$.)

- V **II.1.44** Suppose for contradiction that f is not one-to-one. Then there are $x, \hat{x} \in D$ such that $f(x) = f(\hat{x})$ but $x \neq \hat{x}$. But one or the other of these two is smaller; without loss of generality suppose that $x < \hat{x}$. Then $x < \hat{x}$ but $f(x) = f(\hat{x})$, which contradicts that the function is strictly increasing.

Yes, the same argument works for $D \subseteq \mathbb{N}$.

- V **II.1.45**

- (A) Let $\mathcal{E} = \{2n \mid n \in \mathbb{N}\}$. The natural correspondence is $f: \mathbb{N} \rightarrow \mathcal{E}$ given by $n \mapsto 2n$. This map is one-to-one because if $f(n) = f(\hat{n})$ then $2n = 2\hat{n}$ and so $n = \hat{n}$. This map is onto because if $y \in \mathcal{E}$ then $y = 2k$ for some $k \in \mathbb{N}$ and setting $n = k$ gives that $y = f(n)$.
- (B) Let $T = \{n \in \mathbb{N} \mid n > 4\}$. Consider $f: \mathbb{N} \rightarrow T$ given by $n \mapsto n + 5$. This is one-to-one because if $f(n) = f(\hat{n})$ then $n + 5 = \hat{n} + 5$ and $n = \hat{n}$. It is onto because if $y \in T$ then $y = f(x)$ where $x = y - 5$, and observe that $y \in T$ implies that $y - 5 \in \mathbb{N}$. So f is a correspondence.

- V **II.1.46** We will show that no matter what is the cardinality of the finite set $S \subset \mathbb{N}$, there is a correspondence $f: \mathbb{N} \rightarrow \mathbb{N} - S$. We will use induction, at each step taking the finite set to have one element more than at the prior step.

Before the proof we give an illustration. Suppose that we have a five-element set $S_5 = \{2, 5, 8, 10, 11\}$ and this correspondence $f_5: \mathbb{N} \rightarrow \mathbb{N} - S_5$.

n	0	1	2	3	4	5	6	7	...
$f_5(n)$	0	1	3	4	6	7	9	12	...

Now we add a sixth element, so let $S = S_5 \cup \{9\}$. We want a correspondence $f: \mathbb{N} \rightarrow \mathbb{N} - S$. The natural approach is find that $9 = f_5(6)$ and define f in this way.

$$f(n) = \begin{cases} f_5(n) & \text{-- if } n < 6 \\ f_5(n + 1) & \text{-- if } n > 6 \end{cases} \quad \begin{array}{c|cccccccc} n & 0 & 1 & 2 & 3 & 4 & 5 & 6 & \dots \\ f(n) & 0 & 1 & 3 & 4 & 6 & 7 & 9 & 12 & \dots \end{array}$$

The induction argument's base case is $S = \emptyset$ and here the identity function $f(n) = n$ is clearly a one-to-one and onto map from $\mathbb{N}$ to $\mathbb{N} - S$.

For the inductive step take $k \in \mathbb{N}$, assume that the statement is true for any subset of $\mathbb{N}$ whose cardinality is less than or equal to k , and consider $S \subset \mathbb{N}$ with $|S| = k + 1$. Fix any $s \in S$ and let $S_k = S - \{s\}$. By the inductive hypothesis there is a correspondence $f_k: \mathbb{N} \rightarrow \mathbb{N} - S_k$. Set $\hat{n}$ to be such that $f_k(\hat{n}) = s$ and define $f: \mathbb{N} \rightarrow \mathbb{N} - S$ in this way.

$$f(n) = \begin{cases} f_k(n) & \text{-- if } n < \hat{n} \\ f_k(n + 1) & \text{-- if } n > \hat{n} \end{cases}$$

The function $f: \mathbb{N} \rightarrow \mathbb{N} - S$ is one-to-one because it is a restriction of f_k , which is a one-to-one map from $\mathbb{N}$ to $\mathbb{N} - \hat{S}$. Similarly, f is onto because f_k is onto, from $\mathbb{N}$ to $\mathbb{N} - S_k$. Thus f is a correspondence between $\mathbb{N}$ and $\mathbb{N} - S$, and so the two have the same cardinality.

- V **II.1.47**

- (A) The inverse function $f^{-1}: C \rightarrow D$ is given by $f^{-1}(\text{Spades}) = 0$, $f^{-1}(\text{Hearts}) = 1$, $f^{-1}(\text{Clubs}) = 2$, and $f^{-1}(\text{Diamonds}) = 3$. By inspection it is both one-to-one and onto.
- (B) Suppose that $f: D \rightarrow C$ is a correspondence. Consider the binary relation, the set of ordered pairs, $f^{-1} = \{\langle y, x \rangle \mid f(x) = y\}$. We will show that f^{-1} is a function, which means that we will show that it is well-defined, that each input $y \in C$ is associated with one and only one output $x \in D$.

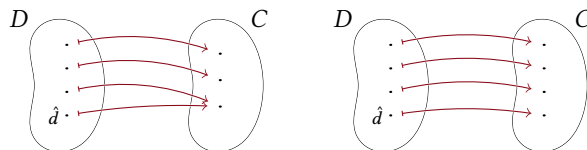


FIGURE 13, FOR QUESTION II.1.49: A finite function's domain has at least as many elements as its range.

Each $y \in C$ is associated with at least one x because the function f is onto. Each $y \in C$ is associated with at most one x because f is one-to-one.

- (c) To show that $f^{-1}: C \rightarrow D$ is one-to-one suppose that $f^{-1}(y) = f^{-1}(\hat{y})$. Then there is a $x \in D$ such that $f(x) = y$ and $f(x) = \hat{y}$. This contradicts that f is a function, because a function must be well-defined.

To show that f^{-1} is onto consider a member of its codomain, $\hat{x} \in D$. By the definition of a function there is a $\hat{y} \in C$ with $f(\hat{x}) = \hat{y}$, and then $\hat{x} = f^{-1}(\hat{y})$.

- V II.1.48 One direction is straightforward: no finite set has the same cardinality as a proper subset of itself, by Lemma 1.5.

For the other direction, fix an infinite set S , to show that it has the same cardinality as a proper subset of itself. (Remark: as elsewhere, we feel free here to use the Axiom of Choice.) Choose any element $s_0 \in S$. Then $S - \{s_0\}$ is not empty (or else S is not infinite) so choose $s_1 \in S - \{s_0\}$. Continuing in this way gives $\{s_0, s_1, s_2, \dots\}$. Then this is a correspondence from S to $S - \{s_0\}$.

$$f(x) = \begin{cases} s_{i+1} & \text{-- if } x = s_i \text{ for some } i \in \mathbb{N} \\ x & \text{-- otherwise} \end{cases}$$

- V II.1.49 Denote the function $f: D \rightarrow C$, and assume that D is finite.

- (A) We will argue that the statement $|\text{dom}(f)| \geq |\text{ran}(f)|$ holds for all functions on finite domains by induction on the number of elements in D .

The base step is that the domain has no elements at all, $\text{dom}(f) = \emptyset$. The only function with an empty domain is the empty function. The empty function's range is empty, $\text{ran}(f) = \emptyset$, and thus $|\text{dom}(f)| \geq |\text{ran}(f)|$.

For the inductive step, fix $k \in \mathbb{N}$, assume that the statement is true for any function with a domain having 0 elements, or 1 elements, $\dots$ or k elements, and consider the $|D| = k + 1$ case.

Because $k + 1 > 0$ there is a $\hat{d} \in D$. Let $\hat{D} = D - \{\hat{d}\}$ and consider the restriction $f|_{\hat{D}}: \hat{D} \rightarrow C$. Its domain has k many elements, so the induction hypothesis applies, giving $|\text{dom}(f|_{\hat{D}})| \geq |\text{ran}(f|_{\hat{D}})|$.

To add back the element $\hat{d}$ there are two cases; see Figure 13 on page 52. The first case, on the left in the figure, is that the range of f is the same as the range of the restriction $f|_{\hat{D}}$. The other case is that the range of f has one additional element. In either case, adding 1 to both sides ends the argument.

$$|\text{dom}(f)| = |\text{dom}(f|_{\hat{D}})| + 1 \geq |\text{ran}(f|_{\hat{D}})| + 1 \geq |\text{ran}(f)|$$

- (B) Assume that the function is one-to-one. We will show that the domain and range have the same size by induction on the number of elements in the domain. This argument is like the prior item's except that it has only the single case shown on the right of Figure 13 on page 52.

The base step is that the function's domain has no elements at all, $\text{dom}(f) = \emptyset$. This function's range is empty and $|\text{dom}(f)| = |\text{ran}(f)|$.

For the inductive step, fix $k \in \mathbb{N}$, assume that the domain and range have the same size for any one-to-one function where the domain D has 0 elements, or 1 element, $\dots$ or k elements, and suppose that $|D| = k + 1$.

Because $k + 1 > 0$ there is a $\hat{d} \in D$. Let $\hat{D} = D - \{\hat{d}\}$. Consider the restriction function $f|_{\hat{D}}: \hat{D} \rightarrow C$. The function f is one-to-one so the restriction is also one-to-one. The induction hypothesis applies, giving $|\text{dom}(f|_{\hat{D}})| = |\text{ran}(f|_{\hat{D}})|$. Adding 1 to both sides gives $|\text{dom}(f)| = |\text{dom}(f|_{\hat{D}})| + 1 = |\text{ran}(f|_{\hat{D}})| + 1 = |\text{ran}(f)|$.

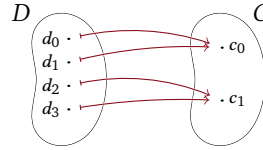


FIGURE 14, FOR QUESTION II.1.49: A finite function that is not one-to-one has more elements in its domain than its range.

(c) Figure 14 on page 53 illustrates the argument.

For the proof, let f be a function with a finite domain that is not one-to-one. Fix some numbering of its domain, $D = \{d_0, d_1, \dots, d_n\}$. Consider

$$\hat{D} = \{d_i \in D \mid i \text{ is minimal among the indices } j \text{ where } f(d_j) = f(d_i)\}$$

(in the illustration, $f(d_0) = f(d_1)$ and $f(d_2) = f(d_3)$) so this is the set $\{d_0, d_2\}$.

Because f is not one-to-one, $\hat{D}$ is a proper subset of D . The restriction $f|_{\hat{D}}: \hat{D} \rightarrow C$ is one-to-one. By the prior item, its range has the same number of elements as its domain. The range of f is the same as the range of $f|_{\hat{D}}$. But the domain of f has more elements than the domain of $f|_{\hat{D}}$. Thus the domain of f contains more elements than its range.

(d) Let D and C be finite sets. If there is a correspondence $f: D \rightarrow C$ then the prior items show that the sets have the same number of elements.

For the other direction assume that the two have the same number of elements. Fix an ordering of each, $D = \{d_0, \dots, d_k\}$ and $C = \{c_0, \dots, c_k\}$. Then the map $f: D \rightarrow C$ defined by $f(d_i) = c_i$ is clearly a correspondence.

V II.2.20 The table alternates between column 0 and column 1.

$n \in \mathbb{N}$	6	7	8	9	10	11	12
$f(n) \in \{0, 1\} \times \mathbb{N}$	$\langle 0, 3 \rangle$	$\langle 1, 3 \rangle$	$\langle 0, 4 \rangle$	$\langle 1, 4 \rangle$	$\langle 0, 5 \rangle$	$\langle 1, 5 \rangle$	$\langle 0, 6 \rangle$

The formulas are $x(n) = (1 + (-1)^{n+1})/2$ and $y(n) = \lfloor n/2 \rfloor$.

V II.2.21

- (A) The pair before $\langle 50, 50 \rangle$ is $\langle 49, 51 \rangle$ and the pair after is $\langle 51, 49 \rangle$. As a check they correspond to $\text{cantor}(50, 50) = 5\,100$, $\text{cantor}(49, 51) = 5\,099$, and $\text{cantor}(51, 49) = 5\,101$.
- (B) The pair before $\langle 100, 4 \rangle$ is $\langle 99, 5 \rangle$ and the pair after is $\langle 101, 3 \rangle$. As a check, the three correspond to $\text{cantor}(100, 4) = 5\,560$, $\text{cantor}(99, 5) = 5\,559$, and $\text{cantor}(101, 3) = 5\,561$.
- (C) The pair before $\langle 4, 100 \rangle$ is $\langle 3, 101 \rangle$ and the pair after is $\langle 5, 99 \rangle$. The three give $\text{cantor}(4, 100) = 5\,464$, $\text{cantor}(3, 101) = 5\,463$, and $\text{cantor}(5, 99) = 5\,465$.
- (D) Before $\langle 0, 200 \rangle$ is $\langle 199, 0 \rangle$ while after is $\langle 1, 199 \rangle$. These three correspond to $\text{cantor}(0, 200) = 20\,100$, $\text{cantor}(199, 0) = 20\,099$, and $\text{cantor}(1, 199) = 20\,101$.
- (E) Before $\langle 200, 0 \rangle$ is $\langle 199, 1 \rangle$ while after is $\langle 0, 201 \rangle$. They correspond to $\text{cantor}(200, 0) = 20\,300$, $\text{cantor}(199, 1) = 20\,299$, and $\text{cantor}(0, 201) = 20\,301$.

V II.2.22

(A) The natural correspondence $f_T: \mathbb{N} \rightarrow T$ is $f_T(n) = 2n$. The natural correspondence $f_F: \mathbb{N} \rightarrow T$ is given by $f_F(m) = 5m$.

To verify that f_T is one-to-one start with $f_T(n_0) = f_T(n_1)$, which is $2n_0 = 2n_1$, and cancelling gives $n_0 = n_1$. To verify that f_T is onto take $y \in T$. Then $y = 2k$ by definition, and so $y = f_T(k)$. Checking that f_N is one-to-one and onto works the same way.

$n \in \mathbb{N}$	0	1	2	3	4	5	6	7	8	9
$f_T(n) \in T$	0	2	4	6	8	10	12	14	16	18
$f_F(n) \in F$	0	5	10	15	20	25	30	35	40	45

(B) A natural correspondence gives outputs in ascending order.

$n \in \mathbb{N}$	0	1	2	3	4	5	6	7	8	9
$f(n) \in T \cup F$	0	2	4	5	6	8	10	12	14	15

- V II.2.23 This gives the first few associations for a function $f: \mathbb{N} \rightarrow \mathbb{N} \times \{0, 1\}$.

n	0	1	2	3	4	5	...
$f(n)$	$\langle 0, 0 \rangle$	$\langle 0, 1 \rangle$	$\langle 1, 0 \rangle$	$\langle 1, 1 \rangle$	$\langle 2, 0 \rangle$	$\langle 2, 1 \rangle$	...

From that table a description sufficient for computing f is easy.

$$f(n) = \begin{cases} \langle \lfloor n/2 \rfloor, 0 \rangle & \text{if } n \text{ is even} \\ \langle \lfloor n/2 \rfloor, 1 \rangle & \text{if } n \text{ is odd} \end{cases}$$

That gives $f(0) = \langle 0, 0 \rangle$, $f(10) = \langle 5, 0 \rangle$, $f(100) = \langle 50, 0 \rangle$, and $f(101) = \langle 50, 1 \rangle$.

The above table makes it easy to find the formula for the inverse, $f^{-1}(i, j) = 2i + j$. Then $f^{-1}(2, 1) = 5$, $f^{-1}(20, 1) = 41$, and $f^{-1}(200, 1) = 401$.

- V II.2.24 Members of $\{0, 1, 2\} \times \mathbb{N}$ are pairs $\langle x, y \rangle$ where the first entry is either 0, or 1, or 2, and the second entry is a natural number. This gives the first few of the associations under the obvious correspondence.

Number	0	1	2	3	4	5	6	7	8	...
Pair	$\langle 0, 0 \rangle$	$\langle 1, 0 \rangle$	$\langle 2, 0 \rangle$	$\langle 0, 1 \rangle$	$\langle 1, 1 \rangle$	$\langle 2, 1 \rangle$	$\langle 0, 2 \rangle$	$\langle 1, 2 \rangle$	$\langle 2, 2 \rangle$	...

The formula for the top to bottom association is this.

$$f(n) = \begin{cases} \langle 0, \lfloor n/3 \rfloor \rangle & \text{if 3 divides } n \\ \langle 1, \lfloor n/3 \rfloor \rangle & \text{if } n \bmod 3 \text{ is 1} \\ \langle 2, \lfloor n/3 \rfloor \rangle & \text{if } n \bmod 3 \text{ is 2} \end{cases}$$

In short: $f(n) = \langle n \bmod 3, \lfloor n/3 \rfloor \rangle$. Thus $f(0) = \langle 0, 0 \rangle$, $f(10) = \langle 1, 3 \rangle$, $f(100) = \langle 1, 33 \rangle$, and $f(1000) = \langle 1, 333 \rangle$. The formula for the bottom to top function $f^{-1}: \{0, 1, 2\} \times \mathbb{N} \rightarrow \mathbb{N}$ is also easy from the above table: $f^{-1}(x, y) = x + 3y$. With that, $f^{-1}(1, 2) = 7$, $f^{-1}(1, 20) = 61$, and $f^{-1}(1, 200) = 601$.

- V II.2.25 This is the initial part of an enumeration f of $\{0, 1, 2, 3\} \times \mathbb{N}$.

Number	0	1	2	3	4	5	6	7	8	...
Pair	$\langle 0, 0 \rangle$	$\langle 1, 0 \rangle$	$\langle 2, 0 \rangle$	$\langle 3, 0 \rangle$	$\langle 0, 1 \rangle$	$\langle 1, 1 \rangle$	$\langle 2, 1 \rangle$	$\langle 3, 1 \rangle$	$\langle 0, 2 \rangle$	...

The formula for the top to bottom association is $f(n) = \langle n \bmod 4, \lfloor n/4 \rfloor \rangle$, so $x(n) = n \bmod 4$ and $y(n) = \lfloor n/4 \rfloor$.

In general, for $k \in \mathbb{N}$ an enumeration f of $\{0, 1, 2, \dots, k\} \times \mathbb{N}$ is $f(n) = \langle n \bmod (k+1), \lfloor n/(k+1) \rfloor \rangle$.

- V II.2.26 Example 2.4's table shows up to $n = 6$.

Number	7	8	9	10	11	12	13	14	15	16
Pair	$\langle 1, 2 \rangle$	$\langle 2, 1 \rangle$	$\langle 3, 0 \rangle$	$\langle 0, 4 \rangle$	$\langle 1, 3 \rangle$	$\langle 2, 2 \rangle$	$\langle 3, 1 \rangle$	$\langle 4, 0 \rangle$	$\langle 0, 5 \rangle$	$\langle 1, 4 \rangle$

- V II.2.27 One way to compute this function is by brute force. Given n , a program can generate the pairs in ascending order $\langle 0, 0 \rangle$, $\langle 0, 1 \rangle$, $\langle 1, 0 \rangle$, $\langle 0, 2 \rangle$, $\dots$, keeping count until it generates the matching one. Figure 15 on page 55 has Racket code to do this.
- V II.2.28 Corollary 2.14 gives that, because the set $A - B$ is a subset of A , it is countable. Similarly $B - A \subseteq B$ is countable. Then, by the same result, their union is also countable.
- V II.2.29 Where $S = \{a + bi \mid a, b \in \mathbb{Z}\}$, obviously the map $g: \mathbb{Z} \times \mathbb{Z} \rightarrow S$ given by $g(a, b) = a + bi$ is a correspondence. Lemma 2.9 says that $\mathbb{Z} \times \mathbb{Z}$ is countable so there is a correspondence $f: \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{Z}$. Then the composition $g \circ f: \mathbb{N} \rightarrow S$ is a correspondence.
- V II.2.30 The table below walks through the algorithm of that example. The second column gives the pairs in ascending order. The third column lists the rational number result of the search loop. For example, loop number $n = 1$ first generates $\langle 1, 0 \rangle$ and rejects it since the denominator is zero, then it generates $\langle 0, 2 \rangle$ and rejects it as the rational number $0 = 0/2$ has already been enumerated, and then it generates $\langle 1, 1 \rangle$ and since $1 = 1/1$ is a rational that is so-far not enumerated, we have $f(1) = 1$.

```
;; Return the pair following p in Cantor's correspondence
(define (next-pair p)
  (let ((x (first p))
        (y (second p)))
    (if (= y 0)
        (list 0 (+ x 1))
        (list (+ x 1) (- y 1)))))

;; Find pair n by brute force
(define (brute-force-pairing n)
  (define (bfp-helper j pr)
    (if (zero? j)
        pr
        (bfp-helper (sub1 j) (next-pair pr))))
  (bfp-helper n (list 0 0)))
```

FIGURE 15, FOR QUESTION II.2.27: Racket code that inputs a number and finds the pair by brute force.

n	pairs	$f(n) \in \mathbb{Q}$
0	$\langle 0, 0 \rangle, \langle 0, 1 \rangle$	0
1	$\langle 1, 0 \rangle, \langle 0, 2 \rangle, \langle 1, 1 \rangle$	1
2	$\langle 2, 0 \rangle, \langle 0, 3 \rangle, \langle 1, 2 \rangle$	$1/2$
3	$\langle 2, 1 \rangle$	2
4	$\langle 3, 0 \rangle, \langle 0, 4 \rangle, \langle 1, 3 \rangle$	$1/3$
5	$\langle 2, 2 \rangle, \langle 3, 1 \rangle$	3
6	$\langle 4, 0 \rangle, \langle 0, 5 \rangle, \langle 1, 4 \rangle$	$1/4$
7	$\langle 2, 3 \rangle$	$2/3$
8	$\langle 3, 2 \rangle$	$3/2$
9	$\langle 4, 1 \rangle$	4
10	$\langle 5, 0 \rangle, \langle 0, 6 \rangle, \langle 1, 5 \rangle$	$1/5$
11	$\langle 2, 4 \rangle, \langle 3, 3 \rangle, \langle 4, 2 \rangle, \langle 5, 1 \rangle$	5

V II.2.31

- (A) The set $S - T$ is a subset of a countable set, so it is countable.
 (B) The set $S - T$ cannot be finite, or else $T \cup (S - T) = S$ would be the union of two finite sets and therefore finite.
 (C) Yes, the set $\mathbb{N}$ is countably infinite and the set of even numbers $2\mathbb{N} \subseteq \mathbb{N}$ is also infinite, and $\mathbb{N} - 2\mathbb{N}$, the set of odd numbers, is infinite.

V II.2.32 An infinite set contains some a_0 . Because it is infinite it has more than one element so it also contains some $a_1 \neq a_0$. Proceeding in this way we get the countably infinite set $\{a_0, a_1, \dots\}$. (If this is a subset of $\mathbb{N}$ then we could make the choice of a_1 definite by taking it to be the smallest element that is larger than a_0 , and proceed in that way.)**V II.2.33**

- (A) The map $f: \mathbb{Z}^{n+1} \rightarrow \mathbb{Z}_n[x]$ given by $f(a_0, \dots, a_n) = a_n x^n + \dots + a_1 x + a_0$ is clearly a correspondence. (We are not claiming that it is clear that different polynomials are different as functions, we are only claiming that they are different as polynomials.) Since Corollary 2.10 says that there is a correspondence $g: \mathbb{N} \rightarrow \mathbb{Z}^{n+1}$, we have a correspondence $f \circ g$ from $\mathbb{N}$ to $\mathbb{Z}_n[x]$.
 (B) The set $\mathbb{Z}[x]$ is the union of the $\mathbb{Z}_i[x]$ over $i \in \mathbb{N}$, and the union of countably many countable sets is countable.

II.2.34 Let $f: \mathbb{N} \rightarrow S$ be a correspondence. Where we write $f(n)$ as s_n , the map $g: S \rightarrow S$ given by $s_n \mapsto s_{n+1}$ is one-to-one but not onto.

V II.2.35 We will use that the function $\text{cantor}_3: \mathbb{N}^2 \rightarrow \mathbb{N}$ is a correspondence.

To verify that cantor_3 is onto, fix a codomain element $n \in \mathbb{N}$. Because cantor is onto there is a pair $\langle w, z \rangle \in \mathbb{N}^2$ such that $\text{cantor}(w, z) = n$. Also because cantor is onto, there is a pair $\langle x, y \rangle \in \mathbb{N}^2$ so that $\text{cantor}(x, y) = w$. Then $\text{cantor}_3(x, y, z) = \text{cantor}(\text{cantor}(x, y), z) = n$.

What's left is to verify that cantor_3 is one-to-one. Suppose that $\text{cantor}_3(a_0, b_0, c_0) = \text{cantor}_3(a_1, b_1, c_1)$. Then $\text{cantor}(\text{cantor}(a_0, b_0), c_0) = \text{cantor}(\text{cantor}(a_1, b_1), c_1)$. The function cantor is one-to-one so the prior sentence implies that $\text{cantor}(a_0, b_0) = \text{cantor}(a_1, b_1)$ and $c_0 = c_1$. Again applying that cantor is one-to-one gives that $a_0 = a_1$, $b_0 = b_1$, along with the $c_0 = c_1$.

V **II.2.36** A way to be confident about these is to write a small script.

```
sage: def cantor(x,y):
.....:     return x+ ((x+y)*(x+y+1)/2 )
.....:
sage: def cantor3(x,y,z):
.....:     return cantor(cantor(x,y),z)
.....:
```

(A) $c_3(0, 0, 0) = 0$, $c_3(1, 2, 3) = 62$, $c_3(3, 3, 3) = 402$

(B) $c_3(0, 0, 0) = 0$, $c_3(0, 0, 1) = 1$, $c_3(0, 1, 0) = 2$, $c_3(0, 0, 2) = 3$, $c_3(0, 1, 1) = 4$, $c_3(1, 0, 0) = 5$

(C) Substituting gives this.

$$\begin{aligned} c_3(x, y, z) &= \left(x + \frac{(x+y)(x+y+1)}{2} \right) + \left(\frac{\left(x + \frac{(x+y)(x+y+1)}{2} + z \right) \left(x + \frac{(x+y)(x+y+1)}{2} \right) + z + 1}{2} \right) \\ &= \frac{1}{8} \cdot ((x+y+1)(x+y) + 2x + 2z + 2) \cdot ((x+y+1)(x+y) + 2x + 2z) \\ &\quad + \frac{1}{2} \cdot (x+y+1)(x+y) + x \end{aligned}$$

That last is computed by machine, not by hand.

```
sage: x,y,z = var('x,y,z')
sage: cantor3(x,y,z)
1/8*((x + y + 1)*(x + y) + 2*x + 2*z + 2)*((x + y + 1)*(x + y) + 2*x + 2*z) + 1/2*(x + y + 1)*(x + y) + x
```

V **II.2.37** Plug $x = y = i$ into $\text{cantor}(x, y) = x + (x+y)(x+y+1)/2$ to get $\text{cantor}(i, i) = i + (2i)(2i+1)/2 = 2i^2 + 2i$. That's $2i(i+1)$. It is obviously a multiple of 2 but because the numbers i and $i+1$ are consecutive, one of them contributes another factor of 2. Hence the entire expression is a multiple of 4.

V **II.2.38** Fix b . For $n \in \mathbb{N}$, let S_N be the set of sequences that are equivalent to b because the two are equal from place N on. This set is finite because the only way some $\hat{b} \in S_N$ varies from b is in the first N places, so the set S_N has 2^N elements. (For example, if $b = \langle 0, 1, 0, 1, 0, 1, \dots \rangle$ and $N = 1$ then the only element of S_N other than b is $\hat{b} = \langle 1, 1, 0, 1, 0, 1, \dots \rangle$.) Therefore the total number of equivalent sequences is the union of the finite sets S_N over the countably many N .

V **II.2.39** Let S_0, S_1 , and S_2 be countably infinite and suppose that $f_0: \mathbb{N} \rightarrow S_0$, $f_1: \mathbb{N} \rightarrow S_1$ and $f_2: \mathbb{N} \rightarrow S_2$ are onto. Then $\hat{f}: \mathbb{N} \rightarrow S_0 \cup S_1 \cup S_2$ is given by

$$\hat{f}(n) = \begin{cases} f_0(n/3) & \text{-- if } n \text{ is a multiple of 3, that is, } n \bmod 3 = 0 \\ f_1((n-1)/3) & \text{-- if } n \bmod 3 = 1 \\ f_2((n-2)/3) & \text{-- if } n \bmod 3 = 2 \end{cases}$$

(briefly, $\hat{f}(n) = f_{n \bmod 3}(\lfloor n/3 \rfloor)$). This tabular illustration

n	0	1	2	3	4	5	6	...
$\hat{f}(n)$	$f_0(0)$	$f_1(0)$	$f_2(0)$	$f_0(1)$	$f_1(1)$	$f_2(1)$	$f_3(0)$	...

shows why the function is onto. Lemma 2.13 applies, and so the union is countable.

V **II.2.40** A function $f: \{0, 1\} \rightarrow \mathbb{N}$ is two pieces of information, $f(0)$ and $f(1)$. So we can take it to be a pair $\langle f(0), f(1) \rangle$. Restated, there is a natural correspondence between the set of functions $\{f \mid f: \{0, 1\} \rightarrow \mathbb{N}\}$ and the set of pairs $\mathbb{N} \times \mathbb{N}$, the latter of which we know to be countable.

V **II.2.41** Because S is countable, Lemma 2.13 says that either it is empty or there is an onto function $g: \mathbb{N} \rightarrow S$. If S is empty then the range set is empty, so it is countable. Otherwise, the function $f \circ g: \mathbb{N} \rightarrow \text{ran}(f)$ is onto.

- V **II.2.42** The cross product of two finite sets is finite because the number of elements in the cross product equals the product of the number of elements in the sets.

Next consider a finite set, $S = \{s_0, \dots, s_{m-1}\}$, and a countably infinite set, $T = \{t_0, t_1, \dots\}$. We will show that $S \times T$ is countable (the $T \times S$ argument is similar). Enumerate it row-wise, so that where $n = d \cdot m + r$ is the usual quotient-remainder relationship then $f(n) = \langle s_r, t_d \rangle$. (For illustration, this initial portion of the $m = 3$ case

n	0	1	2	3	4	5	6	...
$f(n)$	$\langle s_0, t_0 \rangle$	$\langle s_1, t_0 \rangle$	$\langle s_2, t_0 \rangle$	$\langle s_0, t_1 \rangle$	$\langle s_1, t_1 \rangle$	$\langle s_2, t_1 \rangle$	$\langle s_0, t_2 \rangle$	...

has the first entries cycle $s_0, s_1, s_2, s_0, \dots$ while the second entries ratchet up through the infinite set.) Clearly this function is a correspondence.

- V **II.2.43**

- (A) The set of length 5 strings $\Sigma^5 = \{\langle s_0, \dots, s_4 \rangle \mid s_i \in \Sigma\}$ is the cross product of five finite sets, each with 40 characters. So Σ^5 has 40^5 many members. (By the way, $40^5 = 102\,400\,000$.) Thus it is countable.
 (B) The set $\Sigma^0 \cup \Sigma^1 \cup \dots \cup \Sigma^5$ is the union of finitely many finite sets. So it is finite and therefore countable.
 (C) The set $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \dots$ is the union of countably many finite sets. By Corollary 2.14 it is countable.
 (D) A program is a finite string. So the set of all programs is a subset of Σ^* . Corollary 2.14 says that this set is countable.

- V **II.2.44**

- (A) These are easy to calculate by hand or with a script.

	$m = 0$	$m = 1$	$m = 2$	$m = 3$
$n = 0$	0	2	4	6
$n = 1$	1	5	9	13
$n = 2$	3	11	19	27
$n = 3$	7	23	39	55

- (B) Every natural number greater than zero is a unique product of primes, $2^{e_2} 3^{e_3} \dots p^{e_p}$. So every such number is the product of a maximal power of two, 2^n , with an odd number, $2m + 1$. Thus the pair $\langle n, m \rangle$ corresponds to $2^n \cdot (2m + 1)$, and therefore the map g with domain $\mathbb{N} \times \mathbb{N}$ given by $g(n, m) = 2^n \cdot (2m + 1) - 1$ is a correspondence, with range $\mathbb{N}$.
 (C) This table shows that under the box enumeration $\langle 3, 4 \rangle$ corresponds with 19.

$y = 4$	16	17	18	19	20
$y = 3$	9	10	11	12	21
$y = 2$	4	5	6	13	22
$y = 1$	1	2	7	14	23
$y = 0$	0	3	8	15	24
	$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$

- II.2.45** Here is an onto map from $\mathbb{N}$ to $\mathbb{Q}$. Given a natural number input n , factor out the first three primes to get $n = 2^{e_2} 3^{e_3} 5^{e_5} \cdot k$ where k is not divisible by 2, 3, or 5. If e_2 is odd then the output is $-e_3/(e_5 + 1)$ while if e_2 is even then the output is $e_3/(e_5 + 1)$.

- V **II.2.46**

- (A) $\text{cantor}(2, 1) = 8$
 (B) Solving $8 = 1 + [(1 + y)(1 + y + 1)]/2$ leads to the quadratic equation $0 = y^2 - 3y - 12$. The quadratic formula gives $(-3 \pm \sqrt{57})/2$, two different solutions, $y \approx 2.27$ and $y \approx -5.27$. Figure 16 on page 58 shows the graph.

- V **II.3.10** Infinite is different than exhaustive. The list 0, 2, 4, ... is infinite but does not exhaust all of the natural numbers.

- V **II.3.11** The union is not the entire set of real numbers. It does not include irrational numbers. For instance, $\pi = 3.14\dots$ is not in any of the sets and so is not in the union. Your classmate is counting only numbers that end in zeros.

- V **II.3.12** Cantor's Theorem says that the power set has more elements than the set.

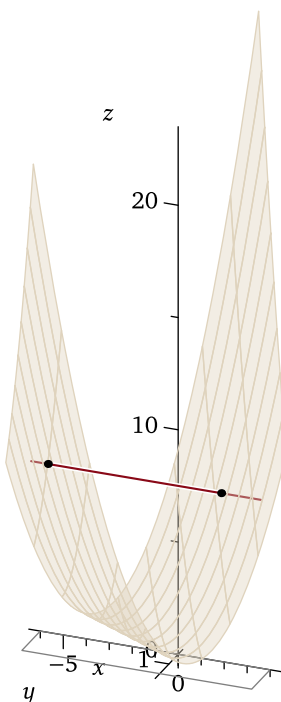


FIGURE 16, FOR QUESTION II.2.46: Cantor's unpairing function, with the line where $x = 1$ and $z = 8$.

- (A) The set $\{0, 1, 2\}$ has three elements. Its power set has eight.

$$\mathcal{P}(\{0, 1, 2\}) = \{\{0, 1, 2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0\}, \{1\}, \{2\}, \emptyset\}$$

- (B) The set $\{0, 1\}$ has two elements while its power set $\mathcal{P}(\{0, 1\}) = \{\{0, 1\}, \{0\}, \{1\}, \emptyset\}$ has four.

- (C) The set $\{0\}$ only has one element but its power set $\mathcal{P}(\{0\}) = \{\{0\}, \emptyset\}$ has two.

- (D) The empty set has zero elements, while its power set $\mathcal{P}\emptyset = \{\emptyset\}$ has one.

V II.3.13 The definition of ' $\leq$ ' for cardinality is that $|A| \leq |B|$ if there is a one-to-one function from A to B .

- (A) The function $f: S \rightarrow \hat{S}$ given by $f(1) = 11$, $f(2) = 12$, and $f(3) = 13$ is one-to-one by inspection.

- (B) The function $f: T \rightarrow \hat{T}$ given by $f(1) = 11$, $f(2) = 12$, and $f(3) = 13$ is one-to-one by inspection.

- (C) Again by inspection, the function $f: U \rightarrow \mathbb{O}$ where $f(1) = 1$, $f(2) = 3$, and $f(3) = 5$ is one-to-one.

- (D) The function $f: \mathbb{E} \rightarrow \mathbb{O}$ given by $f(x) = x+1$ is a one-to-one function. The verification is easy: $f(x_0) = f(x_1)$ implies that $x_0 + 1 = x_1 + 1$, and so $x_0 = x_1$.

II.3.14 The two differ on the set from which the elements are drawn.

- (A) The set $S = \{n \in \mathbb{Z} \mid n+3 < 5\} = \{\dots, -2, -1, 0, 1\}$ is countable. The function $f: \mathbb{N} \rightarrow S$ defined by $n \mapsto -n+1$ is a correspondence, as is easy to check.

- (B) Uncountable; it is the set of real numbers $(-\infty .. 2)$. (This set corresponds to $\mathbb{R}$ under the map $x \mapsto \ln(2-x)$.)

V II.3.15

- (A) Uncountable. (The function $x \mapsto \ln(x-5)$ is a correspondence with $\mathbb{R}$.)

- (B) Countable. (Because it is given as a subset of the naturals, and also because this set is $\{1, 2, 3\}$, which is finite.)

- (C) This set is uncountable. (The set $S = [1 .. 4) \subset \mathbb{R}$ has at least the same cardinality as $T = (1 .. 4) \subset \mathbb{R}$ since the inclusion map $\iota: T \rightarrow S$ given by $x \mapsto x$ is one-to-one. The interval T has the same cardinality as $\mathbb{R}$. One way to see that is to start with the correspondence between the interval $(-\pi/2 .. \pi/2)$ and $\mathbb{R}$ given by $x \mapsto \tan(x)$. Adapt it to the case of domain T by resizing that interval by a factor of $\pi/3$ and then shifting it to get that $x \mapsto \tan((\pi/3) \cdot (x-1) - (\pi/2))$ is a correspondence.)

- (D) Countable. (Any subset of $\mathbb{N}$ is countable.)

- V II.3.16** For the two element set $A_2 = \{0, 1\}$, the power set, $\mathcal{P}(A_2) = \{\{0, 1\}, \{0\}, \{1\}, \{\}\}$, has four elements. Making a function from A_2 to $\mathcal{P}(A_2)$ involves picking where to send 0 and where to send 1. So there are $4^2 = 16$ different functions.

function f_i	$f_i(0)$	$f_i(1)$	function f_i	$f_i(0)$	$f_i(1)$
f_0	$\{\}$	$\{\}$	f_8	$\{1\}$	$\{\}$
f_1	$\{\}$	$\{0\}$	f_9	$\{1\}$	$\{0\}$
f_2	$\{\}$	$\{1\}$	f_{10}	$\{1\}$	$\{1\}$
f_3	$\{\}$	$\{0, 1\}$	f_{11}	$\{1\}$	$\{0, 1\}$
f_4	$\{0\}$	$\{\}$	f_{12}	$\{0, 1\}$	$\{\}$
f_5	$\{0\}$	$\{0\}$	f_{13}	$\{0, 1\}$	$\{0\}$
f_6	$\{0\}$	$\{1\}$	f_{14}	$\{0, 1\}$	$\{1\}$
f_7	$\{0\}$	$\{0, 1\}$	f_{15}	$\{0, 1\}$	$\{0, 1\}$

The three element set A_3 has $|\mathcal{P}(A_3)| = 8$. The number of functions from A_3 to $\mathcal{P}(A_3)$ is $8^3 = 512$. In general, where $|A| = n$ the power set has $|\mathcal{P}(A_3)| = 2^n$ -many elements. The number of functions is $(2^n)^n = 2^{(n^2)}$. (As a check, $n = 2$ gives $2^4 = 16$ and $n = 3$ gives $2^9 = 512$.)

V II.3.17

(A) These are the functions from S to T .

function f_i	$f_i(0)$	$f_i(1)$
f_0	10	10
f_1	10	11
f_2	11	10
f_3	11	11

The functions that are one-to-one are f_1 and f_2 .

(B) These are the functions from S to T .

function f_i	$f_i(0)$	$f_i(1)$	function f_i	$f_i(0)$	$f_i(1)$
f_0	10	10	f_5	11	12
f_1	10	11	f_6	12	10
f_2	10	12	f_7	12	11
f_3	11	10	f_8	12	12
f_4	11	11			

The functions that are one-to-one are f_1, f_2, f_3, f_5, f_6 , and f_7 .

V II.3.18

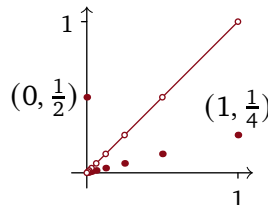
- (A) The answer is (iii) finite, since the union of two finite sets is finite. (All of these are true: (ii) countable or uncountable, (iii) finite, (iv) countable, and (v) finite, countably infinite, or uncountable. But the sharpest one, the one that is most specific, is (iii).)
- (B) The sharpest is (iv).
- (C) The sharpest is (i).
- (D) The sharpest is (iv). (The intersection could be finite but there are cases where it is countable, such as $A = \mathbb{N}$ and $B = \mathbb{R}$. So that is the sharpest statement we can make in general is (iv).)

V II.3.19 The question doesn't require a proof but here the parenthetical comments give a justification.

- (A) They are all possible except for the last one. (An example of (i) and (ii) is $\text{id}: \{0, 1\} \rightarrow \{0, 1\}$. An example of (iii) and (iv) is $\text{id}: \mathbb{N} \rightarrow \mathbb{N}$. Lemma 2.13 rules out (v).)
- (B) They are all possible. (An example of (i) and (ii) is $\text{id}: \{0, 1\} \rightarrow \{0, 1\}$. An example of (iii) and (iv) is $\text{id}: \mathbb{N} \rightarrow \mathbb{N}$. An example of (v) is $\iota: \mathbb{N} \rightarrow \mathbb{R}$ given by $\iota(x) = x$.)

V II.3.20 Following the hint, consider this function.

$$f(x) = \begin{cases} 1/2 & \text{-- if } x = 0 \\ 1/2^{n+2} & \text{-- if } x = 1/2^n \text{ for } n \in \mathbb{N} \\ x & \text{-- otherwise} \end{cases}$$



One way to argue that the function f is one-to-one and onto is to see that it satisfies the horizontal line test, that every horizontal line $y = t$ for $t \in (0..1)$ has one and only one associated input x .

We can expand. Consider first the case that the output $y \in (0..1)$ satisfies that $y \neq 1/2^n$ for $n \in \mathbb{N}^+$. Then y has exactly one associated input, namely $x = y$. The other case is that $y = 1/2^n$ for some $n \in \mathbb{N}^+$. Here also there is one and only one associated input x . Specifically, if $n = 1$ then that input is $x = 0$, which is in the domain $[0..1]$. If $n > 1$ then that input is $x = 1/2^{n-2}$, which is also an element of $[0..1]$.

V **II.3.21** Theorem 3.7 says that for every set, its cardinality is strictly less than that of its power set. So one set with a cardinality larger than $\mathbb{R}$ is $\mathcal{P}(\mathbb{R})$. (Another is $\mathcal{P}(\mathcal{P}(\mathbb{R}))$.)

V **II.3.22**

(A) The set of finite bit strings, $\mathbb{B}^*$, is the union of the sets $\mathbb{B}^k = \{ \langle b_0 b_1 \dots b_{k-1} \rangle \mid k \in \mathbb{N} \}$. The set $\mathbb{B}^k$ is finite since it has 2^k many elements (when $k = 0$ it contains only the empty string), and so this is a countable union of countable sets.

(B) Suppose that we can count the set of infinite bit strings, meaning that it has an exhaustive enumeration $f_0, f_1, \dots$. Let $g: \mathbb{B} \rightarrow \mathbb{B}$ be given by $g(0) = 1$ and $g(1) = 0$. Then the infinite bit string $F: \mathbb{N} \rightarrow \mathbb{B}$ defined by $F(i) = g(f_i(i))$ is not equal to any of the f_i , which contradicts that the enumeration is exhaustive.

V **II.3.23** Let S and T be such that $S \subseteq T$. Then the inclusion function, the map $\iota: S \rightarrow T$ given by $\iota(s) = s$, is clearly one to one. So $|S| \leq |T|$ by Definition 3.4.

V **II.3.24** We follow the hint. Call the collection of all such functions $S = \{ f: \mathbb{N} \rightarrow \mathbb{N} \mid \text{ran}(f) \text{ is finite} \}$. Assume that S is countable so that there is a correspondence $h: \mathbb{N} \rightarrow S$. For all natural numbers i , the value of $h(i)$ is a function, which we will denote f_i .

Define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = 0$ if $x \neq 0$ and $g(x) = 1$ otherwise. Consider the function $k: \mathbb{N} \rightarrow \mathbb{N}$ given by $k(i) = g(f_i(i))$. This map has a finite range, namely $\text{ran}(k) = \{0, 1\}$, but k is not equal to any $f_i \in S$ because they differ on the input i .

V **II.3.25** Example 2.11 shows that the set of rational numbers is countable. Suppose that the irrational numbers were also countable. Then because the union of two countable sets is countable, the real numbers would be countable. But the set of reals is not countable and so the set of irrationals is not countable either.

V **II.3.26**

(A) Write the decimal expansion of z as $z = 0.z_0 z_1 z_2 \dots$. It does not end in 9's as the output of g is never a 9. So z has a unique decimal expansion. For each q_i the decimal expansion of z differs from q_i 's in that $z_i \cdot 10^{-(i+1)} \neq q_{i,i} \cdot 10^{-(i+1)}$, by the definition of g . Because their decimal expansions differ, and because neither ends in 9's, the two are different numbers.

Since z does not equal any rational number, it is irrational.

(B) If $d = \sum_{n \in \mathbb{N}} d_{n,n} \cdot 10^{-(n+1)}$ is rational then its decimal expansion repeats. That is, there is a decimal place such that past that place the expansion consists of a sequence of digits $d_j d_{j+1} \dots d_{j+k}$ repeated again and again. But then the prior item's z would also repeat, in that past that same decimal place its expansion would consist of a sequence of digits $g(d_j) g(d_{j+1}) \dots g(d_{j+k})$ repeated again and again. That would make z rational, which it isn't.

(C) In Theorem 3.1 we start with what purports to be a list of all reals and then produce a real that is not on that list. Here we start with what is a list of all rationals and produce something that is not a rational. That makes it still a real, just not a rational real.

V **II.3.27** Before the proof, as a motivation for the inductive step consider $S = \{0, 1, 2, 3\}$. These are the sixteen elements of $\mathcal{P}(S)$. The top line has the sets that do not contain 3 while the second line has the sets that do.

$$\begin{array}{cccccccc} \{\}, & \{0\}, & \{1\}, & \{2\}, & \{0, 1\}, & \{0, 2\}, & \{1, 2\}, & \{0, 1, 2\}, \\ \{3\}, & \{0, 3\}, & \{1, 3\}, & \{2, 3\}, & \{0, 1, 3\}, & \{0, 2, 3\}, & \{1, 2, 3\}, & \{0, 1, 2, 3\} \end{array}$$

Each set on the bottom line is the same as the matching set on top, with an added 3.

We proceed by induction on $|S|$. The base step is that $|S| = 0$, so that S is empty. Then $\mathcal{P}(S) = \{\emptyset\}$ and $|\mathcal{P}(S)| = 1 = 2^0$, as required.

For the inductive step fix $k \in \mathbb{N}^+$, assume that the statement is true for $n = 0, n = 1, \dots, n = k$, and consider a set S where $|S| = k + 1$. Because $|S| = k + 1$, the set S has at least one element, $\hat{s} \in S$. The inductive hypothesis applies to $S - \{\hat{s}\}$, so $\mathcal{P}(S - \{\hat{s}\})$ has 2^k elements. As illustrated above, the elements of $\mathcal{P}(S)$ that do not contain $\hat{s}$ are in correspondence with the elements of $\mathcal{P}(S)$ that do contain $\hat{s}$. But the sets from $\mathcal{P}(S)$ that do not contain $\hat{s}$ are exactly the sets in $\mathcal{P}(S - \{\hat{s}\})$. Thus the total number of elements of $\mathcal{P}(S)$ is $2 \cdot 2^k$, which equals 2^{k+1} .

- V **II.3.28** Suppose that $f(s_i) = H$. We will argue that neither is $s_i \in H$ nor is $s_i \notin H$. If $s_i \in H$ then student s_i is a member of their own list, so s_i is not humble by the definition of humble, which violates the assumption of this case. On the other hand, if $s_i \notin H$ then s_i is not a member of their own list, so s_i is humble, which violates this case's assumption.
- V **II.3.29** Suppose otherwise, that $\mathcal{C}$ is the set of all sets. Then $\mathcal{P}(\mathcal{C})$ would be a collection of sets so we would have $\mathcal{P}(\mathcal{C}) \subseteq \mathcal{C}$. That would give $|\mathcal{P}(\mathcal{C})| \leq |\mathcal{C}|$, which would contradict Cantor's Theorem.
- V **II.3.30** One way is to take the diagonal bits two at a time; for instance changing 01 to 10, and changing any other two-bit sequence to 01.
- V **II.3.31** In the prototype $1.000\dots = 0.999\dots$ we will write the two representations as $x_0 = 1, x_{-1} = 0, x_{-2} = 0, \dots$, and $y_0 = 0, y_{-1} = 9, y_{-2} = 9, \dots$. Observe that the subscripts, the decimal places, are integers such as 0 and -1 . They are not natural numbers.
- (A) In $a + ar + ar^2 + ar^3 + \dots = a/(1 - r)$, take $a = 9$ and $r = 0.1$. The left side is $9 + 0.9 + 0.09 + 0.009 + \dots = 9.99999\dots$. The right side is $9/(1 - (0.1)) = 9/0.9 = 10$. Subtract 9 from both sides to get $0.99999\dots = 1$.
- (B) Consider a real number with two representations. Among the two there is a furthest left nonzero decimal place, $N \in \mathbb{Z}$. Consequently, write the two representations as $\vec{x} = \langle x_N, x_{N-1}, \dots \rangle$ and $\vec{y} = \langle y_N, y_{N-1}, \dots \rangle$. These are subject to the condition that $x_i, y_i \in \{0, 1, \dots, 9\}$ and to the condition that they add to the same number.

$$\sum_{i \leq N} x_i \cdot 10^i = \sum_{i \leq N} y_i \cdot 10^i \quad (*)$$

Rewrite $(*)$ as a difference.

$$0 = \left(\sum_{i \leq N} x_i \cdot 10^i \right) - \left(\sum_{i \leq N} y_i \cdot 10^i \right) = \sum_{i \leq N} (x_i - y_i) \cdot 10^i$$

Let the leftmost decimal place where $\vec{x}$ and $\vec{y}$ differ be the M -th place, so that $x_i = y_i$ for $i \in \{N, N-1, \dots, M+1\}$ and $x_M \neq y_M$. One of x_M or y_M is larger; without loss of generality suppose that $x_M > y_M$. Throwing away the initial zeroes, now the difference starts at the M -th place.

$$0 = \sum_{i \leq M} (x_i - y_i) \cdot 10^i = (x_M - y_M) \cdot 10^M + \sum_{i \leq (M-1)} (x_i - y_i) \cdot 10^i \quad (**)$$

On the right we've broken out the first term because we must show that $x_M = y_M + 1$ (as it does for the prototype $\vec{x} = \langle 1, 0, 0, \dots \rangle$ and $\vec{y} = \langle 0, 9, 9, \dots \rangle$). Consider the tail, $\sum_{i \leq (M-1)} (x_i - y_i) \cdot 10^i$. To make the sum come out to be zero this tail must offset $(x_M - y_M) \cdot 10^M$, which is a positive value since $x_M > y_M$. The largest offset that the tail can provide is if each of its $x_i - y_i$'s equals -9 . The geometric series formula gives this.

$$-9 \cdot 10^{M-1} - 9 \cdot 10^{M-2} + \dots = -9 \cdot 10^{M-1} \cdot \left[1 + \frac{1}{10} + \frac{1}{100} + \dots \right] = \frac{-9 \cdot 10^{(M-1)}}{1 - (1/10)} = \frac{-9 \cdot 10^{(M-1)}}{9/10} = -10^M$$

Therefore $x_M - y_M = 1$, since the tail cannot offset a larger difference.

- (c) Consider again the prior item's $(**)$, now that we have shown $x_M - y_M = 1$.

$$0 = 1 \cdot 10^M + \sum_{i \leq (M-1)} (x_i - y_i) \cdot 10^i$$

We want to show that every $x_i - y_i$ equals -9 . We will just do the $i = M - 1$ case, as the others are very similar.

If $x_{M-1} - y_{M-1} \geq -8$ then $(**)$ becomes this.

$$\begin{aligned} 0 &= 1 \cdot 10^M + (x_{M-1} - y_{M-1}) \cdot 10^{(M-1)} + \sum_{i \leq (M-2)} (x_i - y_i) \cdot 10^i \\ &\geq 10 \cdot 10^{(M-1)} - 8 \cdot 10^{(M-1)} + \sum_{i \leq (M-2)} (x_i - y_i) \cdot 10^i \\ &= 2 \cdot 10^{(M-1)} + \sum_{i \leq (M-2)} (x_i - y_i) \cdot 10^i \end{aligned}$$

As in the prior item the tail must offset the first term for the entire sum to come out negative. We get the largest tail by taking each $x_i - y_i$ to be -9 . Again apply the sum of an infinite geometric series.

$$-9 \cdot 10^{M-2} - 9 \cdot 10^{M-3} + \dots = -9 \cdot 10^{M-2} \cdot \left[1 + \frac{1}{10} + \frac{1}{100} + \dots \right] = \frac{-9 \cdot 10^{(M-2)}}{1 - (1/10)} = \frac{-9 \cdot 10^{(M-2)}}{9/10} = -10^{M-1}$$

The tail is not enough to offset the $2 \cdot 10^{(M-1)}$ so $x_{M-1} - y_{M-1} \geq -8$ is not possible.

V II.3.32

(A) These are the chains for S .

$s \in S$	Chain
0	... 0, a, 0, a ...
1	... 1, b, 1, b ...
2	... 2, d, 3, c, 2, d ...
3	— same as the chain for 2 —

These are the chains for T .

$t \in T$	Chain
a	... a, 0, a, 0 ...
b	... b, 1, b, 1 ...
c	... c, 2, d, 3, c, 2 ...
d	— same as the chain for c —

(B) The function f is one-to-one because if $f(s) = f(\hat{s})$ then $s + 1 = \hat{s} + 1$, and so $s = \hat{s}$. The function g is one-to-one in the same way.

The chain associated with $0 \in S$ is $0, 1, 2, 3 \dots$. It contains all of the elements of both sets, and so the chain is unique. It also has a first element, that is, there is no element of T that is $g^{-1}(0)$.

(C) If they are not already disjoint, if $S \cap T \neq \emptyset$, then we can just color all the elements of S green and all the elements of T gold and that will make them disjoint. More precisely, consider the set $\{0\} \times S$ of pairs $\langle 0, s \rangle$ along with the set $\{1\} \times T$ of pairs $\langle 1, t \rangle$. These are disjoint sets because any two elements differ in the first entry.

Obviously they correspond to the original sets S and T . So we can rephrase the result to a disjoint version that if both $|\{0\} \times S| \leq |\{1\} \times T|$ and $|\{1\} \times T| \leq |\{0\} \times S|$ then $|\{0\} \times S| = |\{1\} \times T|$.

(D) First we show that every element is in a unique chain. Fix some $x \in S \cup T$. Because we assume that the two sets are disjoint, the function operation that applies to x is determined, meaning that if $x \in S$ then for the chain we next consider $f(x)$ while if $x \in T$ then we are next considering $g(x)$. With that, all the elements of the chain to the right of x are determined. And, because the two functions are one-to-one, all of the element to the left are likewise also determined.

As to the four possibilities, either a chain is finite, that is, it repeats, or it is infinite. If it repeats then it is case (i). If it is infinite then either it does not have an initial element, and so is case (ii), or else it does. If it does have an initial element then either that element is from the set S , as in case (iii), or else that element is from the set T , as in case (iv).

(E) We will consider the correspondence for each chain type below. But first note that because each case defines h by using restrictions of the functions f and g , in each case h is well-defined. That is, never does any of these maps associate two different outputs with the same input.

A type (i) chain repeats after some number n of elements $s_0, t_0, s_1, t_1, \dots, s_{n-1}, t_{n-1}, s_n = s_0$ where $n > 0$. (It cannot repeat after a single element—it cannot be that $s_0 = t_0$ —because the sets S and T are disjoint.) We can therefore take the first element in this loop to be a member of S . The function f makes these associations.

$$s_0 \mapsto t_0 \quad s_1 \mapsto t_1, \quad \dots \quad s_{n-1} \mapsto t_{n-1}$$

This is clearly a correspondence among the elements of the chain. It is manifestly onto the set $\{t_0, t_1, \dots, t_{n-1}\}$ and it is one-to-one because the t_j 's are distinct, or else the chain would be shorter.

A type (iii) chain

$$s, f(s), g(f(s)), f(g(f(s))), \dots$$

associates even-indexed entries with the odd-indexed ones.

$$s \mapsto f(s) \quad g(f(s)) \mapsto f(g(f(s))), \quad \dots$$

That is, we are defining h in a way that takes members of S to members of T . Clearly this is a correspondence.

A type (ii) chain is similar.

$$\dots f^{-1}(g^{-1}(s)), g^{-1}(s), s, f(s), g(f(s)), f(g(f(s))), \dots$$

The function h associates these, following the action of f .

$$\dots f^{-1}(g^{-1}(s)) \mapsto g^{-1}(s) \quad s \mapsto f(s) \quad g(f(s)) \mapsto f(g(f(s))), \quad \dots$$

This association also is clearly a correspondence.

The one that takes some thought is a chain of type (iv).

$$t, g(t), f(g(t)), g(f(g(t))), \dots$$

We want the correspondence function h to go from S to T . Consider these associations.

$$g(t) \mapsto t \quad g(f(t)) \mapsto f(t), \quad \dots$$

Because the function g is one-to-one, its inverse is defined over its range. So taking these associations to define h gives a well-defined function. Like the chain types above, this also is clearly a correspondence.

V II.4.6 A Universal Turing machine is a regular machine, in that it is just another machine in the list $\mathcal{P}_0, \mathcal{P}_1, \dots$. Perhaps your friend meant something more like, “What does a Universal Turing machine do that a non-universal machine does not do?” Every machine does a job. Some machines expect a single input and then double it, some interpret their input as two numbers and then add them. A Universal Turing machine expects two inputs, e and x , and the output of this machine is the same as the output of the machine $\mathcal{P}_e$ on input x , including failing to halt if that machine fails to halt. It is universal in that any input-output behavior that can be mechanically produced can be produced by this one device.

V II.4.7 Every general purpose computer, including any standard laptop or cell phone, is essentially a Universal Turing machine.

Comment: Some observers object to the contention that a general purpose computer is essentially a Turing machine by saying that a laptop does not have unboundedly much memory. That's true. But also true is that such a machine can be hooked up to an external memory, perhaps the Cloud or perhaps a device allowing it to make marks on a tape, so that it is not restricted to the installed RAM.

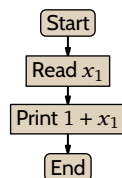
A person can go on to object that there are only so many elementary particles in the physical universe with which to make bits, and thus memory is not in the end unbounded. Perhaps that is true—it seems to match

current Physics, although we could worry back that we have not got enough time to exhaust what bits there are before the Big Bang becomes a Big Crunch, or whatever will happen.

However, in the section on Church's Thesis we noted that the definition of Turing machine makes precise the notion of intuitively computable. That is, the point of 'Turing machine' is that it is the right model in which to think about algorithms; it is a mathematical idealization. A laptop is a physical instantiation of the same notion.

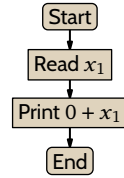
It is true that in a sense they cannot be the same because that would be a type error — the type of one is 'mathematical object' while the other is 'physical object'. In the same way, a carpenter's T-square is an instantiation of the mathematical ideal of a right angle. Asserting that T-squares are illegitimate in that no one ever will make one that is a perfect right angle to more decimal places than there are elementary particles in the universe seems to have lost track of the reasons why we find it useful to think about models. The essence of the instantiation is the idealization.

- V **II.4.8** Yes. If $\mathcal{P}_{e_0}$ and $\mathcal{P}_{e_1}$ are universal then giving the first one the inputs e_1 and x will result in an input-output behavior exactly like the second. Likewise, giving $\mathcal{P}_{e_0}$ the inputs e_0 and x will result in the machine behaving just like itself.
- V **II.4.9** The universal machine does not use its states to simulate the states of the other machine. Rather, it has codes for those and it manipulates those codes. In short, it does the simulation in software so the number of states in the universal machine is not relevant.
- V **II.4.10** Yes, by Lemma 2.18 every computable function has infinitely many indices. So if $\mathcal{P}_e$ is universal, then the associated function ϕ_e has infinitely many unequal indices: $\phi_e = \phi_{e_0} = \phi_{e_1} = \dots$. The machines $\mathcal{P}_{e_0}, \mathcal{P}_{e_1}, \dots$ are different but they have the same input/output behavior so all are universal.
- V **II.4.11** We write $f_{i,a}$ for the function produced by freezing x_i to the value a .
- (A) The resulting one-variable function is $f_{0,4}(x_1) = 12 + 4x_1$.
- (B) The one-variable function is $f_{0,5}(x_1) = 15 + 5x_1$.
- (C) The one-variable function is $f_{1,0}(x_0) = 3x_0$.
- V **II.4.12** We use $\hat{f}$ for the name of each function.
- (A) The resulting two-variable function is $\hat{f}(x_1, x_2) = 1 + 2x_1 + 3x_2$.
- (B) The result is the two-variable function $\hat{f}(x_1, x_2) = 2 + 2x_1 + 3x_2$.
- (C) This gives $\hat{f}(x_1, x_2) = a + 2x_1 + 3x_2$.
- (D) This results in the one-variable function $\hat{f}(x_2) = 11 + 3x_2$.
- (E) The result is $\hat{f}(x_2) = a + 2b + 3x_2$.
- V **II.4.13** From the machine's flowchart we have the associated function $\phi_{e_0}(x_0, x_1) = x_0 + x_1$.
- (A) The two subscripts on $s_{1,1}$ mean that we will freeze the first input and leave the second one as a variable. So $\phi_{s_{1,1}(e_0,1)}$ will be a one-input function. The arguments e_0 and 1 indicate that we are working on Turing machine e_0 , the one whose flow chart is given in the problem statement, and that the input is frozen to be $x_0 = 1$. This flowchart gives a sense of the result.



The function is $\phi_{s_{1,1}(e_0,1)}(x_1) = 1 + x_1$.

- (B) The values are $\phi_{s_{1,1}(e_0,1)}(0) = 1$, $\phi_{s_{1,1}(e_0,1)}(1) = 2$, and $\phi_{s_{1,1}(e_0,1)}(2) = 3$.
- (C) The notation $s_{1,1}(e_0, 0)$ means that we work on Turing machine e_0 , that we freeze the first input to $x_0 = 0$, and that we leave the second input as a variable. This flowchart gives the idea.

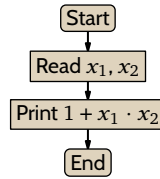


The function is $\phi_{s_{1,1}(e_{0,1})}(x_1) = 0 + x_1 = x_1$.

(D) We have $\phi_{s_{1,1}(e_{0,0})}(0) = 0$, $\phi_{s_{1,1}(e_{0,0})}(1) = 1$, and $\phi_{s_{1,1}(e_{0,0})}(2) = 2$.

V II.4.14 The machine's flowchart gives that the associated function is $\phi_{e_0}(x_0, x_1, x_2) = x_0 + x_1 \cdot x_2$.

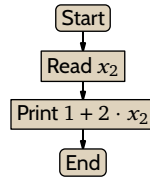
(A) The notation $s_{1,2}(e_0, 1)$ indicates that we work on $\mathcal{P}_{e_0}$, that we freeze the first input to $x_0 = 1$, and that we leave the second and third inputs as variables. This flowchart sketches the behavior.



The function is $\phi_{s_{1,2}(e_{0,1})}(x_1, x_2) = 1 + x_1 \cdot x_2$.

(B) We have $\phi_{s_{1,2}(e_{0,1})}(0, 1) = 1$, $\phi_{s_{1,2}(e_{0,1})}(1, 0) = 1$, and $\phi_{s_{1,2}(e_{0,1})}(2, 3) = 7$.

(C) The $s_{2,1}(e_0, 1, 2)$ says that we start with $\mathcal{P}_{e_0}$, that we freeze the first input to $x_0 = 1$ and the second input to $x_1 = 2$, and leave the third input as a variable. This outlines the result.



The function is $\phi_{s_{2,1}(e_{0,1,2})}(x_2) = 1 + 2 \cdot x_2$.

(D) The values are $\phi_{s_{2,1}(e_{0,1,2})}(0) = 1$, $\phi_{s_{2,1}(e_{0,1,2})}(1) = 3$, and $\phi_{s_{2,1}(e_{0,1,2})}(2) = 5$.

V II.4.15 The machine's flowchart says that the associated function is this.

$$\phi_{e_0}(x_0, x_1) = \begin{cases} x_1 & \text{-- if } x_0 \leq 1 \\ \uparrow & \text{-- otherwise} \end{cases}$$

(A) $\phi_{s_{1,1}(e_{0,0})}(x_1) = x_1$

(B) $\phi_{s_{1,1}(e_{0,0})}(5) = 5$

(C) $\phi_{s_{1,1}(e_{0,1})}(x_1) = x_1$

(D) $\phi_{s_{1,1}(e_{0,1})}(5) = 5$

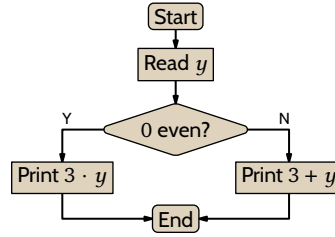
(E) $\phi_{s_{1,1}(e_{0,2})}(x_1) \uparrow$

(F) $\phi_{s_{1,1}(e_{0,2})}(5) \uparrow$

II.4.16 From the machine's flowchart we get that this is the associated function.

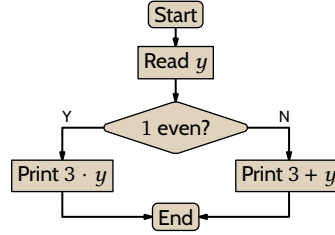
$$\phi_{e_0}(x_0, x_1, y) = \begin{cases} x_1 \cdot y & \text{-- if } x_0 \text{ is even} \\ x_1 + y & \text{-- otherwise} \end{cases}$$

(A) This function $\mathcal{P}_{s(e_{0,0,3})}$ has $x_0 = 0$ so it is even.



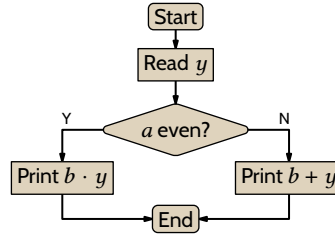
Thus the function is $\phi_{s(e_0,0,3)}(y) = 3y$. With that, $\phi_{s(e_0,0,3)}(5) = 15$.

(B) This function has $x_0 = 1$, which is odd. This flowchart sketches $\mathcal{P}_{s(e_0,1,3)}$.



The computed function is $\phi_{s(e_0,1,3)}(y) = 3 + y$ and consequently $\phi_{s(e_0,1,3)}(5) = 8$.

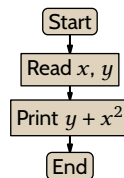
(c) This is the flowchart sketching $\mathcal{P}_{s(e_0,a,b)}$.



The distinction between this and the flowchart in the question statement is that here the machine does not read in x_0 or x_1 . The two a and b are parameters, which are hard-coded into the program body. This is a family of functions parametrized by a and b , and the indices of these functions are uniformly computable from e_0 , a , and b , using the s - m - n function s .

$$\phi_{s(e_0,a,b)}(y) = \begin{cases} b \cdot y & \text{-- if } a \text{ is even} \\ b + y & \text{-- otherwise} \end{cases}$$

- V **II.4.17** Start with the function $\psi: \mathbb{N}^2 \rightarrow \mathbb{N}$ defined by $\psi(x, y) = y + x^2$. It is intuitively computable because it is computed by the machine sketched by this flowchart.

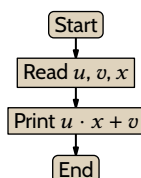


By Church's Thesis there is a Turing machine that computes it. Let that machine have index e_0 , that is, suppose that $\psi(x, y) = \phi_{e_0}(x, y)$. By the s - m - n theorem there is a family of computable functions parametrized by n , $\phi_{s_{1,1}(e_0,n)}$, with the desired behavior, $\phi_{s_{1,1}(e_0,n)}(y) = y + n^2$. Take $g(n)$ to be $s_{1,1}(e_0, n)$ (the e_0 is not a variable, it is the fixed index of the above machine).

- V **II.4.18** Start with the function $\psi(u, v, x) = ux + v$. It is intuitively computable as sketched by this flowchart.

```
;; Return the first character of the file (raise exception if file empty)
(define (first-char fn)
  (let ([contents (port->string (open-input-file fn) #:close? #t)])
    (string-ref contents 0)))
```

FIGURE 17, FOR QUESTION II.4.21: Racket code that returns the first character of a file.



By Church's Thesis there is a Turing machine that computes it. Let that machine have index e_0 , that is, suppose that $\psi(u, v, x) = \phi_{e_0}(u, v, x)$. The s - m - n theorem gives a family of computable functions $\phi_{s_{2,1}(e_0, m, b)}$ parametrized by m and b with the desired behavior, $\phi_{s_{2,1}(e_0, m, b)}(x) = m \cdot x + b$. Take $g(m, b)$ to be $s_{2,1}(e_0, m, b)$ (note that e_0 is not a variable, it is the fixed index of the Turing machine from the flowchart).

V II.4.19

- (A) This gives the same result as $\phi_{e_1}(5)$, that is, it converges and gives 20.
- (B) This returns 4 times the value of $\text{cantor}(e_0, 5)$. We don't know the value of e_0 so that's the most that we can say.
- (C) By the first item, this returns $\phi_{e_0}(20)$. Because $\text{pair}(20) = \langle 5, 0 \rangle$ this is the value of $\phi_5(0)$, whatever that is.

V II.4.20

- (A) Because $\text{pair}(4) = \langle 1, 1 \rangle$, this returns the same as $\phi_1(1)$, whatever that is.
- (B) This gives the same result as $\phi_{e_1}(4)$, that is, it gives 6.
- (C) This returns the value of $\phi_4(e_1)$, whatever that is.
- (D) This returns $2 + \text{cantor}(e_0, 4)$. The value of $\text{cantor}(e_0, 4)$ depends on e_0 , which we don't know. The most we can say is that $\text{cantor}(e_0, 4) = e_0 + (e_0 + 4)(e_0 + 5)/2$.

V II.4.21 Figure 17 on page 67 has Racket code. Here is what it does when run on its own source.

```
> (first-char "first-char.rkt")
#\#
```

V II.4.22 Figure 18 on page 68 has some Racket code. Here is what it does when run on its own source.

```
jim@millstone:~/computing/src/scheme/background$ ./self-modifying.rkt
Counter=11
jim@millstone:~/computing/src/scheme/background$ ls -l self-modifying.rkt
-rw-rw-r-- 1 jim jim 1463 Aug 28 10:17 self-modifying.rkt
jim@millstone:~/computing/src/scheme/background$ chmod u+x self-modifying.rkt
jim@millstone:~/computing/src/scheme/background$ ./self-modifying.rkt
Counter=12
```

And, after being run, the third line of the source now says `(define COUNTER 12)`.

V II.5.8 We have shown that there are tasks that no computer can do. The Halting problem is a concrete example of such a task. (We will see many more.)

As to why this task is of any interest, the point of the Turing machine definition is to give a place on which to model running algorithms. Whether the algorithms we are modeling will halt, even when given an unbounded chance to do so, is an important question.

V II.5.9 Wrong. There is such a function. But it is not computable; there is no Turing machine whose behavior is that function.

V II.5.10

- (A) The first is right. If we can find the store then we can get bread.
- (B) The first is right. If we can exhibit a factor then we can show that the number is nonprime.
- (C) The second.

```

#!/usr/bin/env racket
#lang racket
(define COUNTER 11)

;; Get list of file lines, where the counter defined in the third line is incremented
(define (intake fn)
  (let* ([contents (port->string (open-input-file fn) #:close? #t)]
        [lines-in (string-split contents "\n")]
        [third-line-in (string-split (third lines-in))]
        [later-lines-in (cdr (cdr (cdr lines-in)))]
        [counter (add1 (string->number (string-trim (third third-line-in) ")))])
    [third-line-out (string-join (list "(define"
                                       "COUNTER"
                                       (number->string counter)
                                       ")")
                                " "
                                #:before-last "")]
    [lines-out (append (list (first lines-in)
                             (second lines-in)
                             third-line-out)
                       later-lines-in)])
  lines-out))

;; Output the strings in a list to the named file
(define (update fn str-list)
  (let ([out-port (open-output-file fn #:exists 'replace #:permissions #o664)]) ; permit rw-rw-r--
    (write-string (string-join str-list "\n") out-port)
    (close-output-port out-port)))

;; When this file is called from the command line it runs these commands
(update "self-modifying.rkt" (intake "self-modifying.rkt"))
(displayln (~a "Counter=" COUNTER))

```

FIGURE 18, FOR QUESTION II.4.22: Racket code that is self-modifying.

- V **II.5.11** The unsolvability results do not say that you cannot prove that a particular machine does a particular thing. The given machine does indeed fail to halt on all inputs and it is indeed obvious. Instead the results are about uniformity: they say that no machine can take in any index e and correctly determine whether $\mathcal{P}_e$ does that thing.
- V **II.5.12** First, to detect a loop we must look for a repeated configuration, not just a repeated state-character pair. But beyond that, failing to halt and repeating a configuration are not the same thing. Machines can run forever without repeating a configuration. Here is one.

$$\mathcal{P} = \{q_0B1q_0, q_01Rq_0\}$$

It keeps writing a 1 and then shifting right.

The Halting problem is not the problem of detecting whether or not a machine has an infinite loop. Rather, the Halting problem is to detect whether the machine will halt or fail to halt. It could fail to halt because there is a loop but there are other, more complicated, ways to fail to halt.

- V **II.5.13** Sure, we could get the runtime environment to do that. However, ‘fails to halt’ and ‘in an infinite loop’ are not the same. Consider the code below.

```

;; Run forever without repeating a configuration
(define (go-up)
  (do ([i 0] (add1 i)])
      (#f (displayln "routine halt"))
      (displayln (~a "i=" i))))

```

The runtime environment described would not flag this routine as being in an infinite loop but it does not halt.

```

> i=0
> i=1
> i=2
> :

```

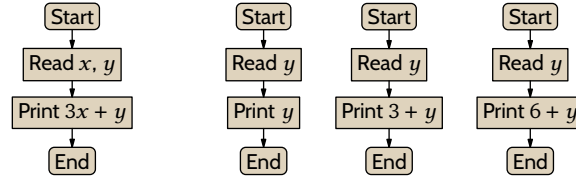


FIGURE 19, FOR QUESTION II.5.17: The machine associated with the function $f = \phi_{e_0}$ and the machines associated with $\phi_{s(e_0, x)}$ for $x = 0$, $x = 1$, and $x = 2$.

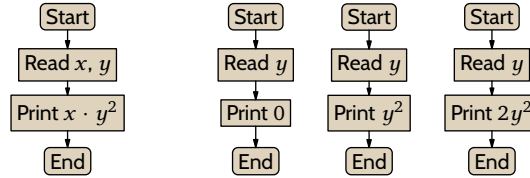


FIGURE 20, FOR QUESTION II.5.17: The machine associated with the function $f = \phi_{e_0}$ and machines associated with $\phi_{s(e_0, x)}$ for $x = 0$, $x = 1$, and $x = 2$.

It only stops when the user forces it to stop.

V II.5.14 No, it is not an unsolvable problem. Either f does halt on all inputs or else it doesn't. That's two possibilities. So, as in the book's discussion of the Halting problem, overall the problem is solved by some computer program, either the one that prints 'yes' or the one that prints 'no'. (It is perhaps a unsettling that we don't know which of the two is true but nonetheless both are computable.)

V II.5.15

- (A) False. Such a function exists. Rather, we say that the problem is unsolvable because there is no such mechanically computable function.
- (B) False. Church's Thesis asserts that Turing machines are a maximally-powerful model of computation. The existence of unsolvable problems is inherent in effective computation.

II.5.16 Any finite set is computable, so yes it is computable. There are subtleties in conflating 'computable' with 'knowable'.

V II.5.17

- (A) The function $f(x, y) = 3x + y$ is intuitively computable, as sketched on the left of Figure 19 on page 69. Church's Thesis gives that there is a Turing machine with that behavior; let that machine have index e_0 , so that $f(x, y) = \phi_{e_0}(x, y)$ for all $x, y \in \mathbb{N}$. Applying the s - m - n Theorem to parametrize x gives this computable family: $\phi_{s(e_0, x)}(y) = 3x + y$, which for any x is a one-input function. Flowcharts sketching some associated machines are on the right side of Figure 19 on page 69.
- (B) The function $f(x, y) = xy^2$ is intuitively computable, as in Figure 20 on page 69. Church's Thesis gives that there is a Turing machine $\mathcal{P}_{e_0}$ with that behavior, so that $f(x, y) = \phi_{e_0}(x, y)$ for all $x, y \in \mathbb{N}$. Applying the s - m - n Theorem to parametrize x gives $\phi_{s(e_0, x)}(y) = xy^2$, which is a family of one-input functions. Flowcharts sketching some associated machines are on the right side of Figure 20 on page 69.
- (C) The function f is intuitively computable, as sketched on the left of Figure 21 on page 70. Church's Thesis gives that there is a Turing machine $\mathcal{P}_{e_0}$ with that behavior, so that $f(x, y) = \phi_{e_0}(x, y)$ for all $x, y \in \mathbb{N}$. Apply the s - m - n Theorem to parametrize x . The resulting family of one-input functions is this.

$$\phi_{s(e_0, x)}(y) = \begin{cases} x & \text{-- if } x \text{ is odd} \\ 0 & \text{-- otherwise} \end{cases}$$

Flowcharts sketching associated machines are on the right side of Figure 21 on page 70.

V II.5.18 All three have the same answer. Running a Turing machine for a fixed number of steps is decidable (and in the third item, for any given input e , the number of steps is fixed).

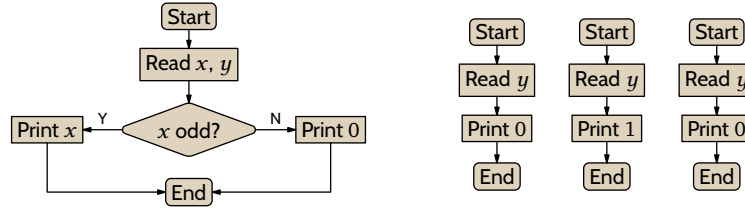


FIGURE 21, FOR QUESTION II.5.17: A sketch of the machine for the function $f = \phi_{e_0}$ and of the machines associated with $\phi_{s(e, x_0)}$ when $x = 0$, $x = 1$, and $x = 2$.

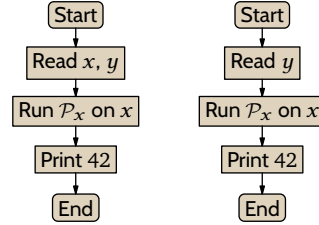


FIGURE 22, FOR QUESTION II.5.19: The machine associated with the function $\psi = \phi_e$ and the machine for $\phi_{s(e, x)}$.

V II.5.19 We will show that this function is not mechanically computable.

$$\text{total_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(y) \downarrow \text{ for all } y \\ 0 & \text{-- otherwise} \end{cases}$$

The function

$$\psi(x, y) = \begin{cases} 42 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

is intuitively computable by the flowchart on the left of Figure 22 on page 70. Thus Church's Thesis says that there is a Turing machine that computes it; let that machine have index e_0 , so that $\psi(x, y) = \phi_{e_0}(x, y)$. The s - m - n Theorem gives a family of one-input functions parametrized by x . The generic such machine is sketched on the right of the figure. Notice that $\phi_x(x) \downarrow$ if and only if $\text{total_decider}(s(e_0, x)) = 1$.

Since s is computable, if we could mechanically compute total_decider then we could mechanically solve the Halting problem. We cannot mechanically solve the Halting problem, so we cannot mechanically compute total_decider .

V II.5.20 We will show that this function is not mechanically computable.

$$\text{square_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(y) = y^2 \text{ for all } y \\ 0 & \text{-- otherwise} \end{cases}$$

First consider this function.

$$\psi(x, y) = \begin{cases} y^2 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

The flowchart on the left of Figure 23 on page 71 shows that ψ is intuitively computable. So by Church's Thesis there is a Turing machine that computes it. Let that machine have index e_0 . Apply the s - m - n Theorem to get a family of functions parametrized by x . The matching machine is on the right of the figure. Then $\phi_x(x) \downarrow$ if and only if $\text{square_decider}(s(e_0, x)) = 1$.

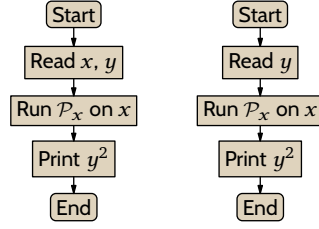


FIGURE 23, FOR QUESTION II.5.20: The machine associated with the function $\psi = \phi_{e_0}$ and the machine associated with $\phi_{s(e_0, x)}$.

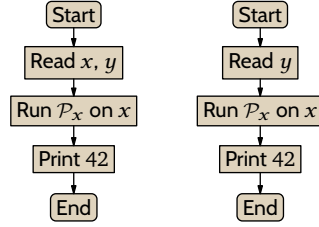


FIGURE 24, FOR QUESTION II.5.21: The machine associated with the function $\psi = \phi_{e_0}$ and the machine for $\phi_{s(e_0, x)}$.

Because of that equivalence, since the function s is computable, if we could mechanically compute `square_decider` then we could mechanically solve the Halting problem. But we can't mechanically solve the Halting problem and hence we can't mechanically compute `square_decider`.

V II.5.21 We will show that this is not computable.

$$\text{same_value_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(y) = \phi_e(y+1) \text{ for some } y \\ 0 & \text{-- otherwise} \end{cases}$$

The function

$$\psi(x, y) = \begin{cases} 42 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

is intuitively computable by the flowchart on the left of Figure 24 on page 71. So by Church's Thesis there is a Turing machine that computes ψ ; let that machine have index e_0 , so $\psi(x, y) = \phi_{e_0}(x, y)$. Apply the s - m - n Theorem to get a family of functions parametrized by x , the $\phi_{s(e_0, x)}$, where the associated machines are sketched on the right of the figure. These machines produce the function with constant output 42 if and only if the machine gets through the middle box. That is, $\phi_x(x) \downarrow$ if and only if $\text{same_value_decider}(s(e_0, x)) = 1$.

Since the function s is computable, by the prior sentence if we could mechanically compute `same_value_decider` then we could mechanically solve the Halting problem. But that we cannot do.

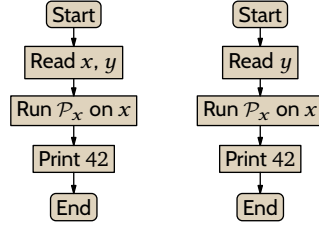
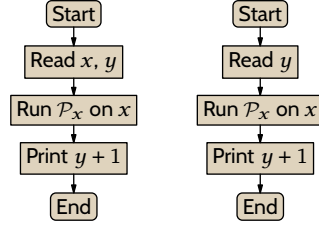
V II.5.22 We will show that this function is not computable; no Turing machine has this input-output behavior.

$$\text{diverges_on_five_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(5) \uparrow \\ 0 & \text{-- otherwise} \end{cases}$$

The function

$$\psi(x, y) = \begin{cases} 42 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

is intuitively computable by the flowchart on the left of Figure 25 on page 72. So Church's Thesis says that there is a Turing machine that computes it. Let that machine have index e_0 , meaning that $\psi(x, y) = \phi_{e_0}(x, y)$.

FIGURE 25, FOR QUESTION II.5.22: The machine associated with $\psi = \phi_{e_0}$ and the machine for $\phi_{s(e_0, x)}$.FIGURE 26, FOR QUESTION II.5.24: Machine associated with $\psi = \phi_{e_0}$ and machine associated with $\phi_{s(e_0, x)}$.

The s - m - n Theorem gives a family of functions parametrized by x , whose machines are sketched on the right of the same figure. Then $\phi_x(x) \downarrow$ if and only if $\text{diverges_on_five_decider}(s(e_0, x)) = 0$. (We could also state it as: $\phi_x(x) \downarrow$ if and only if $1 - \text{diverges_on_five_decider}(s(e_0, x)) = 1$.) Because the function s is mechanically computable, this means that if we could mechanically compute $\text{diverges_on_five_decider}$ then we could mechanically solve the Halting problem. But we cannot mechanically solve the Halting problem, and therefore we cannot mechanically compute $\text{diverges_on_five_decider}$.

V II.5.23 We want to show that this function is not computable.

$$\text{diverge_on_odds_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(y) \uparrow \text{ for all odd } y \in \mathbb{N} \\ 0 & \text{-- otherwise} \end{cases}$$

The argument given in the prior item will do again here.

V II.5.24 We will show that no Turing machine has this input-output behavior.

$$\text{successor_decider}(e) = \begin{cases} 1 & \text{-- if } \phi_e(y) = y + 1 \text{ for all } y \in \mathbb{N} \\ 0 & \text{-- otherwise} \end{cases}$$

The function

$$\psi(x, y) = \begin{cases} y + 1 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

is intuitively computable by the flowchart on the left of Figure 26 on page 72. So Church's Thesis says that there is a Turing machine that computes it. Let that machine have index e_0 , meaning that $\psi(x, y) = \phi_{e_0}(x, y)$. The s - m - n Theorem gives a family of functions parametrized by x , with machines sketched on the right of that figure. Then $\phi_x(x) \downarrow$ if and only if $\text{successor_decider}(s(e_0, x)) = 1$. The function s is mechanically computable, so if we could mechanically compute successor_decider then we could mechanically solve the Halting problem. But we can't do that and hence we cannot mechanically compute successor_decider .

V II.5.25 We will show that no Turing machine has this input-output behavior.

$$\text{diverges_on_twice_input_decider}(e) = \begin{cases} 1 & \text{-- when for some } y \in \mathbb{N}, \text{ both } \phi_e(y) \uparrow \text{ and } \phi_e(2y) \uparrow \\ 0 & \text{-- otherwise} \end{cases}$$

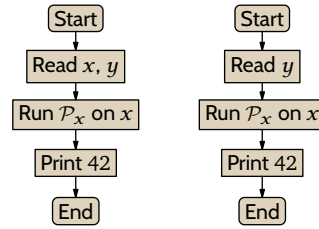


FIGURE 27, FOR QUESTION II.5.25: Outline of the machine for $\psi = \phi_{e_0}$ and the machine for $\phi_{s(e_0, x)}$.

The function

$$\psi(x, y) = \begin{cases} 42 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

is intuitively computable by the flowchart on the left of Figure 27 on page 73. So Church's Thesis says that there is a Turing machine that computes it. Let that machine have index e_0 . Apply the s - m - n Theorem to get a family of functions parametrized by x , whose machines are sketched on the right of the figure. Then $\phi_x(x) \downarrow$ if and only if $\text{diverges_on_twice_input_decider}(s(e_0, x)) = 0$. (We could instead write $\phi_x(x) \downarrow$ if and only if $1 - \text{diverges_on_twice_input_decider}(s(e_0, x)) = 1$.) So if we could effectively compute $\text{converges_on_twice_input_decider}$ then we could effectively solve the Halting problem. But we cannot effectively solve the Halting problem, so we cannot effectively compute $\text{converges_on_twice_input_decider}$.

- V II.5.26 The second is solvable: just run the machine for a thousand steps. The first is unsolvable. For a proof we can use the blanks of the prior exercise: (1) halts_on_153 , (2) $\phi_e(153) \downarrow$, (3) 42, (4) Print 42.
- V II.5.27 This problem is solvable. Because x and y can't be bigger than c , we just write a program that inputs $\text{cantor}(x, y)$ for $0 \leq x, y \leq c$, applies the unpairing function to get x and y , and then plugs them into $ax + by$, to see if the result equals c . That's a bounded search and it halts.
- V II.5.28
 - (A) Unsolvable. (Verification is like many other exercises in this section.)
 - (B) Unsolvable. (Verification is like many other of these exercises.)
 - (C) Cannot tell; whether this is solvable or not depends on the specifics of the way Turing machines are counted, that is, it depends on the details of $\mathcal{P}_4$.
 - (D) Solvable. From e we can determine the set of five-tuples $\mathcal{P}_e$. It is finite so we can just search for the desired instruction.
- V II.5.29
 - (A) (1) $\text{same_output_as_input}$, (2) there exists $y \in \mathbb{N}$ such that $\phi_e(y) = y$, (3) y , (4) Print y .
 - (B) (1) outputs_42 , (2) there exists $y \in \mathbb{N}$ such that $\phi_e(y) = 42$, (3) 42, (4) Print 42.
 - (C) (1) ever_adds_2 , (2) there exists $y \in \mathbb{N}$ such that $\phi_e(y) = y + 2$, (3) $y + 2$ (4) Print $y + 2$.
- V II.5.30 Suppose that we could compute K_0 . That is, suppose that there is a Turing machine $\mathcal{P}_{e_0}$ such that $\phi_{e_0}(\langle e, x \rangle) = 1$ if $\phi_e(x) \downarrow$ and $\phi_{e_0}(\langle e, x \rangle) = 0$ if $\phi_e(x) \uparrow$. Then we could solve the Halting problem, we could determine membership in K , just by applying $\mathcal{P}_{e_0}$ to the input $\langle e, e \rangle$.
- V II.5.31
 - (A) One such machine takes input e , uses a Universal Turing machine to run $\mathcal{P}_e$ on input e , and then exits. Figure 28 on page 74 has a flowchart. To see that computing the domain of the associated computable function is not mechanically computable, observe that if we could compute its domain then we could compute the solution to the Halting problem.
 - (B) For any Turing machine $\mathcal{P}_e$ and any fixed input y , The answer is either "yes it halts on y " or "no it does not." We can write two programs, one for each, and one of the programs is right.
- V II.5.32
 - (A) This is solvable. The index e is computationally equivalent to the source because we are using machine numberings that are acceptable in the sense defined on page 71. So given e , start by translating it to the

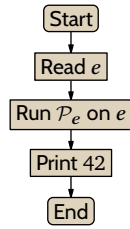


FIGURE 28, FOR QUESTION II.5.31: A single machine whose domain is undecidable.

set of Turing machine instructions, $\mathcal{P}_e$. Finish by counting the number of states that appear in that Turing machine source.

(B) This is not solvable; this function

$$\text{halts_on_empty_decider}(e) = \begin{cases} 1 & \text{-- if } \mathcal{P}_e \text{ halts when started on an empty tape} \\ 0 & \text{-- otherwise} \end{cases}$$

is not mechanically computable. For, consider this function.

$$\psi(x) = \begin{cases} 42 & \text{-- if } \phi_x(x) \downarrow \\ \uparrow & \text{-- otherwise} \end{cases}$$

It is intuitively mechanically computable and so by Church's Thesis there is a Turing machine that computes it. Let the index of that machine be e_0 . Apply the s - m - n theorem to parametrize x , giving a uniformly computable family of functions $\phi_{s(e_0, x)}$. (The $\phi_{s(e_0, x)}$'s are functions of no input; that's fine as the associated Turing machine simply does not read any input from the tape.)

Then $\phi_x(x) \downarrow$ if and only if $\text{halts_on_empty_decider}(s(e_0, x)) = 1$. If $\text{halts_on_empty_decider}$ were mechanically computable then we could mechanically solve the Halting problem. Therefore $\text{halts_on_empty_decider}$ is not mechanically computable.

(c) This is solvable. From the index e , use a Universal Turing machine to run $\mathcal{P}_e$ on input e for one hundred steps. By the end either it has halted or it hasn't.

V **II.5.33** Yes. If K were finite then it would be computable, since any finite set is computable. For one thing, we can decide membership in a finite set with a sequence of if-then-else statements.

V **II.5.34** False; it is indeed infinite but it is uncountably infinite. There are uncountably infinitely many problems, that is, functions from $\mathbb{N}$ to $\mathbb{N}$. But there are only countably many Turing machines and hence countably many solvable problems. Therefore there are uncountably many unsolvable problems.

V **II.5.35** Either $\mathcal{P}_e$ does halt on all inputs, or it doesn't. So there are two possibilities, both of which are computable (either the answer is 'yes' or the answer is 'no').

Note the contrast with the unsolvable problem: given $e \in \mathbb{N}$ decide whether $\mathcal{P}_e$ halts on all inputs. As asked, this exercise does not look for an algorithm that is uniform in the index e .

V **II.5.36** Consider a Turing machine that, for any input, ignores the input and just loops, searching for an even number greater than two that is not the sum of two primes. If on any input, that Turing machine halts then the conjecture is false. Otherwise the conjecture is true. So if we could solve the Halting problem then we could apply it to this machine and settle the question.

XKCD's take on Goldbach is Figure 29 on page 75.

V **II.5.37** We can write a program that takes input, ignores it, and performs a loop that searches for a solution larger than 7, halting if it finds one. With a solution to the Halting problem we could check whether that program halts, and thereby settle the problem.

V **II.5.38** The set of functions $f: \mathbb{N} \rightarrow \mathbb{N}$ is the disjoint union of two subsets: the functions that are computable and the functions that are not. We know that the set of computable functions is countable. Because the union of two countable sets is countable, if the set of non-computable functions were also countable then there

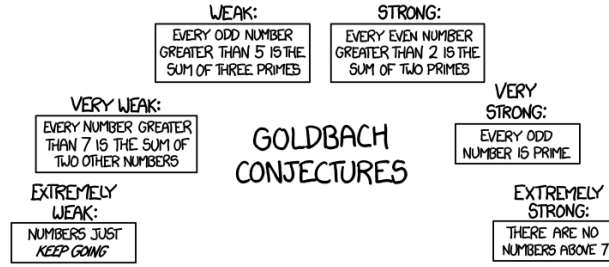


FIGURE 29, FOR QUESTION II.5.36: Goldbach's conjectures. (Courtesy xkcd.com)

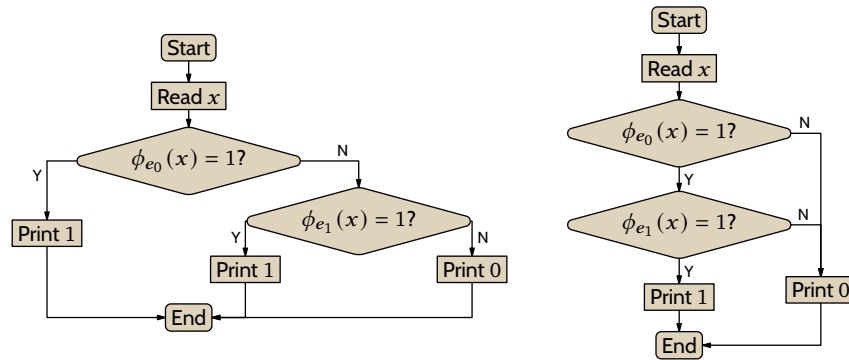


FIGURE 30, FOR QUESTION II.5.40: Sketches of machines to decide union and intersection of decidable languages.

would be countably many functions overall, which there are not. Thus the set of non-computable functions is not countable.

II.5.39 Fix an element $k \in K$. The range of the following function $f: \mathbb{N}^2 \rightarrow \mathbb{N}$ is K .

$$f(x, n) = \begin{cases} x & \text{-- if } \mathcal{P}_x \text{ with input } x \text{ halts in } n \text{ steps} \\ k & \text{-- otherwise} \end{cases}$$

Clearly the function is computable and converges on all input pairs $x, n \in \mathbb{N}^2$. For all elements of K there is an n where x appears in the range. And, for the same reason, only the elements of K appear in the range.

V II.5.40 Start with decidable languages $\mathcal{L}_0$ and $\mathcal{L}_1$. Because they are decidable, there are Turing machines $\mathcal{P}_{e_0}$ and $\mathcal{P}_{e_1}$ that decide membership.

$$\phi_{e_0}(x) = \begin{cases} 1 & \text{-- if } x \in \mathcal{L}_0 \\ 0 & \text{-- if } x \notin \mathcal{L}_0 \end{cases} \quad \phi_{e_1}(x) = \begin{cases} 1 & \text{-- if } x \in \mathcal{L}_1 \\ 0 & \text{-- if } x \notin \mathcal{L}_1 \end{cases}$$

Note that these machines halt on all inputs.

- A sketch of the machine that decides membership in $\mathcal{L}_0 \cup \mathcal{L}_1$ is on the left in Figure 30 on page 75. A number is an element of the union if it is a member of either set. So, given x , this machine first runs $\mathcal{P}_{e_0}$ on x . If that returns 1 then x is a member of the union and so the new machine outputs 1. Otherwise the new machine runs $\mathcal{P}_{e_1}$ on x and returns the result.
- A sketch of the machine that decides membership in $\mathcal{L}_0 \cap \mathcal{L}_1$ is on the right in the same figure. A number is an element of the intersection if it is a member of both sets. So, given x , this machine first runs $\mathcal{P}_{e_0}$ on x . If that returns 0 then x is not a member of the intersection, so the new machine outputs 0. Otherwise the new machine runs $\mathcal{P}_{e_1}$ on x and returns the result.
- Complement is straightforward. For the complement of $\mathcal{L}_0$ the new machine just runs $\mathcal{P}_{e_0}$ and subtracts the result from 1.

- V **II.6.10** Rice's Theorem does not say that everything is impossible. For instance, it does not say that writing a routine that implements $f(x) = x^2$ is impossible.

What it does say is impossible is to write a routine that, given source code, will correctly decide whether that source, no matter how clever or for that matter obfuscated, implements f . (Such a decider is of interest because we could apply it to the routine in the first paragraph to know for sure that it works correctly in all cases.)

- V **II.6.11** No, it is not an index set. We can write two Turing machines with the same behavior, say, on any input x they return output x . One of them takes only a few steps to do this (the trivial Turing machine $\mathcal{P} = \{ \}$ does it in no steps), while the other takes 101 steps. The index of the second set is a member of $\mathcal{I}$ but the index of the first set is not.
- V **II.6.12** The set of indices $e \in \mathbb{N}$ such that $\emptyset \subseteq \{x \mid \phi_e(x) \downarrow\}$ is all of $\mathbb{N}$. This problem is solvable, although the solution is trivial in that for any e , the answer is 'yes'.

V **II.6.13**

- (A) True, this is about the input-output behavior of the machine. If machines have the same input/output behavior and one halts on input 5 then the other must also.
- (B) False, we can make two machines with the same behavior but where one has four instructions and the other does not. For instance we can take the four-instruction machine and make a new one by adding unreachable code, dummy instructions that start with states not referenced by the original four instructions.
- (C) True. If two machines compute the same function and the function of the first is $x \mapsto 2x$ then that is also the function of the second.

- V **II.6.14** In each case we will argue that there are two machines, $\mathcal{P}$ and $\hat{\mathcal{P}}$, with the same input/output behavior, $\phi = \hat{\phi}$, but where one machine has the property and the other does not.

- (A) The trivial Turing machine $\mathcal{P} = \{ \}$ halts in zero steps and has the behavior $\phi(x) = x$. We can easily make another machine with the same behavior but which takes more than one hundred steps.

$$\hat{\mathcal{P}} = \{q_0BBq_1, q_011q_1, q_1BBq_2, q_111q_2, \dots, q_{100}BBq_{101}, q_{100}11q_{101}\}$$

- (B) As with the prior item, the trivial machine uses zero tape cells on input 0. This machine visits one hundred and one cells on input 0.

$$\hat{\mathcal{P}} = \{q_0BLq_1, q_011q_{102}, q_1BLq_2, q_11Lq_2, \dots, q_{100}BLq_{101}, q_{100}1Lq_{101}\}$$

Both machines have the same behavior, $\phi(x) = x$.

- (C) The trivial machine does not contain any instruction with state q_{10} . The machine

$$\hat{\mathcal{P}} = \{q_0BBq_1, q_011q_1, q_1BBq_2, q_111q_2, \dots, q_{10}BBq_{11}, q_{10}11q_{11}\}$$

has the same behavior and contains an instruction with that state.

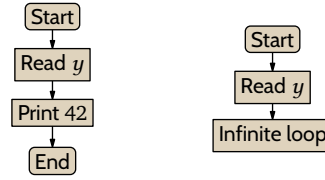
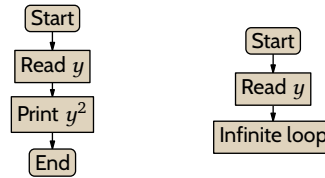
- V **II.6.15** There are many ways to describe the empty set. One is $\{e \mid \mathcal{P}_e \text{ solves the Halting problem}\}$.
- V **II.6.16** One way to describe the set of all indices, $\mathbb{N}$, is $\{e \mid \text{on input 3 machine } \mathcal{P}_e \text{ either halts or does not}\}$.
- V **II.6.17** Notation like $\phi_e(x) = y$ means that $\phi_e(x)$ converges and gives the output value y .

- (A) $\{e \in \mathbb{N} \mid \phi_e(7) = 7\}$
- (B) $\{e \in \mathbb{N} \mid \phi_e(e) = e\}$
- (C) $\{e \in \mathbb{N} \mid \text{there is a } y \text{ where } \phi_{2e}(y) = 7\}$
- (D) $\{e \in \mathbb{N} \mid \phi_e(7) \downarrow \text{ and is prime}\}$

- V **II.6.18** Let $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e \text{ is total}\}$. We will show that it is a nontrivial index set.

We first show that $\mathcal{I}$ is not the empty set. Consider the program sketched on the left of Figure 31 on page 77. By Church's Thesis, there is a Turing machine fitting this sketch. It has an index and that index is an element of $\mathcal{I}$. So $\mathcal{I}$ is not empty.

Next we show that $\mathcal{I}$ is not all of $\mathbb{N}$. Consider the program sketched on the right of the same figure. By Church's Thesis, there is a Turing machine fitting this sketch. Its index is not an element of $\mathcal{I}$ and so $\mathcal{I}$ is not all of $\mathbb{N}$.

FIGURE 31, FOR QUESTION II.6.18: The machines used to show that the set $\mathcal{I}$ is not trivial.FIGURE 32, FOR QUESTION II.6.19: The machines used to show that the set $\mathcal{I}$ of indices of the squarer function is not trivial.

To finish, we show that $\mathcal{I}$ is an index set. So suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, we know that $\phi_e(a) \downarrow$ for all $a \in \mathbb{N}$. Because $\phi_e \simeq \phi_{\hat{e}}$ we therefore know that $\phi_{\hat{e}}(a) \downarrow$ for all $a \in \mathbb{N}$. Thus, $\hat{e} \in \mathcal{I}$ and consequently $\mathcal{I}$ is an index set.

Therefore, Rice's theorem shows that the problem of determining totality, the problem of computing whether, given e , the Turing machine $\mathcal{P}_e$ halts on all inputs, is unsolvable.

V II.6.19 Let $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(y) = y^2 \text{ for all } y\}$. We will show that it is a nontrivial index set.

We first show that $\mathcal{I}$ is not the empty set. Consider the program sketched on the left of Figure 32 on page 77. By Church's Thesis there is a Turing machine matching this sketch. It has an index, and that index is an element of $\mathcal{I}$. So $\mathcal{I}$ is not empty.

Next we show that $\mathcal{I}$ is not all of $\mathbb{N}$. Consider the program sketched on the right of the same figure. By Church's Thesis there is a Turing machine fitting this sketch. Its index is not an element of $\mathcal{I}$ and so $\mathcal{I}$ is not all of $\mathbb{N}$.

To finish we show that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ we know that $\phi_e(y) = y^2$ for all $y \in \mathbb{N}$. Because $\phi_e \simeq \phi_{\hat{e}}$ we therefore know that $\phi_{\hat{e}}(y) = y^2$ for all $y \in \mathbb{N}$. Thus, $\hat{e} \in \mathcal{I}$ and consequently $\mathcal{I}$ is an index set.

With those, Rice's theorem shows that the problem of deciding, given e , whether the Turing machine $\mathcal{P}_e$ squares its input, is unsolvable.

V II.6.20 We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(y) = \phi_e(y+1) \text{ for some } y \in \mathbb{N}\}$ is a nontrivial index set.

We first show that $\mathcal{I}$ is not empty. Consider the program sketched on the left of Figure 33 on page 78. It is intuitively computable so by Church's Thesis there is a Turing machine fitting this sketch. That Turing machine's index is an element of $\mathcal{I}$. So $\mathcal{I}$ is not empty.

Next we argue that $\mathcal{I} \neq \mathbb{N}$. Consider the program outlined on the right of the figure. By Church's Thesis, there is a Turing machine fitting this. Its index is not an element of $\mathcal{I}$ and so $\mathcal{I}$ is not all of $\mathbb{N}$.

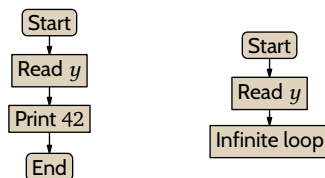
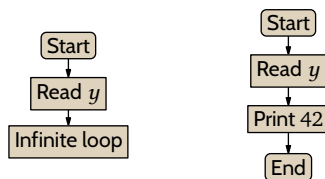
To finish, we show that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, there exists $y \in \mathbb{N}$ so that $\phi_e(y) = \phi_e(y+1)$. Because $\phi_e \simeq \phi_{\hat{e}}$, we know that $\phi_{\hat{e}}(y) = \phi_{\hat{e}}(y+1)$ for the same y . Thus, $\hat{e} \in \mathcal{I}$ also. Therefore $\mathcal{I}$ is an index set.

Consequently, by Rice's theorem, the problem of determining membership in $\mathcal{I}$ is unsolvable.

V II.6.21 We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(5) \uparrow\}$ is a nontrivial index set.

First we show that $\mathcal{I} \neq \emptyset$. Consider the program sketched on the left of Figure 34 on page 78. It is intuitively computable, so by Church's Thesis there is a Turing machine fitting this sketch. That machine's index is an element of $\mathcal{I}$.

Next we argue that $\mathcal{I} \neq \mathbb{N}$. Consider the program outlined on the right of that figure. By Church's Thesis,

FIGURE 33, FOR QUESTION II.6.20: The machines used to show that the set $\mathcal{I}$ is not trivial.FIGURE 34, FOR QUESTION II.6.21: Outlines of the Turing machines used to show that $\mathcal{I}$ is not trivial.

there is a Turing machine fitting this. Its index is not an element of $\mathcal{I}$.

To finish, we show that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, we know that $\phi_e(5) \uparrow$. But $\phi_e \simeq \phi_{\hat{e}}$ so we have that $\phi_{\hat{e}}(5) \uparrow$ also. Thus, $\hat{e} \in \mathcal{I}$. Therefore $\mathcal{I}$ is an index set.

With those, Rice's theorem says that the problem of determining membership in $\mathcal{I}$ is unsolvable.

V II.6.22 We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(y) \uparrow \text{ for all odd } y\}$ is a nontrivial index set.

First we show that $\mathcal{I} \neq \emptyset$. Consider the program sketched on the left of Figure 35 on page 79. It is intuitively computable, so by Church's Thesis there is a Turing machine fitting this sketch. That machine's index is an element of $\mathcal{I}$.

Next we verify that $\mathcal{I} \neq \mathbb{N}$. Consider the program outlined on the right of the same figure. By Church's Thesis, there is a Turing machine fitting this. Its index is not an element of $\mathcal{I}$.

To finish we show that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ we know that $\phi_e(y) \uparrow$ for all odd inputs y . Because $\phi_e \simeq \phi_{\hat{e}}$ we know also that $\phi_{\hat{e}}(y) \uparrow$ for all odd y . Thus, $\hat{e} \in \mathcal{I}$, and therefore $\mathcal{I}$ is an index set.

With those, Rice's theorem gives that the problem of determining membership in $\mathcal{I}$ is unsolvable.

V II.6.23 We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(x) = x + 1\}$ is a nontrivial index set.

First we show that $\mathcal{I} \neq \emptyset$. Consider the program sketched on the left of Figure 36 on page 79. It is intuitively computable, so by Church's Thesis there is a Turing machine fitting this sketch. That machine's index is an element of $\mathcal{I}$.

Next we show that $\mathcal{I} \neq \mathbb{N}$. Consider the program outlined on the right of that figure. By Church's Thesis, there is a Turing machine matching this. Its index is not an element of $\mathcal{I}$.

We finish by showing that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, we know that $\phi_e(x) = x + 1$ for all inputs x . Because $\phi_e \simeq \phi_{\hat{e}}$ we also know that $\phi_{\hat{e}}(x) = x + 1$ for all x . Thus, $\hat{e} \in \mathcal{I}$. Therefore $\mathcal{I}$ is an index set.

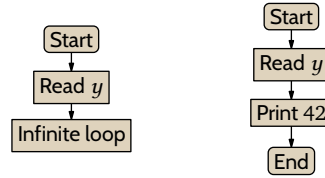
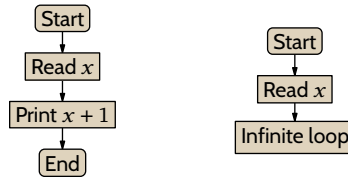
Consequently, Rice's theorem gives that the problem of determining membership in $\mathcal{I}$ is unsolvable.

V II.6.24 We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \text{both } \phi_e(x) \uparrow \text{ and } \phi_e(2x) \uparrow \text{ for some } x\}$ is a nontrivial index set.

First we show that $\mathcal{I} \neq \emptyset$. Consider the program sketched on the left of Figure 37 on page 80. It is intuitively computable, so by Church's Thesis there is a Turing machine fitting this sketch. That machine's index is an element of $\mathcal{I}$.

Next we argue that $\mathcal{I} \neq \mathbb{N}$. Consider the program outlined on the right of Figure 37 on page 80. By Church's Thesis, there is a Turing machine fitting this. Its index is not an element of $\mathcal{I}$.

To finish, we show that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, we know that for some input x we have both $\phi_e(x) \uparrow$ and $\phi_e(2x) \uparrow$. Because $\phi_e \simeq \phi_{\hat{e}}$ we know that $\phi_{\hat{e}}(x) \uparrow$ and $\phi_{\hat{e}}(2x) \uparrow$, for the same x . Thus, $\hat{e} \in \mathcal{I}$. Therefore $\mathcal{I}$ is an index set.

FIGURE 35, FOR QUESTION II.6.22: The machines used to show that $\mathcal{I}$ is not trivial.FIGURE 36, FOR QUESTION II.6.23: The machines used to show that $\mathcal{I}$ is not trivial.

Consequently, by Rice's theorem, the problem of determining membership in $\mathcal{I}$ is unsolvable.

- V II.6.25** The index set in the first item, is the complement of the one from 6.18. A later exercise, 6.35, shows that the complement of an index set is also an index set. But we will verify that the given problem is unsolvable without reference to those exercises.

- (A) The problem is: given an index e , decide whether $\phi_e(y) \uparrow$ for some $y \in \mathbb{N}$. To show that this is unsolvable we will verify that $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(y) \uparrow \text{ for some } y \in \mathbb{N}\}$ is a nontrivial index set.

The set $\mathcal{I}$ is not empty because we can write a program that does not halt on any input and then by Church's Thesis there is a Turing machine with that behavior. That machine's index is a member of $\mathcal{I}$.

The set is not equal to $\mathbb{N}$ because we can write a program that returns its input and then by Church's Thesis there is a Turing machine with that behavior. That machine's index is not a member of $\mathcal{I}$.

We finish verifying that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ we know that $\phi_e(y) \uparrow$ for at least one $y \in \mathbb{N}$. Because the two have the same behavior, $\phi_{\hat{e}}(y) \uparrow$ for at least one y also. Therefore $\hat{e} \in \mathcal{I}$ and $\mathcal{I}$ is an index set.

- (B) Let $\mathcal{I} = \{e \in \mathbb{N} \mid \phi_e(y) \downarrow \text{ for some } y \in \mathbb{N}\}$. We will verify that it is a nontrivial index set.

The set $\mathcal{I}$ is not empty because we can write a program that returns 0 on all inputs. Church's Thesis gives a Turing machine with that behavior. That machine's index is a member of $\mathcal{I}$.

The set $\mathcal{I}$ is not equal to $\mathbb{N}$ because we can write a program that fails to halt on any input, and by Church's Thesis there is a Turing machine with that behavior, and that machine's index is not a member of $\mathcal{I}$.

With $\mathcal{I}$ as nontrivial, we finish by verifying that it is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ we know that $\phi_e(y) \downarrow$ for at least one $y \in \mathbb{N}$. The two have the same behavior and thus $\phi_{\hat{e}}(y) \downarrow$ for at least one y also. Therefore $\mathcal{I}$ is an index set.

- V II.6.26** Each answer just gives the numbered fill-ins.

- (A) (1) $\phi_e(x) \downarrow$ for all inputs x that are a multiple of 5. (2) and (3) Figure 38 on page 80. (4) $\phi_e(5k) \downarrow$ for all $k \in \mathbb{N}$. (4) $\phi_{\hat{e}}(5k) \downarrow$ for all $k \in \mathbb{N}$ also.

- (B) (1) $\phi_e(x) = 7$ for some input x . (2) and (3) Figure 39 on page 81. (4) $\phi_e(x) = 7$ for some input $x \in \mathbb{N}$. (4) $\phi_{\hat{e}}(x) = 7$ for the same x .

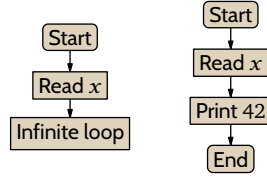
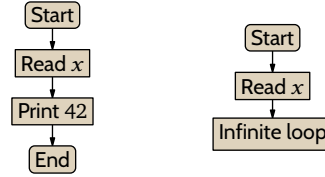
- V II.6.27** Every finite set is computable. With a finite set we can easily write a routine that acts as its characteristic function; there are just a finite number of cases. (Church's Thesis says that if there is such a program then there is such a Turing machine.)

- V II.6.28**

- (A) We will show that this is a nontrivial index set: $\mathcal{I} = \{e \in \mathbb{N} \mid \mathcal{P}_e \text{ accepts an infinite language}\}$.

First we verify that $\mathcal{I}$ is nonempty. We can write a Turing machine to output 1 on all inputs and this machine accepts $\mathcal{L} = \mathbb{B}$, which is an infinite set. Thus the index of this Turing machine is an element of $\mathcal{I}$.

Second we verify that $\mathcal{I}$ does not equal $\mathbb{N}$. We can write a Turing machine to output 0 on all inputs. This

FIGURE 37, FOR QUESTION II.6.24: The machines used to show that $\mathcal{I}$ is not trivial.FIGURE 38, FOR QUESTION II.6.26: The machines used to argue that $\mathcal{I}$ is not trivial.

machine accepts the empty subset of $\mathbb{B}$, which is not an infinite set. Thus the index of this Turing machine is not an element of $\mathcal{I}$.

Finally, to verify that $\mathcal{I}$ is an index set, suppose that $e \in \mathcal{I}$ and that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ the language that $\mathcal{P}_e$ accepts is infinite. Because $\phi_e \simeq \phi_{\hat{e}}$ the set of bit strings for which $\mathcal{P}_{\hat{e}}$ outputs 1 equals the set for which $\mathcal{P}_e$ outputs 1, and so $\mathcal{P}_{\hat{e}}$ also accepts an infinite language. Therefore $\hat{e} \in \mathcal{I}$, and $\mathcal{I}$ is an index set.

- (B) We will show that $\mathcal{I} = \{e \in \mathbb{N} \mid \mathcal{P}_e \text{ accepts } 101\}$ is a nontrivial index set. First, $\mathcal{I}$ is nonempty because we can write a Turing machine to output 1 on all inputs, and that machine accepts the string 101. The index of that machine is an element of $\mathcal{I}$.

Second, $\mathcal{I}$ does not equal $\mathbb{N}$ because we can write a Turing machine to output 0 on all inputs, and that machine does not accept the string 101. The index of that machine is not an element of $\mathcal{I}$.

Finally, we verify that $\mathcal{I}$ is an index set. Suppose that $e \in \mathcal{I}$ and that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ the machine $\mathcal{P}_e$ accepts 101. Because $\phi_e \simeq \phi_{\hat{e}}$, the machine $\mathcal{P}_{\hat{e}}$ also accepts 101. Thus $\hat{e} \in \mathcal{I}$ and $\mathcal{I}$ is an index set.

V II.6.29 We will use Rice's Theorem so consider $\mathcal{I} = \{e \mid \mathcal{P}_e \text{ accepts every } \sigma \in \mathbb{B}^*\}$.

First, we can write a Turing machine that halts on all inputs and outputs 1, and the index of this machine is a member of $\mathcal{I}$, so that is not the empty set.

Second, we can write a Turing machine that halts on all inputs and outputs 0, and the index of that machine is not a member of $\mathcal{I}$, so $\mathcal{I} \neq \mathbb{N}$.

To show that $\mathcal{I}$ is an index set, suppose that $e \in \mathcal{I}$ and that the number $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, machine $\mathcal{P}_e$ accepts all bitstrings. Because $\phi_e \simeq \phi_{\hat{e}}$, machine $\mathcal{P}_{\hat{e}}$ also accepts all bitstrings. Thus $\hat{e} \in \mathcal{I}$, and $\mathcal{I}$ is a nontrivial index set.

V II.6.30 We can produce two Turing machines, $\mathcal{P}_e$ and $\mathcal{P}_{\hat{e}}$ that have the same behavior, $\phi_e \simeq \phi_{\hat{e}}$, but $e \in \mathcal{I}$ while $\hat{e} \notin \mathcal{I}$. Here are two such machines.

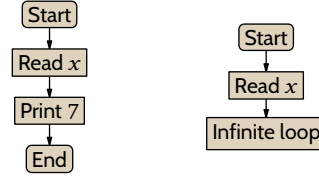
$$\mathcal{P}_e = \{q_0 \text{BB}q_0, q_011q_0, q_1 \text{BB}q_1, q_111q_1\} \quad \mathcal{P}_{\hat{e}} = \{q_0 \text{BB}q_0, q_011q_0\}$$

The first has the unreachable state of q_1 .

V II.6.31 This problem is not solvable. We cannot tell anything by whether the machine 'goes on', that is, does not stop immediately. It can go on to do quite complex things and then stop eventually, or do complex things and never stop.

For more, we can reduce the Halting problem to this problem. Consider this $\psi: \mathbb{N}^2 \rightarrow \mathbb{N}$.

$$\psi(x, y) = \begin{cases} 42 & \text{-- if machine } \mathcal{P}_x \text{ halts on } x \text{ and } y = \epsilon \\ \uparrow & \text{-- otherwise} \end{cases}$$

FIGURE 39, FOR QUESTION II.6.26: The machines used to argue that $\mathcal{I}$ is not trivial.

(We take this machine to have tape alphabet $\Sigma = \{B, 1\}$.) This is intuitively computable so Church's Thesis gives that there is a Turing machine with this behavior, and we can assume it has index e_0 .

Apply the s - m - n Theorem to parametrize x , giving a family of machines and associated computable functions as here.

$$\phi_{s(e_0, x)}(y) = \begin{cases} 42 & \text{-- if machine } \mathcal{P}_x \text{ halts on } x \text{ and } y = \varepsilon \\ \uparrow & \text{-- otherwise} \end{cases}$$

Then $\phi_x(x) \downarrow$ if and only if the function $\phi_{s(e_0, x)}$ halts only on $y = \varepsilon$. So if we could check the latter mechanically then we could mechanically solve the Halting problem, which of course we cannot.

V II.6.32 The Padding Lemma, I.2.18, says that every computable function has infinitely many indices. So if the set contains an $e \in \mathbb{N}$ then it must contain the infinitely many indices $\hat{e}$ such that $\phi_e \simeq \phi_{\hat{e}}$.

V II.6.33 We will call each set $\mathcal{I}$.

- (A) Suppose that $e \in \mathcal{I}$ and suppose that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$ the machine $\mathcal{P}_e$ halts on at least five inputs, that is, the function ϕ_e converges on at least five inputs. Because $\phi_e \simeq \phi_{\hat{e}}$, the function $\phi_{\hat{e}}$ also halts on at least five inputs, namely, the same five. Thus $\hat{e} \in \mathcal{I}$ also.
- (B) Suppose that $e \in \mathcal{I}$ and suppose also that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, the function ϕ_e is one-to-one. Because $\phi_e \simeq \phi_{\hat{e}}$, the function $\phi_{\hat{e}}$ is also one-to-one. Thus $\hat{e} \in \mathcal{I}$.
- (C) Suppose that $e \in \mathcal{I}$ and also that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, the function ϕ_e is either total or else $\phi_e(3) \uparrow$. In the case that ϕ_e is total, because $\phi_e \simeq \phi_{\hat{e}}$, the function $\phi_{\hat{e}}$ is also total. In the case that $\phi_e(3) \uparrow$, because they have the same behavior, $\phi_{\hat{e}}(3) \uparrow$ also. In either case, $\hat{e} \in \mathcal{I}$.

V II.6.34

- (A) The relation $\simeq$ is reflexive because a Turing machine $\mathcal{P}_e$ has the same behavior as itself: $\phi_e \simeq \phi_e$. It is symmetric because if $\mathcal{P}_e$ has the same behavior as $\mathcal{P}_{\hat{e}}$ then the converse also holds. Transitivity is just as clear.
- (B) Each equivalence class is a set of integers $e \in \mathbb{N}$. Two integers are together in a class, $e, \hat{e} \in \mathcal{C}$ if the input-output behaviors of the associated Turing machines $\mathcal{P}_e$ and $\mathcal{P}_{\hat{e}}$ are the same, $\phi_e \simeq \phi_{\hat{e}}$. (As an example, one class contains the indices of all Turing machines that double the input, $\mathcal{C} = \{e \in \mathbb{N} \mid \phi_e(x) = 2x \text{ for all } x\}$.)
- (C) We follow the hint. Suppose that $\mathcal{I}$ is an index set, that e is an element of $\mathcal{I}$, and that it belongs to the equivalence class $\mathcal{C}$ of the relation $\simeq$. Let $\hat{e}$ be another element of the class $\mathcal{C}$, so that $\phi_{\hat{e}} \simeq \phi_e$. Because $\mathcal{I}$ is an index set, $\hat{e} \in \mathcal{I}$ also. Thus having one element of $\mathcal{C}$ as an element of $\mathcal{I}$ implies that every element of $\mathcal{C}$ is in $\mathcal{I}$.

V II.6.35

- (A) Let $\mathcal{I}$ be an index set and consider the complement, $\mathcal{I}^c$. Suppose that $e \in \mathcal{I}^c$ and suppose also that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. We will show that $\hat{e} \in \mathcal{I}^c$.

Either $\hat{e} \in \mathcal{I}$ or $\hat{e} \in \mathcal{I}^c$. If it were to be that $\hat{e} \in \mathcal{I}$ then because $\phi_e \simeq \phi_{\hat{e}}$ and because $\mathcal{I}$ is an index set, we would have that $e \in \mathcal{I}$ also. This contradicts the assumption that $e \in \mathcal{I}^c$. Therefore $\hat{e} \in \mathcal{I}^c$.

- (B) Fix a collection $\mathcal{I}_j$ of index sets and consider the union $\mathcal{I} = \cup_j \mathcal{I}_j$. Suppose that $e \in \mathcal{I}$ and suppose also that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. We will show that $\hat{e} \in \mathcal{I}$.

The number e is an element of the union $\cup_j \mathcal{I}_j = \mathcal{I}$ so it is an element of some $\mathcal{I}_{j_0}$. Because $\phi_e \simeq \phi_{\hat{e}}$ and because $\mathcal{I}_{j_0}$ is an index set, the number $\hat{e}$ is an element of $\mathcal{I}_{j_0}$, and is therefore an element of the union $\mathcal{I}$. Thus $\mathcal{I}$ is an index set.

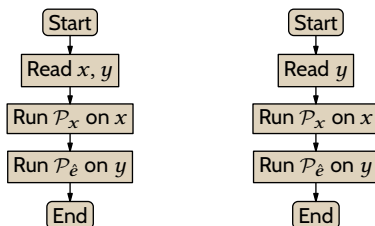


FIGURE 40, FOR QUESTION II.6.36: Sketches of machines for the proof of half of Rice's Theorem.

(c) It is closed under intersection. Let $\mathcal{I}_j$ be a collection of index sets and let $\mathcal{I} = \cap_j \mathcal{I}_j$. Suppose that $e \in \mathcal{I}$, and also that $\hat{e} \in \mathbb{N}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. We will show that $\hat{e} \in \mathcal{I}$.

Since e is an element of the intersection, $\cap_j \mathcal{I}_j = \mathcal{I}$, it is an element of each set $\mathcal{I}_j$. Because $\phi_e \simeq \phi_{\hat{e}}$ and because $\mathcal{I}_j$ is an index set, the number $\hat{e}$ is an element of each $\mathcal{I}_j$ also. As $\hat{e}$ is an element of each $\mathcal{I}_j$, it is an element of the intersection $\mathcal{I}$. Thus $\mathcal{I}$ is an index set.

V **II.6.36** The proof half in the section fixes an index $e_0 \in \mathbb{N}$ so that $\phi_{e_0}(y) \uparrow$ for all inputs $y \in \mathbb{N}$. It then does the $e_0 \notin \mathcal{I}$ case, so what's left to do here is the $e_0 \in \mathcal{I}$ case.

Because $\mathcal{I}$ is nontrivial it does not equal $\mathbb{N}$. So there is some $\hat{e} \notin \mathcal{I}$. Since $\mathcal{I}$ is an index set, $\phi_{e_0} \neq \phi_{\hat{e}}$. Thus there is a y such that $\phi_{\hat{e}}(y) \downarrow$.

Consider the flowchart on the left of Figure 40 on page 82. By Church's Thesis there is a Turing machine with that behavior; call it $\mathcal{P}_e$. Apply the s - m - n theorem to parametrize x , resulting in the uniformly computable family of functions $\phi_{s(e,x)}$, whose computation is outlined on the right of Figure 40 on page 82. We've constructed the machines so that if $\phi_x(x) \uparrow$ then $\phi_{s(e,x)} \simeq \phi_{e_0}$ and thus $s(e,x) \in \mathcal{I}$. Further, if $\phi_x(x) \downarrow$ then $\phi_{s(e,x)} \simeq \phi_{\hat{e}}$ and thus $s(e,x) \notin \mathcal{I}$. Therefore if $\mathcal{I}$ were mechanically computable, so that we could effectively check whether $s(e,x) \in \mathcal{I}$, then we could mechanically solve the Halting problem. Of course, that's impossible.

V **II.7.7** Half right. It is right that a set is computably enumerable if it can be enumerated by a Turing machine. However, the "but is not computable" is wrong. For instance, the set of even numbers is computably enumerable, since it is clearly effectively listable, and is also computable in that its characteristic function is computable.

V **II.7.8** The idea of 'effectively listable' is that a machine process gives this as its set of outputs. Some Turing machines never halt, no matter the input, so these machines list the empty set.

V **II.7.9** For each part we will produce a computable total function $f: \mathbb{N} \rightarrow \mathbb{N}$ that has the given set as range.

(A) The identity function, $f(x) = x$, works.

(B) The function $f(x) = 2x$ comes to mind.

(C) $f(x) = x^2$

(D) This will do.

$$f(x) = \begin{cases} 5 & \text{-- if } x = 0 \\ 7 & \text{-- if } x = 1 \\ 11 & \text{-- otherwise} \end{cases}$$

V **II.7.10** For each item we produce an effective function $f: \mathbb{N} \rightarrow \mathbb{N}$.

(A) The association $n \mapsto n$ -th prime is a function. Clearly we can write a program that computes it, so by Church's Thesis it is effective.

(B) To compute this function we can use brute force: given n , searching through the natural numbers, $0, 1, \dots$, checking each to see if its digits appear in non-increasing order. Output the n -th such number.

V **II.7.11** Yes to all four. For the first and fourth, we can write a Turing machine whose input/output behavior is to act as the identity function $x \mapsto x$, and this machine lists all natural numbers. So the set of all natural numbers is computably enumerable.

For the second and third, there are Turing machines that halt on no inputs, so the empty set is computably enumerable.

- V **II.7.12** The first is computable, since to decide if a number e is a member of this set we can run a simulation for twenty steps. The second is computably enumerable but not computable. It is computably enumerable because we can dovetail computations for the indices $e \in \mathbb{N}$. It is not computable because clearly if we could solve it then we could solve the problem of deciding membership in $\{e \mid \phi_e(4) \downarrow\}$, but we cannot solve that problem by Rice's Theorem.
- V **II.7.13** Recall that for sets 'decidable' means the same as 'computable', namely that the characteristic function is a total computable function. Also, 'semidecidable' means the same as 'computably enumerable'.
- (A) This set is computable, since we can just run the machine for a hundred steps. Because it is computable, it is also computably enumerable.
- (B) This set is also computable, again since we can just run the machine for a hundred steps. Because it is computable, it is also computably enumerable.
- V **II.7.14** The second is true by Lemma 7.5. The first is false; an example is that the set $\mathbb{N}$ is computably enumerable but its proper subset K is not computable.
- V **II.7.15** The function is an enumeration but unless it is computable, it is not a computable enumeration. An example of a countable set that is not computably enumerable is K^c . It is countable because it is a subset of the countable set $\mathbb{N}$, and it is not computably enumerable by Corollary 7.6.
- V **II.7.16** Briefly, we dovetail. To get the enumeration, first run $\mathcal{P}_0$ on input 5 for one step. Then run $\mathcal{P}_0$ on input 5 for a second step and $\mathcal{P}_1$ on input 5 for one step. Then run $\mathcal{P}_0$ for a third step, along with $\mathcal{P}_1$ for its second step and $\mathcal{P}_2$ on input 5 for one step. Continue in this way. The enumerating function $f: \mathbb{N} \rightarrow \mathbb{N}$ is: $f(0)$ is the index of the first such computation to halt, $f(1)$ is the index of the second, etc.
- V **II.7.17** There are only countably many Turing machines to do the enumeration.
- V **II.7.18** No. The empty set is computably enumerable but finite.
- V **II.7.19**
- (A) The set is computable. We are given an e and want to decide whether $\mathcal{P}_e$ contains such an instruction. Indices are source-equivalent, meaning that there is a program that takes in the index e and puts out the set of instructions for $\mathcal{P}_e$. We can then decide whether e is in the set by scanning the set of instructions for a q_4 .
- (B) This set is computably enumerable but not computable. To show it is computably enumerable, use dovetailing on the indices $e \in \mathbb{N}$ to run $\mathcal{P}_e$'s on input 4. That is, first run $\mathcal{P}_0$ on input 4 for one step. Next run $\mathcal{P}_0$ on input 4 for one more step and also run $\mathcal{P}_1$ on input 4 for one step. In continuing this process, when a computation halts, enumerate its e into the set. As to this set being not computable, Rice's Theorem easily shows that this is not a computable set.
- (C) Computable. Given e , decide if it is in the set by running $\mathcal{P}_e$ on 4 for the hundred steps.
- V **II.7.20** To do the enumeration, dovetail. Start by running $\mathcal{P}_0$ on input 2 for one step. Then run $\mathcal{P}_0$ on input 2 for a second step and $\mathcal{P}_1$ on input 2 for one step. Next, run $\mathcal{P}_0$ for a third step, along with $\mathcal{P}_1$ for its second step and $\mathcal{P}_2$ on input 2 for one step. Continue in this way. Then let $f(0)$ be the index of the first such computation to halt and output 4, let $f(1)$ be the index of the second, etc.
- V **II.7.21** There are of course many correct answers. One answer is to take K as the first set. For the second, take K with its smallest element omitted and for the third, take K with the smallest two elements omitted.
- A perhaps less wiseguy-ish answer is to take K along with some sets that we used as examples in the prior two sections, such as $\{e \in \mathbb{N} \mid \phi_e(3) \downarrow\}$ and $\{e \in \mathbb{N} \mid \text{there is } x \in \mathbb{N} \text{ so that } \phi_e(x) = 7\}$. The latter two are computably enumerable via a dovetailing process.
- V **II.7.22**
- (A) Enumerate it by dovetailing. The difference from some of the dovetail examples is that here there are two numbers, e and x . So recall Cantor's pairing function.

Number	0	1	2	3	4	5	6	...
Pair	$\langle 0, 0 \rangle$	$\langle 0, 1 \rangle$	$\langle 1, 0 \rangle$	$\langle 0, 2 \rangle$	$\langle 1, 1 \rangle$	$\langle 2, 0 \rangle$	$\langle 0, 3 \rangle$	...

For the initial time through the dovetailing loop find the initial pair in the table, $\langle 0, 0 \rangle$ (we take the first number to be e and the second to be x), and run $\mathcal{P}_0$ on input 0, for a single step. In the next time through the loop, run $\mathcal{P}_0$ on input 0 for a second step and also run (because $\langle 0, 1 \rangle$ is the second pair) $\mathcal{P}_0$ on input 1 for one step. For the next time through, run $\mathcal{P}_0$ on input 0 for its third step, and $\mathcal{P}_0$ on input 1 for its second

step, and $\mathcal{P}_1$ on input 0 for one step. The enumerating function outputs $\langle e, x \rangle$ when $\mathcal{P}_e$ halts on x .

(b) This is immediate from Lemma 7.3.

V II.7.23 No, for two reasons.

First, it is not the case that the collection of computable sets and the collection of computably enumerable sets are disjoint. Rather, the collection of computable sets is a subset of the collection of computably enumerable sets, by Lemma 7.5.

Furthermore, the collection $\{W_e \mid e \in \mathbb{N}\}$ is countable, since the elements in that collection, the W_e 's are indexed by the natural numbers. Since the number of subsets of $\mathbb{N}$ is uncountable (because the power set of $\mathbb{N}$ has cardinality greater than $\mathbb{N}$), there are more subsets of $\mathbb{N}$ than there are countably enumerable sets. Therefore there are subsets of $\mathbb{N}$ that are not computably enumerable.

V II.7.24 The computable sets are also computably enumerable so we will have shown that Tot is neither computable nor computably enumerable if we have shown that it is not computably enumerable.

Looking for a contradiction, assume that $\text{Tot} = \{e \mid \phi_e(x) \downarrow \text{ for all } x\}$ is computably enumerable, enumerated by the function $f: \mathbb{N} \rightarrow \mathbb{N}$. Following the hint, that gives a table like the one starting Section 5 on the Halting problem, because the i, j entry $\phi_{f(i)}(j)$ is sure to halt. Diagonalize: consider $h: \mathbb{N} \rightarrow \mathbb{N}$ given by $h(i) = 1 + \phi_{f(i)}(i)$. That function is effective, total, and unequal to any member of Tot, which is a contradiction.

V II.7.25 Rice's theorem easily shows that the set $S = \{e \mid \phi_e(3) \uparrow\}$ is not computable. Clearly the complement, $S^c = \{e \mid \phi_e(3) \downarrow\}$, is computably enumerable—just dovetail and enumerate elements as they converge. If S were also computably enumerable then by Lemma 7.5 it would be computable, which would be a contradiction.

V II.7.26 Let the infinite computably enumerable set be W and fix a computable enumeration $\phi(0), \phi(1), \dots$. We will define an infinite $S \subseteq W$ that is computable.

Let the first element of S be $s_0 = \phi(0)$. Take the second element s_1 to be the first member of the enumeration that is larger than s_0 ; such a number must eventually appear because W is infinite. In general s_{i+1} is the first element of the enumeration that is larger than s_i (as with s_1 , one must eventually appear because W is infinite).

The set S is computably enumerable because we have just described a mechanical procedure to enumerate it. It is computable because, given a number $n \in \mathbb{N}$, to decide whether it is an element of S just wait until either it is enumerated into S or a number larger than it is enumerated into S , which shows $n \notin S$.

V II.7.27

(A) We can argue by Church's Thesis: compute $\text{steps}(e)$ by running $\mathcal{P}_e$ on input e (using a Universal Turing machine), and if it ever halts then output the number of steps.

(B) We claim that if steps were total in addition to being computable then we could solve the Halting problem. The reason is that to decide whether $e \in K$, first compute $\text{steps}(e)$. Then run $\mathcal{P}_e$ on input e for $\text{steps}(e)$ -many steps. If this computation has not halted after that many steps then it will never halt.

(c) This is the same as the prior argument, but with f in place of steps.

V II.7.28 Suppose that S is computable, so that its characteristic function

$$\mathbb{1}_S(x) = \begin{cases} 1 & \text{-- if } x \in S \\ 0 & \text{-- if not} \end{cases}$$

is a computable function. To enumerate S in increasing order, run $\mathbb{1}_S(0), \mathbb{1}_S(1), \dots$, each of which must converge because the function is computable, and then the first element of S is the smallest, the second element is the second smallest, etc.

More formally, define the enumerating function by: where s_0 is the smallest element of S then $f(0) = s_0$, and where we have values for $f(0), \dots, f(k)$ then the value for $f(k+1)$ is the smallest x such that $x > f(k)$ and $\mathbb{1}_S(x) = 1$. Because S is infinite, there is always such an x .

For the other direction, assume that S is an infinite set that is enumerable in increasing order, by the computable function h . Given x , to decide whether $x \in S$, compute $h(0), h(1), \dots$ until the output is x (and so $x \in S$) or the output is larger than x (and so $x \notin S$).

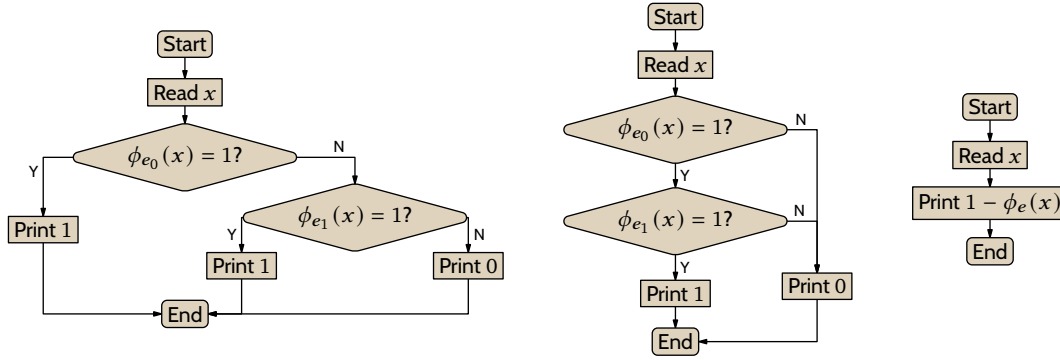


FIGURE 41, FOR QUESTION II.7.30: Machines to decide union, intersection, and complement of computable sets.

- V **II.7.29** Consider the set of ordered pairs $S = \{ \langle a, b \rangle \mid a \in K \text{ and } b \in K^c \}$. It is not computably enumerable, or enumerating it would enumerate its second entries, but K^c is not computably enumerable. Similarly its complement is not computably enumerable or enumerating it would enumerate the complement of S 's first entries, again impossible because K^c is not computably enumerable.
- V **II.7.30**
- (A) No. The set $\mathbb{N}$ is computable but its subset K is not.
 - (B) Yes. Let $S_0 \subseteq \mathbb{N}$ and $S_1 \subseteq \mathbb{N}$ be computable via the functions ϕ_{e_0} and ϕ_{e_1} . A sketch of the machine to compute the union is in Figure 41 on page 85.
 - (C) Yes. A sketch of the machine to compute the intersection is in the same figure.
 - (D) Yes. A sketch of the machine to compute the complement is also in that figure.
- V **II.7.31**
- (A) No. The set $\mathbb{N}$ is computably enumerable. The complement of the Halting problem set, K^c , is a subset of $\mathbb{N}$ and is not computably enumerable.
 - (B) Yes. Let $W_{e_0}, W_{e_1} \subseteq \mathbb{N}$ be computably enumerable, via $\phi_{e_0}, \phi_{e_1} : \mathbb{N} \rightarrow \mathbb{N}$. Dovetail enumerations of the two sets. That is, run $\mathcal{P}_{e_0}$ and $\mathcal{P}_{e_1}$ on input 0 for a step each. Then, run $\mathcal{P}_{e_0}$ and $\mathcal{P}_{e_1}$ on input 0 for a second step, and also run $\mathcal{P}_{e_0}$ and $\mathcal{P}_{e_1}$ on input 1 for a step each. Continue in this way, and when a number is enumerated into either of W_{e_0} or W_{e_1} , enumerate it into their union.
 - (C) Yes. As in the prior item, dovetail enumerations of the two sets. When a number has been enumerated into both of W_{e_0} and W_{e_1} then enumerate it into their intersection.
 - (D) No. The Halting problem set K is computably enumerable but its complement is not.
- V **II.7.32** We must show that if a set S is the range of a total computable function or is empty, then S is the domain of a partial computable function.

We first do the left to right implication. If S is empty then it is the domain of the partial computable function that diverges on all inputs. Otherwise S is the range of a total computable f ; we will describe a partial computable g with domain S . To compute g : given the input $x \in \mathbb{N}$, enumerate $f(0), f(1), \dots$. If x ever appears in this stream then halt the computation of g (with some nominal output such as 42). If x never appears then the computation of $g(x)$ never halts. The domain of g is S .

For the other implication, assume that S is the domain of a partial computable function $\hat{g}$. We must show that S is either empty or is the range of a total computable function. The domain of a partial computable function can be empty; in the case that S is empty then we are done. Otherwise fix some $s_0 \in S$ and we will produce a total computable $\hat{f}$ whose range is S . Given $n \in \mathbb{N}$, run the computations of $\hat{g}(0), \hat{g}(1), \dots, \hat{g}(n)$, each for n -many steps. Possibly some of these computations halt. Define $\hat{f}(n)$ to be the least k where the computation of $\hat{g}(k)$ halts within n steps and where $k \notin \{\hat{f}(0), \hat{f}(1), \dots, \hat{f}(n-1)\}$. If no such k exists then define $\hat{f}(n) = s_0$ (this makes $\hat{f}$ a total function).

We finish by verifying that $\hat{f}$ is the desired enumeration. If $t \notin S$ then $\hat{g}(t)$'s computation never halts and so t is never enumerated by $\hat{f}$. If $t \in S$ then eventually the computation of $\hat{g}(t)$ must halt, in some number of

steps, n_t . The number t is then queued for output by $\hat{f}$ in that it will be enumerated as, at most, $\hat{f}(n_t + t)$.

- V **II.8.14** It depends on what “assume the existence of” means. We do not assume that there exists an in-principle physically realizable device that can correctly answer ‘yes’ or ‘no’ in a finite time to questions of the form, “Does $\mathcal{P}_e$ halt on input e ?” But it is OK to have an argument with the phrase, “Assuming that we had a Halting problem oracle, then we could . . .”

Comment. It is worth a mention that the concept of an oracle, whether as described in this section or as extended to ideas such as oracles that can solve cryptographic problems, or oracles that can sort number lists in linear time, is very commonly used to address in-practice problems.

- V **II.8.15** A decider is a Turing machine (whose computed function is the characteristic function of some set). An oracle is a set, used as an addition to a Turing machine that the machine can query.
- V **II.8.16** First of all, they are fascinating and fun. Is that not enough?

Beyond that, they help us understand the relationships among problems. This will include trying to understand which are easy and which are hard.

In the Complexity chapter we will expand on the idea to help understand how many everyday, practical, problems are related.

- V **II.8.17** Their answer captures the spirit of the reason for considering oracle computations, but it is lacking in two ways. The first is that we can use a computable set, such as the empty set, as an oracle. With a computable oracle we cannot solve any problems we could not solve without that oracle.

The second lack is that their answer doesn’t refer to the mechanism at all. The oracle set doesn’t solve the problems. We might informally speak of an oracle as solving problems but for a definition we should say that we give a Turing machine access to answers about membership in the oracle and then the input-output behavior of the machine is the solution to the problem.

- V **II.8.18** We take the first question to ask: is there a most-powerful oracle, one that solves every problem? The answer is no. Given an oracle $X \subseteq \mathbb{N}$, the problem of deciding membership in K^X , the relativized Halting problem for that oracle, is not solvable from X .

The second question asks whether, given a set $S \subseteq \mathbb{N}$, there is an oracle that suffices to determine membership in S . The answer is yes: the oracle S will do.

- V **II.8.19** Each oracle can only compute answers to some problems. (There are countably many $\mathcal{P}_e^X$ but uncountably many subsets of $\mathbb{N}$.) So for each oracle set there are problems that it cannot solve. In particular, it cannot solve the relativized Halting problem for that set.

- V **II.8.20** Recall that the notation for ‘ A is Turing-reducible to B ’ is $A \leq_T B$.

- (A) This is false. For instance, the empty set is reducible to the Halting problem set, $\emptyset \leq_T K$, but while a decider for $\emptyset$ is trivial, there is no computable decider for K .

The correct statement is the opposite: if $A \leq_T B$ then we can use a decider for B to decide questions about membership in the set A .

- (B) False. As in the prior item, if A is the empty set then it is computable. Then $\emptyset \leq_T K$ but K is not decidable. (What’s right is the other way around: if B is decidable then A is decidable also.)

- (C) True. If $A \leq_T B$ then for some index $e \in \mathbb{N}$, the function ϕ_e^B is the characteristic function of A , $\phi_e^B = \mathbb{1}_A$. With this function, if we could compute B then we could use it to compute A . So if A is uncomputable then B must be uncomputable also.

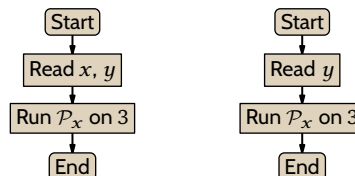
- V **II.8.21** See Figure 42 on page 87.

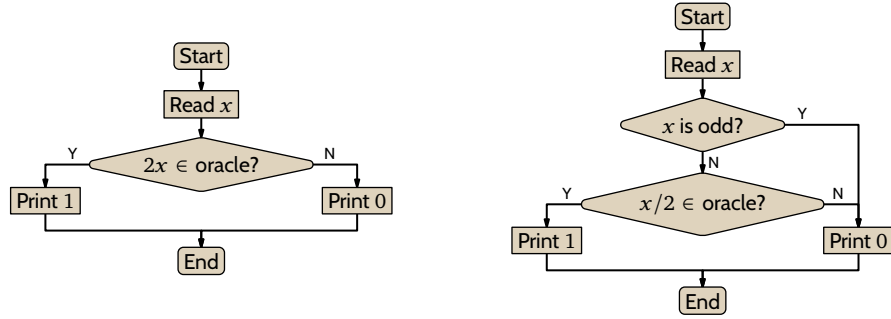
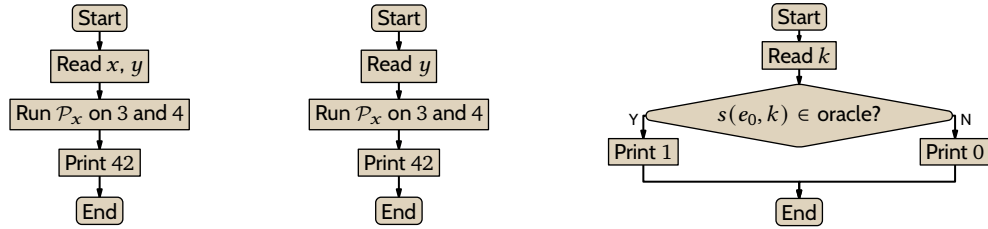
- (A) The flowchart on the left is straightforward.

- (B) For the flowchart on the right, notice that if the input x is odd then this machine must reply “no,” since no odd number can be an element of $2B$.

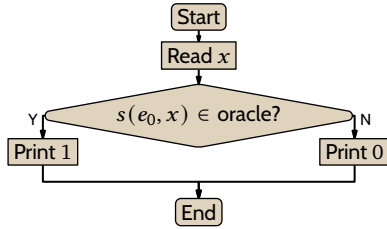
- V **II.8.22** Start with the function $\psi: \mathbb{N} \rightarrow \mathbb{N}$ below.

$$\psi(x, y) = \begin{cases} \mathcal{P}_x(3) & \text{– if } \mathcal{P}_x(3) \downarrow \\ \uparrow & \text{– otherwise} \end{cases}$$



FIGURE 42, FOR QUESTION II.8.21: Showing that $B \leq_T 2B$ and $2B \leq_T B$.FIGURE 43, FOR QUESTION II.8.23: The set $\{x \mid \phi_x(3) \downarrow \text{ and } \phi_x(4) \downarrow\}$ Turing-reduces to K .

By Church's Thesis the first flowchart gives that ψ is mechanically computable. Let it have index e_0 . Apply the s - m - n theorem to parametrize x , resulting in the machine sketched on the right, with index $s(e_0, x)$. The behavior of this machine does not depend on its input y so $\phi_{s(e_0, x)}(s(e_0, x)) \downarrow$ if and only if $\phi_x(3) \downarrow$. Thus, given x , we can test whether $x \in A$ via this machine by using a K oracle.



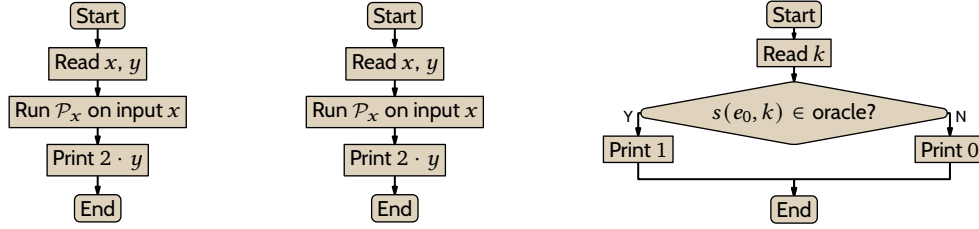
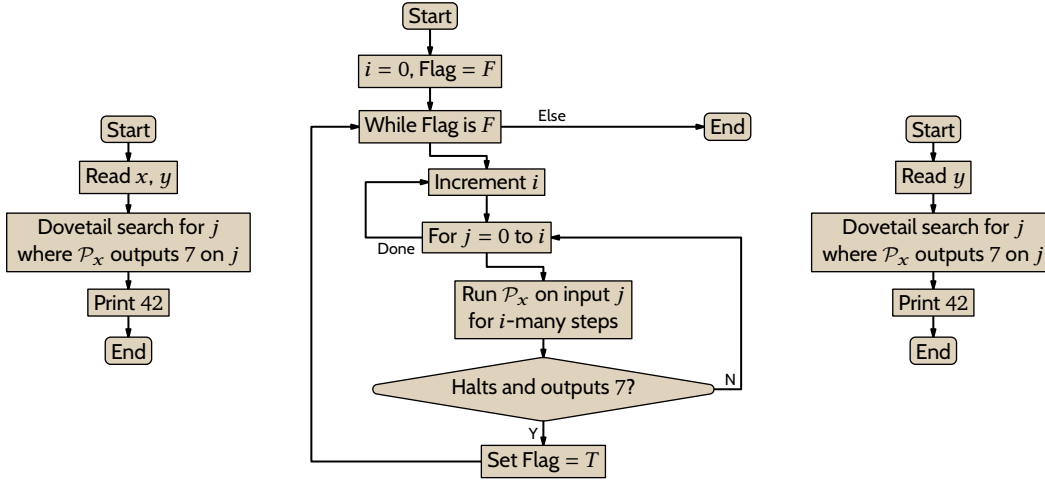
- V II.8.23** Mimicking Example 8.2, we go in two steps. First, on the left of Figure 43 on page 87 is a flowchart sketching a program. By Church's Thesis there is a Turing machine that computes it. Let the index of that machine be e_0 . Apply the s - m - n theorem to get a family of machines parametrized by x . The x -th member of that family is shown in the middle. Clearly if it halts for any input then it halts for all inputs. Hence $\mathcal{P}_x$ halts on inputs 3 and 4 if and only if $s(e_0, x) \in K$. Restated, there is a computable function, namely $x \mapsto s(e_0, x)$, so that $x \in \{x \mid \phi_x(3) \downarrow\}$ if and only if $s(e_0, x) \in K$. Therefore $\{x \mid \phi_x(3) \downarrow\} \leq_T K$.

The second step uses that computable function $x \mapsto s(e_0, x)$ for the oracle machine. That machine is on the right of the figure.

- V II.8.24** By definition, $K_0 = \{\langle e, x \rangle \mid \phi_e(x) \downarrow\}$. So $e \in S$ if and only if $\langle e, 3 \rangle \in K_0$.
- V II.8.25** Let $D = \{x \mid \phi_x(y) = 2y \text{ for all inputs } y\}$. Consider this function.

$$\psi(x, y) = \begin{cases} 2y & \text{-- if } \mathcal{P}_x \text{ halts on input } x \\ \uparrow & \text{-- otherwise} \end{cases}$$

We can produce this behavior with a program that takes two inputs, $x, y \in \mathbb{N}$, simulates running $\mathcal{P}_x$ on input x , prints $2y$, and then stops. See the flowchart on the left of Figure 44 on page 88. Church's Thesis gives that there is such a Turing machine. Let it have index e_0 , so that $\phi_{e_0} = \psi$.

FIGURE 44, FOR QUESTION II.8.25: Showing that $K \leq_T \{x \mid \phi_x(y) = 2y \text{ for all input } y\}$.FIGURE 45, FOR QUESTION II.8.26: Computing $\{x \mid \phi_x(j) = 7 \text{ for some } j\}$ with a K oracle.

Apply the s - m - n theorem to get a family of functions $\phi_{s(e_0, x)}$ that take in one input y and output $2y$ if $\phi_x(x) \downarrow$, or diverges otherwise. These are sketched in the middle of the figure. Then $x \in K$ if and only if $s(e, x) \in D$. On the right of the figure is an oracle machine giving that $K \leq_T D$, as required.

V II.8.26

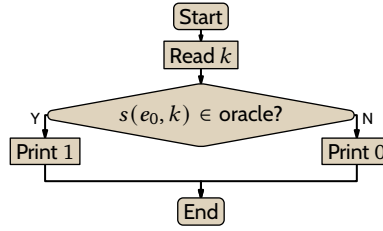
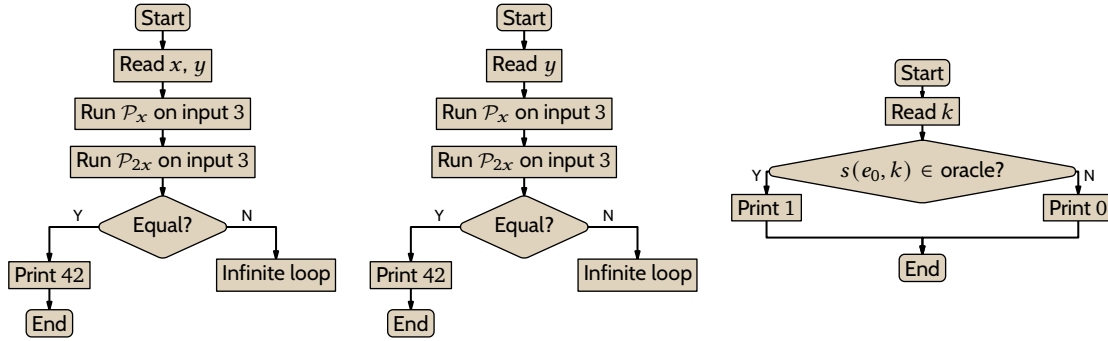
- (A) Let $\mathcal{I} = \{e \mid \phi_e(j) = 7 \text{ for some } j\}$. This set is not empty because one element is an index of a Turing machine that outputs 7 for all inputs. This set is not all of $\mathbb{N}$ because one non-element is an index of a Turing machine that fails to halt on all inputs.

To see that $\mathcal{I}$ is an index set, suppose that $e \in \mathcal{I}$ and that $\hat{e}$ is such that $\phi_e \simeq \phi_{\hat{e}}$. Because $e \in \mathcal{I}$, there is an input j so that $\phi_e(j) = 7$. Because $\phi_e \simeq \phi_{\hat{e}}$, for the same j we have $\phi_{\hat{e}}(j) = 7$, and therefore $\hat{e} \in \mathcal{I}$ also.

- (B) We use Example 8.2 as a basis. On the left of Figure 45 on page 88 is a flowchart sketching a program. The second chart gives extra detail on what's involved in 'dovetail'. By Church's Thesis there is a Turing machine that computes this flowchart. Let the index of that machine be e_0 . Apply the s - m - n theorem to get a family of machines parametrized by x , shown in the third flowchart. Clearly it halts for all inputs y if and only if P_x outputs a 7 for any input. Thus, P_x ever outputs a 7 if and only if $s(e_0, x) \in K$. That means there is a computable function, $x \mapsto s(e_0, x)$, where $x \in \{x \mid \phi_x(j) = 7 \text{ for some } j\}$ if and only if $s(e_0, x) \in K$.

The second step is to use the computable function $x \mapsto s(e_0, x)$ to get the oracle machine in Figure 46 on page 89.

- V II.8.27 On the left of Figure 47 on page 89 is a sketch of a program. By Church's Thesis there is a Turing machine with the same behavior. Let the index of that machine be e_0 . Apply the s - m - n theorem to get a family of machines parametrized by x , whose x -th member, $P_{s(e_0, x)}$, is shown in the middle. Clearly it halts for all inputs y if and only if $x \in S$. In particular, it halts for $y = s(e_0, x)$ if and only if $x \in S$. Thus, we have a computable function $x \mapsto s(e_0, x)$ so that $x \in S$ if and only if $s(e_0, x) \in K$. With that, hooking up the oracle machine on the right to oracle K gives that $S \leq_T K$.

FIGURE 46, FOR QUESTION II.8.26: Oracle machine for computing $\{x \mid \phi_x(j) = 7 \text{ for some } j\}$ from K .FIGURE 47, FOR QUESTION II.8.27: The set $\{x \mid \phi_x(3) \downarrow = \phi_{2x}(3) \downarrow\}$ is T -equivalent to K .

✓ **II.8.28** Consider the routine sketched on the left of Figure 48 on page 90. By Church's Thesis there is a Turing machine with that behavior. Let its index be e_0 . Apply the s - m - n -theorem to get, as shown in the middle, the machine $\mathcal{P}_{s(e_0, x)}$ that computes the partial function $\phi_{s(e_0, x)}$. This shows that there is a computable function, namely $x \mapsto s(e_0, x)$, so that $k \in K$ if and only if $s(e_0, k) \in \text{Tot}$. On the right is an oracle machine using that computable function, which when we plug in a Tot oracle will act as the characteristic function of K . Thus $K \leq_T \text{Tot}$.

✓ **II.8.29**

- (A) For contradiction suppose that the function steps has a computable extension g , which by the definition of extensible must be total. Then we could solve the Halting problem by running $\mathcal{P}_x$ on input x for $g(x)$ -many steps. If that computation hasn't halted by then, it will never halt.
- (B) Consider the routine sketched on the left of Figure 49 on page 90. By Church's Thesis there is a Turing machine with that behavior. Let its index be e_0 . In the middle is $\mathcal{P}_{s(e_0, x)}$, computing the partial function $\phi_{s(e_0, x)}$. If $x \notin K$ then $\phi_{s(e_0, x)}(y)$ is undefined for all inputs y , and this function is a member of Ext, since for instance it is extended by the function that always returns 0, which is obviously computable. If $x \in K$ then $\phi_{s(e_0, x)}(y) = \text{steps}(y)$ for all inputs y , and by the first item this is not a member of Ext. This gives a computable function, namely $x \mapsto s(e_0, x)$, so that by the first part of this question, $k \in K$ if and only if $s(e_0, k) \notin \text{Ext}$. The oracle machine on the right, when hooked up to an Ext oracle will act as the characteristic function of K . Therefore $K \leq_T \text{Ext}$.

✓ **II.8.30** The computation will proceed in exactly the same way, so it will come to the same result.

✓ **II.8.31** The oracle machine is in Figure 50 on page 90. That machine, when hooked up to an A^c oracle, will act as the characteristic function of A .

✓ **II.8.32** Suppose that $K \leq_T \emptyset$ were true. Then there would be an oracle machine whose computed function $\phi^\emptyset$ is the characteristic function of K . But because membership in $\emptyset$ is itself computable we could replace oracle calls with the computation itself. Specifically, in the case of the empty set, any time that the machine asks whether x is an element of the oracle, we replace that with the response of 'no'. That would mechanically compute K , which is impossible.

✓ **II.8.33** Any subset of $\mathbb{N}$ can be an oracle so there are uncountably many oracles.

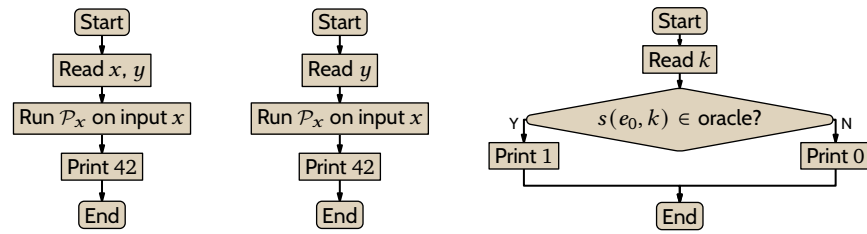


FIGURE 48, FOR QUESTION II.8.28: Computing K with a Tot oracle.

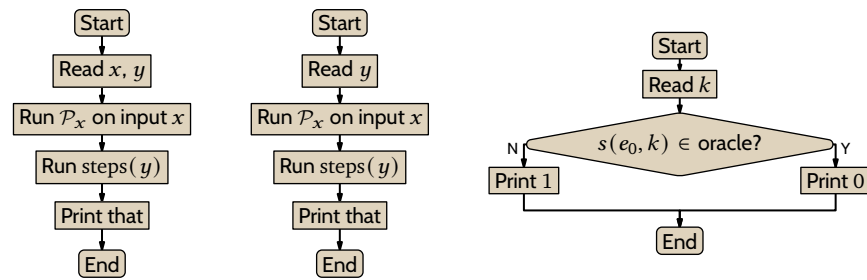


FIGURE 49, FOR QUESTION II.8.29: Computing K with a Ext oracle.

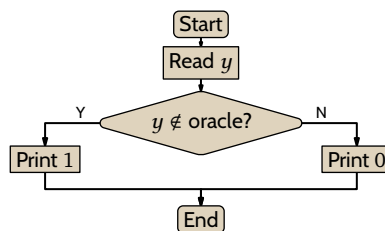


FIGURE 50, FOR QUESTION II.8.31: Computing A from an A^c oracle.

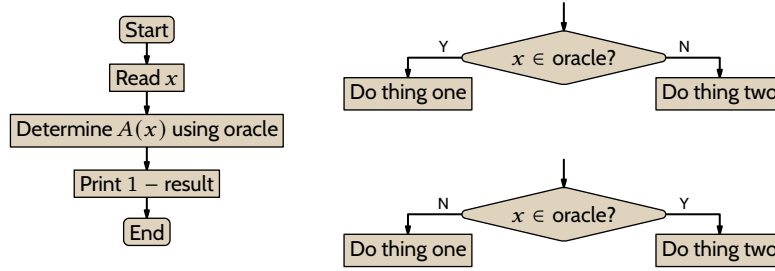


FIGURE 51, FOR QUESTION II.8.37: Oracle machines relating to the complement.

- V **II.8.34** There are only countably many programs that can involve an `oracle` call. That is, there are only countably many oracle-enhanced Turing machines, enumerated by the index e in $\mathcal{P}_e^X$.
- V **II.8.35** One is $C = \{\text{cantor}(0, a) \mid a \in A\} \cup \{\text{cantor}(1, b) \mid b \in B\}$. Clearly from this set we can answer questions about membership in each of A and B .
- V **II.8.36**
- (A) No. The Halting problem set K is a subset of the natural numbers $\mathbb{N}$ but $K \not\leq_T \mathbb{N}$, because the Halting problem is unsolvable.
 - (B) No. The Halting problem set K is not Turing reducible to $\mathbb{N}$, but $\mathbb{N} = K \cup K^c$.
 - (C) No. The Halting problem set K is not Turing reducible to $\emptyset$, but $\emptyset = K \cap K^c$.
 - (D) Yes. We can write this oracle-using program: read the input x and if `oracle(x)` then output 0, otherwise output 1.
- V **II.8.37** Refer to Figure 51 on page 91.
- (A) True. This is shown in the flowchart on the left. Basically, interchange outputs of ‘yes’ and ‘no’.
 - (B) True. This is shown in the middle flowchart. Replace all oracle calls like those on top with the ones on the bottom (the only difference is interchanging the ‘Y’ and ‘N’).
 - (C) True. Combine the first two items.
- V **II.8.38**
- (A) The natural approach is to relativize Definition 7.1 and define that a set B is computably enumerable in the set A if B is the range of an A -computable function. That is, B is c.e. in A if $B = \{\phi_e^A(x) \mid x \in \mathbb{N}\}$ for some index e (where that function may be a partial function).
 - (B) As in Lemma 8.4, we can output the elements of $\mathbb{N}$ without even referring to the oracle. The function $\phi^A(x) = x$ is computable, and does not need to ever refer to the oracle.
 - (C) Dovetail. First run the computation of $\mathcal{P}_0^A$ on input 0 for one step. Then run the computation of $\mathcal{P}_0^A$ on input 0 for a second step, and the computation of $\mathcal{P}_1^A$ on input 1 for one step. Next run the computation of $\mathcal{P}_0^A$ on input 0 for its third step, the computation of $\mathcal{P}_1^A$ on input 1 for its second step, and the computation of $\mathcal{P}_2^A$ on input 2 for one step. Continuing in this way, over time some of these halt. When a computation $\mathcal{P}_e^A(e)$ halts, enumerate the index e into K^A . (Here in the enumerating function, $\phi^A(i)$ is the i -th index that is listed in this way.)
- V **II.9.6** In everyday programming we can easily write two different source codes with the same input/output behavior, that is, computing the same function. Similarly, saying that the two are the same function, that $\phi_{g(v)} = \phi_{\phi_v(v)}$, is different than saying that they are produced by the same Turing machine, that the indices are the same, $g(v) = \phi_v(v)$.
- V **II.9.7**
- (A) Apply the Fixed Point Theorem to the computable function $f(x) = x + 7$.
 - (B) Apply the Fixed Point Theorem to $f(x) = 2x$.
- V **II.9.8** No matter what numbering scheme we use, (as long as it is acceptable) there is a Turing machine that computes the same function as the machine whose index is five more, so that there exists $k \in \mathbb{N}$ with $\phi_k = \phi_{k+5}$.
- The natural generalization is that for any acceptable numbering scheme and any constant $c \in \mathbb{N}$, there is a

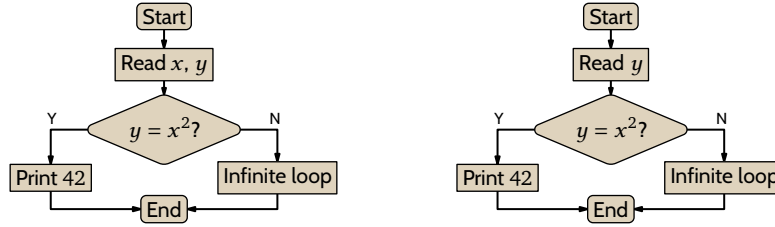


FIGURE 52, FOR QUESTION II.9.10: Flowcharts sketching the machine for $\psi = \phi_{e_0}$ and the machine for $\phi_{s(e_0, x)}$.

Turing machine that computes the same function as the machine whose index is c greater. That is, there exists a $k \in \mathbb{N}$ so that $\phi_k = \phi_{k+c}$.

V II.9.9

- (A) For any (acceptable) numbering scheme for Turing machines, there is one that computes the same function as the machine with three times its index. That is, there exists a $k \in \mathbb{N}$ so that $\phi_k = \phi_{3k}$.
- (B) For any acceptable numbering scheme, there is a Turing machine that computes the same function as the machine whose index is the square. That is, there exists a $k \in \mathbb{N}$ so that $\phi_k = \phi_{k^2}$.
- (C) Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be this function.

$$f(x) = \begin{cases} 1 & \text{-- if } x = 5 \\ 0 & \text{-- otherwise} \end{cases}$$

For any acceptable numbering for Turing machines there will be a $k \in \mathbb{N}$ so that $\phi_k = \phi_{f(k)}$.

- (D) If the function is $f(x) = 42$ then the Fixed Point Theorem gives that for any acceptable numbering there is a $k \in \mathbb{N}$ so that $\phi_k = \phi_{f(k)} = \phi_{42}$. This is not interesting because the number $k = 42$ comes to mind, but the Padding Lemma shows that there are infinitely many such indices k .

V II.9.10

- (A) Consider the function on the left below. It is computed by the program sketched on the left of Figure 52 on page 92. Church's Thesis then gives that there is a Turing machine with this behavior. Let that machine's index be e_0 .

$$\psi(x, y) = \begin{cases} 42 & \text{-- if } y = x^2 \\ \uparrow & \text{-- otherwise} \end{cases} \quad \phi_{s(e_0, x)}(y) = \begin{cases} 42 & \text{-- if } y = x^2 \\ \uparrow & \text{-- otherwise} \end{cases}$$

Apply the s - m - n Theorem to get the family of functions on the right.

- (B) The number e_0 is fixed because it is the index of a Turing machine that computes ψ . So define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = s(e_0, x)$.
- (C) The function g is total and computable because s is total and computable. Apply the Fixed Point Theorem with g to get an $m \in \mathbb{N}$ with $\phi_m = \phi_{g(m)} = \phi_{s(e_0, m)}$. By construction that last function halts only on input $y = m^2$. Thus $W_m = \{m^2\}$.

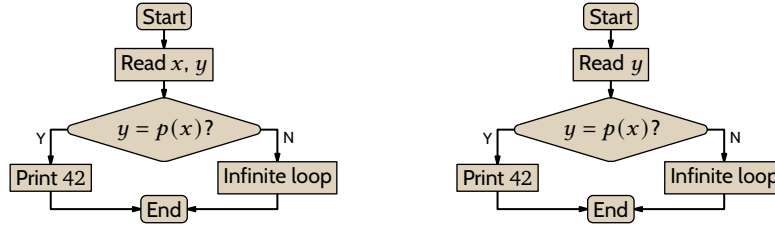
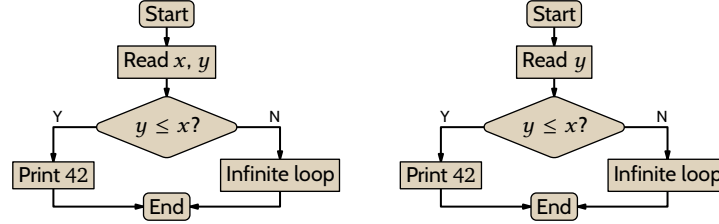
V II.9.11

- (A) We can write a program in a standard programming language, such as Racket, that inputs n and outputs the n -th prime.

```
(require math/number-theory)
> (nth-prime 5)
13
> (nth-prime 666)
4987
```

So by Church's Thesis the function $p: \mathbb{N} \rightarrow \mathbb{N}$ such that $p(x)$ is the x -th prime number is computable.

- (B) Consider the function on the left side below, computed by the machines sketched on the left of Figure 53 on page 93. By Church's Thesis that function has an index; let that index be e_0 . Apply the s - m - n theorem to get

FIGURE 53, FOR QUESTION II.9.11: Flowcharts outlining the machine for $\psi = \phi_{e_0}$ and the machine for $\phi_{s(e_0, x)}$.FIGURE 54, FOR QUESTION II.9.14: A sketch of the machine for $\psi = \phi_{e_0}$, and the machine for $\phi_{s(e_0, x)}$.

the family of functions on the right, parametrized by x .

$$\psi(x, y) = \begin{cases} 42 & \text{if } y = p(x) \\ \uparrow & \text{otherwise} \end{cases} \quad \phi_{s(e_0, x)}(y) = \begin{cases} 42 & \text{if } y = p(x) \\ \uparrow & \text{otherwise} \end{cases}$$

Apply the s - m - n Theorem to get the uniformly computable family of functions on the right.

- (c) The number e_0 is fixed so $s(e_0, x)$ is a function only of the variable x . To emphasize that, define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = s(e_0, x)$. The Fixed Point Theorem gives a $m \in \mathbb{N}$ with $\phi_m = \phi_{g(m)} = \phi_{s(e_0, m)}$, which halts only when the input is the m -th prime.

- V II.9.12 Consider the function on the left. By Church's Thesis it is computable and so has an index; let that index be e_0 .

$$\psi(x, y) = \begin{cases} 42 & \text{if } y = 10^x \\ \uparrow & \text{otherwise} \end{cases} \quad \phi_{s(e_0, x)}(y) = \begin{cases} 42 & \text{if } y = 10^x \\ \uparrow & \text{otherwise} \end{cases}$$

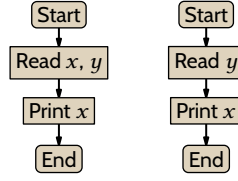
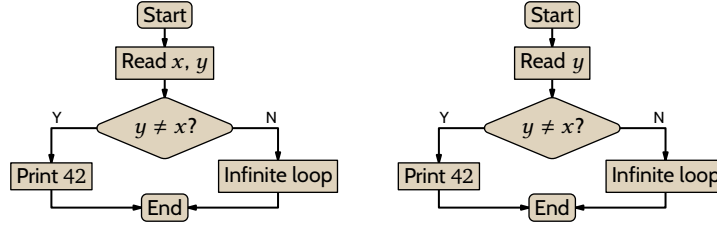
Apply the s - m - n Theorem to get the uniformly computable family of functions on the right. The number e_0 is fixed so define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = s(e_0, x)$. The Fixed Point Theorem gives $m \in \mathbb{N}$ with $\phi_m = \phi_{g(m)} = \phi_{s(e_0, m)}$, and so $W_m = \{10^m\}$.

- V II.9.13 Consider the function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = y$ where all of the digits in y are one greater than the digits in x , except that a 9 in x changes to a 0 in y . Thus $f(35) = 46$, and $f(29) = 30$, and $f(99) = 0$. Clearly this function is computable and total. Apply the Fixed Point theorem.
- V II.9.14 Consider the function on the left below. It is computed by the machine sketched on the left in Figure 54 on page 93, so Church's Thesis gives that there is a Turing machine index for this machine. Let that index be e_0 .

$$\psi(x, y) = \begin{cases} 42 & \text{if } y \leq x \\ \uparrow & \text{otherwise} \end{cases} \quad \phi_{s(e_0, x)}(y) = \begin{cases} 42 & \text{if } y \leq x \\ \uparrow & \text{otherwise} \end{cases} \quad (*)$$

Apply the s - m - n theorem to get the family of machines sketched on the right, which compute the family of functions given on the right of (*).

Because e_0 is the index of the machine on the left, it is fixed. So $s(e_0, x)$ is not a function of two variables, it is a function of the single variable x . Let $g: \mathbb{N} \rightarrow \mathbb{N}$ be defined by $g(x) = s(e_0, x)$. Then g is a computable

FIGURE 55, FOR QUESTION II.9.15: Flowcharts for ϕ_{e_0} and $\phi_{s(e_0,x)}$, for the machine that outputs its own index.FIGURE 56, FOR QUESTION II.9.17: Sketches for $\psi = \phi_{e_0}$ and for $\phi_{s(e_0,x)}$.

and total function of one variable. By the Fixed Point Theorem Theorem 9.1, the function g has a fixed point so that there is a $n \in \mathbb{N}$ with $\phi_n = \phi_{g(n)}\phi_{s(e_0,n)}$. Then $W_n = \{0, 1, \dots, n\}$.

V II.9.15 See the flowcharts in Figure 55 on page 94. On the left is a function that outputs its first input. By Church's Thesis there is a Turing machine with that behavior; let it have index e_0 so that $\phi_{e_0}(x, y) = x$. On the right we've applied the s - m - n theorem to parametrize that first input, resulting in $\phi_{s(e_0,x)}(y) = x$. Now let $f(x) = s(e_0, x)$. It is total and computable so the Fixed Point theorem applies. Hence there is a natural number k such that $\phi_k = \phi_{f(k)}$. Since we have $\phi_{f(k)}(y) = \phi_{s(e_0,k)}(y) = k$ we are done.

V II.9.16 Yes, we can take $f(x) = x$.

V II.9.17

(A) The function on the left below is computed by the machine sketched on the left of Figure 56 on page 94. Church's Thesis gives that it has an index; let that index be e_0 .

$$\psi(x, y) = \begin{cases} 42 & \text{if } y \neq x \\ \uparrow & \text{otherwise} \end{cases} \quad \phi_{s(e_0,x)}(y) = \begin{cases} 42 & \text{if } y \neq x \\ \uparrow & \text{otherwise} \end{cases}$$

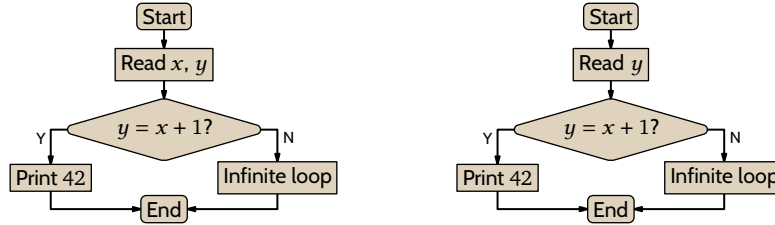
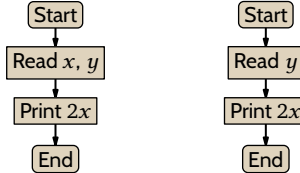
The s - m - n theorem parametrizes x to get the family of functions on the right. The number e_0 is fixed so $s(e_0, x)$ is a function of one variable. Consequently, define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = s(e_0, x)$. The Fixed Point Theorem gives $m \in \mathbb{N}$ with $\phi_m = \phi_{g(m)} = \phi_{s(e_0,m)}$, and so $W_m = \mathbb{N} - \{m\}$.

(B) No such computably enumerable W_m exists. Fix m . By definition W_m is the set $\{x \mid \phi_m(x) \downarrow\}$. Any $x \in \mathbb{N}$ has either $\phi(x)_m \downarrow$ or $\phi(x)_m \uparrow$. So x is in one of the two sets $\{x \mid \phi_m(x) \downarrow\}$ and $\{x \mid \phi_m(x) \uparrow\}$ and not the other. Therefore the two sets are not equal.

V II.9.18

(A) Consider the function $\psi: \mathbb{N}^2 \rightarrow \mathbb{N}$ given by: $\psi(x, y) = 42$ if $y = x + 1$ and $\psi(x, y) \uparrow$ otherwise. It is computed by the machine on the left of Figure 57 on page 95. So by Church's Thesis there is a Turing machine with that behavior. Let that be machine $\mathcal{P}_{e_0}$. Apply the s - m - n Theorem to get the machine on the right, which computes the function $\phi_{s(e_0,x)}(y)$ with the property that it converges only on $x + 1$. Let $g: \mathbb{N} \rightarrow \mathbb{N}$ be given by $g(x) = s(e_0, x)$. By the Fixed Point Theorem there is an $m \in \mathbb{N}$ where $\phi_m = \phi_{g(m)} = \phi_{s(e_0,m)}$, which has domain $\{m + 1\}$.

(B) Consider the function $\psi(x, y) = 2x$. It is computed by the machine sketched on the left of Figure 58 on page 95. So Church's Thesis gives that there is a machine with this behavior. Let its index be e_0 . Apply the s - m - n theorem to get the family of machines on the right of that figure. Define $g: \mathbb{N} \rightarrow \mathbb{N}$ by $g(x) = s(e_0, x)$ (the number e_0 is a constant so $s(e_0, x)$ is a function of one variable). The Fixed Point Theorem gives a number m such that $\phi_m = \phi_{g(m)}$, and since the range of $\phi_{g(m)}$ is a singleton set, $\{2m\}$, we are done.

FIGURE 57, FOR QUESTION II.9.18: Flowcharts for a function ϕ_m whose domain is $\{m + 1\}$.FIGURE 58, FOR QUESTION II.9.18: Flowcharts for a function ϕ_m whose range is $\{2m\}$.

V **II.9.19** By Corollary 9.3 there is an index e satisfying this.

$$\phi_e(y) = \begin{cases} \downarrow & \text{-- if } y = e \\ \uparrow & \text{-- otherwise} \end{cases} \quad (*)$$

This means that $e \in K$.

Now, by the Padding Lemma there exists $\hat{e} \neq e$ such that $\phi_{\hat{e}} \simeq \phi_e$. Then $(*)$ gives that $\phi_{\hat{e}}(\hat{e}) \uparrow$, and so $\hat{e} \notin K$. Therefore K is not an index set.

V **II.9.20**

(A) The set F is finite so the set of functions computed by indices that are members of F is finite. But there are infinitely many partial computable functions, so there must be ones that are not computed by any index in F .

(B) If $x \in F$ then $h(x) = e_0$. Since e_0 is an index for g , which has no indices in F , it cannot be that $\phi_{h(x)} = \phi_{e_0} = \phi_x$.

The other case is $x \notin F$. Then $\phi_{h(x)} = \phi_{f(x)}$, but $\phi_{f(x)} \neq \phi_x$ because $x \notin F$, and so $\phi_{h(x)} \neq \phi_x$.

V **II.A.1** One is to put the passengers from bus B_0 in rooms 0, 100, 200, etc. Then put the passengers from B_1 in rooms 1, 101, 201, etc. In general, put $b_{i,j}$ in room $i + 100j$.

V **II.A.2** First move the current guests, assigning the person now in room n to room $2n$. That leaves free the odd-numbered rooms. Then Person j from bus B_i goes into $f(p_{i,j}) = 2 \cdot \text{cantor}(i, j) + 1$.

V **II.A.3** Each person is a triple, $\langle i, j, k \rangle = \langle \text{floor number, space number for the bus, person number on that bus} \rangle$. The natural way to assign them rooms is to clear the odd-numbered rooms and then put $p_{i,j,k}$ into $2 \cdot \text{cantor}(\text{cantor}(i, j), k) + 1$.

V **II.A.4** No, the power set $\mathcal{P}(\mathbb{R})$ has more members than does $\mathbb{R}$.

V **II.D.3** The Δ function has $n \cdot 2$ inputs and $(n + 1) \cdot 2 \cdot 2$ outputs. So there are $(2n)^{4(n+1)}$ machines.

V **II.D.4** The Racket code in Figure 59 on page 96 computes it. The values are 6, 16, 34, 64, 114, 196, 334, 564, 946, 1584, 2646, 4416, 7366, and 12284.

V **II.D.5** The exercise does not ask us to produce a computable function; indeed, there is no such computable function.

Define $f(0)$ to be $1 + \phi_0(0)$ if $\phi_0(0)$ converges, otherwise define $f(0) = 1$. (Note that f is not a computable function.) Similarly define $f(1)$ to be the maximum of $1 + \phi_0(1)$ and $1 + \phi_1(1)$ where if either diverges then we don't include it in the maximum and if both diverge then we take $f(1) = 1$. Do the same for all n .

More formally, define $F(e, x)$ to be $\phi_e(x)$ if it converges and to be 0 otherwise. Then $f(i)$ is the maximum of $\{F(0, i), \dots, F(i, i)\}$ plus 1.

```

(define (g n)
  (let ([i (modulo n 3)])
    (cond
      [(= i 0) (/ (+ (* n 5) 18) 3)]
      [(= i 1) (/ (+ (* n 5) 22) 3)]
      [else -1])))

; Show values computed while finding Busy Beaver 5.
; r values; usually initially use r=0
; N cap on number of values to compute; use N=10 for instance
(define (BB5 r N)
  (let ([val (g r)])
    (if (> val 0)
      (begin
        (writeln (number->string val))
        (BB5 val (- N 1)))
      (writeln "done"))))

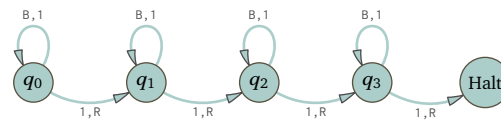
```

FIGURE 59, FOR QUESTION II.D.4: Racket code producing values found during computation of BB(5).

II.D.6 Following the hint, note that there are uncountably many functions with output in the set $\{0, 1\}$, that is, where every output is less than or equal to 1. There are only countably many Turing machines, so there are only countably many computable functions, and so there are only countably many computable functions with output bounded by 1. Therefore there is a uncomputable function with output bounded by 1.

V II.D.7

(A) This is best answered with an example: here is $\mathcal{M}_4$.



- (B) The first is true because $f(m)$ is a summand in the expression for $F(m)$, and similarly the second is true because m^2 is a summand. The third holds because all of the summands for $m > 0$ are positive.
- (C) If started on a blank tape this machine will first produce j -many 1's, then produce $F(j)$ -many 1's, and finally will leave the tape with $F(F(j))$ -many 1's. Thus its productivity is $F(F(j))$.
- (D) Since F is strictly increasing, $F(j + 2n_F) < F(F(j))$. Combining gives $f(j + 2n_F) < F(j + 2n_F) < F(F(j)) \leq \Sigma(j + 2n_F)$, as required.

V II.E.1

```

> (cantor-pairing 42)
'(6 2)

```

V II.E.2

```

> (cantor-pairing 666)
'(0 36)

```

V II.E.3

```

> (cantor-unpairing 3 0)
9
> (for ([i '(0 1 2 3 4 5 6 7 8 9)])
  (display (cantor-pairing-omega i)) (newline))
()
(0)
(0 0)
(1)
(0 1)
(0 0 0)
(2)
(1 0)
(0 0 1)
(0 0 0 0)

```


Chapter III: Languages and graphs

V III.1.17

(A) Five are 000, 001, 010, 011, and 100.

(B) These five work: 0, 1, 00, 11, and 000. (Is the empty string in that language? It is an arguable point.)

V III.1.18 No. By definition, a language is a set of strings and also by definition, a string is of finite length. For some real numbers the decimal representation does not make a finite string; one is $\pi = 3.14 \dots$ and another is $1/3 = 0.33 \dots$

III.1.19 It is both.

V III.1.20 If $\beta = \langle b_0, \dots, b_{i-1} \rangle$ is a string then $\beta^\sim \beta^R = \langle b_0, \dots, b_{i-1} \rangle^\sim \langle b_{i-1}, \dots, b_0 \rangle$ is clearly a palindrome. There are palindromes not of that form. One is $\gamma = 010$, which cannot be decomposed into $\gamma = \beta^\sim \beta^R$.

V III.1.21

(A) These are the members of $\mathcal{L}_0 \cap \mathcal{L}_1$.

$\varepsilon, b, bb, bbb,$
 $a, ab, abb, abbb$
 $aa, aab, aabb, aabbb$
 $aaa, aaab, aaabb, aaabbb$

(B) These are the members of $\mathcal{L}_1 \cap \mathcal{L}_0$.

$\varepsilon, a, aa, aaa,$
 $b, ba, baa, baaa$
 $bb, bba, bbba, bbaaa$
 $bbb, bbba, bbbba, bbbbaa$

(C) We have this.

$$\mathcal{L}_0^2 = \mathcal{L}_0 \cap \mathcal{L}_0 = \{ \emptyset, a, aa, aaa, aaaa, aaaaa, aaaaaa \}$$

(D) There are many different correct answers, but a natural ten are $\varepsilon, a, aa, \dots, a^9$.

V III.1.22

(A) The language consists of any number of a's followed by a single b. Thus five members are b, ab, aab, aaab, and aaaab.

(B) The language consists of a number of a's followed by the same number of b's. Thus five members are $\varepsilon, ab, aabb, aaabbb, \text{ and } aaaabbbb$.

(C) The language is a set of strings of 1's followed by 0's, where there is one more 0 than 1. Five members are 0, 100, 11000, 1110000, and 111100000.

(D) The language consists of strings from $\mathbb{B}^*$ where there is a string of 1's, then a string of 0's, then a single 1. The number of 0's is twice the number of leading 1's. Five members are 1, 1001, 1100001, $1^3 0^6 1$, and $1^4 0^8 1$.

V III.1.23

(A) $\mathcal{L}^2 = \{ aa, aab, aba, abab \}$

(B) $\mathcal{L}^3 = \{ aaa, aaab, aaba, aabab, abaa, abaab, ababa, ababab \}$

(C) $\mathcal{L}^1 = \mathcal{L} = \{ a, ab \}$

(D) $\mathcal{L}^0 = \{ \varepsilon \}$

III.1.24

- (A) $\mathcal{L}_0 \cap \mathcal{L}_1 = \{ab, abb, abb, abbb\}$
- (B) $\mathcal{L}_1 \cap \mathcal{L}_0 = \{ba, bab, bba, bbab\}$
- (C) $\mathcal{L}_0^2 = \{aa, aab, aba, abab\}$
- (D) $\mathcal{L}_1^2 = \{bb, bbb, bbbb\}$
- (E) $\mathcal{L}_0^2 \cap \mathcal{L}_1^2 = \{aabb, aabbb, aabbbb, aabbbbb, ababb, ababbb, ababbbb, ababbbbb\}$

III.1.25

- (A) The minimum is three, which happens when the second is a subset of the first. The maximum is five, which happens when the two are disjoint.
- (B) The minimum is zero, which happens when the two are disjoint. The maximum is two, which happens when the second is a subset of the first.
- (C) The maximum is six, as with $\mathcal{L}_0 = \{a, b, c\}$ and $\mathcal{L}_1 = \{x, y\}$, which gives $\mathcal{L}_0 \cap \mathcal{L}_1 = \{ax, ay, bx, by, cx, cy\}$. The minimum is four, as with $\mathcal{L}_0 = \{\varepsilon, a, aa\}$ and $\mathcal{L}_1 = \{\varepsilon, a\}$, which gives $\mathcal{L}_0 \cap \mathcal{L}_1 = \{\varepsilon, a, aa, aaa\}$.
- (D) The minimum is three, as when $\mathcal{L}_1 = \{\varepsilon, a\}$ gives $\mathcal{L}_1^2 = \{\varepsilon, a, aa\}$. The maximum is four, an example of which is $\mathcal{L}_1 = \{a, b\}$ giving $\mathcal{L}_1^2 = \{aa, ab, ba, bb\}$.
- (E) The intersection can be infinite, as with $\mathcal{L}_0 = \{\varepsilon, a, b\}$ and $\mathcal{L}_1 = \{\varepsilon, a\}$, where $a^n \in \mathcal{L}_0^* \cap \mathcal{L}_1^*$ for all $n \in \mathbb{N}$. The smallest the intersection can be is size one, as when $\mathcal{L}_0 = \{a, b, c\}$ and $\mathcal{L}_1 = \{x, y\}$ and the only element of $\mathcal{L}_0^* \cap \mathcal{L}_1^*$ is the empty string.

III.1.26 It is the empty set, $\emptyset^* = \emptyset$. By definition, $\mathcal{L}^* = \{\sigma_0 \cap \dots \cap \sigma_{k-1} \mid k \in \mathbb{N} \text{ and } \sigma_0, \dots, \sigma_{k-1} \in \mathcal{L}\}$. If the language is empty then the condition ' $\sigma_0, \dots, \sigma_{k-1} \in \mathcal{L}$ ' is never met.

V III.1.27 No. Let $\mathcal{L} = \{a, b\}$ and let $k = 2$. Then the k -th power of the language is $\mathcal{L}^k = \{aa, ab, ba, bb\}$, while the language of k -th powers is $\{\sigma^2 \mid \sigma \in \mathcal{L}\} = \{aa, bb\}$.

III.1.28 Yes, but only in an edge case. If $\mathcal{L} = \emptyset$ then $\mathcal{L}^* = \emptyset$ while $(\mathcal{L} \cup \{\varepsilon\})^* = \{\varepsilon\}$.

We will show that if $\mathcal{L} \neq \emptyset$ then the two sets are equal, $\mathcal{L}^* = (\mathcal{L} \cup \{\varepsilon\})^*$. The containment $\mathcal{L}^* \subseteq (\mathcal{L} \cup \{\varepsilon\})^*$ is clear.

For the other direction, $(\mathcal{L} \cup \{\varepsilon\})^* \subseteq \mathcal{L}^*$, assume that $\sigma \in (\mathcal{L} \cup \{\varepsilon\})^*$. Then $\sigma = \sigma_0 \cap \dots \cap \sigma_{n-1}$ for some $n \in \mathbb{N}$ and $\sigma_i \in \mathcal{L} \cup \{\varepsilon\}$. If for all $i \in \{0, \dots, n-1\}$ the string σ_i equals ε then $\sigma = \varepsilon$, and is an element of $\mathcal{L}^*$ because $\mathcal{L} \neq \emptyset$ and $\varepsilon = \rho^0$ for any $\rho \in \mathcal{L}$. Otherwise there is at least one $i \in \{0, \dots, n-1\}$ such that $\sigma_i \neq \varepsilon$. In this case, omitting from the string σ those σ_j that equal the empty string gives that $\sigma \in \mathcal{L}^k$ for some $k \leq n$, and so $\sigma \in \mathcal{L}^*$.

III.1.29 Let the alphabet be Σ .

- (A) Consider unequal strings $\sigma_0 \neq \sigma_1$. They can't both be the empty string, and if one of them is the empty string but the other is not then clearly $\sigma_0^2 \neq \sigma_1^2$ because one of those two is empty while the other is not. So now suppose that neither of the two strings σ_0 and σ_1 is empty. If they are different lengths then their squares are also different lengths. The case remaining is that they are nonempty equal-length strings, $\sigma_0 = \langle s_{0,0}, s_{0,1}, \dots, s_{0,n} \rangle$ and $\sigma_1 = \langle s_{1,0}, s_{1,1}, \dots, s_{1,n} \rangle$ for some $n \in \mathbb{N}$. Fix the smallest index i where $s_{0,i} \neq s_{1,i}$. For that same i we have that σ_0^2 and σ_1^2 differ on index i , and are therefore unequal.

The prior paragraph implies that if $\mathcal{L}$ has k -many different elements $\mathcal{L} = \{\sigma_0, \dots, \sigma_{k-1}\}$ then the elements of the set $\{\sigma_0^2, \dots, \sigma_{k-1}^2\} \subseteq \mathcal{L}^2$ are different.

- (B) The language $\mathcal{L} = \{\varepsilon\}$ has $\mathcal{L}^2 = \{\varepsilon\}$.
- (C) Let $\mathcal{L} = \{\sigma_0, \sigma_1, \dots, \sigma_{k-1}\}$ for $k \in \mathbb{N}^+$ (we take these strings to all be different from each other). Then the list $\sigma_0 \cap \sigma_0, \sigma_0 \cap \sigma_1, \dots, \sigma_{k-1} \cap \sigma_{k-1}$ has length k^2 . Hence the largest number of elements that $\mathcal{L}^2$ can have is k^2 .
- (D) Consider the language $\mathcal{L} = \{\sigma_0, \sigma_1, \dots, \sigma_{k-1}\}$ where the k -many strings have length one and are unequal, so each consists of a single symbol that is different from the symbols used in any of the others. Clearly all of the members of the list $\sigma_0 \cap \sigma_0, \sigma_0 \cap \sigma_1, \dots, \sigma_{k-1} \cap \sigma_{k-1}$ are unequal, and so $\mathcal{L}^2$ has k^2 -many elements.

III.1.30 Recall that $\mathcal{L}^k = \{\sigma_0 \cap \dots \cap \sigma_{k-1} \mid \sigma_i \in \mathcal{L}\}$ and that $\mathcal{L}^* = \{\sigma_0 \cap \dots \cap \sigma_{k-1} \mid k \in \mathbb{N} \text{ and } \sigma_0, \dots, \sigma_{k-1} \in \mathcal{L}\}$. If $\mathcal{L} = \emptyset$ then both sets are empty, so we can assume $\mathcal{L} \neq \emptyset$.

We first show that $\mathcal{L}^k \subseteq \mathcal{L}^*$. If $k = 0$ then because we are assuming $\mathcal{L} \neq \emptyset$, we can take some $\sigma \in \mathcal{L}$ and find $\sigma^0 = \varepsilon$. Thus $\mathcal{L}^0 = \{\varepsilon\}$ and $\varepsilon \in \mathcal{L}^*$. Now suppose that $k > 0$ and fix $\sigma \in \mathcal{L}^k$, so that $\sigma = \sigma_0 \cap \dots \cap \sigma_{k-1}$. Then the definition of $\mathcal{L}^*$ gives that $\sigma \in \mathcal{L}^*$.

We finish by showing that $\mathcal{L}^* \subseteq \mathcal{L}^0 \cup \mathcal{L}^1 \cup \dots$. Fix $\sigma \in \mathcal{L}^*$ (recall that $\mathcal{L} \neq \emptyset$), so that $\sigma = \sigma_0 \cap \dots \cap \sigma_{k-1}$

for some $k \in \mathbb{N}$. Note that $\sigma \in \mathcal{L}^k$ (even if $k = 0$). Thus $\mathcal{L}^* \subseteq \mathcal{L}^0 \cup \mathcal{L}^1 \cup \dots$.

III.1.31 The set of concatenations $\sigma_1 \frown \sigma_0$ where $\sigma_1 \in \mathcal{L}_1$ and $\sigma_0 \in \mathcal{L}_0$ is empty, because there are no σ_0 members of $\mathcal{L}_0$.

III.1.32 The intersection of a language over Σ_0 with a language over Σ_1 is a language over $\Sigma_0 \cap \Sigma_1$. The reason is that to be in the intersection of the languages a string σ must involve only characters that are in both alphabets.

V III.1.33

- (A) Yes, but it has nothing to do with the language being finite. An alphabet is finite by definition. (Appendix A has a review.)
- (B) Because the language is finite, there is a longest string and for B we can take the length of that string (or else the language is empty and we can take $B = 0$).
- (C) Let $\mathcal{L}_0, \dots, \mathcal{L}_k$ be finite languages over Σ^* for some $k \in \mathbb{N}$. Then the union is also finite because it contains at most $|\mathcal{L}_0| + \dots + |\mathcal{L}_k|$ elements.
- (D) Let $\mathcal{L}_0, \dots, \mathcal{L}_k$ be finite languages over Σ^* . Then the intersection contains at most $\min(|\mathcal{L}_0|, \dots, |\mathcal{L}_k|)$ -many elements, so it is finite also. The number of strings in the concatenation $\mathcal{L}_0 \frown \mathcal{L}_1 \frown \dots \frown \mathcal{L}_k$ is at most the product $|\mathcal{L}_0| \cdot |\mathcal{L}_1| \cdot \dots \cdot |\mathcal{L}_k|$.
- (E) If $\Sigma = \{a, b\}$ then the complement of the finite language $\mathcal{L} = \{\}$ is $\mathcal{L}^c = \Sigma^*$, which is infinite since it contains a^n for all n . For the same alphabet, $\mathcal{L} = \{a, b\}$ is finite but $\mathcal{L}^*$ is infinite.

V III.1.34 In $\hat{\mathcal{L}}$ all strings are of even length. The language $\mathcal{L}$ contains all palindromes, of any length.

V III.1.35 The prefixes are $\varepsilon, a, b, ab, bb, aba, bba, abaa, abaab$, and $abaaba$.

III.1.36

- (A) Concatenating $\sigma_0 \frown \dots \frown \sigma_{m-1} \in \mathcal{L}^m$ with $\tau_0 \frown \dots \frown \tau_{n-1} \in \mathcal{L}^n$ gives $\sigma_0 \frown \dots \frown \sigma_{m-1} \frown \tau_0 \frown \dots \frown \tau_{n-1} \in \mathcal{L}^{m+n}$.
- (B) By definition $\mathcal{L}_0 \frown \mathcal{L}_1 = \{\sigma_0 \frown \sigma_1 \mid \sigma_0 \in \mathcal{L}_0 \text{ and } \sigma_1 \in \mathcal{L}_1\}$. If one of the two is the empty set then the condition is never satisfied, so the result is the empty set.
- (C) For any string $\sigma \in \mathcal{L}^*$ we have that $\sigma^0 = \varepsilon$. Thus, if there are any strings in the language then when raised to the zero power they all give the empty string, so the result is $\mathcal{L}^0 = \{\varepsilon\}$.
- (D) By the second item, if $\mathcal{L} = \emptyset$ gave that $\mathcal{L}^0 = \emptyset$ for $\mathcal{L} = \emptyset$, then for any language $\hat{\mathcal{L}}$, extending the formula for the prior item results in $\hat{\mathcal{L}} = \hat{\mathcal{L}}^1 = \hat{\mathcal{L}}^{1+0} = \hat{\mathcal{L}} \frown \emptyset = \emptyset$, which is nonsense.

III.1.37

- (A) Consider the set $S = \Sigma \times \dots \times \Sigma$ (we are avoiding calling this Σ^n because the two differ, in that the cross product is set of n -tuples while Σ^n is a subset of Σ^*). It is the cross product of countably many countable sets and so is countable by Chapter Two's Corollary 2.10. So there is a one-to-one and onto function $g: \mathbb{N} \rightarrow S$. The function $\text{concat}: S \rightarrow \Sigma^n$ that concatenates its arguments together $f(\sigma_0, \dots, \sigma_{n-1}) = \sigma_0 \frown \dots \frown \sigma_{n-1}$ is clearly onto. So the composition $f \circ g: \mathbb{N} \rightarrow \Sigma^n$ is onto. Then by Chapter Two's Lemma 2.13, Σ^n is countable.
- (B) The set $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \dots$ is the countable union of countable sets and so by Chapter Two's Corollary 2.14 is a countable set.

III.1.38 To prevent confusion between the two forms, denote $\{\sigma_0 \frown \dots \frown \sigma_{k-1} \mid \sigma_i \in \mathcal{L}\}$ as ${}^k\mathcal{L}$, for the purpose of this argument.

We first show that any element of ${}^k\mathcal{L}$ is an element of $\mathcal{L}^k$. We do a simple induction argument. The base cases are that ${}^0\mathcal{L}$ and $\mathcal{L}^0$ both equal to $\{\varepsilon\}$ by definition, and that $\sigma_0 \in \mathcal{L}^1$ and $\sigma_0 \in {}^1\mathcal{L}$ for any $\sigma_0 \in \mathcal{L}$. The inductive step is that from the hypothesis that $\sigma_0 \frown \dots \frown \sigma_{i-1} \in {}^i\mathcal{L}$ implies that $\sigma_0 \frown \dots \frown \sigma_{i-1} \in \mathcal{L}^i$ we get that $\sigma_0 \frown \dots \frown \sigma_{i-1} \frown \sigma_i \in \mathcal{L}^{i+1}$, because it is $(\sigma_0 \frown \dots \frown \sigma_{i-1}) \frown \sigma_i$.

The other direction is that any element of any $\mathcal{L}^k$ is an element of ${}^k\mathcal{L}$. Again we do this by a simple induction. Again the base cases are that ${}^0\mathcal{L}$ and $\mathcal{L}^0$ both equal to $\{\varepsilon\}$ by definition, and that $\sigma_0 \in \mathcal{L}^1$ and $\sigma_0 \in {}^1\mathcal{L}$ for any $\sigma_0 \in \mathcal{L}$. The inductive step is that if $\sigma \in \mathcal{L}^i$ then $\sigma \in {}^i\mathcal{L}$ and so $\sigma = \sigma_0 \frown \dots \frown \sigma_{i-1}$. Then $\sigma \frown \tau$ has the form $\sigma_0 \frown \dots \frown \sigma_{i-1} \frown \tau$ and so is an element of ${}^{i+1}\mathcal{L}$.

V III.1.39 The statement is true. As motivation for the argument, the natural language to try for a counterexample is $\mathcal{L} = \{a, aa, \dots\}$. However, $\mathcal{L} \frown \mathcal{L}$ is a proper subset of $\mathcal{L}$ since it does not contain a .

With that, suppose $\mathcal{L} \neq \emptyset$. Then there is a string $\sigma \in \Sigma^*$ such that $\sigma \in \mathcal{L}$. From among all language members, take σ to have minimal length. If $\varepsilon \neq \mathcal{L}$ then $\sigma \frown \tau \in \mathcal{L} \frown \mathcal{L}$ is strictly longer than σ for any $\tau \in \mathcal{L}$,

and by the minimality of σ , it is not an element of $\mathcal{L} \cap \mathcal{L}$. This means that $\mathcal{L} \cap \mathcal{L}$ is not equal to $\mathcal{L}$.

III.1.40 Every language is countable, since it is a subset of Σ^* , which is countable. But there are uncountably many reals.

III.1.41

- (A) For any sets, languages or not, union and intersection are commutative.
- (B) We have $\mathcal{L} \cap \{\varepsilon\} = \{\sigma \cap \varepsilon \mid \sigma \in \mathcal{L}\} = \{\sigma \mid \sigma \in \mathcal{L}\} = \mathcal{L}$.
- (C) Where $\Sigma = \{a, b\}$ take $\mathcal{L}_0 = \{a\}$ and $\mathcal{L}_1 = \{b\}$. Then $\mathcal{L}_0 \cap \mathcal{L}_1 = \{ab\}$ while $\mathcal{L}_1 \cap \mathcal{L}_0 = \{ba\}$.
- (D) This is immediate from the fact that string concatenation is associative: the i -th entry of $(\sigma_0 \cap \sigma_1) \cap \sigma_2$ is the same as the i -th entry of $\sigma_0 \cap (\sigma_1 \cap \sigma_2)$.
- (E) This is immediate from the fact that for strings $(\sigma_0 \cap \sigma_1)^R$ equals $\sigma_1^R \cap \sigma_0^R$.
- (F) A string is an element of $(\mathcal{L}_0 \cup \mathcal{L}_1) \cap \mathcal{L}_2$ if and only if it has the form $\sigma \cap \tau$ where $\tau \in \mathcal{L}_2$, and $\sigma \in \mathcal{L}_0$ or $\sigma \in \mathcal{L}_1$. That's true if and only if both $\tau \in \mathcal{L}_2$ and $\sigma \in \mathcal{L}_0$, or both $\tau \in \mathcal{L}_2$ and $\sigma \in \mathcal{L}_1$. In turn that holds if and only if $\sigma \in (\mathcal{L}_0 \cap \mathcal{L}_2) \cup (\mathcal{L}_1 \cap \mathcal{L}_2)$. The right-distributive argument goes the same way.
- (G) For any language $\mathcal{L}$ the definition gives $\emptyset \cap \mathcal{L} = \{\sigma_0 \cap \sigma_1 \mid \sigma_0 \in \emptyset \text{ and } \sigma_1 \in \mathcal{L}\}$. The condition ' $\sigma_0 \in \emptyset$ ' is never satisfied so this set is empty. The same goes for $\mathcal{L} \cap \emptyset$.
- (H) For any strings, no matter how the concatenation is parenthesized we can remove the parentheses. For instance, $((\sigma_0 \cap \sigma_1) \cap \sigma_2) \cap \sigma_3$ equals $((\sigma_0 \cap (\sigma_1 \cap \sigma_2)) \cap \sigma_3$ because the i -th elements of the two are equal. (We can easily show this statement by induction.) From this, the result is immediate.

III.2.12

- (A) We've adopted the convention that the start symbol is the head of the first listed rule so it is $\langle \text{expr} \rangle$.
- (B) The terminals are a, b, etc.
- (C) The nonterminals are $\langle \text{expr} \rangle$, $\langle \text{term} \rangle$, and $\langle \text{factor} \rangle$.
- (D) There are twenty seven rewrite rules on the final line. The other two lines have a total of four, so that makes thirty one.
- (E) Two are $a+b$ and $j*h+k*m$.
- (F) Three are $a+$, and $+$, and $**$.

III.2.13

- (A) The start symbol, by our convention that it is the head of the first listed rule, is $\langle \text{sentence} \rangle$.
- (B) The terminals are the, young, caught, man, and ball.
- (C) The nonterminals are $\langle \text{sentence} \rangle$, $\langle \text{noun phrase} \rangle$, $\langle \text{verb phrase} \rangle$, $\langle \text{article} \rangle$, $\langle \text{noun} \rangle$, $\langle \text{adjective} \rangle$, and $\langle \text{verb} \rangle$.
- (D) Three are the ball caught the young man, the ball caught the man, and the man caught the man.
- (E) Three are ball, man man man, and ε .

III.2.14

- (A) The alphabet is $\Sigma = \{0, \dots, 9\}$. The terminals are $0, \dots, 9$. The nonterminals are $\langle \text{natural} \rangle$ and $\langle \text{digit} \rangle$. The start symbol is $\langle \text{natural} \rangle$.
- (B) For the production

$$\langle \text{natural} \rangle \rightarrow \langle \text{digit} \rangle$$

the head is $\langle \text{natural} \rangle$ while the body is $\langle \text{digit} \rangle$. For the production

$$\langle \text{natural} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{natural} \rangle$$

the head is $\langle \text{natural} \rangle$ while the body is $\langle \text{digit} \rangle \langle \text{digit} \rangle$. For the production

$$\langle \text{digit} \rangle \rightarrow 0$$

the head is $\langle \text{digit} \rangle$ and the body is 0 . The remaining eight productions are similar.

- (C) The two metacharacters are $\rightarrow$ and $|$.
- (D) Here is a derivation.

$$\begin{aligned} \langle \text{natural} \rangle &\Rightarrow \langle \text{digit} \rangle \langle \text{natural} \rangle \\ &\Rightarrow 4 \langle \text{natural} \rangle \\ &\Rightarrow 4 \langle \text{digit} \rangle \\ &\Rightarrow 42 \end{aligned}$$

For the associated parse tree, see Figure 60 on page 101.

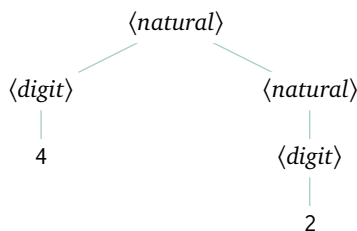


FIGURE 60, FOR QUESTION III.2.14: Parse tree for 42.

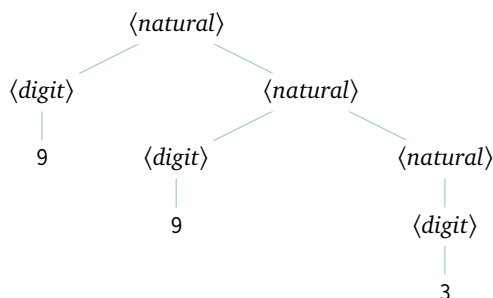


FIGURE 61, FOR QUESTION III.2.14: Parse tree associated with the derivation of 993.

(E) Here is a derivation.

$$\begin{aligned}
 \langle natural \rangle &\Rightarrow \langle digit \rangle \langle natural \rangle \\
 &\Rightarrow \langle digit \rangle \langle digit \rangle \langle natural \rangle \\
 &\Rightarrow \langle digit \rangle \langle digit \rangle \langle digit \rangle \\
 &\Rightarrow \langle digit \rangle 9 \langle digit \rangle \\
 &\Rightarrow \langle digit \rangle 93 \\
 &\Rightarrow 993
 \end{aligned}$$

The parse tree is Figure 61 on page 101.

(F) We want to start at the start symbol and end at the desired string. Just willy-nilly expanding is not the game.

(G) Here is one answer

$$\begin{aligned}
 \langle integer \rangle &\rightarrow +\langle natural \rangle \mid -\langle natural \rangle \mid \langle natural \rangle \\
 \langle natural \rangle &\rightarrow \langle digit \rangle \mid \langle digit \rangle \langle natural \rangle \\
 \langle digit \rangle &\rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9
 \end{aligned}$$

(H) Yes, we can derive both $+0$ and -0 .

III.2.15

(A) Here is a derivation of 'the car hit a wall'.

$$\begin{aligned}
 \langle sentence \rangle &\Rightarrow \langle subject \rangle \langle predicate \rangle \Rightarrow \langle article \rangle \langle noun \rangle \langle predicate \rangle \\
 &\Rightarrow \langle article \rangle \langle noun \rangle \langle verb \rangle \langle direct object \rangle \Rightarrow \langle article \rangle \langle noun \rangle \langle verb \rangle \langle article \rangle \langle noun \rangle \\
 &\Rightarrow \text{the} \langle noun \rangle \langle verb \rangle \langle article \rangle \langle noun \rangle \Rightarrow \text{the car} \langle verb \rangle \langle article \rangle \langle noun \rangle \\
 &\Rightarrow \text{the car hit} \langle article \rangle \langle noun \rangle \Rightarrow \text{the car hit a} \langle noun \rangle \\
 &\Rightarrow \text{the car hit a wall}
 \end{aligned}$$

(B) Here is a derivation of 'the car hit the wall'.

$\langle \text{sentence} \rangle \Rightarrow \langle \text{subject} \rangle \langle \text{predicate} \rangle \Rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \langle \text{predicate} \rangle$
 $\Rightarrow \text{the} \langle \text{noun} \rangle \langle \text{predicate} \rangle \Rightarrow \text{the car} \langle \text{predicate} \rangle$
 $\Rightarrow \text{the car} \langle \text{verb} \rangle \langle \text{direct object} \rangle \Rightarrow \text{the car hit} \langle \text{direct object} \rangle$
 $\Rightarrow \text{the car hit} \langle \text{article} \rangle \langle \text{noun} \rangle \Rightarrow \text{the car hit the} \langle \text{noun} \rangle$
 $\Rightarrow \text{the car hit the wall}$

(c) This is one for ‘the wall hit a car’.

$\langle \text{sentence} \rangle \Rightarrow \langle \text{subject} \rangle \langle \text{predicate} \rangle \Rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \langle \text{predicate} \rangle$
 $\Rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \langle \text{verb} \rangle \langle \text{direct object} \rangle \Rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \langle \text{verb} \rangle \langle \text{article} \rangle \langle \text{noun} \rangle$
 $\Rightarrow \langle \text{article} \rangle \langle \text{noun} \rangle \text{hit} \langle \text{article} \rangle \langle \text{noun} \rangle \Rightarrow \langle \text{article} \rangle \text{wall hit} \langle \text{article} \rangle \langle \text{noun} \rangle$
 $\Rightarrow \langle \text{article} \rangle \text{wall hit} \langle \text{article} \rangle \text{car} \Rightarrow \langle \text{article} \rangle \text{wall hit a car}$
 $\Rightarrow \text{the wall hit a car}$

III.2.16

(A) This is a derivation for ‘dog bites man’.

$\langle \text{sentence} \rangle \Rightarrow \langle \text{subject} \rangle \langle \text{predicate} \rangle \Rightarrow \langle \text{article} \rangle \langle \text{noun}_1 \rangle \langle \text{predicate} \rangle$
 $\Rightarrow \langle \text{article} \rangle \langle \text{noun}_1 \rangle \langle \text{verb} \rangle \langle \text{direct object} \rangle \Rightarrow \langle \text{article} \rangle \langle \text{noun}_1 \rangle \langle \text{verb} \rangle \langle \text{article} \rangle \langle \text{noun}_2 \rangle$
 $\Rightarrow \varepsilon \langle \text{noun}_1 \rangle \langle \text{verb} \rangle \langle \text{article} \rangle \langle \text{noun}_2 \rangle \Rightarrow \varepsilon \text{dog} \langle \text{verb} \rangle \langle \text{article} \rangle \langle \text{noun}_2 \rangle$
 $\Rightarrow \varepsilon \text{dog bites} \langle \text{article} \rangle \langle \text{noun}_2 \rangle \Rightarrow \varepsilon \text{dog bites} \varepsilon \langle \text{noun}_2 \rangle$
 $\Rightarrow \varepsilon \text{dog bites} \varepsilon \text{man} = \text{dog bites man}$

(B) The start symbol $\langle \text{subject} \rangle$ expands to $\langle \text{article} \rangle \langle \text{noun}_1 \rangle$, but man is derived only from $\langle \text{noun}_2 \rangle$. So man cannot be the first word of the sentence.

III.2.17 There is no nonterminal $\langle \text{dog} | \text{flea} \rangle$ or $\langle \text{man} | \text{dog} \rangle$. This is confusing nonterminals with terminals.

III.2.18 The outermost operation is the $*$, so we go for that first. Next we get the parentheses and the $+$. Here is a derivation.

$\langle \text{expr} \rangle \Rightarrow \langle \text{term} \rangle$
 $\Rightarrow \langle \text{term} \rangle * \langle \text{factor} \rangle$
 $\Rightarrow \langle \text{factor} \rangle * \langle \text{factor} \rangle$
 $\Rightarrow (\langle \text{expr} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (\langle \text{term} \rangle + \langle \text{expr} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (\langle \text{term} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (\langle \text{factor} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (a + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (a + \langle \text{factor} \rangle) * \langle \text{factor} \rangle$
 $\Rightarrow (a + b) * \langle \text{factor} \rangle$
 $\Rightarrow (a + b) * c$

III.2.19

(A) First the leftmost derivation.

$S \Rightarrow T b U \Rightarrow a T b U \Rightarrow a a T b U \Rightarrow a a \varepsilon b U = a a b U$
 $\Rightarrow a a b a U \Rightarrow a a b a b U \Rightarrow a a b a b \varepsilon = a a b a b$

Next the rightmost derivation.

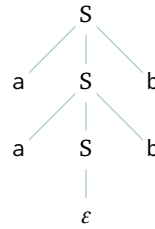
$S \Rightarrow T b U \Rightarrow T b a U \Rightarrow T b a b U \Rightarrow T b a b \varepsilon = T b a b$
 $\Rightarrow a T b a b \Rightarrow a a T b a b \Rightarrow a a \varepsilon b a b = a a b a b$

(B) This is the leftmost derivation.

$S \Rightarrow T b U \Rightarrow \varepsilon b U = b U \Rightarrow b a U$
 $\Rightarrow b a a U \Rightarrow b a a b U \Rightarrow b a a b \varepsilon = b a a b$

Here is the rightmost.

$S \Rightarrow T b U \Rightarrow T b a U \Rightarrow T b a a U \Rightarrow T b a a b U$
 $\Rightarrow T b a a b \varepsilon = T b a a b \Rightarrow \varepsilon b a a b = b a a b$

FIGURE 62, FOR QUESTION III.2.22: Parse tree for a^2b^2 .

(c) The start symbol expands to something that includes a b.

III.2.20

(A) Here are the derivations of aabb,

$$S \Rightarrow aABb \Rightarrow aaBb \Rightarrow aabb$$

of aaabb,

$$S \Rightarrow aABb \Rightarrow aaABb \Rightarrow aaaBb \Rightarrow aabb$$

and of aabbb

$$S \Rightarrow aABb \Rightarrow aABb \Rightarrow aaBb \Rightarrow aaBbb \Rightarrow aabbb$$

(B) The production starting with the start symbol S shows that every derived string must begin with a and end with b. So three strings that cannot be derived are a, ba, and ϵ (the empty string).

(c) $\mathcal{L} = \{a^n b^m \mid n, m \geq 2\}$

III.2.21 Here is one.

$$S \rightarrow TU$$

$$T \rightarrow aTb \mid \epsilon$$

$$U \rightarrow bUa \mid \epsilon$$

III.2.22 Figure 62 on page 103 shows the tree.

III.2.23 We will use induction to show that in the course of a derivation all of the results are of either the form $a^k S b^k$ or of the form $a^k b^k$. By the definition of the language generated by a grammar, only strings without any nonterminals are in the language so that will give the verification.

The induction is on the number of derivation steps, that is, on the number of rule applications. The base step is that there are zero-many rule applications. In this case the string is 'S', which is the $k = 0$ instance of the desired form.

For the inductive step assume that the statement is true where the number of rule applications is $n = 0$, $n = 1$, $\dots$ $n = k$ and consider the $n = k + 1$ case. We get the string for the $k + 1$ case by applying a rule to a string after the k case. The inductive hypothesis is that such a string has one of two forms, $a^k S b^k$ or $a^k b^k$. The rules don't apply to the latter string so consider the former one.

Applied to $a^k S b^k$ the first rule gives $a^{k+1} S b^{k+1}$, which has the right form. The second rule gives $a^{k+1} b^{k+1}$, again of the right form.

III.2.24 Bit of a trick question: this grammar has no terminating derivations so it generates the empty language $\mathcal{L} = \emptyset$.

III.2.25 Here is one.

$$\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \mid \langle \text{letter-or-digit-string} \rangle$$

$$\langle \text{letter-or-digit-string} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{letter-or-digit-string} \rangle$$

$$\mid \langle \text{digit} \rangle \langle \text{letter-or-digit-string} \rangle$$

$$\mid \epsilon$$

$$\langle \text{letter} \rangle \rightarrow a \mid \dots \mid z \mid A \mid \dots \mid Z$$

$\langle \text{digit} \rangle \rightarrow 0 \mid \dots \mid 9$

III.2.26

$\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{second} \rangle$

$\langle \text{second} \rangle \rightarrow \langle \text{letter-or-digit} \rangle \langle \text{third} \rangle \mid \varepsilon$

$\langle \text{third} \rangle \rightarrow \langle \text{letter-or-digit} \rangle \langle \text{fourth} \rangle \mid \varepsilon$

$\langle \text{fourth} \rangle \rightarrow \langle \text{letter-or-digit} \rangle$

$\langle \text{letter-or-digit} \rangle \rightarrow \langle \text{letter} \rangle \mid \langle \text{digit} \rangle$

$\langle \text{letter} \rangle \rightarrow a \mid \dots \mid z \mid A \mid \dots \mid Z$

$\langle \text{digit} \rangle \rightarrow 0 \mid \dots \mid 9$

III.2.27 It is the empty language, $\mathcal{L} = \emptyset$.

III.2.28

(A) This is one derivation.

$$\begin{aligned}
 \langle \text{CNF} \rangle &\Rightarrow (\langle \text{Disjunction} \rangle) \wedge \langle \text{CNF} \rangle \\
 &\Rightarrow (\langle \text{Disjunction} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (\langle \text{Literal} \rangle \vee \langle \text{Disjunction} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (\langle \text{Literal} \rangle \vee \langle \text{Literal} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (\langle \text{Variable} \rangle \vee \langle \text{Literal} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (x_0 \vee \langle \text{Literal} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (x_0 \vee \neg \langle \text{Variable} \rangle) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (x_0 \vee \neg x_1) \wedge (\langle \text{Disjunction} \rangle) \\
 &\Rightarrow (x_0 \vee \neg x_1) \wedge (\langle \text{Literal} \rangle \vee \langle \text{Disjunction} \rangle) \\
 &\Rightarrow (x_0 \vee \neg x_1) \wedge (\langle \text{Literal} \rangle \vee \langle \text{Literal} \rangle) \\
 &\Rightarrow (x_0 \vee \neg x_1) \wedge (x_1 \vee \langle \text{Literal} \rangle) \\
 &\Rightarrow (x_0 \vee \neg x_1) \wedge (x_1 \vee x_2)
 \end{aligned}$$

(B) In the grammar the only way to have a multiple clause expression is to join them with $\wedge$'s.

III.2.29

(A) $\langle \text{string} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{string} \rangle \mid \varepsilon$

$\langle \text{letter} \rangle \rightarrow a \mid \dots \mid z$

(B) $\langle \text{string}_1 \rangle \rightarrow \langle \text{digit} \rangle \langle \text{string}_2 \rangle$

$\langle \text{string}_2 \rangle \rightarrow \langle \text{digit} \rangle \langle \text{string}_2 \rangle \mid \varepsilon$

$\langle \text{digit} \rangle \rightarrow 0 \mid \dots \mid 9$

III.2.30 Here is the start of a derivation.

$$\begin{aligned}
 \langle \text{postal address} \rangle &\Rightarrow \langle \text{name} \rangle \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{personal part} \rangle \langle \text{last name} \rangle \langle \text{opt suffix} \rangle \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{first name} \rangle \langle \text{last name} \rangle \langle \text{opt suffix} \rangle \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{last name} \rangle \langle \text{opt suffix} \rangle \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{opt suffix} \rangle \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \varepsilon \langle \text{EOL} \rangle \langle \text{street address} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &= \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{house num} \rangle \langle \text{street name} \rangle \langle \text{apt num} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{street name} \rangle \langle \text{apt num} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \langle \text{apt num} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \varepsilon \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &= \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{town} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{town name} \rangle, \langle \text{state or region} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{char string} \rangle, \langle \text{state or region} \rangle \\
 &\Rightarrow \langle \text{char string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{digit string} \rangle \langle \text{char string} \rangle \langle \text{EOL} \rangle \langle \text{char string} \rangle, \langle \text{char string} \rangle
 \end{aligned}$$

(A) From the above derivation steps we can easily get to the address.

- (B) The above derivation steps get us most of the way there. The problem is the 'B' in 221B; it is not part of a digit string.
- (C) Some are that you may need more than one line for the address, that some addresses do not have a house number, and that some names or addresses require non-ASCII letters.

V III.2.31

- (A) These two rules suffice: $S \rightarrow 11S \mid \epsilon$.
- (B) As with the prior item, this is straightforward: $S \rightarrow 111S \mid \epsilon$.

III.2.32

- (A) This is a sentence of length one.

$$\langle \text{sentence} \rangle \Rightarrow \text{buffalo} \langle \text{sentence} \rangle \Rightarrow \text{buffalo } \epsilon = \text{buffalo}$$

Here is a sentence of length one.

$$\begin{aligned} \langle \text{sentence} \rangle &\Rightarrow \text{buffalo} \langle \text{sentence} \rangle \Rightarrow \text{buffalo buffalo} \langle \text{sentence} \rangle \\ &\Rightarrow \text{buffalo buffalo } \epsilon = \text{buffalo buffalo} \end{aligned}$$

And, a sentence of length three.

$$\begin{aligned} \langle \text{sentence} \rangle &\Rightarrow \text{buffalo} \langle \text{sentence} \rangle \Rightarrow \text{buffalo buffalo} \langle \text{sentence} \rangle \\ &\Rightarrow \text{buffalo buffalo buffalo} \langle \text{sentence} \rangle \Rightarrow \text{buffalo buffalo buffalo } \epsilon \\ &= \text{buffalo buffalo buffalo} \end{aligned}$$

- (B) In English 'buffalo' is a noun referring to the American bison, or to a city in New York State, and it is a verb referring to fooling or confusing someone. So, the sentence of length one could be a command instructing the listener to fool someone. The sentence of length two could be a command instructing that same listener to fool all of a city in upstate New York. The length three sentence could exhort American bison in particular to confuse that entire city.

III.2.33

- (A) This is a derivation of $(a \cdot b)$.

$$\begin{aligned} \langle s \text{ expression} \rangle &\Rightarrow (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle) \Rightarrow (\langle \text{atomic symbol} \rangle \cdot \langle s \text{ expression} \rangle) \\ &\Rightarrow (\langle \text{atomic symbol} \rangle \cdot \langle \text{atomic symbol} \rangle) \Rightarrow (\langle \text{letter} \rangle \langle \text{atom part} \rangle \cdot \langle \text{atomic symbol} \rangle) \\ &\Rightarrow (\langle \text{letter} \rangle \epsilon \cdot \langle \text{atomic symbol} \rangle) = (\langle \text{letter} \rangle \cdot \langle \text{atomic symbol} \rangle) \\ &\Rightarrow (a \cdot \langle \text{atomic symbol} \rangle) \Rightarrow (a \cdot \langle \text{letter} \rangle \langle \text{atom part} \rangle) \\ &\Rightarrow (a \cdot \langle \text{letter} \rangle \epsilon) = (a \cdot \langle \text{letter} \rangle) \Rightarrow (a \cdot b) \end{aligned}$$

- (B) This is a derivation of $(a \cdot (b \cdot c))$.

$$\begin{aligned} \langle s \text{ expression} \rangle &\Rightarrow (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle) \\ &\Rightarrow (\langle s \text{ expression} \rangle \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (\langle s \text{ expression} \rangle \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (\langle \text{atomic symbol} \rangle \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (\langle \text{letter} \rangle \langle \text{atomic part} \rangle \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (a \langle \text{atomic part} \rangle \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (a \epsilon \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) = (a \cdot (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle)) \\ &\Rightarrow (a \cdot (\langle \text{atomic symbol} \rangle \cdot \langle s \text{ expression} \rangle)) \Rightarrow (a \cdot (\langle \text{atomic symbol} \rangle \cdot \langle \text{atomic symbol} \rangle)) \\ &\Rightarrow (a \cdot (\langle \text{letter} \rangle \langle \text{atomic part} \rangle \cdot \langle \text{atomic symbol} \rangle)) \\ &\Rightarrow (a \cdot (\langle \text{letter} \rangle \langle \text{atomic part} \rangle \cdot \langle \text{letter} \rangle \langle \text{atomic part} \rangle)) \\ &\Rightarrow (a \cdot (\langle \text{letter} \rangle \epsilon \cdot \langle \text{letter} \rangle \langle \text{atomic part} \rangle)) = (a \cdot (\langle \text{letter} \rangle \cdot \langle \text{letter} \rangle \langle \text{atomic part} \rangle)) \\ &\Rightarrow (a \cdot (\langle \text{letter} \rangle \cdot \langle \text{letter} \rangle \epsilon)) = (a \cdot (\langle \text{letter} \rangle \cdot \langle \text{letter} \rangle)) \\ &\Rightarrow (a \cdot (b \cdot \langle \text{letter} \rangle)) \Rightarrow (a \cdot (b \cdot c)) \end{aligned}$$

- (C) This is the first few steps of a derivation of $((a \cdot b) \cdot c)$.

$$\begin{aligned} \langle s \text{ expression} \rangle &\Rightarrow (\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle) \\ &\Rightarrow ((\langle s \text{ expression} \rangle \cdot \langle s \text{ expression} \rangle) \cdot \langle s \text{ expression} \rangle) \end{aligned}$$

The rest proceeds as does the prior answer.

III.2.34

The steps are tedious but routine.

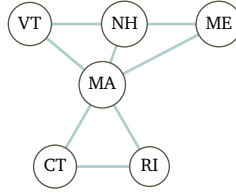


FIGURE 63, FOR QUESTION III.3.16: Adjacency relation among the New England states.

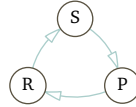


FIGURE 64, FOR QUESTION III.3.16: Relation of 'beats' among Rock, Paper, and Scissors.

$$\begin{aligned}
 \langle \text{expr} \rangle &\Rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle \Rightarrow \langle \text{expr} \rangle + \langle \text{factor} \rangle \\
 &\Rightarrow \langle \text{expr} \rangle + (\langle \text{expr} \rangle) \Rightarrow \langle \text{expr} \rangle + (\langle \text{term} \rangle) \\
 &\Rightarrow \langle \text{expr} \rangle + (\langle \text{term} \rangle * \langle \text{factor} \rangle) \Rightarrow \langle \text{expr} \rangle + (\langle \text{factor} \rangle * \langle \text{factor} \rangle) \\
 &\Rightarrow \langle \text{term} \rangle + (\langle \text{factor} \rangle * \langle \text{factor} \rangle) \Rightarrow \langle \text{factor} \rangle + (\langle \text{factor} \rangle * \langle \text{factor} \rangle) \\
 &\Rightarrow a + (\langle \text{factor} \rangle * \langle \text{factor} \rangle) \Rightarrow a + (b * \langle \text{factor} \rangle) \\
 &\Rightarrow a + (b * c)
 \end{aligned}$$

III.2.35

- (A) It is $\mathcal{L} = \{ \varepsilon \}$.
 (B) This is one leftmost derivation.

$$S \Rightarrow \varepsilon$$

and here is another.

$$S \Rightarrow S \Rightarrow \varepsilon$$

III.2.36 One leftmost derivation is this.

$$\begin{aligned}
 \langle \text{bit-string} \rangle &\Rightarrow \langle \text{bit-string} \rangle \langle \text{bit-string} \rangle \Rightarrow \emptyset \langle \text{bit-string} \rangle \\
 &\Rightarrow \emptyset \langle \text{bit-string} \rangle \langle \text{bit-string} \rangle \Rightarrow \emptyset \emptyset \langle \text{bit-string} \rangle \\
 &\Rightarrow \emptyset \emptyset \emptyset
 \end{aligned}$$

Here is another.

$$\begin{aligned}
 \langle \text{bit-string} \rangle &\Rightarrow \langle \text{bit-string} \rangle \langle \text{bit-string} \rangle \Rightarrow \langle \text{bit-string} \rangle \emptyset \\
 &\Rightarrow \langle \text{bit-string} \rangle \langle \text{bit-string} \rangle \emptyset \Rightarrow \emptyset \langle \text{bit-string} \rangle \emptyset \\
 &\Rightarrow \emptyset \emptyset \emptyset
 \end{aligned}$$

III.2.37

- (A) One leftmost derivation is $E \Rightarrow E - E \Rightarrow a - E \Rightarrow a - E - E \Rightarrow a - b - E \Rightarrow a - b - a$. A second one is $E \Rightarrow E - E \Rightarrow E - E - E \Rightarrow a - E - E \Rightarrow a - b - E \Rightarrow a - b - a$.
 (B) Here we go: $E \Rightarrow E - T \Rightarrow E - T - T \Rightarrow T - T - T \Rightarrow a - T - T \Rightarrow a - b - T \Rightarrow a - b - a$.

III.2.38 $S \Rightarrow aBSc \Rightarrow aBaBScc \Rightarrow aBaBabccc \Rightarrow aaBBabccc \Rightarrow aaBaBbccc \Rightarrow aaaBBbccc \Rightarrow aaaBbbccc \Rightarrow aaabbbccc$ **III.3.16**

- (A) Figure 63 on page 106 shows the graph.
 (B) The graph is a loop, Figure 64 on page 106.
 (C) The graph is Figure 65 on page 107.
 (D) This graph is Figure 66 on page 107.

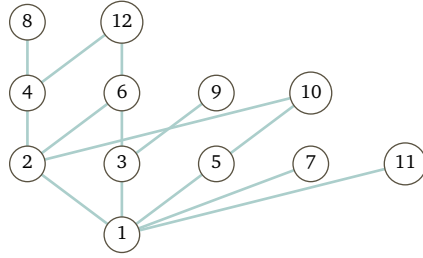


FIGURE 65, FOR QUESTION III.3.16: Divisibility relation among small numbers.

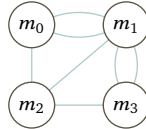


FIGURE 66, FOR QUESTION III.3.16: Bridge connection relation between land masses.

(E) This prerequisite structure is Figure 67 on page 108.

III.3.17

	Vertices can repeat?	Edges can repeat?	Can be closed?	Can be open?
Walk	Y	Y	Y	Y
Trail	Y	N	Y	Y
Circuit	Y	N	Y	N
Path	N	N	Y	Y
Cycle	N	N	Y	N

III.3.18

(A) One possible picture is Figure 68 on page 108.

(B) Here is the matrix.

$$\mathcal{M}(\mathcal{G}) = \begin{matrix} & \begin{matrix} v_0 & v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}$$

(c) From the picture, Figure 68 on page 108, there are clearly only two: $G_0 = \{v_0, v_1, v_4, v_5\}$ and $G_1 = \{v_0, v_3, v_4, v_5\}$.

(D) It is the same two as in the prior item.

III.3.19

(A) See Figure 69 on page 108.

(B) See Figure 70 on page 108.

(c) It is $\binom{n}{2} = n(n+1)/2$.

III.3.20

See Figure 71 on page 108.

III.3.21

(A) There are ten vertices, $\mathcal{N} = \{v_0, \dots, v_9\}$. There are fifteen edges.

$$\mathcal{E} = \text{set } v_0v_1, v_0v_4, v_0v_5, v_1v_2, v_1v_6, v_2v_3, v_2v_7, v_3v_4, v_3v_8, v_4v_9, v_5v_7, v_5v_8, v_6v_8, v_6v_9, v_7v_9$$

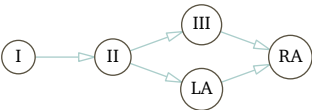


FIGURE 67, FOR QUESTION III.3.16: Prerequisite structure among courses.

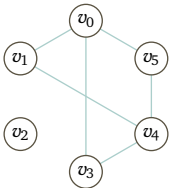


FIGURE 68, FOR QUESTION III.3.18: A graph with some edges.

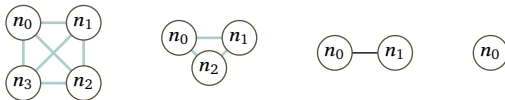


FIGURE 69, FOR QUESTION III.3.19: Complete graphs K_4 , K_3 , K_2 , and K_1 .

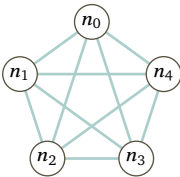


FIGURE 70, FOR QUESTION III.3.19: Complete graph K_5 .

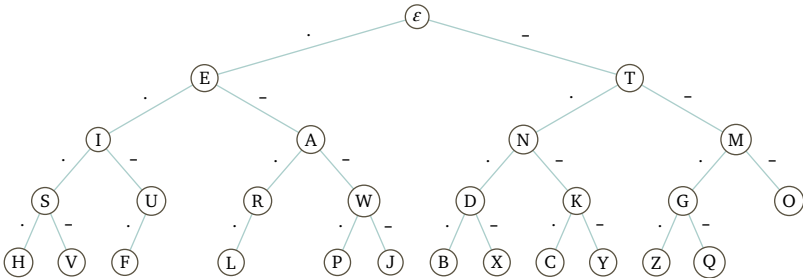


FIGURE 71, FOR QUESTION III.3.20: Prefix relation for Morse code representations of ASCII letters.

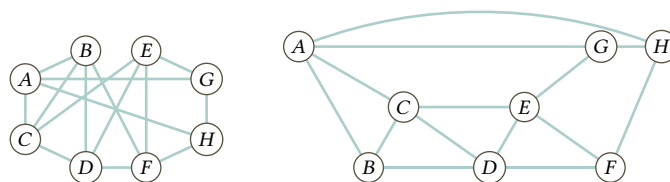


FIGURE 72, FOR QUESTION III.3.22: Drawings of the same graph, with edge crossings and without.

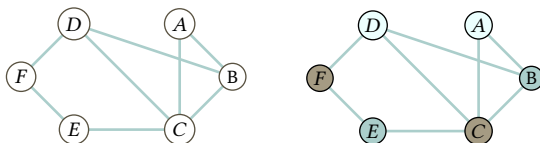


FIGURE 73, FOR QUESTION III.3.24: Relation of incompatibility between fish species.

- (B) The walk $w_1 = \langle v_0v_5, v_5v_7 \rangle$ has length two. The walk $w_2 = \langle v_0v_1, v_1v_2, v_2v_7 \rangle$ has length three.
 (C) A closed walk is $\langle v_4v_0, v_0v_1, v_1v_2, v_2v_3, v_3v_4 \rangle$. An open one is $\langle v_4v_9, v_9v_6, v_6v_8, v_8v_3, v_3v_2 \rangle$.
 (D) One cycle is $\langle v_5v_0, v_0v_1, v_1v_6, v_6v_8, v_8v_5 \rangle$.
 (E) Yes, this graph is connected since we can find a path between any two vertices.

III.3.22

- (A) On the left of Figure 72 on page 109 is the drawing with the vertices connected by the given edges.
 (B) See the right side of Figure 72 on page 109 for the planar drawing.

III.3.23 Fix a vertex (if the tree is empty then the problem is trivial). Give it one color, and give its neighbors the second color. Assign the first color to all the vertices that are two away from the initial vertex. In general, where a vertex is k away from the initial vertex, give it the first color if k is even and the second color if k is odd. Trees don't have loops, so this assignment is well-defined.

III.3.24

- (A) See Figure 73 on page 109.
 (B) The chromatic number is three; see the right-hand graph in the same figure.
 (C) It is the number of fish tanks needed to safely keep all the fish species.

III.3.25 The minimal number of frequencies is three. Figure 74 on page 110 gives a three-coloring. Starting with v_6 and tracing around shows that no two-coloring will suffice.

III.3.26 The degree sequence is $\langle 3, 3, 2, 2, 2, 2, 2, 1 \rangle$.

III.3.27 The directed graph is Figure 75 on page 110. Every type is compatible with itself so every vertex is shown with a loop. The type O^- can donate to everyone, as we see from its row below, so it is a universal donor. The type AB^+ is a universal receptor, as we see from its column.

Here is the graph's adjacency matrix.

	O^-	O^+	A^-	A^+	B^-	B^+	AB^-	AB^+
O^-	1	1	1	1	1	1	1	1
O^+	0	1	0	1	0	1	0	1
A^-	0	0	1	1	0	0	1	1
A^+	0	0	0	1	0	0	0	1
B^-	0	0	0	0	1	1	1	1
B^+	0	0	0	0	0	1	0	1
AB^-	0	0	0	0	0	0	1	1
AB^+	0	0	0	0	0	0	0	1

- V III.3.28** The graph in Example 3.2 has this degree sequence: $\langle 4, 4, 3, 3, 2 \rangle$. The vertices v_1 and v_2 have degree 4, the vertices v_3 and v_4 have degree 3, and the vertex v_0 has degree 2.

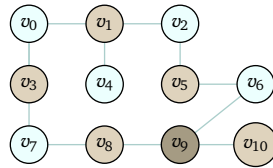


FIGURE 74, FOR QUESTION III.3.25: Cell tower three-coloring.

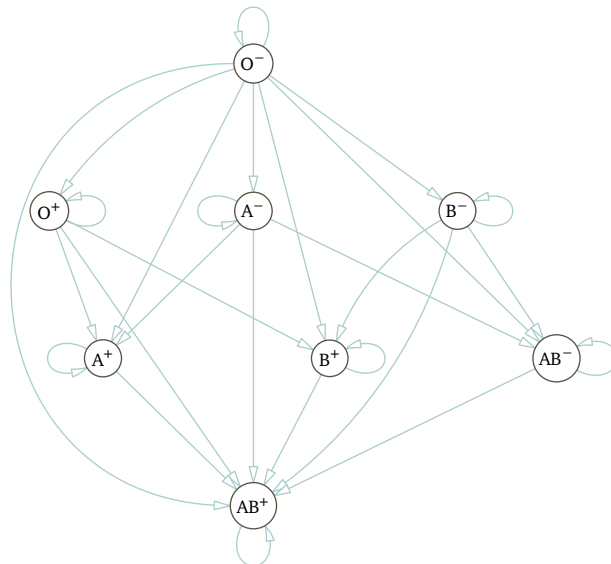


FIGURE 75, FOR QUESTION III.3.27: Compatibility for blood types.

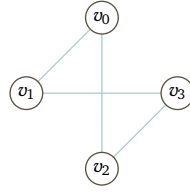


FIGURE 76, FOR QUESTION III.3.30: Graph associated with the given matrix.

In Example 3.4 the graph on the left has degree sequence $\langle 3, 3, 3, 3 \rangle$. The graph on the right has $\langle 3, 3, 3, 3, 3, 3, 3, 3 \rangle$ (there are eight 3's, one for each corner of the cube).

III.3.29 Both of these graphs are complete in the sense that every possible node-to-node connection is there (except that there are no loops). Here is the adjacency matrix for the graph on the left.

$$\begin{array}{c} u_0 \quad u_1 \quad u_2 \quad u_3 \\ \begin{array}{c} u_0 \\ u_1 \\ u_2 \\ u_3 \end{array} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \end{array}$$

And this is the adjacency matrix for the cube graph on the right.

$$\begin{array}{c} v_0 \quad v_1 \quad v_2 \quad v_3 \quad v_4 \quad v_5 \quad v_6 \quad v_7 \\ \begin{array}{c} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} \end{array}$$

III.3.30 The simple graph is Figure 76 on page 111.

III.3.31

- (A) The function is suggested by the names of the vertices: $a \mapsto A$, $b \mapsto B$, etc. It is one-to-one and onto by inspection so it is a correspondence.
- (B) The edges on the left are: ax , ay , az , bx , by , bz , cx , cy , and cz . The edges on the right are: AX , AY , AZ , BX , BY , BZ , CX , CY , and CZ . Under the mapping the edge ax is associated with AX , the edge ay is associated with AY , etc., and this exhausts the edges on the right.

III.3.32 Both graphs have six nodes of degree 3. So for both graphs the degree sequence is $\{3, 3, 3, 3, 3, 3\}$.

III.3.33

- (A) A path is a trail with distinct vertices (except that possibly the starting vertex equals the ending vertex). This trail meets that criteria.
- (B) The vertex B appears twice, not both at the start and end.
- (C) Figure 77 on page 112 illustrates. By definition every tree is connected, so there is a path. Suppose for contradiction that there are two unequal paths from v_0 to v_n . Then there is some vertex v_i where the paths diverge and also a vertex v_k where they come together again. But that makes a cycle from v_i , around the one path to v_k , and back by the other path to return to v_i . The contradiction is that trees do not have cycles.

III.3.34 There are three nodes of degree 2 and four nodes of degree 1. The degree sequence is $\{2, 2, 2, 1, 1, 1, 1\}$.

III.3.35

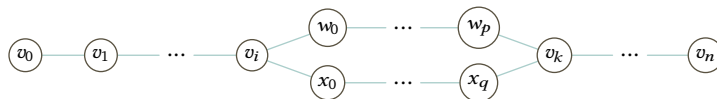


FIGURE 77, FOR QUESTION III.3.33: Two distinct paths makes a cycle.

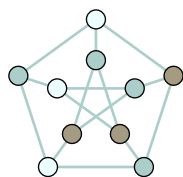


FIGURE 78, FOR QUESTION III.3.35: A 3-coloring of the Petersen graph.

- (A) The outside is a ring of five nodes. With only two colors, in a path around it we must alternate colors but then because it has an odd number of vertices, the fourth node and the fifth are then the same color.

(B) See Figure 78 on page 112.

III.3.36

(A) A breadth first traversal is the sequence $\langle A, B, C, D, E, F, G \rangle$.

(B) A depth-first traversal is $\langle A, B, D, E, C, F, G \rangle$.

III.3.37

(A) For each edge we choose two vertices. Thus the number of potential edges is $\binom{n}{2} = n \cdot (n-1)/2$.

(B) For each edge, we either include it or leave it out. So with n vertices there are $2^{n \cdot (n-1)/2}$ graphs. (Some of these may be isomorphic, but they are different in that they involve different vertices connected.)

(C)

n	0	1	2	3	4	5	6
$2^{n \cdot (n-1)/2}$	1	1	2	8	64	1024	32768

III.3.38

(A) By the definition of graph isomorphism, Definition 3.15, if two graphs $\mathcal{G}$ and $\hat{\mathcal{G}}$ are isomorphic then there is a correspondence between their vertices. They therefore have the same number of vertices.

(B) As in the prior item, the definition of graph isomorphism states that the edges correspond.

(C) Suppose that $\mathcal{G} = \langle \mathcal{N}, \mathcal{E} \rangle$ and $\hat{\mathcal{G}} = \langle \hat{\mathcal{N}}, \hat{\mathcal{E}} \rangle$ are isomorphic via the function f . Let vertex $v \in \mathcal{N}$ have degree k . Then in the set $\mathcal{E}$ the vertex v occurs k times (counting once for edges in which the vertex occurs once, and twice for loops on v).

Apply the correspondence. That is, consider the vertex $f(v)$ and the set $\hat{\mathcal{E}}$. Because f gives a correspondence among edges, the vertex must occur k times in the set. So $f(v)$ has degree k .

(D) This is the same as the prior item, but instead of a single vertex v there is a sequence of vertices $v_0, \dots, v_n$.

(E) Clearly they have the same degree sequence, $\langle 3, 1, 1, 1, 1 \rangle$. Suppose that the two are isomorphic via the correspondence f . Then f must associate the two degree-3 vertices. The graph on the left has two unequal length 2 paths that start with its degree 3 vertex. The graph on the right does not have two.

(F) The degree sequences differ. On the left the degree sequence is $\langle 3, 3, 3, 3 \rangle$ while on the right it is $\langle 4, 4, 4, 4, 4, 4, 4 \rangle$.

III.3.39

(A) There is a length 2 walk from v_i to v_j if and only if there is some v_k where both $v_i v_k$ and $v_k v_j$ are edges. Thus the number of such walks is $m_{i,1}m_{1,j} + m_{i,2}m_{2,j} + \dots + m_{i,n}m_{n,j}$, where each $m_{a,b}$ is an entry from $\mathcal{M}(\mathcal{G})$. But that is the i, j entry of $(\mathcal{M}(\mathcal{G}))^2$.

(B) For induction assume the hypothesis that the statement is true for $n = 2, \dots, n = p$ and consider $n = p + 1$. There is a length $p + 1$ walk from v_i to v_j if and only if there is some v_k where there is a length p walk from v_i to v_k , and also $v_k v_j$ is an edge. By the inductive hypothesis the number of such walks is

$q_{i,1}m_{1,j} + q_{i,2}m_{2,j} + \cdots q_{i,n}m_{n,j}$, where each $q_{i,b}$ is an entry from $(\mathcal{M}(\mathcal{G}))^p$ and $m_{b,j}$ is an entry from $\mathcal{M}(\mathcal{G})$. We recognize that as the i, j entry of $(\mathcal{M}(\mathcal{G}))^{p+1}$.

III.3.40

(A) The vertices are $\mathcal{N}_0 = \{v_0, \dots, v_7\}$. These are the edges.

$$\mathcal{E}_0 = \{v_0v_1, v_0v_2, v_0v_4, v_1v_3, v_2v_3, v_2v_6, v_4v_5, v_4v_6, v_5v_7, v_6v_7\}$$

The vertices of $\mathcal{G}_1$ are $\mathcal{N}_1 = \{n_0, \dots, n_7\}$. These are the edges.

$$\mathcal{E}_1 = \{n_0n_4, n_0n_5, n_1n_4, n_1n_5, n_2n_3, n_2n_6, n_3n_4, n_3n_7, n_5n_6, n_6n_7\}$$

(B) The degree sequences are equal. Each is 3, 3, 3, 3, 2, 2, 2, 2.

(C) These are the images of edges from the left-hand graph.

edge of $\mathcal{G}_0$	v_0v_1	v_0v_2	v_0v_4	v_1v_3	v_2v_3	v_2v_6	v_4v_5	v_4v_6	v_5v_7	v_6v_7
image edge	n_6n_2	n_6n_7	n_6n_5	n_2n_3	n_7n_3	n_7n_0	n_5n_1	n_5n_0	n_1n_4	n_0n_4

That list has no n_3n_4 , although $\mathcal{G}_1$ has such an edge. Further, that list shows n_7n_0 , although $\mathcal{G}_1$ has no such edge. So the given correspondence between vertices does not induce a correspondence between edges. This map is not a graph isomorphism.

(D) The graph $\mathcal{G}_1$ has a pair of degree 3 vertices, n_6 and n_3 , that are separated by a degree 2 vertex, n_2 . But $\mathcal{G}_0$ has no such pair (there, the degree 3 vertices are separated by two degree 2 vertices).

III.3.41

(A) We will keep track of a set R of reachable vertices. To start, put q_0 into a set R_0 . For the first step find all vertices connected to q_0 , and enter them into the set $R_1 \supseteq R_0$. At step $k + 1$ put all vertices connected to any vertex in R_k into a set $R_{k+1} \supseteq R_k$. When there are no new vertices then we are done (this must happen after finitely many steps because there are finitely many vertices, by our definition of a graph). The result is the set R . The unreachable vertices are those that are not members of R .

(B) For the graph on the left these are the steps.

step k	R_k
0	$\{w_0\}$
1	$\{w_0, w_1\}$
2	$\{w_0, w_1, w_2\} = R$

Here are the steps for the graph on the right.

step k	R_k
0	$\{w_0\}$
1	$\{w_0, w_1\}$
2	$\{w_0, w_1, w_3\} = R$

III.A.4

$\langle \text{zip} \rangle ::= \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle [- \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle]$

$\langle \text{digit} \rangle ::= 0 \mid 1 \dots \mid 9$

III.A.5

Here we stick to lower-case ASCII letters.

$\langle \text{palindrome} \rangle ::= a \langle \text{palindrome} \rangle a \mid b \langle \text{palindrome} \rangle b \mid \dots \mid z \langle \text{palindrome} \rangle z \mid \langle \text{letter} \rangle \mid \varepsilon$

$\langle \text{letter} \rangle ::= a \mid b \mid \dots \mid z$

You could omit ε if you don't like it as a palindrome.

III.A.6

Here is one way to do it.

$\langle \text{course code} \rangle ::= \langle \text{dept} \rangle \langle \text{space} \rangle \langle \text{course-number} \rangle$

$\langle \text{dept} \rangle ::= \langle \text{two-letter} \rangle \mid \langle \text{three-letter} \rangle$

$\langle \text{two-letter} \rangle ::= \langle \text{letter} \rangle \langle \text{letter} \rangle$

$\langle \text{three-letter} \rangle ::= \langle \text{two-letter} \rangle \langle \text{letter} \rangle$

$\langle \text{course-number} \rangle ::= \langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle$

$\langle \text{digit} \rangle ::= 0 \mid \dots 9$

The $\langle \text{space} \rangle$ produces a hard-to-show blank space.

You could also put the department names in on a case-by-case basis, as here.

$\langle \text{dept} \rangle ::= \text{MA} \mid \text{PSY} \mid \dots$

This has the disadvantage that if a department code changes then this part of the description also changes. It has the advantage of being more correct, of not allowing department names that are nonsensical.

III.A.7

(A) This will do.

$\langle \text{pointfloat} \rangle ::= \langle \text{intpart} \rangle \langle \text{fraction} \rangle \mid \langle \text{intpart} \rangle . \mid \langle \text{fraction} \rangle$

(B)

$\langle \text{intpart} \rangle ::= \langle \text{intpart} \rangle \langle \text{digit} \rangle \mid \langle \text{digit} \rangle$

(C) This is just a list of cases

$\langle \text{exponent} \rangle ::= e \langle \text{intpart} \rangle \mid e+ \langle \text{intpart} \rangle \mid e- \langle \text{intpart} \rangle \mid E \langle \text{intpart} \rangle \mid E+ \langle \text{intpart} \rangle \mid E- \langle \text{intpart} \rangle$

III.A.8

(A) This is a grammar for standard notation.

$\langle \text{start} \rangle ::= \langle \text{thousands} \rangle \langle \text{hundreds} \rangle \langle \text{tens} \rangle \langle \text{ones} \rangle$

$\langle \text{thousands} \rangle ::= M^*$

$\langle \text{hundreds} \rangle ::= C \mid CC \mid CCC \mid CCCC \mid D \mid DC \mid DCC \mid DCCC \mid DCCCC \mid \epsilon$

$\langle \text{tens} \rangle ::= X \mid XX \mid XXX \mid XXXX \mid L \mid LX \mid LXX \mid LXXX \mid LXXXX \mid \epsilon$

$\langle \text{ones} \rangle ::= I \mid II \mid III \mid IIII \mid V \mid VII \mid VIII \mid VIIII \mid \epsilon$

(B) This is a grammar for subtractive notation.

$\langle \text{start} \rangle ::= \langle \text{thousands} \rangle \langle \text{hundreds} \rangle \langle \text{tens} \rangle \langle \text{ones} \rangle$

$\langle \text{thousands} \rangle ::= M^*$

$\langle \text{hundreds} \rangle ::= CM \mid CD \mid (D \mid \epsilon) (\epsilon \mid C \mid CC \mid CCC)$

$\langle \text{tens} \rangle ::= XC \mid XL \mid (L \mid \epsilon) (\epsilon \mid X \mid XX \mid XXX)$

$\langle \text{ones} \rangle ::= IX \mid IV \mid (V \mid \epsilon) (\epsilon \mid I \mid II \mid III)$

III.A.9

(A) Here is a derivation. We feel free to use the BNF notation to, for instance, put in multiple $\langle \text{statement} \rangle$'s on the third line instead of putting them in one at a time.

$\langle \text{program} \rangle \Rightarrow \{ \langle \text{statement-list} \rangle \}$
 $\Rightarrow \{ \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \langle \text{data-type} \rangle \langle \text{identifier} \rangle ; \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean} \langle \text{identifier} \rangle ; \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean} \langle \text{letter} \rangle ; \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; \langle \text{statement} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; \langle \text{identifier} \rangle = \langle \text{expression} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; \langle \text{letter} \rangle = \langle \text{expression} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = \langle \text{expression} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = \langle \text{number} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = \langle \text{digit} \rangle ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = 1 ; \langle \text{statement} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = 1 ; \text{print } \langle \text{identifier} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = 1 ; \text{print } \langle \text{letter} \rangle ; \}$
 $\Rightarrow \{ \text{boolean } A ; A = 1 ; \text{print } A ; \}$

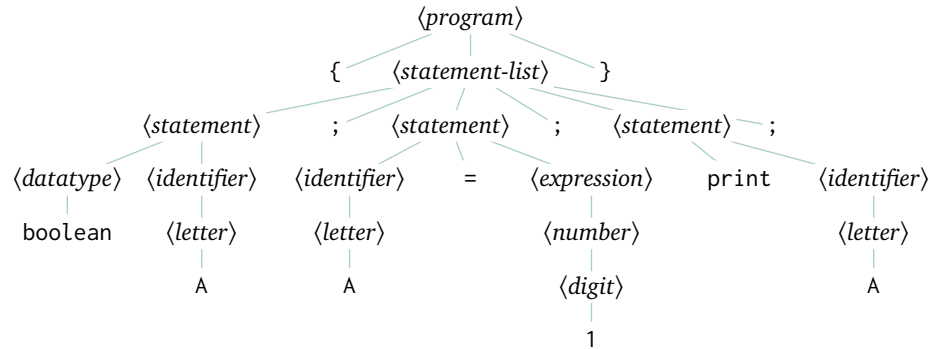


FIGURE 79, FOR QUESTION III.A.9: Parse tree for C-like language.

Figure 79 on page 115 shows the tree.

- (B) Yes, because all derivations must go from $\langle program \rangle$ through $\langle statement-list \rangle$, which forces them to have curly braces.

III.A.10 In this partial derivation we feel free to leverage the BNF notation to, for instance, put in multiple $\langle s-expression \rangle$'s for a $\langle list \rangle$ at one time, instead of putting them in one at a time.

$$\begin{aligned}
 \langle s-expression \rangle &\Rightarrow \langle list \rangle \\
 &\Rightarrow (\langle s-expression \rangle \langle s-expression \rangle \langle s-expression \rangle) \\
 &\Rightarrow (\langle s-expression \rangle \langle list \rangle \langle s-expression \rangle) \\
 &\Rightarrow (\langle s-expression \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (\langle atomic-symbol \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (\langle letter \rangle \langle atomic-part \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (\langle letter \rangle \langle letter \rangle \langle atomic-part \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (\langle letter \rangle \langle letter \rangle \langle letter \rangle \langle atomic-part \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (\langle letter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle) \\
 &\Rightarrow (c \langle letter \rangle \langle letter \rangle \langle letter \rangle (\langle s-expression \rangle \langle s-expression \rangle) \langle s-expression \rangle)
 \end{aligned}$$

Expanding the remaining nonterminals to letters is tedious, so we will stop here.

III.A.11

- (A) Here we will leverage the BNF notation to, for instance, substitute for multiple nonterminals, instead of putting them in one at a time.

$$\begin{aligned}
 \langle format-spec \rangle &\Rightarrow 0 \langle width \rangle \langle type \rangle \\
 &\Rightarrow 0 \langle integer \rangle \langle type \rangle \\
 &\Rightarrow 0 \langle digit \rangle \langle type \rangle \\
 &\Rightarrow 0 \ 3 \langle type \rangle \\
 &\Rightarrow 0 \ 3 \ f
 \end{aligned}$$

- (B) $\langle format-spec \rangle \Rightarrow \langle sign \rangle \# 0 \langle width \rangle \langle type \rangle$
- $$\begin{aligned}
 &\Rightarrow + \# 0 \langle width \rangle \langle type \rangle \\
 &\Rightarrow + \# 0 \langle integer \rangle \langle type \rangle \\
 &\Rightarrow + \# 0 \langle digit \rangle \langle type \rangle \\
 &\Rightarrow + \# 0 \ 2 \langle type \rangle \\
 &\Rightarrow + \# 0 \ 2 \ X
 \end{aligned}$$

III.B.1

- (A) Here is the code.

```

(define (exercise-tree-make)
  (let* ([t (tree-create "a")]
        [nb (node-add-child! t "b")]
        [nc (node-add-child! t "c")])

```

```

[nd (node-add-child! nb "d")]
[ne (node-add-child! nb "e")]
[nf (node-add-child! nc "f")]
[ng (node-add-child! nc "g")]
[nh (node-add-child! ng "h")]
[ni (node-add-child! ng "i")]
)
t))

```

(B) From the interpreter after running the definitions we get this.

```

> (define t (exercise-tree-make))
> (tree-bfs t show-node-name)
a
c
b
  e
  d
  g
  f
    i
    h

```

(c) We get this.

```

> (tree-dfs t 0 show-node-name)
a
b
  d
  e
c
  f
  g
    h
    i

```

III.B.2

(A) The difference with this graph is that some nodes have more than one parent. This will do.

```

(define (cantor-DAG-make)
  (let* ([t (tree-create "0,0")]
        [nb (node-add-child! t "0,1")]
        [nc (node-add-child! t "1,0")]
        [nd (node-add-child! nb "0,2")]
        [ne (node-add-child! nb "1,1")]
        [v0 (set-add! (node-children nc) ne)]
        [nf (node-add-child! nb "2,0")]
        [ng (node-add-child! nd "0,3")]
        [nh (node-add-child! nd "1,2")]
        [v1 (set-add! (node-children ne) nh)]
        [ni (node-add-child! ne "2,1")]
        [v2 (set-add! (node-children nf) ni)]
        [nj (node-add-child! nf "3,0")]
        ])
    t))

```

(B) In the result, nodes with multiple parents are visited multiple times.

```

> (define t (cantor-DAG-make))
> (tree-bfs t show-node-name)
0,0
  1,0
    0,1
      2,0
        1,1
          0,2
            1,1
              2,1
                1,2
                  0,3
                    1,2
                      2,1
                        1,2
                          2,1
                            3,0

```

(c) The issue is that next-level is a list. When nodes get added to that list multiple times, they then show up multiple times. We can fix this by making it a set, and adjusting the related code.

```
(define (tree-bfs-set node fcn #:maxrank [maxrank MAXIMUM-RANK])
  (tree-bfs-set-helper (mutable-set node) 0 fcn #:maxrank maxrank))

(define (tree-bfs-set-helper level rank fcn #:maxrank [maxrank MAXIMUM-RANK])
  (when (< rank maxrank)
    (let ([next-level (mutable-set)])
      (for ([node level])
        (fcn node rank)
        (for ([child-node (node-children node)])
          (set-add! next-level child-node)
          ))
      (when (not (set-empty? next-level))
        (tree-bfs-set-helper next-level (+ 1 rank) fcn))))))
```

Here is the result at the interpreter.

```
> (tree-bfs-set t show-node-name)
0,0
  0,1
    1,0
      0,2
        1,1
          2,0
            3,0
              1,2
                2,1
                  0,3
```


Chapter IV: Automata

IV.1.18

(A)

Step	Configuration	Step	Configuration
0	abba q₀	3	a q₂
1	bba q₁	4	q₁
2	ba q₂		

(B)

Step	Configuration	Step	Configuration
0	bab q₀	2	b q₁
1	ab q₀	3	q₂

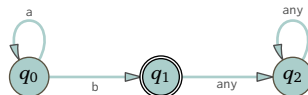
(C)

Step	Configuration	Step	Configuration	Step	Configuration
0	bbbaabbaa q₀	3	abbaa q₁	6	aa q₂
1	baabbaa q₀	4	bbbaa q₁	7	a q₁
2	aaabbaa q₀	5	baa q₂	8	q₁

V IV.1.19 False. An example is that Example 1.8's machine recognizes the set consisting of valid decimal representations of integers and thus containing each of these: 0, 1, 2, ...

V IV.1.20 First, " n is infinite" is wrong. That set is a language $\{b, ab, a^2b, \dots\}$ and in each string from that language the number of a 's is finite.

Second, this Finite State machine recognizes the language.



V IV.1.21 Languages do not recognize strings, machines recognize strings. Maybe they mean that they have a language that contains the empty string? Or maybe they mean that they have a machine that accepts the empty string?

V IV.1.22 Each input character consumed causes one transition. So an input string of length n causes the machine to undergo n transitions. (This includes that the empty string ϵ causes no transitions.) With n transitions, the machine will have visited $n + 1$ many not necessarily distinct states, including the start state.

V IV.1.23

(A) A string is in this language when it has some number of a 's, followed by a single b , followed by the same number of a 's (that is, the two strings of a 's have the same length). Five elements of the language are aba , $aabaa$, b , $aaabaaa$, and $aaaabaaaa$. Five non-elements are ba , $aaba$, a , $abba$, and $bbabb$.

- (B) The strings in this language have some number of a's followed by a single b, followed by a number of a's. The two numbers of a's need not be the same, and may be zero. Five language elements are abaa, aabaa, ab, baaaa, and b. Five non-elements are bab, aabb, bb, aabaab, and a.
- (C) The criteria for a string to be in this language is that it must start with b and be followed by some number of a's. That could be zero-many a's. Five elements of the language are baa, ba, b, ba^3 , and ba^4 . Five non-elements are a, ϵ , abaaa, baaab, and bb.
- (D) A string is in this language if and only if it consists of some number of a's followed by a b, followed by some number of a's, where the second string of a's has two more than the first string. Five members of the language are abaaa, aabaaaa, a^3ba^5 , a^4ba^6 , and baa. Five non-members are aba, abaaaa, ba, b, and bab.
- (E) This language has members that are doubles in that a member consists of the concatenation of two copies of some string. Five members of the language are aaabaaab, babbab, aa, bbbbabbbba, and abbaabba. Five non-members are aba, b, aaabaab, babaa, and bbb.

V IV.1.24

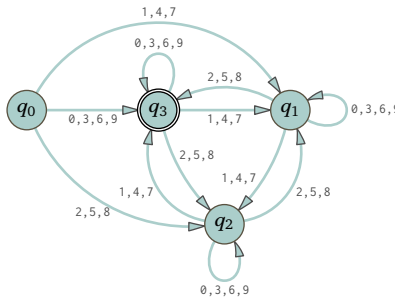
- (A) For Example 1.6 the only accepting state is q_2 . In Example 1.8 also, the only accepting state is q_2 . For Example 1.9 the accepting states are q_3 , q_6 , and q_8 . For Example 1.10 it is q_0 .
- (B) Only Example 1.10 accepts the empty string ϵ because only in this machine is q_0 an accepting state.
- (C) For Example 1.6 the shortest accepted string is the length two string 00 . For Example 1.8 it is any single-digit length one string, such as 6. For Example 1.9 any of the length three strings jpg, png, and pdf. For Example 1.10 it is the empty string ϵ .

IV.1.25 This is the computation.

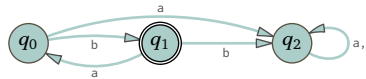
$$\langle q_0, 2332 \rangle \vdash \langle q_2, 332 \rangle \vdash \langle q_2, 32 \rangle \vdash \langle q_2, 2 \rangle \vdash \langle q_1, \epsilon \rangle$$

The state q_1 is not an accepting state so the machine rejects 2332.

- V IV.1.26 We must change the machine so that its initial state is not an accepting state. Below there there are two states, q_0 and q_3 , whose intuitive meaning is that the sum of the digits seen so far is congruent to 0 modulo 3. But the initial state q_0 is not accepting.



- V IV.1.27 This is the graph picture.



The language is $\{\sigma \in \{a, b\}^* \mid \sigma = b(ab)^n \text{ for } n \in \mathbb{N}\}$.

- V IV.1.28 The language is the set of strings over $\Sigma = \{a, b\}$ the contain exactly one b.

IV.1.29 In the section body is the advice to design machines by explicating an intuitive meaning for each state. Although that is not a requirement and the question does not ask for it, for each item here we give those.

- (A) Five members are ab, baab, ba, b^7a^3 , and ab^6a . Five nonmembers are aaaa, bbbb, a, ϵ , and b. We take the meaning of the states to be as here.

State	Description
q_0	start state
q_1	have seen at least one a, no b's
q_2	have seen at least one b, no a's
q_3	at least one of each

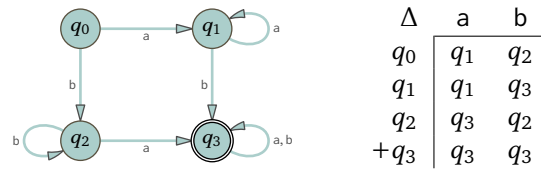


FIGURE 80, FOR QUESTION IV.1.29: Finite State machine accepting strings with at least one a and b.

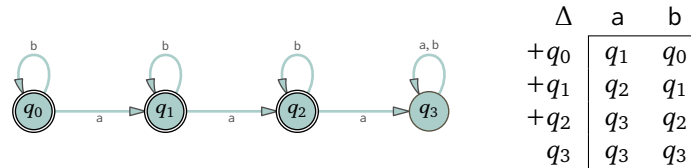


FIGURE 81, FOR QUESTION IV.1.29: Finite State machine accepting strings with fewer than three a's.

The circle drawing and transition table are Figure 80 on page 121.

- (B) Five members are: bb , ϵ , aba , a , and $bbbaab$. Five nonmembers are aaa , $baaa$, a^5 , $bababa$, and a^{10} . We take the states with this meaning.

State	Description
q_0	start state, no a's seen yet
q_1	have seen one a, and maybe some number of b's
q_2	have seen two a's
q_3	have seen three or more a's

For the circle diagram and transition table, see Figure 81 on page 121.

- (C) Five members are ab , bab , aab , $baab$, and $abab$. Five nonmembers: a , bbb , ϵ , b , and $abababb$. We use the states with this meaning.

State	Description
q_0	start state, have not just seen an a
q_1	have seen an a, waiting for a b
q_2	have just seen ab

The transition diagram and table are Figure 82 on page 122.

- (D) Five members are $aabb$, $aaabb$, $aabbb$, a^7b^9 , and a^2b^{200} . Five nonmembers are ϵ , a , ab , ba , and b . We use the states with this meaning.

State	Description
q_0	start state, have not just seen anything
q_1	have seen one character, an a
q_2	have seen at least two a's and no b's
q_3	have seen at least two a's and one b
q_4	have seen at least two a's and at least two b's
$q_5 = e$	error

See Figure 83 on page 122 for the circle diagram and the table.

- (E) Five members are $aabba$, $aabb$, abb , bb , and a^5bb^7 . Five nonmembers are ϵ , bab , $aabaa$, $aabbbbaa$, and $aabbaab$. We take the states with this meaning.

State	Description
q_0	have so far seen some number of a's, including zero-many
q_1	have seen some a's followed by one b
q_2	seen some a's, two b's, and some a's
e	error

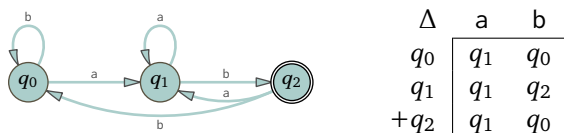


FIGURE 82, FOR QUESTION IV.1.29: Finite State machine accepting strings that end in ab.

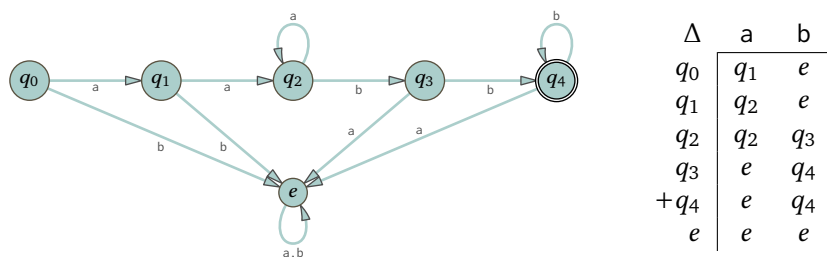
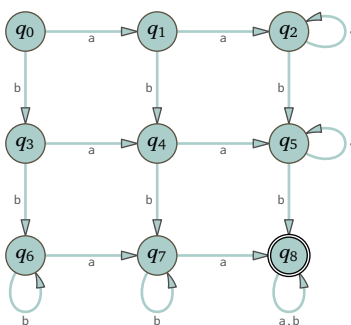


FIGURE 83, FOR QUESTION IV.1.29: Finite State machine for strings of at least two a's and then at least two b's.

The two representations of the transition function are Figure 84 on page 123.

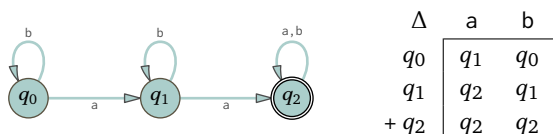
- V IV.1.30 Five elements of the language are aabb, bbaa, abab, aababb, and a^3b^4 . Five non-elements are ϵ , a, b, aab, and bbbbabbbb. Here is the machine.



The intuitive meaning of the states in the first row, q_0 , q_1 , and q_2 , is that when the machine is in those states it has so far seen zero-many b's. States in the second row are where the machine has so far seen one b. The third row contains the states where the machine has seen two b's. Similarly, the first column contains the states where the machine has so far seen no a's, in the second column are states where the machine has seen a single a, and in the final column the machine has seen two a's.

V IV.1.31

- (A) Five strings that are elements of the language are aa, aaa, aab, aba, and baa. Five that are not are ϵ , a, b, abbbbbb, and ba. This machine accepts only strings in that language.



- (B) Five elements of the language are aa, aab, aba, baa, and aabb. Five that are not in the language are ϵ , a, ba, aaa, and baaa.

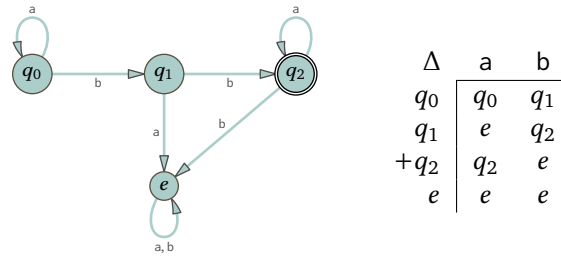
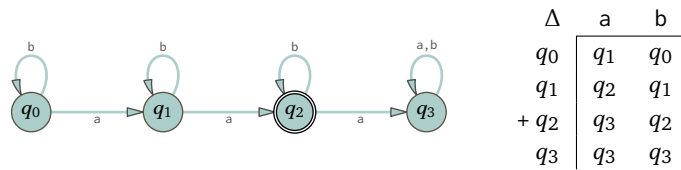
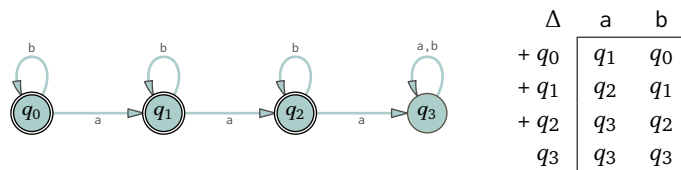


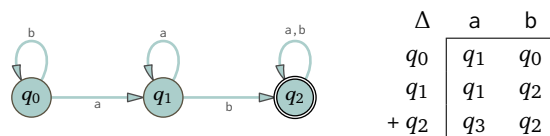
FIGURE 84, FOR QUESTION IV.1.29: Finite State machine for strings with some a's, then two b's, then some a's.



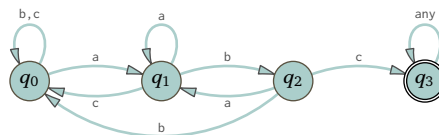
- (c) Five strings that are elements of the language are ϵ , a, b, aa, and ab. Five that are not are aaa, aaab, aaba, abaa, and baaa.



- (d) Five from the language are ab, aab, aba, abb, and bab. Five that are not in the language are ϵ , a, b, ba, and bbbbaaaa.



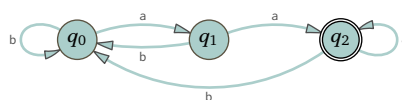
- V IV.1.32 To accept strings containing the substring abc the machine must have a state whose intuitive meaning is “have just seen a, next looking for bc.” That state is q_1 . The state q_2 means that the machine has just processed a substring ab and is looking for c. And q_3 means the machine has seen the substring abc. Finally, q_0 is the state the machine is in if it has so far not seen even the first character in a potential substring abc. Here is the machine.



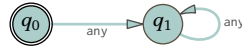
Note that if the machine processes ab followed by b then it returns to q_0 , but if it processes ab followed by a then it does not return to q_0 , it passes to q_1 . This follows from the intuitive meaning of q_1 .

V IV.1.33

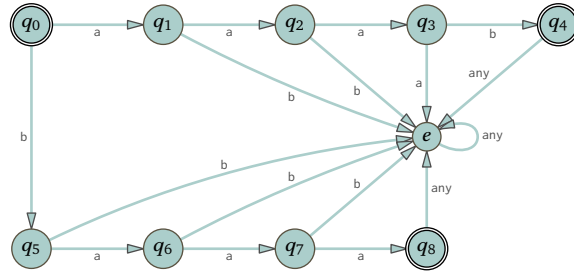
- (A) Five that are in the language are aa, aaa, baa, aaaa, and abaa. Five that are not are ϵ , a, b, ab ba, and aba. Here is the machine.



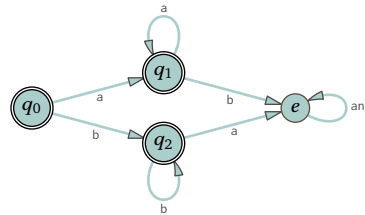
- (B) The only string in the language is ϵ . That is, $\mathcal{L} = \{\epsilon\}$. Five not in the language are a, b, aa, ab, and ba. The machine is this.



- (c) There are only two strings in the language: $\mathcal{L} = \{aaab, baaa\}$. Five strings not in the language are ϵ , a , b , aa , and ab . The machine is this.

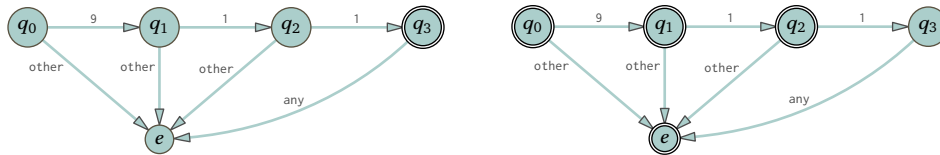


- (d) Five strings in the language are ϵ , a , b , aa , and bb . Five not in the language are ab , ba , aab , aba , and baa . This is the machine.



Note that q_0 is an accepting state.

- V IV.1.34 The first machine accepts only 911 while the second accepts anything but 911. Note that in the two machines the sets of accepting states are complementary.



- V IV.1.35 Here are the length three strings.

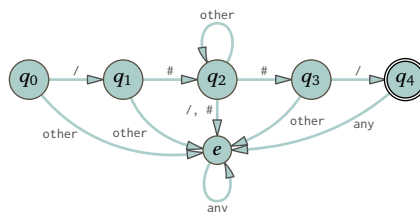
Input σ	aaa	aab	aba	abb	baa	bab	bba	bbb
$\hat{\Delta}(\sigma)$	q_1	q_2	q_1	q_2	q_1	q_2	q_1	q_0

And here are the length four strings.

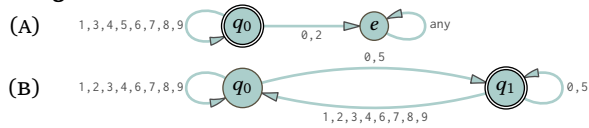
Input σ	aaaa	aaab	aaba	aabb	abaa	abab	abba	abbb
$\hat{\Delta}(\sigma)$	q_1	q_2	q_1	q_2	q_1	q_2	q_1	q_2
	baaa	baab	baba	babb	bbaa	bbab	bbba	bbbb
	q_1	q_2	q_1	q_2	q_1	q_2	q_1	q_0

- V IV.1.36 It outputs the start state, $\hat{\Delta}(\epsilon) = q_0$.

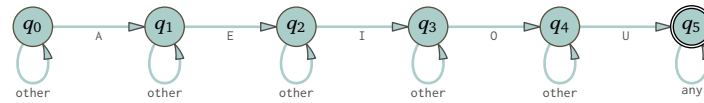
- V IV.1.37



V IV.1.38

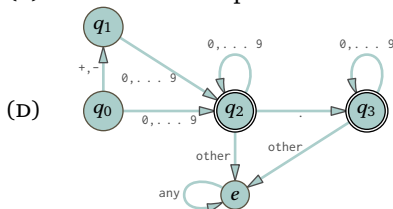


V IV.1.39 Note that although the string $\sigma = \text{AEAI} \text{AOAUA}$ has an A that occurs after an E, nonetheless the machine should accept it because there exists a subsequence of string elements consisting of the ordered vowels.



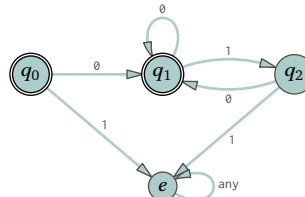
V IV.1.40

- (A) Five that are: +1, 123, -123.45, -0.0, and 0. Five that are not: +, +-, . . ., 12.3.4, and 2+3
 (B) No, before the period must come at least one digit.
 (C) It is the set of representations of real numbers with integer part, a decimal point, and a decimal part.

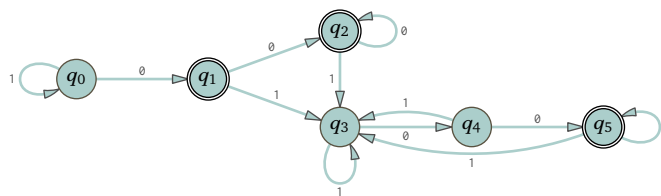


V IV.1.41

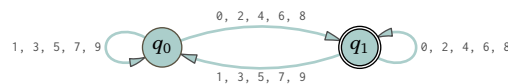
- (A) Five that are in the language are 010, 0010, 0, 0101000, and ϵ . Five that are not are 10, 0110, 0101, 01001, and 1.



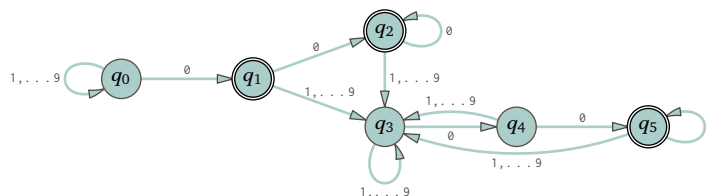
- (B) A string is a binary representation of a multiple of 4 if and only if it ends in two zero's, 00, or it is 0. Five in the language are 100, 0100, 1000, 10100, and 0. Five not in the language are 10, 1, 1001, 001, and ϵ .



- (C) Five strings in the language are 0, 20, 12, 88, and 08. Five that are not are 1, 01, 2221, 17, and ϵ .



- (D) This is basically the same as the second item. A string is a decimal representation of a multiple of 100 if and only if it ends in two zero's, 00, or it is 0. Five in the language are 100, 0100, 2000, 60300, and 0. Five not in the language are 10, 8, 1009, 001, and ϵ .



- V IV.1.42 A number is divisible by four if and only if its representation ends in 00, 04, ... 96, or the single-digit cases where the representation is 0, 4, or 8. That's 28 cases. A picture would be cluttered but this is a finite language and for any finite language there is a Finite State machines that recognizes that language. (We shall see an alternative approach that makes a clearer picture in the section on nondeterminism.)
- V IV.1.43 Recall the transition function for that example.

Δ	0	1
q_0	q_1	q_3
q_1	q_2	q_4
$+q_2$	q_2	q_5
q_3	q_4	q_0
q_4	q_5	q_1
q_5	q_5	q_2

- (A) If the input string is $\sigma = \emptyset$ then peel off the final character with $\sigma = \sigma_0 \frown \emptyset = \varepsilon \frown \emptyset$ and the definition gives this.

$$\hat{\Delta}(\emptyset) = \Delta(\hat{\Delta}(\varepsilon), \emptyset) = \Delta(q_0, \emptyset) = q_1$$

(Note that in the left-most expression $\emptyset$ is a length one string, while in the rest of the above equation $\emptyset$ is a character, and we are ignoring this uninteresting point.) Similarly, $\hat{\Delta}(1) = q_3$.

- (B) For $\sigma = \emptyset\emptyset$ use the prior item's computation of $\hat{\Delta}(\emptyset)$.

$$\hat{\Delta}(\emptyset\emptyset) = \Delta(\hat{\Delta}(\emptyset), \emptyset) = \Delta(q_1, \emptyset) = q_2$$

The other three are similar.

$$\hat{\Delta}(\emptyset 1) = q_4 \quad \hat{\Delta}(1\emptyset) = q_4 \quad \hat{\Delta}(11) = q_0$$

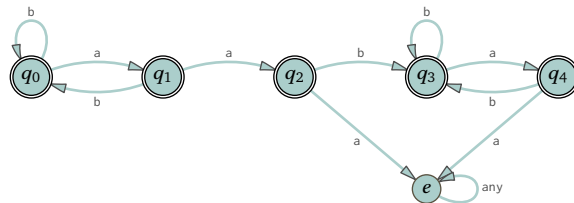
- (c) For each of these use the prior item's computation. Here is the first.

$$\hat{\Delta}(\emptyset\emptyset\emptyset) = \Delta(\hat{\Delta}(\emptyset\emptyset), \emptyset) = \Delta(q_2, \emptyset) = q_2$$

The others are computed in the same way.

$\hat{\Delta}(\emptyset\emptyset\emptyset)$	$\hat{\Delta}(\emptyset\emptyset 1)$	$\hat{\Delta}(\emptyset 1\emptyset)$	$\hat{\Delta}(\emptyset 11)$	$\hat{\Delta}(1\emptyset\emptyset)$	$\hat{\Delta}(1\emptyset 1)$	$\hat{\Delta}(11\emptyset)$	$\hat{\Delta}(111)$
q_2	q_5	q_5	q_1	q_5	q_1	q_1	q_3

IV.1.44



V IV.1.45

- (A) There is one such machine.

Δ	0	1
q_0	q_0	q_0

- (B) This list has 16 machines.

Δ_0	0	1	Δ_1	0	1	Δ_2	0	1	Δ_{15}	0	1
q_0	q_0	q_0	q_0	q_0	q_0	q_0	q_0	q_0	q_0	q_1	q_1
q_1	q_0	q_0	q_1	q_0	q_1	q_1	q_1	q_0	q_1	q_1	q_1

- (c) Fix n states, $Q = \{q_0, \dots, q_{n-1}\}$. Each line of the transition table has two columns so the number of distinct lines is $n \cdot n = n^2$. An overall table chooses n such lines. There are $n^2 \cdot n^2 \cdot n^2 = n^{2n}$ many ways to do that.

state	meaning	with wolf	with goat	with cabbage	with nothing
s_0	n: mwgc, f: –	s_{12}	s_{10}	s_9	s_8
s_1	n: mwg, f: c	s_{13}	s_{11}	e	s_9
s_2	n: mwc, f: g	s_{14}	e	s_{11}	s_{10}
s_3	n: mw, f: gc	e	e	e	e
s_4	n: mgc, f: w	e	s_{14}	s_{13}	s_{12}
s_5	n: mg, f: wc	e	s_{15}	e	s_{13}
s_6	n: mc, f: wg	e	e	e	e
s_7	n: m, f: wgc	e	e	e	e
s_8	n: wgc, f: m–	e	e	e	e
s_9	n: wg, f: mc	e	e	e	e
s_{10}	n: wc, f: mg	e	s_0	e	s_2
s_{11}	n: w, f: mgc	e	s_1	s_{10}	s_3
s_{12}	n: gc, f: mw	e	e	e	e
s_{13}	n: g, f: mwc	s_1	e	s_4	s_5
s_{14}	n: c, f: mwg	s_2	s_4	e	s_6
s_{15}	n: –, f: mwgc	s_{15}	s_{15}	s_{15}	s_{15}
e	error	e	e	e	e

FIGURE 85, FOR QUESTION IV.1.46: Finite state machine for Wolf-Goat-Cabbage.

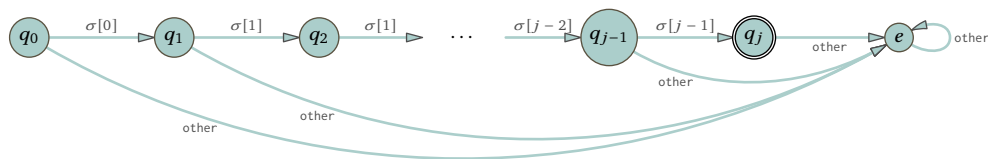
(D) For each of the n states, either it is final or it is not. The additional factor of 2^n makes the total number of distinct tables equal to $2^n n^{2^n}$.

V IV.1.46 We can have the man, the wolf, the goat, and the cabbage, on the near or far side. The allowed transitions are: man crosses with wolf, man crosses with goat, man crosses with cabbage, and man crosses by himself. Here are the states; some of them lead to an error state, as readers of the 700's would have understood.

In the Finite State machine shown in Figure 85 on page 127 the start state is s_0 and the only accepting state is s_{15} . In words: The man first brings the goat to the far side and leaves it there. He goes back and brings the wolf with him, leaving it on the far side, and brings back the goat. He leaves the goat on the initial side, takes the cabbage and brings it to the far side. Finally, he goes back to the original shore, and takes the goat to bring it to the far side.

IV.1.47 This is based on Example 1.9.

We will prove this by induction on the number of strings in the language. If the language is empty then we use any Finite State machine with the right alphabet and no accepting states. If the language has one string σ then we write a Finite State machine whose state transitions track the characters of σ as does the top line in the example. That is, we have this.

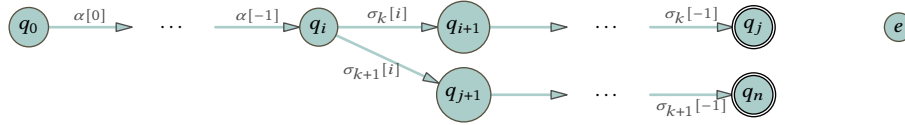


As the diagram shows, the last state in that sequence is an accepting state. Also as shown, all transitions for characters not in the sequence go to e . (If $\sigma = \varepsilon$ then there are two states, with the initial state q_0 an accepting state and all transitions going to the error state e that is not an accepting state.)

For the inductive case, assume that there is a machine for any language of size $n = 1, \dots, n = k$ and consider a language $\mathcal{L}$ with $k + 1$ -many strings. Order the alphabet Σ ; in the example we use the grade school alphabetical ordering. This character ordering gives rise to an ordering of the strings, called lexicographic ordering. List $\mathcal{L}$'s strings lexicographically: $\sigma_0, \dots, \sigma_k, \sigma_{k+1}$.

The induction assumption gives a machine for the language consisting of the strings $\sigma_0, \dots, \sigma_k$. The

example shows what to do next; it has the two final strings png and pdf and the machine splits where the two strings split, after the p. Following that model, find the longest common substring for σ_k and σ_{k+1} , so that $\sigma_k = \alpha \hat{\ } \beta$, and $\sigma_{k+1} = \alpha \hat{\ } \gamma$, and the initial character of β is not equal to the initial character of γ . (It could be that $\beta = \varepsilon$. It could also be that $\alpha = \varepsilon$.) As in the example, split the machine after it has traversed the common string.



(To be more readable this figure omits all the transitions involving e .)

V IV.1.48 Recall that an alphabet is finite, by definition.

- (A) We will show that there are infinitely many different Finite State machines. Fix any Σ and suppose $s \in \Sigma$. The machines below differ in the number of states, and in that they recognize different languages. (If Σ contains characters other than s then use the second column.)

Δ	s	$other$	Δ	s	$other$	Δ	s	$other$
$+ q_0$	q_0	q_0	$+ q_0$	q_1	q_0	$+ q_0$	q_1	q_0
			q_1	q_0	q_0	q_1	q_2	q_0
						q_2	q_0	q_0

That shows, under a reasonable interpretation of what it means for machines to differ, that there are at least countably many Finite State machines. (The technical wording is that these machines are not isomorphic.)

- (B) The argument that there are no more than countably many machines also requires an interpretation. Assume that the set of possible states is countable, $Q = \{q_0, q_1, \dots\}$, so as to not get uncountably many machines by virtue of there being uncountably many possible states. With that, the argument that we can count the Finite State machines is straightforward since for any collection of finitely many states $q_0, \dots, q_k$ there are finitely many transition tables. Thus the number of Finite State machines using those states is countable. And then the union of countably many countable sets is countable, so the number of machines over all collections of finitely many states is countable.
- (C) Because Σ has at least two members, the usual diagonalization construction shows that the power set of Σ is not countable.

V IV.2.24 The main point is that entries in the table are sets of states. This is the machine in Example 2.7.

Δ	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	$\{\}$
q_2	$\{\}$	$\{q_3\}$
$+ q_3$	$\{q_3\}$	$\{q_3\}$

This is Example 2.8's machine.

Δ	a	b	c
$+ q_0$	$\{q_1\}$	$\{\}$	$\{\}$
q_1	$\{\}$	$\{\}$	$\{q_0\}$

V IV.2.25

- (A) Yes. From q_0 a ε transition goes to q_2 , which is an accepting state.
- (B) No. From q_0 the machine can either read the $\emptyset$ and go to q_1 , or take the ε transition to q_2 and then read the $\emptyset$, which sends the machine to no-state. Neither q_2 nor no-state is an accepting state.
- Said more formally, the ε closure of q_0 is $\{q_0, q_2\}$. Neither state, on reading the character $\emptyset$, transitions to a state whose ε closure includes an accepting state.

- (c) Yes. With that input the machine can go from q_0 to q_1 , then to q_2 , and then around the loop to q_2 again. More formally, this is a sequence of allowed transitions that ends in an accepting state.

$$\langle q_0, 011 \rangle \vdash \langle q_1, 11 \rangle \vdash \langle q_2, 1 \rangle \vdash \langle q_2, \varepsilon \rangle$$

- (d) No. Starting at q_0 with the character 0 first on the tape the machine can do two things. If it takes the ε transition then when it reads the 0 it goes to no-state, which is not an accepting state.

Otherwise, the 0 moves the machine to q_1 . Next, on the following tape character 1, the machine moves to $\{q_2\}$. Finally, on the third character 0, the machine moves to no-state.

- (e) There are two: 01111 and 11111.

IV.2.26 Someone in this class saying that a topic is “just mathematical abstraction” is like someone objecting in an aerodynamics class that this is all “just air.” Yes, that’s what we are doing.

Said another way, if someone compares what we are thinking about here with a course on data structures, and holds the latter up as a model of “real” then that is a stretch. “Real” is changing diapers, or grafting fruit trees, or helping the helpless. All of what we are doing in the entire programs of computer science and mathematics is abstraction of one kind of another.

If that person responds, “OK but abstraction can go too far and get away from what brought us into the program” then that is more measured. One answer is that people in a program need to take as the default assumption that the people who put together this program, and such programs across the world, have put in steps that lead somewhere. Nondeterminism is here because it is necessary to understand the material of the fifth chapter, which describes the most important problem in the field today, which will directly impact all practitioners, the P versus NP question.

Besides, it is interesting and fun. Surely those are respectable criteria to merit at least some consideration.

- V IV.2.27** The machine accepts the input string if there is a sequence of legal transitions that allows the machine to process the string and that ends in an accepting state. It does not say there must not be any wrong ways to go, only that there must be a right way.

For instance, the given machine can accept aab by starting in q_0 , transitioning to q_1 on the first a, taking the ε transition to q_0 , then going to q_1 on the second a, then going to q_2 , which is accepting, with the b.

$$\langle q_0, aab \rangle \vdash \langle q_1, ab \rangle \vdash \langle q_0, ab \rangle \vdash \langle q_1, b \rangle \vdash \langle q_2, \varepsilon \rangle$$

This machine recognizes the language $\mathcal{L} = \{a^n b \mid n \in \mathbb{N}\}$.

IV.2.28 To show that a nondeterministic machine accepts the input we need only exhibit an accepting branch of the computation tree. But here we will walk through unproductive branches just for the exercise. (In any case, to show that a nondeterministic machine does not accept we need to check all the branches.)

- (A) After processing the empty string as input, that is, at the moment we press the Start button, the nondeterministic machine is in states q_0 and q_3 . The deterministic machine is in state s_7 . Both result in the input being rejected.
- (B) For the nondeterministic machine, processing the input a will leave the machine in $\{q_2, q_3\}$. The deterministic machine ends in s_{10} . In both cases the machine rejects the input.
- (c) Here is what happens for the nondeterministic machine with input string b. If the machine follows the ε transition before processing the b then it results in a dead end branch because when in state q_3 and reading b, the machine has no next state. However a branch that does not start by following the ε transition will then pass to q_1 on processing the b character. The ε closure of q_1 is $\{q_0, q_1, q_3\}$, so that’s where the nondeterministic machine ends. It accepts the input because q_1 is an accepting state. The deterministic machine passes from s_7 to s_{12} , which is an accepting state.
- (D) As given in an earlier item, on processing the first a the nondeterministic machine is in $\{q_2, q_3\}$. In processing the second a, the branch in q_2 dead ends, while the branch in q_3 returns to q_3 . Thus this machine ends in $\{q_3\}$. The deterministic machine processes aa by passing from s_7 to s_{10} and then to s_4 . Both machines reject the input.
- (E) Again, processing the first a character takes the nondeterministic machine to $\{q_2, q_3\}$. Then the b takes it to $\{q_0, q_3\}$. The deterministic machine passes from the start state of s_7 to s_{10} and then back to $s_7 = \{q_0, q_3\}$. Both nondeterministic and deterministic machines reject the input.

- (F) An earlier item says that in consuming the initial b character the nondeterministic machine ends in states q_0 , q_1 , and q_3 . In processing the a character there is a branch from q_0 that passes to q_2 . There is another branch from q_0 that first takes the ε move to q_3 , and then the a character has the machine loop back to q_3 . Finally there is a branch that starts in q_3 and loops around to q_3 again. Thus, taken together, this nondeterministic machine ends in $\{q_2, q_3\}$. Neither is an accepting state so this machine rejects the input.

The deterministic machine passes from s_7 to s_{12} and then to s_{10} , which is not an accepting state.

- (G) Again, an earlier item says that after processing the b character the nondeterministic machine ends in $\{q_0, q_1, q_3\}$. Then a branch follows the ε transition to q_3 but dead-ends there because in q_3 there is no outward arrow labeled b. From q_1 no outward arrow is labeled b either. So the only branch that does not die is the one passing from q_0 to q_1 , and thus this machine ends the processing of this input in $\{q_1\}$. That means it accepts the input.

The deterministic machine goes from s_7 to s_{12} , and then loops around to s_{12} again, which is an accepting state.

IV.2.29 This is the sequence of allowed transitions.

$$\begin{aligned} \langle q_0, 010101110 \rangle \vdash \langle q_0, 10101110 \rangle \vdash \langle q_0, 0101110 \rangle \vdash \langle q_1, 101110 \rangle \\ \vdash \langle q_2, 01110 \rangle \vdash \langle q_3, 1110 \rangle \vdash \langle q_4, 110 \rangle \vdash \langle q_5, 10 \rangle \vdash \langle q_6, 0 \rangle \vdash \langle q_7, \varepsilon \rangle \end{aligned}$$

IV.2.30

- (A) Here are the ε closures.

state	q_0	q_1	q_2
ε closure	$\{q_0\}$	$\{q_1, q_2\}$	$\{q_2\}$

- (B) It does not accept the empty string because the ε closure of q_0 does not contain any final states.
 (C) It accepts both of the one-character strings a and b. For instance, on the string a the machine can transition from q_0 to q_1 , and then it can make an ε transition to q_2 , which is an accepting state.
 (D) The given machine can accept aab by starting in q_0 , transitioning to q_0 on the first a, then going to q_1 on the second a, taking the ε transition to q_2 , then going to q_2 with the b, thereby ending in an accepting state.

$$\langle q_0, aab \rangle \vdash \langle q_0, ab \rangle \vdash \langle q_1, b \rangle \vdash \langle q_2, b \rangle \vdash \langle q_2, \varepsilon \rangle$$

- (E) Five of the shortest strings are: a, b, aa, ab, and aab.
 (F) Five that it does not accept are: the empty string ε , ba, baa, bab, and baaa.

V IV.2.31

Δ	ε	a	b
q_0	$\{q_1\}$	$\{q_1\}$	$\{\}$
q_1	$\{q_0\}$	$\{\}$	$\{q_2\}$
$+ q_2$	$\{\}$	$\{\}$	$\{\}$

V IV.2.32

- (A) $\langle q_0, 011 \rangle \vdash \langle q_2, 11 \rangle \vdash \langle q_2, 1 \rangle \vdash \langle q_2, \varepsilon \rangle$
 (B) $\langle q_0, 00011 \rangle \vdash \langle q_1, 0011 \rangle \vdash \langle q_1, 011 \rangle \vdash \langle q_1, 11 \rangle \vdash \langle q_2, 1 \rangle \vdash \langle q_2, \varepsilon \rangle$
 (C) Yes, because the state q_0 is accepting.
 (D) It accepts 0 because it can go through these steps that end in an accepting state.

$$\langle q_0, 0 \rangle \vdash \langle q_2, \varepsilon \rangle$$

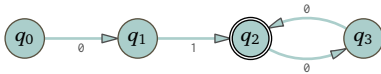
It does not accept 1 because there is no such sequence of steps.

- (E) Five are ε , 0, 01, 011, and 0111.
 (F) Here are five: 1, 10, 11, 010, and 110.
 (G) One description is $\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma = \varepsilon \text{ or } \sigma = 0^i 1^j \text{ for } i \geq 1 \text{ and } j \geq 0\}$.

IV.2.33 Every state reaches its limit after two moves.

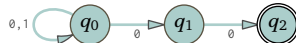
	$m = 0$	1	2	3	$\hat{E}(q)$
q_0	$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
q_1	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_2, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$
q_3	$\{q_3\}$	$\{q_1, q_3\}$	$\{q_1, q_3\}$	$\{q_1, q_3\}$	$\{q_1, q_3\}$

V IV.2.34

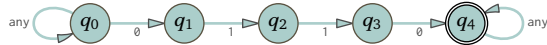


V IV.2.35

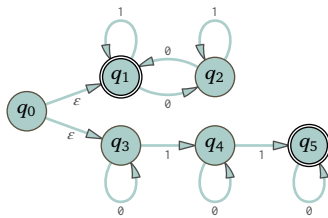
(A)



(B)



(C)



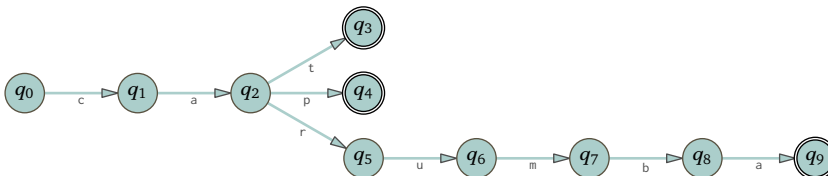
(D)



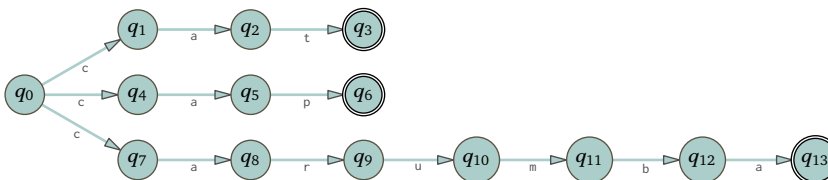
V IV.2.36



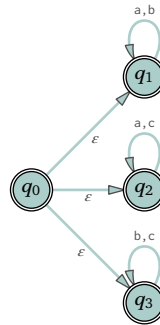
V IV.2.37 There are a number of ways to go. This is one



and this is another.

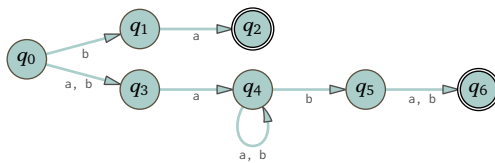


V IV.2.38

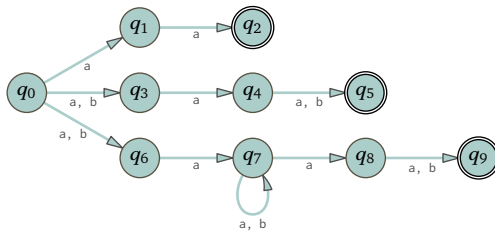


V IV.2.39

(A)

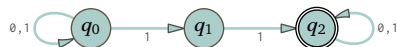


(B)

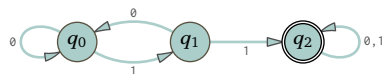


V IV.2.40 It is $\mathcal{L} = \{\sigma \in \{a, b\}^* \mid \sigma \text{ has no substring } bba\}$.

IV.2.41 Here is the nondeterministic machine

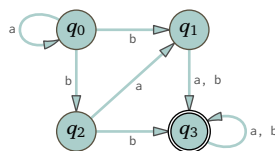


and this is a deterministic version.



IV.2.42

(A)



- (B) A string is in the language if it consists of, first, some number of repetitions of: either a number of repetitions of a or ba. Then it has either bb, or a b followed by an a or b. Finally, the suffix consists of some number of characters, either a or b.
- (c) This is the transition function for the associated deterministic machine.

Δ_D	a	b
$s_0 = \{ \}$	s_0	s_0
$s_1 = \{ q_0 \}$	s_1	s_8
$s_2 = \{ q_1 \}$	s_4	s_4
$+ s_3 = \{ q_2 \}$	s_2	s_4
$s_4 = \{ q_3 \}$	s_4	s_4
$s_5 = \{ q_0, q_1 \}$	s_7	s_{14}
$+ s_6 = \{ q_0, q_2 \}$	s_5	s_{14}
$s_7 = \{ q_0, q_3 \}$	s_7	s_{14}
$+ s_8 = \{ q_1, q_2 \}$	s_9	s_4
$s_9 = \{ q_1, q_3 \}$	s_4	s_4
$+ s_{10} = \{ q_2, q_3 \}$	s_9	s_4
$+ s_{11} = \{ q_0, q_1, q_2 \}$	s_{12}	s_{14}
$s_{12} = \{ q_0, q_1, q_3 \}$	s_7	s_{14}
$+ s_{13} = \{ q_0, q_2, q_3 \}$	s_{12}	s_{14}
$+ s_{14} = \{ q_1, q_2, q_3 \}$	s_9	s_4
$+ s_{15} = \{ q_0, q_1, q_2, q_3 \}$	s_{12}	s_{14}

V IV.2.43



The associated deterministic machine has eight states. Its accepting states are the ones containing q_2 .

Δ_D	$\emptyset$	1
$s_0 = \{ \}$	s_0	s_0
$s_1 = \{ q_0 \}$	s_0	s_2
$s_2 = \{ q_1 \}$	s_3	s_0
$+ s_3 = \{ q_2 \}$	s_3	s_3
$s_4 = \{ q_0, q_1 \}$	s_3	s_2
$+ s_5 = \{ q_0, q_2 \}$	s_3	s_6
$+ s_6 = \{ q_1, q_2 \}$	s_3	s_3
$+ s_7 = \{ q_0, q_1, q_2 \}$	s_3	s_6

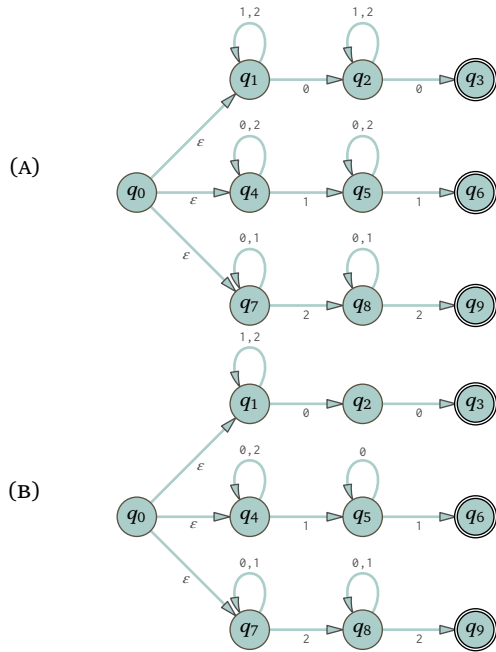
IV.2.44 This is for the machine on the left.

Δ_D	$\emptyset$	1
$s_0 = \{ \}$	s_0	s_0
$s_1 = \{ q_0 \}$	s_4	s_0
$s_2 = \{ q_1 \}$	s_3	s_3
$+ s_3 = \{ q_2 \}$	s_0	s_1
$s_4 = \{ q_0, q_1 \}$	s_7	s_3
$+ s_5 = \{ q_0, q_2 \}$	s_4	s_1
$+ s_6 = \{ q_1, q_2 \}$	s_3	s_5
$+ s_7 = \{ q_0, q_1, q_2 \}$	s_7	s_5

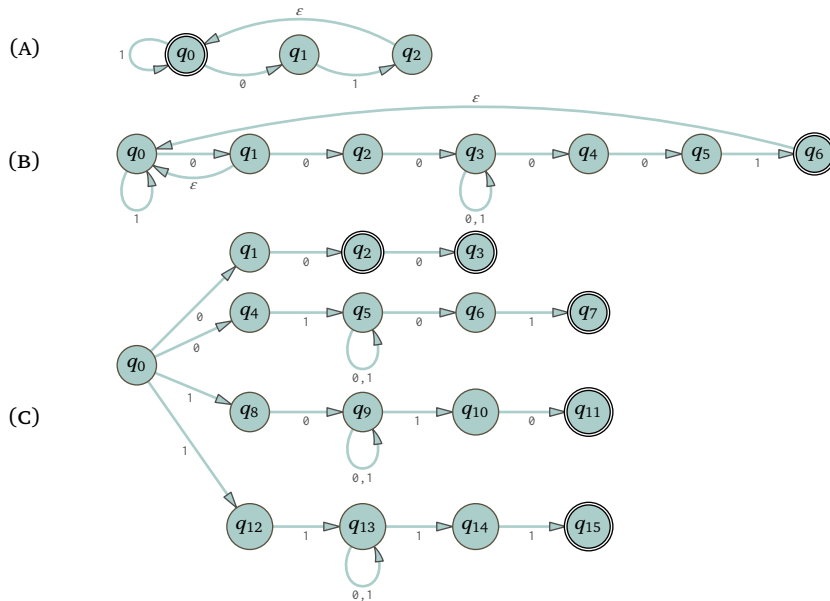
And, this is for the machine on the right.

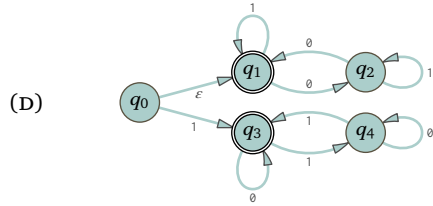
Δ_D	$\emptyset$	1
$s_0 = \{ \}$	s_0	s_0
$s_1 = \{q_0\}$	s_4	s_1
$s_2 = \{q_1\}$	s_3	s_3
$+ s_3 = \{q_2\}$	s_0	s_5
$s_4 = \{q_0, q_1\}$	s_7	s_1
$+ s_5 = \{q_0, q_2\}$	s_4	s_1
$+ s_6 = \{q_1, q_2\}$	s_3	s_5
$+ s_7 = \{q_0, q_1, q_2\}$	s_7	s_5

V IV.2.45



IV.2.46





IV.2.47 Here are the graphs.



The machine on the left, the one that accepts only the empty string, has this table.

Δ	a	b	c
$+ q_0$	$\{\}$	$\{\}$	$\{\}$

The machine on the right has this.

Δ	a	b	c
q_0	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$
$+ q_1$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$

IV.2.48 A nondeterministic machine with ε transitions accepts a string $\tau_0 \in \Sigma^*$ if any of the members of $\hat{\Delta}(q_0, \tau_0)$ is a final state.

V IV.2.49

(A) This is a derivation of abb .

$$S \Rightarrow aA \Rightarrow abB \Rightarrow abb$$

This is a derivation of $aabb$.

$$S \Rightarrow aA \Rightarrow aaA \Rightarrow aabB \Rightarrow aabb$$

And, this is a derivation of $abbb$.

$$S \Rightarrow aA \Rightarrow abB \Rightarrow abbB \Rightarrow abbb$$

Verifying that they are accepted by the machine is routine.

(B) The language of the machine and the grammar is $\mathcal{L} = \{a^i b^j \mid i > 0, j > 1\}$.

V IV.2.50 Each of these is solvable. For each we will sketch an algorithm.

(A) Clearly we can write a simulator for deterministic Finite State machines. By Church's Thesis since we can write it on a modern computer, we can write one for a Turing machine. When given an input σ , the machine will consume it in no more than $|\sigma|$ steps, and so the simulation is sure to halt. Then just check whether the ending state is an accepting state.

(B) The section body discusses how to simulate a nondeterministic computation, by dovetailing (that is, time-slicing). With that we can answer questions about whether this machine accepts a given input σ just as in the prior item.

IV.2.51

(A) Here is the list.

step i	$E(q_0, i)$	$E(q_1, i)$	$E(q_2, i)$	$E(q_3, i)$
0	$\{q_0\}$	$\{q_1\}$	$\{q_2\}$	$\{q_3\}$
1	$\{q_0, q_3\}$	$\{q_0, q_1\}$	$\{q_2\}$	$\{q_3\}$
2	$\{q_0, q_3\}$	$\{q_0, q_1\}$	$\{q_2\}$	$\{q_3\}$
3	$\{q_0, q_3\}$	$\{q_0, q_1\}$	$\{q_2\}$	$\{q_3\}$

So, $\hat{E}(q_0) = \{q_0, q_3\}$, $\hat{E}(q_1) = \{q_0, q_1\}$, $\hat{E}(q_2) = \{q_2\}$, and $\hat{E}(q_3) = \{q_3\}$.

(B) This is similar.

	a*b	a*	$\emptyset$	ε	b(a b)a	(a b)(ε a)a
ε	F	T	F	T	F	F
a	F	T	F	F	F	F
b	T	F	F	F	F	F
aa	F	T	F	F	F	T
ab	T	F	F	F	F	F
ba	F	F	F	F	F	T
bb	F	F	F	F	F	F
aaa	F	T	F	F	F	T
aab	T	F	F	F	F	F
aba	F	F	F	F	F	F
abb	F	F	F	F	F	F
baa	F	F	F	F	T	T
bab	F	F	F	F	F	F
bba	F	F	F	F	T	F
bbb	F	F	F	F	F	F

FIGURE 86, FOR QUESTION IV.3.19: String matching test.

step i	$E(q_0, i)$	$E(q_1, i)$	$E(q_2, i)$
0	$\{q_0\}$	$\{q_1\}$	$\{q_2\}$
1	$\{q_0\}$	$\{q_1, q_2\}$	$\{q_2\}$
2	$\{q_0\}$	$\{q_1, q_2\}$	$\{q_2\}$
3	$\{q_0\}$	$\{q_1, q_2\}$	$\{q_2\}$

So, $\hat{E}(q_0) = \{q_0\}$, $\hat{E}(q_1) = \{q_1, q_2\}$, and $\hat{E}(q_2) = \{q_2\}$.

IV.3.16

(A) Yes. One match is $\emptyset\emptyset 1 \emptyset$.

(B) Yes This works: $1 \emptyset 1$.

(C) No. To match, the $\emptyset$ must be a final character.

(D) Yes. A match is $1 \emptyset 1$.

(E) Yes. Here is a match: $\varepsilon \emptyset 1$.

IV.3.17

(A) Five that match are $\emptyset$, $\emptyset 1$, $\emptyset 11$, $\emptyset 111$, and $\emptyset 1^4 = \emptyset 1111$. Five that do not are ε , 1 , $1\emptyset$, 11 , and $1\emptyset\emptyset$.

(B) Five matching strings are ε , $\emptyset 1$, $\emptyset 1\emptyset 1$, $\emptyset 1\emptyset 1\emptyset 1$, and $(\emptyset 1)^4$. Five that don't match are $\emptyset$, 1 , $\emptyset\emptyset$, $1\emptyset$, and 11 .

(C) Matches are $1\emptyset 1$ and 111 . Five that do not match are ε , $\emptyset$, 1 , $\emptyset\emptyset$, $\emptyset 1$, and $1\emptyset$.

(D) Five matches are $\emptyset$, $\emptyset 1$, $\emptyset\emptyset$, 1 , and $1\emptyset$. Five that don't match are ε , $\emptyset\emptyset 1$, $\emptyset\emptyset\emptyset 1$, $1\emptyset 1$, and $11\emptyset 1$.

(E) Matches: none. Non-matches: ε , $\emptyset$, 1 , $\emptyset\emptyset$, $\emptyset 1$, and $1\emptyset$.

IV.3.18

(A) The language consists of strings with some number of a's (that number may be zero), followed by a single c, followed by some number of b's.

(B) The language consists of strings of at least one a.

(C) Strings in this language start with an a, followed by any number of characters, and end with two b's.

V IV.3.19 Figure 86 on page 136 shows the table. Observe that the regular expression $\emptyset$ never matches.

V IV.3.20 Figure 87 on page 137 shows the table. Note that the regular expression $\emptyset$ never matches.

V IV.3.21 The definition is $\mathcal{L}(R)^* = \{\sigma_0 \frown \dots \frown \sigma_{k-1} \mid k \in \mathbb{N} \text{ and } \sigma_0, \dots, \sigma_{k-1} \in \mathcal{L}(R)\}$. That is, the language of the star is the star of the language: $\mathcal{L}(R^*) = (\mathcal{L}(R))^*$.

	0^*1	1^*0	$\emptyset$	ε	$0(0 1)^*$	$(100)(\varepsilon 1)0^*$
ε	F	F	F	T	F	F
0	F	T	F	F	T	F
1	T	F	F	F	F	F
00	F	F	F	F	T	F
01	T	F	F	F	T	F
10	F	T	F	F	F	F
11	F	F	F	F	F	F
000	F	F	F	F	T	F
001	T	F	F	F	T	F
010	F	F	F	F	T	F
011	F	F	F	F	T	F
100	F	F	F	F	F	T
101	F	F	F	F	F	F
110	F	T	F	F	F	F
111	F	F	F	F	F	F

FIGURE 87, FOR QUESTION IV.3.20: String matching test.

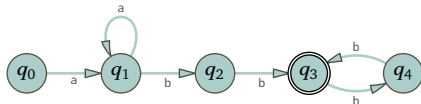
As an example, $(0^*1)^*$ matches 01001 because $01001 = 01 \frown 001$, and both of 01 and 001 matches (0^*1) . Thus, Kleene star means something more like “repeatedly match the inside.”

V IV.3.22 One is $(0|1)^*111(0|1)^*$.

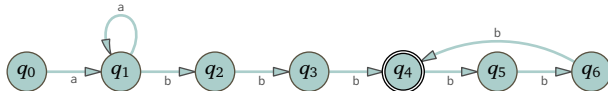
IV.3.23 Regular expressions are not unique. For instance, here are two regular expressions for the language of strings over $\Sigma = \{a, b\}$ that have a substring with at least three a's: $(a|b)^*aaa(a|b)^*$ and $(a|b)^*aaaa(a|b)^*$.

IV.3.24

(A) Of course, many sets of strings are possible answers. Here are five members of $\mathcal{L}_0$: abb, aabb, ab^4 , aab^4 , ab^6 . Here are five that are not: ε , a, b, ab, abba, ba. A regular expression is $aa^*bb(bb)^*$. Here is a (nondeterministic) Finite State machine that accepts this language.



(B) Five members of $\mathcal{L}_1$ are: abbb, aabbb, ab^6 , aab^6 , ab^9 . Five that are not are: ε , a, b, abbb, abbbba, ba. A regular expression is $aa^*bbb(bbb)^*$. This is a nondeterministic Finite State machine that accepts $\mathcal{L}_1$.



IV.3.25 $((a|b)^*)|((a|c)^*)|((b|c)^*)$

IV.3.26

(A) $b(a|b)^*$

(B) $(a|b)^*a(a|b)$

(C) The two characters could come in the order ... a ... b ... or the order ... a ... b So this works: $(a|b)^*((a(a|b)^*b)|(b(a|b)^*a)$

(D) The a's must come in triplets: $(b^*ab^*ab^*ab^*)^*b^*$.

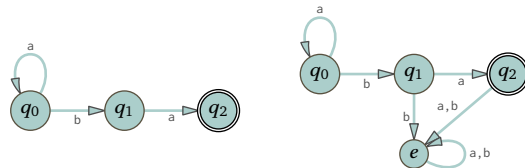
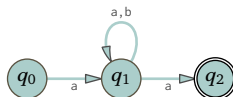
V IV.3.27

(A) $aba(a|b)^*$

(B) $(a|b)^*aba$

(C) $(a|b)^*aba(a|b)^*$

IV.3.28

FIGURE 88, FOR QUESTION IV.3.33: Finite State machine for a^*ba .FIGURE 89, FOR QUESTION IV.3.33: Finite State machine for $ab^*(a|b)^*a$.

- (A) There must be at least one 1, and more 1's must come in pairs: $(0^*10^*1)^*0^*10^*$.
 (B) The regular expression $(\epsilon|1)(01)^*(\epsilon|0)$ will do. Note that it matches the empty string, as well as the string 0 and the string 1.
 (C) A multiple of eight in binary ends in three 0's: $(0|1)^*000$. That expression matches binary numbers with leading 0's and won't match 0. If you don't like that then instead use $(1(0|1)^*000)|0$.

IV.3.29

- (A) This is an answer: $((ba)^*bb^*)^*$. Notice that it matches the empty string, and that it matches babab.
 (B) $(a|b)^*((bba^*bb)|(bbb))(a|b)^*$

IV.3.30

- (A) Here the 0's come in pairs: $(1^*01^*01^*)^*1^*$.
 (B) Here two of the 1's have no Kleene star's: $0^*10^*10^*1^*(0|1)^*$.
 (C) One that will do is to start with the expression from the prior item and replace all the 0's with the expression from the first item.

$$((1^*01^*01^*)^*1^*)^*1((1^*01^*01^*)^*1^*)^*1((1^*01^*01^*)^*1^*)^*1^*((1^*01^*01^*)^*1^*)^*|1)^*$$

IV.3.31

- (A) $(a(a|b)^*a)|(b(a|b)^*b)$
 (B) $((aaa)^*b(aaa)^*)|(a(aaa)^*ba(aaa)^*)|(aa(aaa)^*baa(aaa)^*)$

IV.3.32

- (A) $1(0|1)^*000$
 (B) $1(01)^*0|0(10)^*1$
 (C) $\epsilon|0|1|0(10^*)(1|\epsilon)|1(01)^*(0|\epsilon)$

IV.3.33

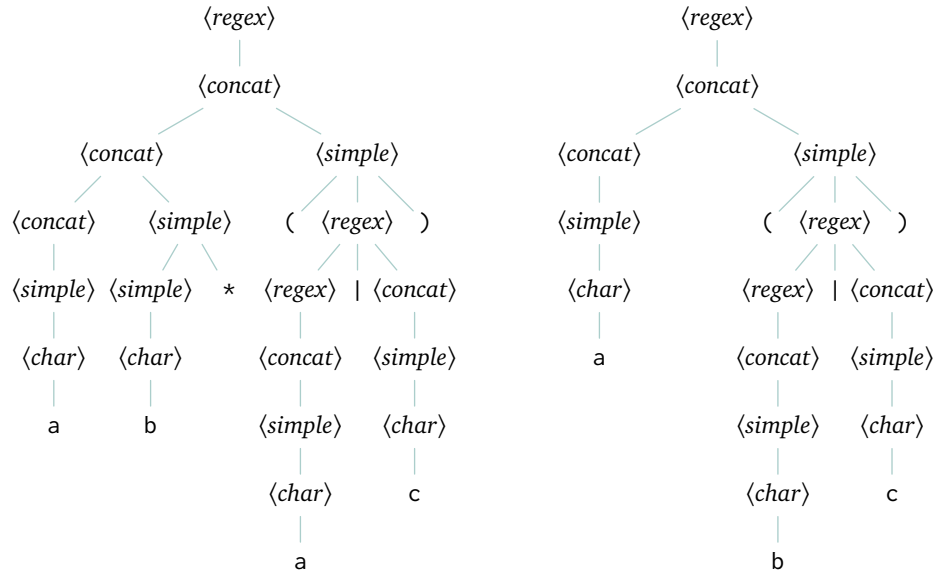
- (A) A nondeterministic Finite State machine is on the left in Figure 88 on page 138. If you prefer a deterministic machine, use the one on the right.
 (B) A nondeterministic machine is in Figure 89 on page 138. Note that $b^*(a|b)^*$ collapses to $(a|b)^*$, so that is what the machine handles. We have given a way to convert a nondeterministic Finite State machine to a deterministic one, so if you prefer a deterministic one then take the above nondeterministic machine as an abbreviation.

IV.3.34 The first will be accepted by the machine with the accepting states switched to nonaccepting, and the nonaccepting states switched to accepting. The second will not work if $\mathcal{L} = \emptyset$ and $k = 1$.

IV.3.35

- (A) Transform the machine $\mathcal{M}$ to a new machine $\hat{\mathcal{M}}$ by making the set of final states of $\hat{\mathcal{M}}$ be S .
 (B) Transform the machine $\mathcal{M}$ to $\hat{\mathcal{M}}$ by making the start state be the given first single state, and also make the set S of the prior item contain only the ending state. Then apply the argument given for the prior item.

IV.3.36 We will construct the reachable states, and then the unreachable ones are the others.

FIGURE 90, FOR QUESTION IV.3.38: Parse trees for $a(b|c)$ and $ab^*(a|c)$.

(A) The set of reachable states is $S_2 = S_3 = \dots = \{q_0, q_1, q_2\}$. So the set of unreachable states is $\{q_3\}$.

i	S_i
0	$\{q_0\}$
1	$\{q_0, q_1\}$
2	$\{q_0, q_1, q_2\}$
3	$\{q_0, q_1, q_2\}$

(B) The set of reachable states is $S_2 = S_3 = \dots = \{q_0, q_1, q_3\}$. Thus the set of unreachable states is $\{q_2, q_4\}$.

i	S_i
0	$\{q_0\}$
1	$\{q_0, q_1\}$
2	$\{q_0, q_1, q_3\}$
3	$\{q_0, q_1, q_3\}$

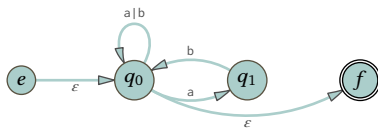
IV.3.37 Alphabets are finite, as defined in Appendix A. So the set of length n strings over Σ is finite, and therefore countable. Chapter II's Corollary 2.14 shows that the countable union of countable sets is countable, and so the union over all n of length n strings is countable.

IV.3.38 Figure 90 on page 139 shows the two trees.

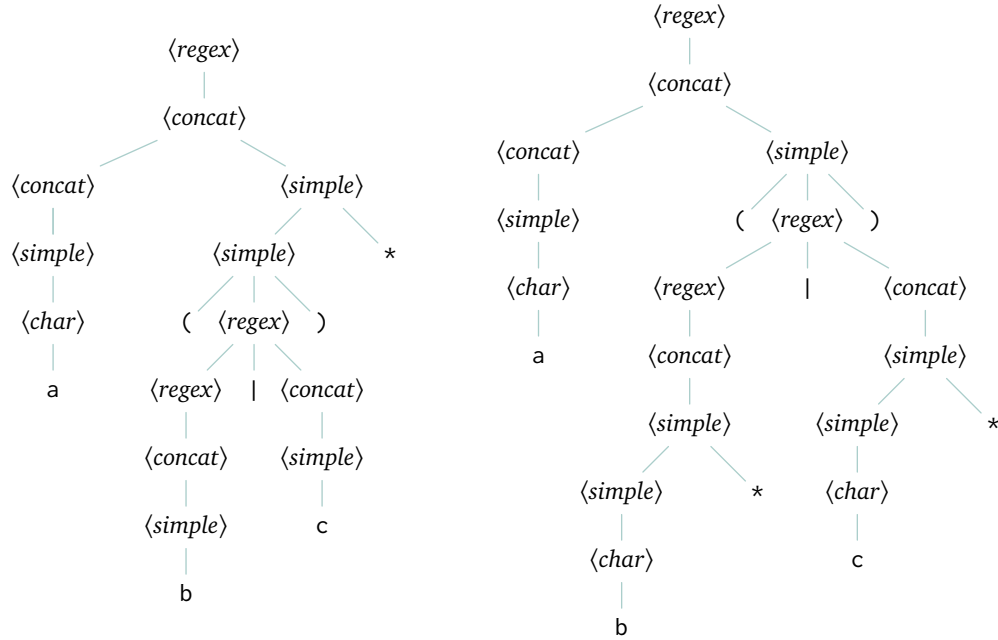
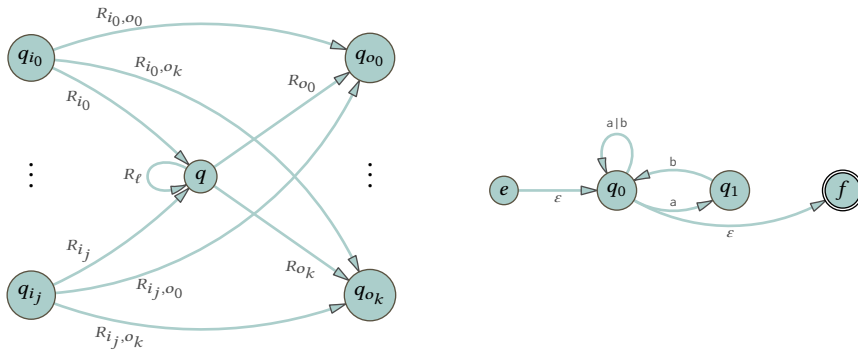
IV.3.39 Figure 91 on page 140 shows the two trees.

IV.3.40

(A) This is $\hat{\mathcal{M}}$.



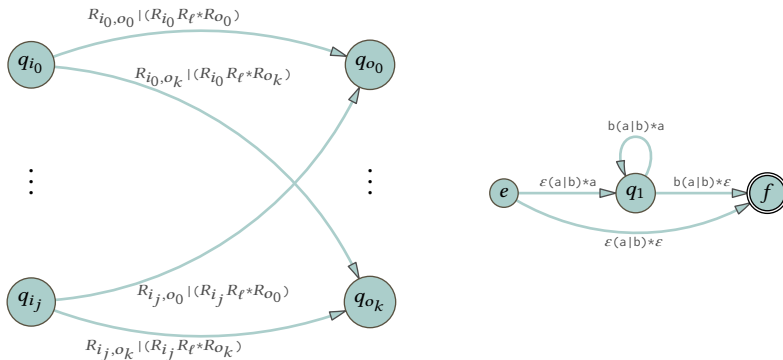
(B) This is the before half of the diagram from Lemma 3.14's proof, with $\hat{\mathcal{M}}$ repeated for convenience.

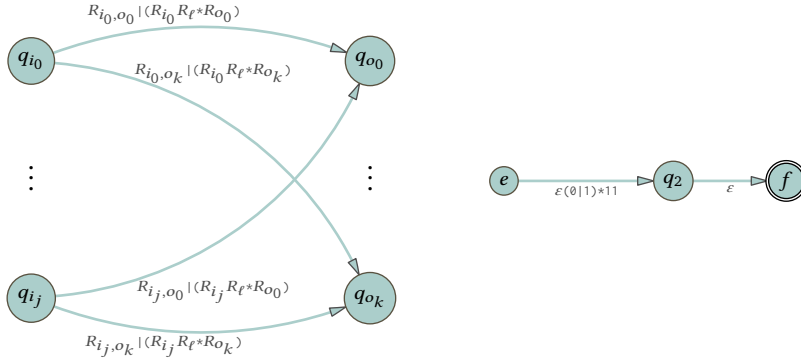
FIGURE 91, FOR QUESTION IV.3.39: Parse trees for $a(b|c)^*$ and $a(b^*|c^*)$.

The state q_0 has two states that originate arrows into it, and two that receive arrows from it. So we get this translation of the diagram's notation.

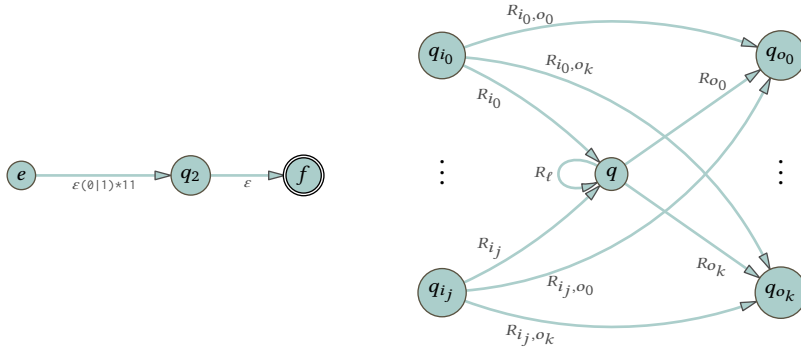
q	q_{i_0}	q_{i_1}	q_{o_0}	q_{o_1}	R_{i_0}	R_{i_1}	R_{i_0,o_0}	R_{i_0,o_1}	R_{i_1,o_0}	R_{i_1,o_1}	R_{ℓ}	R_{o_0}	R_{o_1}
q_0	e	q_1	q_1	f	ϵ	b	$-$	$-$	$-$	$-$	$a b$	a	ϵ

(c) This is the after half diagram of Lemma 3.14's proof, along with the machine $\hat{\mathcal{M}}$ after q_0 is eliminated.





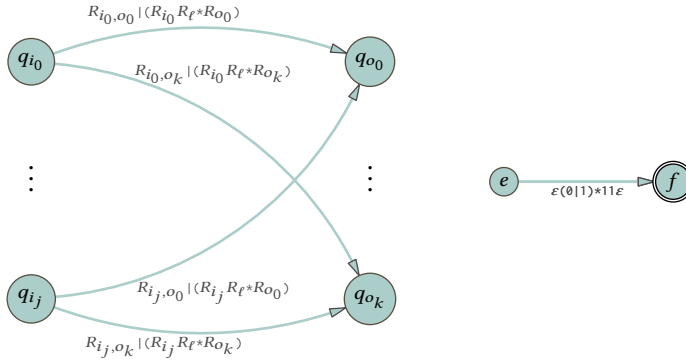
(c) This is the machine from the prior answer, along with the before half diagram.



The state q_2 has one state that originates arrows into it, and one that receives arrows from it. So we get this translation of the diagram's notation.

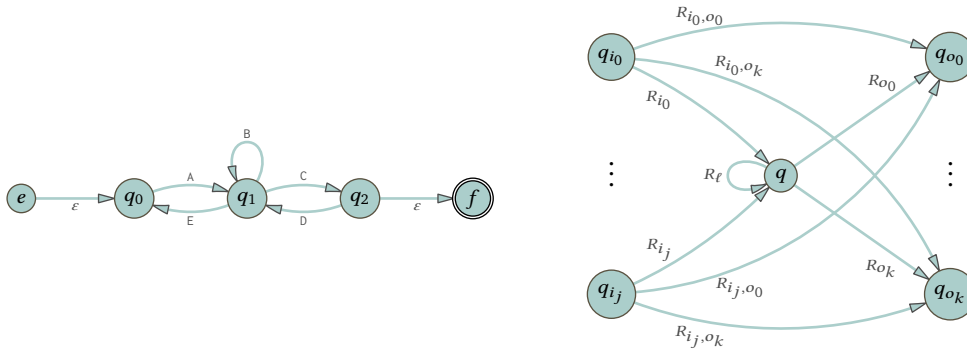
$$\begin{array}{c|ccc} q & q_{i_0} & q_{o_0} & \\ \hline q_2 & e & f & \end{array} \quad \begin{array}{c|cccc} R_{i_0} & R_{i_0,o_0} & R_\ell & R_{o_0} \\ \hline \varepsilon(\emptyset|1)^*11 & - & - & \varepsilon \end{array}$$

This is the after half diagram along with the machine after q_2 is gone.



(D) The regular expression is $\varepsilon(\emptyset|1)^*11\varepsilon$.

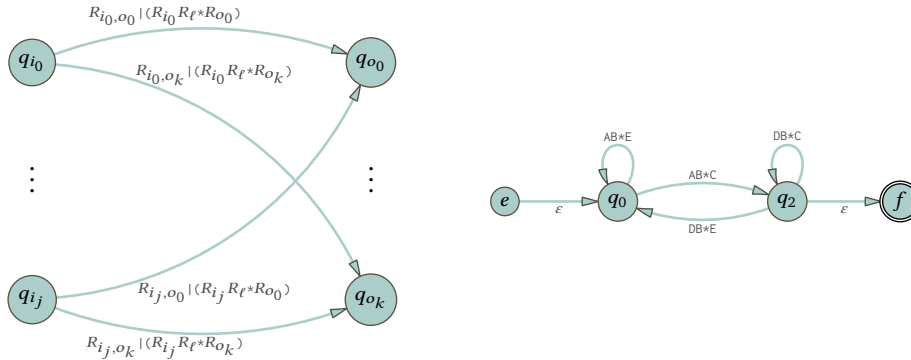
IV.3.42 This is $\hat{\mathcal{M}}$ along with the before half of the diagram from Lemma 3.14's proof.



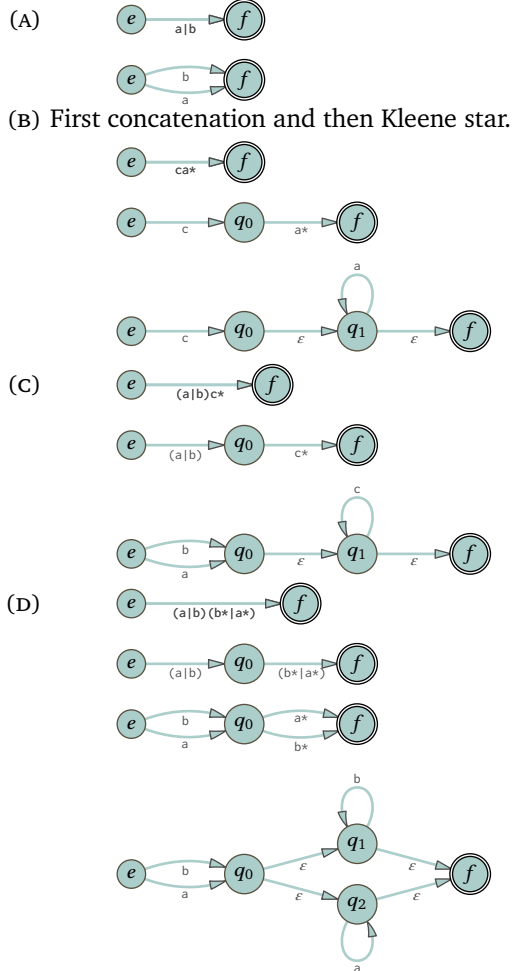
The state q_1 has two states that originate arrows into it, and two that receive arrows from it. So we get this translation of the diagram's notation.

q	q_{i_0}	q_{i_1}	q_{o_0}	q_{o_1}	R_{i_0}	R_{i_1}	R_{i_0,o_0}	R_{i_0,o_1}	R_{i_1,o_0}	R_{i_1,o_1}	R_ℓ	R_{o_0}	R_{o_1}
q_1	q_0	q_2	q_0	q_2	A	D	—	—	—	—	B	E	C

Here is the after diagram, along with the machine $\hat{\mathcal{M}}$ after the elimination of q_1 .



IV.3.43 Each of these gives the progression of steps.



V IV.4.7

- (A) False. For example, the language $\{a^n \mid n \in \mathbb{N}\} = \{\epsilon, a, aa, \dots\}$ is regular and infinite.
- (B) False, the empty language is recognized by any Finite State machine that has no accepting states. It is also the language described by the regular expression $\emptyset$.

- (C) False. Rather, the intersection of two regular languages is regular. For example, where $\mathcal{L}$ is not regular, the intersection $\mathcal{L} \cap \mathbb{B}^* = \mathcal{L}$ is not regular.
- (D) False. The language $\mathbb{B}^*$ is the set of all strings is recognized by any Finite State machine whose states are all accepting states. It is also the language described by the regular expression $(0|1)^*$.
- (E) False, a Finite State machine with no accepting states does not accept any strings.
- (F) False, a minimal machine cannot be further reduced. For instance, a machine with only one state, and that state accepting, is the smallest machine that recognizes $\mathcal{L} = \mathbb{B}^*$.

IV.4.8 Yes. To show this we need to show that this is the language of some Finite State machine. In binary, powers of 2 are represented by strings of bits starting with 1 and ending in 0's. So the regular expression 10^* describes the language. (If you want to allow leading 0's then the regular expression is 0^*10^* .)

IV.4.9 This is not really a sensible question. If we assert that no English sentence has more than a million words (see (Wikipedia contributors 2020)) then we might claim that English is a regular language because we can list the sentences that we declare allowed, and there are finitely many of them.

Otherwise we must choose a grammar. But which one? Is “Colorless green ideas sleep furiously” a legal sentence even though it is not sensible? Is “My god, it is her.” allowed even though some would say it should be “My god, it is she.”?

Really, it is not a sensible question. (See <https://cs.stackexchange.com/a/116178/67754>.)

V IV.4.10 The first is true and the second is false.

- (A) If the language consists of the strings $\sigma_0, \dots, \sigma_{n-1}$ then a regular expression is $\sigma_0 | \dots | \sigma_{n-1}$.
- (B) The language over $\Sigma = \{a, b\}$ described by the regular expression a^* is $\{\epsilon, a, aa, \dots\}$, and is infinite.

V IV.4.11 One way to show this is to give a regular expression for each.

- (A) $a(a|b)^*a$
- (B) $b^*(ab^*a)^*b^*$

IV.4.12 Because the language is regular it is described by a regular expression, R . Then the regular expression $1R$ describes $\hat{\mathcal{L}}$.

IV.4.13 The size of the set $Q_0 \times Q_1$ is $n_0 \cdot n_1$.

IV.4.14 These are the tables for the machines.

Δ_0	a	b	Δ_1	a	b
+ q_0	q_0	q_1	s_0	s_1	s_0
+ q_1	q_2	q_1	+ s_1	s_0	s_0
q_2	q_2	q_0			

Here is the cross product machine, without specifying any accepting states.

Δ	a	b
(q_0, s_0)	(q_0, s_1)	(q_1, s_0)
(q_0, s_1)	(q_0, s_0)	(q_1, s_0)
(q_1, s_0)	(q_2, s_1)	(q_1, s_0)
(q_1, s_1)	(q_2, s_0)	(q_1, s_0)
(q_2, s_0)	(q_2, s_1)	(q_0, s_0)
(q_2, s_1)	(q_2, s_0)	(q_0, s_0)

- (A) To have this machine accept the intersection of the two languages, take the accepting states to be those in $F = \{(q_0, s_1), (q_1, s_1)\}$.
- (B) $F = \{(q_0, s_0), (q_0, s_1), (q_1, s_0), (q_1, s_1)\}$

IV.4.15 Here is the transition table for $\mathcal{M}_1$.

Δ_1	a	b
s_0	s_0	s_1
+ s_1	s_1	s_0

The product of this machine with itself has four states.

Δ_1	a	b
(s_0, s_0)	(s_0, s_0)	(s_1, s_1)
(s_0, s_1)	(s_0, s_1)	(s_1, s_0)
(s_1, s_0)	(s_1, s_0)	(s_0, s_1)
+ (s_1, s_1)	(s_1, s_1)	(s_0, s_0)

Here the accepting state is set for the intersection, to only (s_1, s_1) . We could include the states (s_0, s_1) and (s_1, s_0) for the union construction, but it wouldn't matter because those are unreachable states.

IV.4.16 The machine on the left below accepts a string $\sigma \in \mathbb{B}^*$ if it contains at least two 0's. The one on the right accepts a string if it contains an even number of 1's.

Δ_0	0	1		Δ_1	0	1
q_0	q_1	q_0	+	s_0	s_0	s_1
q_1	q_2	q_1		s_1	s_1	s_0
+ q_2	q_2	q_2				

The product construction gives this. We want the intersection of the two languages.

Δ	0	1
(q_0, s_0)	(q_1, s_0)	(q_0, s_1)
(q_0, s_1)	(q_1, s_1)	(q_0, s_0)
(q_1, s_0)	(q_2, s_0)	(q_1, s_1)
(q_1, s_1)	(q_2, s_1)	(q_1, s_0)
+ (q_2, s_0)	(q_2, s_0)	(q_2, s_1)
(q_2, s_1)	(q_2, s_1)	(q_2, s_0)

IV.4.17

- (A) True. Every language is a subset of the language Σ^* .
- (B) False. Take $\mathcal{L}_0$ to be not regular, and take $\mathcal{L}_1$ to be the empty language, which is regular. Their union is $\mathcal{L}_0$.
- (C) False. A language with one element is regular, and its only subset is the empty language, which is also regular.
- (D) True. This is part of Lemma 4.3.

IV.4.18 It might be regular, or might be not regular. First note that the language of all strings $\mathcal{L} = \Sigma^*$ is regular. Now assume that $\mathcal{L}_1$ is not regular. The concatenation $\Sigma^* \cap \mathcal{L}_1 = \Sigma^*$ gives a language that is regular. Also, where the alphabet is Σ and $s \in \Sigma$ then the concatenation $\{s\} \cap \mathcal{L}_1$ gives a language that is not regular.

IV.4.19 Each is false. The same example suffices for all three. For $\mathcal{L}_0$ use the set of all bitstrings, $\mathbb{B}^*$. For $\mathcal{L}_1$ use any non-regular subset of it, the existence of which is guaranteed by Lemma 4.2.

IV.4.20 By Lemma 4.3, regular languages are closed under Kleene star and concatenation.

IV.4.21 Let Σ be the alphabet of the language $\mathcal{L}$. The collection of even length strings over Σ is regular since there is clearly a Finite State machine that accepts only those strings. The intersection of $\mathcal{L}$ with this collection is an intersection of two regular languages, so it is regular.

IV.4.22 Let $\{a^n b^n \in \{a, b\}^* \mid n \in \mathbb{N}\}$ be $\mathcal{L}_0$. Let the language described by the regular expression $a^* b^*$ be $\mathcal{L}_1$, and let the language $\{\sigma \in \{a, b\}^* \mid \sigma \text{ contains the same number of } a\text{'s as } b\text{'s}\}$ be $\mathcal{L}_2$. If $\mathcal{L}_2$ were regular then because the intersection of regular languages is a regular language, $\mathcal{L}_2 \cap \mathcal{L}_1$ would be regular. But that set is $\mathcal{L}_0$, which was given as not regular.

IV.4.23 The graphical representation of a Finite State machine is a directed graph. So we can naturally extend the definition of one vertex being reachable from another to one state q being reachable from another state $\hat{q}$, if there is a sequence of directed edges that go from the first node to the second.

- (A) Fix a Finite State machine $\mathcal{M}$ that recognizes $\mathcal{L}$. From it derive a new Finite State machine by making an accepting state every state q from which an accepting state of $\mathcal{M}$ is reachable. Clearly that machine recognizes $\text{pref}(\mathcal{L})$.
- (B) Again, fix a Finite State machine $\mathcal{M}$ that recognizes $\mathcal{L}$. Let S be the set of states reachable from q_0 . Get $\hat{\mathcal{M}}$ by introducing a new state r and defining an ε transition to each state in S . Here also, clearly this machine recognizes $\text{suff}(\mathcal{L})$.

(c) We have $\text{allprefs}(\mathcal{L}) = (\mathcal{L}^c \cap \Sigma^*)^c$, so it is regular.

IV.4.24 If there was a Finite State machine $\mathcal{M}$ that accepted this language then to solve the Halting problem, write $\mathcal{M}$ as a subroutine. Then, given $e \in \mathbb{N}$, feed 1^e to the subroutine. If it accepts, then $e \in K$ and otherwise $e \notin K$.

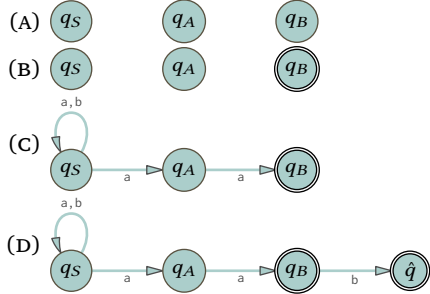
IV.4.25

(A) For every word $\sigma \in \Sigma^*$, the one-word language $\mathcal{L} = \{\sigma\}$ is regular. But any non-regular language is the union of a collection of such languages.

(B) Similarly, for every word $\sigma \in \Sigma^*$, the language $\mathcal{L} = \Sigma^* - \{\sigma\}$ is regular. Any non-regular language $\mathcal{L}$ is the intersection of a collection of such languages, namely the collection over all $\sigma \notin \mathcal{L}$.

IV.4.26 Over $\mathbb{B}$ the language of all strings $\mathcal{L} = \mathbb{B}^*$ is regular, and is uncountable.

IV.4.27 The result is a nondeterministic Finite State machine.



IV.4.28

(A) Lemma 4.3 shows that the collection of regular languages is closed under union.

(B) This identity shows that closure under complement will also demonstrate closure under set difference.

(C) Fix a regular language $\mathcal{L}$ over some alphabet Σ . It is recognized by some deterministic Finite State machine $\mathcal{M}$ with input alphabet Σ .

Define a new machine $\hat{\mathcal{M}}$ with the same states and transition function as $\mathcal{M}$ but whose accepting states are the complement, $F_{\hat{\mathcal{M}}} = Q_{\mathcal{M}} - F_{\mathcal{M}}$. We will show that the language of this machine is $\Sigma^* - \mathcal{L}$.

Because $\mathcal{M}$ is deterministic, each input string τ is associated with a unique last state, the state that the machine is in after it consumes τ 's last character, namely $\hat{\Delta}(\tau)$. Observe that $\hat{\Delta}(\tau) \in F_{\hat{\mathcal{M}}}$ if and only if $\hat{\Delta}(\tau) \notin F_{\mathcal{M}}$. Thus the language of $\hat{\mathcal{M}}$ is the complement of the language of $\mathcal{M}$ and since this language is recognized by a Finite State machine, it too is regular.

IV.4.29 For one direction suppose that the machine accepts at least one string, σ , of length k where $n \leq k < 2n$. In processing σ the machine must go through at least n -many transitions. But the machine has n states and starting at q_0 and visiting all the other states exactly once would require only $n - 1$ many transitions. Thus, by the Pigeonhole principle, in processing the string σ the machine visits some state q twice. That is, in processing σ the machine makes a loop.

Said another way, the accepted string σ decomposes as $\sigma = \alpha \frown \beta \frown \gamma$ where processing the α portion takes the machine from the start to the loop (from state q_0 to q), the β portion takes the machine around the nontrivial loop (so that $\beta \neq \epsilon$ brings the machine from q to q again) and the γ portion brings the machine to some final state. But then the recognized language is infinite, since clearly the machine also accepts $\alpha \frown \beta \frown \beta \frown \gamma$, etc. (Note that we can take β to be of length less than n because otherwise the size of the machine would imply it has a subloop.)

Conversely, suppose the language of accepted strings is infinite. At least one of them is longer than n ; fix one that is longer but of minimal length. As in the prior paragraph, because the machine only has n states, by the Pigeonhole principle the accepted string σ decomposes as $\sigma = \alpha \frown \beta \frown \gamma$ where the β portion is a nontrivial loop so processing α brings the machine from q_0 to some q , then processing $\beta \neq \epsilon$ brings the machine from q to q again, and then processing the γ portion brings the machine to some final state. Because the string is of minimal length, $|\beta| < n$. In total then, $|\sigma| < 2n$.

IV.4.30

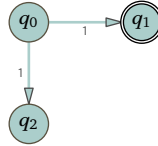
(A) $\hat{h}(\emptyset 1) = \text{aba}$, $\hat{h}(10) = \text{baa}$, $\hat{h}(101) = \text{baaba}$

(B) A regular expression to describe $\hat{\mathcal{L}} = \{\sigma \frown 1 \mid \sigma \in \mathbb{B}^*\}$ is $(\emptyset 1)^* 1$ A regular expression to describe $\hat{h}(\hat{\mathcal{L}})$ is

(a|ba)*ba.

- (c) Let $\mathcal{L}$ be described by the regular expression R . Apply the function h to each non-meta symbol in R . The result is a regular expression that describes $h(\mathcal{L})$.

IV.4.31 For this machine the language of accepted strings contains 1.



If we complement the set of accepting states then the language of the resulting machine also contains 1.

IV.4.32

- (A) We prove this by induction on the length of σ_1 . For the base case consider both $\sigma_1 = \varepsilon$ and $\sigma_1 = a$ for some $a \in \Sigma$. With both, the statement obviously holds.

For the inductive case assume that the statement is true when $|\sigma_1| = 0, |\sigma_1| = 1, \dots, |\sigma_1| = k$ and consider the $|\sigma_1| = k + 1$ case. In this case $\sigma_1 = \alpha \hat{\ } a$ for some character $a \in \Sigma$ and string $\alpha \in \Sigma^*$ of length k . Then $(\sigma_0 \hat{\ } \sigma_1)^R = (\sigma_0 \hat{\ } \alpha \hat{\ } a)^R$. Apply the base case to get $= a^R \hat{\ } (\sigma_0 \hat{\ } \alpha)^R$. The inductive hypothesis gives $= a \hat{\ } \alpha^R \hat{\ } \sigma_0^R$, and finish by recognizing that as $\sigma_1^R \hat{\ } \sigma_0^R$.

- (B) We prove this by induction on the length of the regular expression. The base case is that $|R| = 1$. If $R = \emptyset$ then by condition (i) R^R is the regular expression $\emptyset$, and $\mathcal{L}(R) = \mathcal{L}(R^R)$ because both equal the empty set, as required. If $R = \varepsilon$ then by condition (ii) $R^R = \varepsilon$, and $\mathcal{L}(R) = \{\varepsilon\} = \mathcal{L}(R^R)$, again as required. And, similarly, if $R = x$ for some $x \in \Sigma$ then by condition (iii) $R^R = x$ and $\mathcal{L}(R) = \{x\} = \mathcal{L}(R^R)$.

Now for the inductive step. Assume that the statement is true for regular expressions R of length $n = 1, n = 2, \dots, n = k$, and consider an expression of length $n = k + 1$. We handled the case that it is a concatenation, $R = R_0 \hat{\ } R_1$, in this question's first item.

If $R = R_0 | R_1$ then $\mathcal{L}(R^R) = \mathcal{L}((R_0 | R_1)^R) = \mathcal{L}(R_0^R | R_1^R)$ by condition (v). By the definition of the language associated with a pipe regular expression following Definition 3.4 that gives $= \mathcal{L}(R_0^R) \cup \mathcal{L}(R_1^R)$. The inductive hypothesis gives $= \mathcal{L}(R_0)^R \cup \mathcal{L}(R_1)^R$, which equals $\{\sigma_0^R \mid \sigma_0 \in \mathcal{L}(R_0)\} \cup \{\sigma_1^R \mid \sigma_1 \in \mathcal{L}(R_1)\}$ and again applying the material following Definition 3.4 that equals $\mathcal{L}(R)^R$.

The final form is similar. If $R = R_0^*$ then $\mathcal{L}(R^R) = \mathcal{L}((R_0^*)^R)$. By the definition following Definition 3.4 that equals $\{\sigma_0 \hat{\ } \dots \hat{\ } \sigma_m \mid \sigma_i \in \mathcal{L}(R_0^R)\}$. Rewriting gives $\{\tau_0^R \hat{\ } \dots \hat{\ } \tau_m^R \mid \tau_j \in \mathcal{L}(R_0)\}$. By this exercise's first item that equals $\{(\tau_m \hat{\ } \dots \hat{\ } \tau_0)^R \mid \tau_j \in \mathcal{L}(R_0)\}$, which is the set $(\mathcal{L}(R_0)^*)^R$.

V IV.5.8 There is no such regular expression. Definition 3.4 just does not have the capacity for that construct.

V IV.5.9 Compared with $\sigma = \alpha\beta\gamma$, the string $\alpha\gamma$ has had β omitted. By definition σ has balanced parentheses. Because β consists of at least one open parentheses, $($, its omission of β means that in $\alpha\gamma$ the number of open parentheses is at least one fewer than the number of closed parentheses.

V IV.5.10 If a language is finite, $\mathcal{L} = \{\sigma_0, \dots, \sigma_n\}$, then the conclusion of the Pumping Lemma holds by taking the pumping length to be $p > \max(|\sigma_0|, \dots, |\sigma_n|)$.

IV.5.11 It is not correct.

The first problem is that it is not sensible. There may be many Finite State machines that recognize the language, so there is no well-defined number of states for the language.

Even if you figure that they meant to fix a particular machine, saying something like, “where $\mathcal{M}$ accepts the language, if that language has a string of length greater than the number of states in $\mathcal{M}$ then it cannot be a regular language” then it still isn't sensible because if there is a Finite State machine then the language for sure is regular.

And, even if we stretch figuring out what they meant all the way to taking that they meant something like this, “If in an argument we assume that the language is recognized by a Finite State machine and we go on to show that the language has a string that is longer than the number of states in that machine, then we have the desired contradiction” then this statement, while now technically sensible, is wrong. For example, there is a Finite State machine that accepts strings described by a^* , and there certainly are strings of that form whose length is greater than the number of states in the machine.

- V IV.5.12 The Pumping Lemma only allows us to conclude that $\alpha\beta$ consists of ϵ 's, not that this substring got all the ϵ 's. The substring γ indeed got all of the close parentheses, the ϵ 's, but that we know, it may have gotten some of the open parentheses as well.

IV.5.13 Yes, that is right.

IV.5.14

- (A) Five elements of $\mathcal{L}_0$ are bb , $abbb$, $aabbbb$, a^3b^5 , and a^4b^6 . Five strings that are not elements are ϵ , a , ab , $aabbb$, and $baabbbb$.

Assume for contradiction that $\mathcal{L}_0$ is regular. By the Pumping Lemma it has a pumping length, p . Consider $\sigma = a^p b^{p+2}$. This string is a member of the language that is of length greater than or equal to p so the Pumping Lemma gives a decomposition $\sigma = \alpha\beta\gamma$ subject to the three conditions. Condition (1) is that $|\alpha\beta| \leq p$, so these two substrings contain only a 's. Condition (2) is that β is nonempty, so it consists of at least one a .

Consider the list $\alpha\gamma$, $\alpha\beta^2\gamma$, $\dots$. Its second element $\alpha\beta^2\gamma$ has more a 's, but no more b 's, than does $\sigma = \alpha\beta\gamma$. That means $\alpha\beta^2\gamma$ is not an element of the language $\mathcal{L}_0$, a contradiction to the Pumping Lemma's condition (3).

- (B) In prose, this language consists of a 's followed by b 's and then followed by c 's, with only the restriction that the number of a 's equals the number of c 's. Thus five elements of $\mathcal{L}_1$ are: ϵ , ac , b , $aabcc$, and bb . Five that are not are: aac , acc , ab , bbc , and ca .

Assume for contradiction that $\mathcal{L}_1$ is regular. Then it has a pumping length p . Consider $\sigma = a^p c^p$. Because $\sigma \in \mathcal{L}_1$ and $|\sigma| \geq p$, the Pumping Lemma says that it decomposes, $\sigma = \alpha\beta\gamma$, in a way that satisfies the three conditions. By condition (1) the first two substrings together are short, $|\alpha\beta| \leq p$. Thus these two consist entirely of a 's. By condition (2) the second substring β is nonempty, so it consists of at least one a . Finally, condition (3) gives that the strings $\alpha\gamma$, $\alpha\beta^2\gamma$, $\dots$ are all members of the language $\mathcal{L}_1$. But the string $\alpha\gamma$, when compared with $\sigma = \alpha\beta\gamma$, has at least one fewer a but the same number of c 's. That means it does not have the same number of the two characters, so it is not a member of the language, which is a contradiction.

- (C) A prose description is that in $\mathcal{L}_2$ the strings have a 's followed by b 's, with the number of a 's less than the number of b 's. So five strings that are members of $\mathcal{L}_2$ are abb , $abbb$, $abbbb$, ab^5 , and a^4b^5 . Five strings that are not elements are ϵ , a , ab , aab , and $baabbb$.

For contradiction suppose that $\mathcal{L}_2$ is regular and fix a pumping length, p . The string $\sigma = a^p b^{p+1}$ is an element of the language whose length is greater than or equal to the pumping length. So the Pumping Lemma gives a decomposition, $\sigma = \alpha\beta\gamma$, subject to the three conditions. Condition (1), that $|\alpha\beta| \leq p$, says that these two substrings contain only a 's. Condition (2), that β is nonempty, says that it consists of at least one a .

Consider the strings $\alpha\gamma$, $\alpha\beta^2\gamma$, $\dots$. Compared with $\sigma = \alpha\beta\gamma$, the string $\alpha\beta^2\gamma$ has more a 's, but no more b 's. Thus the number of a 's is at least $p + 1$ while the number of b 's is fixed at $p + 1$. That means that $\alpha\beta^2\gamma$ is not a member of the language $\mathcal{L}_2$, which contradicts the Pumping Lemma's condition (3).

IV.5.15

- (A) Five strings that are members are aaa , $aaab$, $aaabb$, $aaabbb$, and $aaabbbb$. This language is regular, and a regular expression that describes it is $aaab^*$.

- (B) Five strings in this language are bbb , $abbbb$, $aabbbbb$, a^3b^6 , and a^4b^7 . This language is not regular.

To get a contradiction assume that it is regular. The Pumping Lemma says that the language has a pumping length, p . The string $\sigma = a^p b^{p+3}$ is a member of the language and has length greater than or equal to the pumping length. So it decomposes, $\sigma = \alpha\beta\gamma$, in a way that is subject to the three conditions. The first condition is that $|\alpha\beta| \leq p$ and so the two substrings α and β consist only of a 's. The second condition is that $|\beta| > 0$ and so the substring β consists of at least one a .

Finally, consider the list $\alpha\gamma$, $\alpha\beta^2\gamma$, $\dots$. Condition (3) says that any element of that list is a member of the language. However, the list's first element has fewer a 's than σ but the same number of b 's, and therefore the number of a 's is not three less than the number of b 's. This is a contradiction.

- (C) Five members of this language are ϵ , $abab$, $babbab$, $aaaa$, and $aaabaaab$. It is not regular.

Suppose otherwise, that it is regular. By the Pumping Lemma the language has a pumping length, p . Consider $\sigma = a^p b \epsilon a^p b$. It is a member of the language whose length is greater than or equal to p and so it

has a decomposition $\sigma = \alpha\beta\gamma$ that satisfies the three conditions. The first condition, $|\alpha\beta| \leq p$, gives that the two substrings contain only a's. The second condition, $|\beta| > 0$, gives that the second substring consists of at least one a.

Now consider the list from the third condition, $\alpha\gamma, \alpha\beta^2\gamma, \dots$. Compare $\alpha\gamma$ to $\sigma = \alpha\beta\gamma$. Because the β is omitted in $\alpha\gamma$, before its first b it has fewer a's than σ has before its first b. But before the second b the two strings have the same number of a's. So $\alpha\gamma$ is not a member of the language, which contradicts the third condition.

(D) Five strings in this language are ε , aab, b, aaaabbbb, and abbbb. This language is regular. A regular expression that describes it is a^*b^* .

(E) Five strings in this language are b^{13} , ab^{14} , b^{14} , a^2b^{15} , and ab^{15} . This language is not regular.

Assume that it is regular, for contradiction. Then the language has a pumping length, p . Let $\sigma = a^p b^{p+13}$. By the Pumping Lemma it has a decomposition into $\sigma = \alpha\beta\gamma$ that satisfies the three conditions. As a consequence of the first condition the first two substrings consist only of a's. A consequence of the second condition is that β consists of at least one a.

Consider $\alpha\gamma, \alpha\beta^2\gamma, \dots$. Its second element, $\alpha\beta^2\gamma$, has at least one more a than does σ . Thus its number of a's is not more than 12 less than its number of b's, which means that it is not a member of the language. That contradicts the Pumping Lemma's third condition.

IV.5.16 Five strings from the language are cb, acbb, aacbbb, aaacbbbb, and aaaacbbbbb.

To show that this language is not regular, assume for contradiction that it is. Then the language has a pumping length, p . Consider $\sigma = a^p cb^{p+1}$. Because this string is a member of the language and has length at least p , the Pumping Lemma gives that it decomposes into $\sigma = \alpha\beta\gamma$, subject to the three conditions.

The first condition, $|\alpha\beta| \leq p$, implies that the two substrings α and β contain only a's. The second condition, $|\beta| > 0$, implies that β consists of at least one a.

Now, in the list $\alpha\gamma, \alpha\beta^2\gamma, \dots$ consider the first one, $\alpha\gamma$. By the prior paragraph, compared with $\sigma = \alpha\beta\gamma$ this string has at least one fewer a but the same number of b's. So the number of a's is not one less than the number of b's, and this string is not in the language, which is a contradiction to the third condition.

IV.5.17 Let $\mathcal{L} = \{\sigma \in \{a, b\}^* \mid \text{the number of a's is greater than the number of b's}\}$. Assume for contradiction that it is regular. Then the Pumping Lemma applies and this language has a pumping length, p . The string $\sigma = a^{p+1}b^p$ is an element of the language whose length is greater than or equal to p . So there is a decomposition $\sigma = \alpha\beta\gamma$ that satisfies the three conditions. Condition (1) gives that $|\alpha\beta| \leq p$ and so these two substrings contain only a's. Condition (2) gives that β is nonempty and thus it consists of at least one a.

The contradiction comes on considering condition (3)'s list $\alpha\gamma, \alpha\beta^2\gamma, \dots$. The first entry in that list, $\alpha\gamma$, when compared with $\sigma = \alpha\beta\gamma$ will have at least one a fewer, and the same number of b's. It is therefore not a member of the language $\mathcal{L}$, contradicting condition (3). This contradiction shows that $\mathcal{L}$ is not regular.

IV.5.18 The first is not regular while the second is regular.

(A) For contradiction, suppose that this language is regular. Then it has a pumping length, p . Consider the string $\sigma = a^{p^2}b^p$. It is a member of the language whose length is greater than or equal to p so it decomposes into $\sigma = \alpha\beta\gamma$ in a way that satisfies the three conditions. Condition (1) implies that both α and β consist only of a's. Condition (2) implies that β consists of at least one a.

Now consider condition (3)'s list, $\alpha\gamma, \alpha\beta^2\gamma, \dots$. Compared to $\sigma = \alpha\beta\gamma$, the entry $\alpha\gamma$ has at least one fewer a but the same number of b's. Therefore the number of a's is not the square of the number of b's, and so $\alpha\gamma$ is not in the language. That contradicts condition (3).

(B) A regular expression that describes this language is $aaaa^*bbbb^*$.

IV.5.19 The language is regular. It is a trick question: take $\alpha = \varepsilon$ to get that the set equals $\mathbb{B}^*$.

IV.5.20 Assume that it is regular and let the pumping length be p . Take $\sigma = 1^{p!}$. This string is a member of $\mathcal{L}$ and has length at least p , so the Pumping Lemma says it decomposes as $\sigma = \alpha\beta\gamma$ subject to the three conditions. The second condition implies that $|\beta| > 0$. Consider the list $\alpha\gamma, \alpha\beta^2\gamma, \alpha\beta^3\gamma, \dots$. After the first, the difference between the lengths of successive list elements is always $|\beta|$. But, as the hint notes, the difference between successive factorials is not fixed, instead it grows without bound. So there is a list element that is not in the language, which contradicts the third condition. So the language is not regular.

IV.5.21 The first is regular, and is described by the regular expression 0^*10^* . The second is not regular. To

prove that assume for contradiction that it is regular. Then it has a pumping length, p . Consider $\sigma = 0^p 1 0^p$, which is a member of the language of length at least p , so the Pumping Lemma says it decomposes into $\sigma = \alpha\beta\gamma$ in a way that satisfies the three conditions. The first condition implies that the α and β substrings consist only of 0's. The second condition says that β consists of at least one 0.

Consider the strings on the third condition's list $\alpha\gamma, \alpha\beta^2\gamma, \dots$. By the prior paragraph the first, $\alpha\gamma$, has fewer 0's before the 1 than does σ , but the same number of 0's after the 1. Hence it is not a member of the language, contradicting the Pumping Lemma's third condition.

IV.5.22 The language $\mathcal{L}_4 = \{a^i b^j c^k \mid i, j, k \in \mathbb{N} \text{ and } i + j = k \text{ and } k < 4.\}$ is finite, and any finite language is regular.

As to the language if all sums, $\mathcal{L} = \{a^i b^j c^k \mid i, j, k \in \mathbb{N} \text{ and } i + j = k.\}$, suppose that it is regular and fix a pumping length p . Consider $\sigma = a^p b^0 c^p$. Because $|\sigma| \geq p$ it has a decomposition $\sigma = \alpha\beta\gamma$ subject to the three conditions of the Pumping Lemma. By the first condition the length $|\alpha\beta|$ is less than or equal to p , and so consists entirely of a's. By the second condition the string β is nonempty and so consists of at least one a. The third condition says that $\alpha\gamma$ is also a member of the language, but that is a contradiction since this string has fewer a's than c's, and no b's, and thus is not a member of the language.

IV.5.23

(A) It is regular because it is finite, $\{00000, 00001, 00010, \dots, 11111\}$.

(B) The language $\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \text{the number of 0's minus the number of 1's is five}\}$ is not regular. For, suppose otherwise. Fix a pumping length p and consider $\sigma = 0^{p+5} 1^p$. It is a member of $\mathcal{L}$ with length that is at least p , so the Pumping Lemma applies. Decompose it into $\sigma = \alpha\beta\gamma$ subject to the three conditions. Condition (1), that $|\alpha\beta| \leq p$, implies that these two substrings contain only 0's. Condition (2) says that β is not empty so it consists of at least one 0.

Condition (3) says that every string on the list $\alpha\gamma, \alpha\beta^2\gamma, \dots$ is a member of $\mathcal{L}$ but that is not true. By the prior paragraph the string $\alpha\gamma$ has at least one 0 less than the string $\sigma = \alpha\beta\gamma$ but the same number of 1's, and is therefore not an element of $\mathcal{L}$.

IV.5.24 Call this language $\mathcal{L}$. Where $\hat{\mathcal{L}} = \{0^m 1^n \in \mathbb{B}^* \mid m = n\}$, we have $\hat{\mathcal{L}} = \mathbb{B}^* - \mathcal{L}$. By Theorem 4.5 regular languages are closed under set difference. Clearly $\mathbb{B}^*$ is regular—it is described by the regular expression $(0|1)^*$ —so if $\mathcal{L}$ were regular then it would imply that $\hat{\mathcal{L}}$ is regular also. But $\hat{\mathcal{L}}$ is not regular, since it is the language of balanced parentheses, Example 5.2 with 0's substituted for open parentheses and 1's for closed parentheses.

IV.5.25 Following the hint, the language $\mathcal{L}$ is the set of strings that start and end with the same symbol. For instance, starting with a, each ab marks a transition to a block of b's (possibly only one), and each ba marks a transition back to a block of a's. Thus the regular expression is $(a(a|b)^*a) \mid (b(a|b)^*b)$.

IV.5.26 The proof of the Pumping Lemma uses the Pigeonhole Principle to show that there is at least one loop and it then reasons with that loop. It is unaffected by there being a later loop.

More explicitly, the Pumping Lemma says that we can take the pumping length to be the number of states in any machine that recognizes the language. So set $p = 9$. If $\sigma = aabbabbaba$ then because $|\sigma| \geq p$ and $\sigma \in \mathcal{L}$, we know there is a decomposition satisfying the three conditions. One such is $\alpha = a$, $\beta = abbabb$, and $\gamma = aba$. Checking that (1) $|\alpha\beta| \leq p$, that (2) $\beta \neq \epsilon$, and (3) the strings $\alpha\gamma, \alpha\beta^2\gamma, \dots$ are all members of $\mathcal{L}$.

If $\sigma = a(abb)^{200}aba$ then again because $|\sigma| \geq p$ and $\sigma \in \mathcal{L}$, we know there is a decomposition satisfying the three conditions. One such is $\alpha = a$, $\beta = abb$, and $\gamma = (abb)^{199}aba$. Checking the three conditions is routine.

Finally, if $\sigma = aabbbabbaa$ then we can take $\alpha = aab$, $\beta = bbabba$, and $\gamma = a$. Or, we could instead take $\alpha = aab$, $\beta = bba$, and $\gamma = bbaa$. Again, checking the three conditions is routine.

IV.5.27 Let $\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma = 1^n \text{ where } n \text{ is prime}\}$. Assume for contradiction that it is regular and fix a pumping length p . Consider condition (3)'s list.

$$\alpha\gamma, \alpha\beta^2\gamma, \alpha\beta^3\gamma, \dots, \alpha\beta^n\gamma, \dots$$

Except for the first one, all of the gaps between list items have the same width, $|\beta|$. So the length of the general item $\alpha\beta^n\gamma$ is $|\alpha\beta^2\gamma| + (n-2) \cdot |\beta|$, for $n \geq 2$. Take n to be $|\alpha\beta^2\gamma| + 2$ and that length is not prime. So

not every string in condition (3)'s list is a member of $\mathcal{L}$, which is the desired contradiction.

IV.5.28

- (A) The number of c's is the product of the number of a's and the number of b's. Five such strings are abc, abbcc, abbbccc, aabcc, and aabbccccc.
- (B) For contradiction assume that this language, $\mathcal{L}$, is regular. The Pumping Lemma says that $\mathcal{L}$ has a pumping length, p . Consider $\sigma = a^p b^p c^{(p^2)}$. That string has more than p characters so it decomposes as $\sigma = \alpha\beta\gamma$, subject to the three conditions. Condition (1) is that $|\alpha\beta| \leq p$ and so both substrings α and β are composed entirely of a's. Condition (2) is that β is not the empty string and so β consists of at least one a.

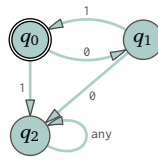
Condition (3) is that the strings in the list $\alpha\gamma, \alpha\beta^2\gamma, \alpha\beta^3\gamma, \dots$ are also members of the language. We will get the contradiction from the first one, $\alpha\gamma$. Compared to $\sigma = \alpha\beta\gamma$, in the first one the β is gone. So the number of a's in $\alpha\gamma$ is less than the number in $\sigma = \alpha\beta\gamma$. But the number of b's and the number of c's is the same. Thus in $\alpha\gamma$ the number of a's times the number of b's does not equal the number of c's.

IV.5.29 We need that $|\alpha\beta| \leq 4$, that $|\beta| > 0$, and that the strings $\alpha\gamma, \alpha\beta^2\gamma, \alpha\beta^3\gamma, \dots$ are members of $\mathcal{L}$.

- (A) $\alpha = \varepsilon$, $\beta = a$, and $\gamma = bbb$
 (B) $\alpha = \varepsilon$, $\beta = b$, and $\gamma = b^{14}$

IV.5.30

- (A) This works.



- (B) The prior item shows that there is a Finite State machine recognizing this language that has three states. So, any non-empty word of the language must cause it to repeat a state, and therefore can be pumped. The minimum length of a non-empty word of the language is 2 so the minimum pumping length is at most 2. However the language contains no word of length 1 so if $\sigma \in \mathcal{L}$ and $|\sigma| \geq 1$ then automatically $|\sigma| \geq 2$, and so σ can be pumped. Therefore the minimum pumping length for the language is 1.

IV.5.31

- (A) Because it accepts the input ε the initial state q_0 must be an accepting state. If q_0 were the only accepting state then this machine accepts a by virtue of a loop from q_0 to itself, which would make the machine accept aa . That is not in the language $\mathcal{L}_1$, so there must be at least one other accepting state.
- (B) Because it accepts the input ε the initial state q_0 must be an accepting state. If q_0 were the only accepting state then this machine accepts a by virtue of a loop from q_0 to itself, which would make the machine accept aaa . That string is not a member of the language $\mathcal{L}_2$ so there must be at least one accepting state other than q_0 .

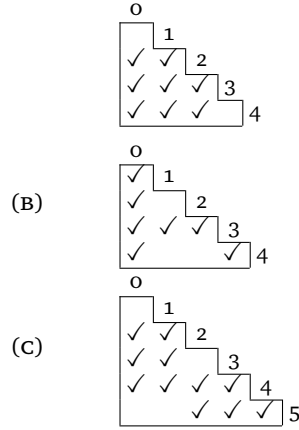
If the machine accepts aa by being in state q_0 then that involves a loop, so the string $aaaa$ would also be accepted by this machine. But that string is not in the language $\mathcal{L}_2$, so the extended transition function on this input, $\hat{E}(aa)$, is not equal to the state q_0 . Similarly, if $\hat{E}(aa)$ were equal to $\hat{E}(a)$ then there would be a loop from this state to itself on input a , and so the machine would also accept aaa . But the machine does not accept aaa because that string is not in the language $\mathcal{L}_2$, so there must be a third accepting state.

- (C) This is the natural extension of the prior two items. Consider $\mathcal{L}_n = \{\varepsilon, a, \dots, a^n\}$. If $\hat{E}(a^n)$ were equal to $\hat{E}(a^i)$ for any $i < n$ then there would be a loop and so the machine would accept additional strings.

IV.6.9 Pairs that are not 1-distinguishable are: q_0, q_1 , and q_2, q_2 , and q_1, q_2 , so one class is $\mathcal{E}_{i,0} = \{q_0, q_1, q_2\}$. Besides that, another pair of states that are not 1-distinguishable is q_3, q_5 so another class is $\mathcal{E}_{i,1} = \{q_3, q_5\}$. The last class is $\mathcal{E}_{i,2} = \{q_4\}$.

IV.6.10

- (A) The given classes say that we can i -distinguish q_0 from q_2, q_3 , and q_4 . So looking down the column headed by 0 we see checkmarks in rows labeled 2, 3, and 4. The other columns work the same way.

**IV.6.11**

(A) These are the checkmarks.

	a	b	a	b
q_0, q_1	q_1, q_1	q_2, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_0, q_2	q_1, q_2	q_2, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_0, q_5	q_1, q_5	q_2, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_2	q_1, q_2	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_1, q_5	q_1, q_5	q_3, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$
q_2, q_5	q_2, q_5	q_4, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$
q_3, q_4	q_3, q_4	q_5, q_5	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$

For example, the first row is checkmarked because on the right side the second entry, the b entry, has two classes that differ, $\mathcal{E}_{0,0} \neq \mathcal{E}_{0,1}$.

(B) The checkmarks show that we can 1-distinguish q_0 from q_1 and q_2 but not from q_5 . And, we can tell q_1 and q_2 from q_5 but not from each other. The matching $\sim_1$ classes are $\mathcal{E}_{1,0} = \{q_0, q_5\}$, $\mathcal{E}_{1,1} = \{q_1, q_2\}$, and $\mathcal{E}_{1,2} = \{q_3, q_4\}$

IV.6.12 First, all the states are reachable, by inspection.

Start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.



The checkmark denote a pair of states that are 0-distinguishable and the blanks denote pairs of states that are 0-indistinguishable. Here are the two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_2\} \quad \mathcal{E}_{0,1} = \{q_1\}$$

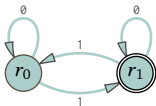
Next we see if any of these classes split.

	0	1	0	1
q_0, q_2	q_0, q_2	q_1, q_1	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$

The answer is that they don't split. The process stops and here are the two $\sim$ -equivalence classes.

$$\mathcal{E}_{1,0} = \{q_0, q_2\} = \mathcal{E}_0 = r_0 \quad \mathcal{E}_{1,1} = \{q_1\} = \mathcal{E}_1 = r_1$$

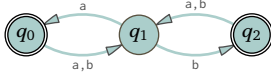
Incorporating the arrow information from the original machine gives this minimized one.

**IV.6.13**

(A) These are the sets.

Step n	R_n
0	$\{q_0\}$
1	$\{q_0, q_1\}$
2	$\{q_0, q_1, q_2\}$
3	$\{q_0, q_1, q_2\}$

Thus the set of reachable states is $R = \{q_0, q_1, q_2\}$. The set of unreachable states is $Q - R = \{q_3\}$. Here is the machine with only the reachable states.

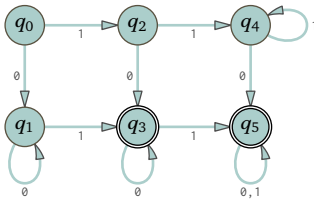


(B) These are the sets.

Step n	R_n
0	$\{q_0\}$
1	$\{q_0, q_1, q_3\}$
2	$\{q_0, q_1, q_3, q_4\}$
3	$\{q_0, q_1, q_3, q_4\}$

The set of reachable states is $R = R_2 = \{q_0, q_1, q_3, q_4\}$. The set of unreachable states is $Q - R = \{q_2, q_5\}$.

IV.6.14 For convenience, here is that machine.



Every state is reachable, by inspection.

Start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

0					
	1				
✓	✓	2			
		✓	3		
✓	✓		✓	4	
				✓	5

There is a checkmark in box i, j if the states q_i and q_j are 0-distinguishable. The blanks denote pairs that are 0-indistinguishable. There are two $\sim_0$ equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_1, q_2, q_4\} \quad \mathcal{E}_{0,1} = \{q_3, q_5\}$$

The next step is to see if any of these classes split.

	0	1	0	1
q_0, q_1	q_1, q_1	q_2, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_0, q_2	q_1, q_3	q_2, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_4	q_1, q_5	q_2, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_2	q_1, q_3	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$
q_1, q_4	q_1, q_5	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$
q_2, q_4	q_3, q_5	q_4, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_3, q_5	q_3, q_5	q_5, q_5	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$

0					
✓	1				
✓	✓	2			
✓	✓	✓	3		
✓	✓		✓	4	
✓	✓			✓	5

The only pairs of states that are 1-distinguishable are q_2, q_4 and q_3, q_5 . These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0, q_2\} \quad \mathcal{E}_{1,1} = \{q_1\} \quad \mathcal{E}_{1,2} = \{q_2, q_4\} \quad \mathcal{E}_{1,3} = \{q_3, q_5\}$$

Now try it again to see if any of these classes split.

	0	1	0	1
q_2, q_4	q_3, q_5	q_4, q_4	$\mathcal{E}_{1,3}, \mathcal{E}_{1,3}$	$\mathcal{E}_{1,2}, \mathcal{E}_{1,2}$
q_3, q_5	q_3, q_5	q_5, q_5	$\mathcal{E}_{1,3}, \mathcal{E}_{1,3}$	$\mathcal{E}_{1,3}, \mathcal{E}_{1,3}$

0	1	2	3	4	5
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓

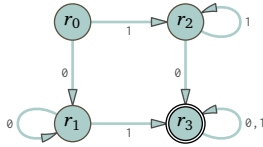
The only pairs of states that are 1-distinguishable are q_2, q_4 and q_3, q_5 . These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0, q_2\} \quad \mathcal{E}_{1,1} = \{q_1\} \quad \mathcal{E}_{1,2} = \{q_2, q_4\} \quad \mathcal{E}_{1,3} = \{q_3, q_5\}$$

There is no splitting; these are the $\sim$ classes.

$$r_0 = \mathcal{E}_0 = \mathcal{E}_{2,0} = \{q_0, q_2\} \quad r_1 = \mathcal{E}_1 = \mathcal{E}_{2,1} = \{q_1\} \quad r_2 = \mathcal{E}_2 = \mathcal{E}_{2,2} = \{q_2, q_4\} \quad r_3 = \mathcal{E} + 3 = \mathcal{E}_{2,3} = \{q_3, q_5\}$$

Bringing in the transition information from the original machine gives this minimized version.



V IV.6.15 Not a lot. You get back the original machine in that you get $r_0 = \mathcal{E}_{n,0} = \{q_0\}$, $r_1 = \mathcal{E}_{n,1} = \{q_1\}$, etc.

IV.6.16 By inspection the states q_6 and q_7 are unreachable, so drop them.

With that, start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

0	1	2	3	4	5
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓

The checkmarks are for pairs of states that are 0-distinguishable, and the blanks denote pairs that are 0-indistinguishable. There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_2\} \quad \mathcal{E}_{0,1} = \{q_1, q_3, q_4, q_5\}$$

Next we see if any of these classes split.

	a	b	a	b
q_1, q_3	q_4, q_0	q_2, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_1, q_4	q_4, q_4	q_2, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_1, q_5	q_4, q_2	q_2, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_3, q_4	q_0, q_4	q_4, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_3, q_5	q_0, q_2	q_4, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_4, q_5	q_4, q_2	q_4, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_0, q_2	q_1, q_1	q_3, q_5	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$

0	1	2	3	4	5
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓

So, q_0 and q_2 are not 1-distinguishable. On the other hand, as shown in the triangular table, q_1 is 1-distinguishable from q_3, q_4 , and q_5 . And, q_3 is 1-distinguishable from q_4 but not from q_5 . Finally, q_4 is 1-distinguishable from q_5 . These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0, q_2\} \quad \mathcal{E}_{1,1} = \{q_1\} \quad \mathcal{E}_{1,2} = \{q_3, q_5\} \quad \mathcal{E}_{1,3} = \{q_4\}$$

Determine which states are 2-distinguishable by computing whether any of those classes split.

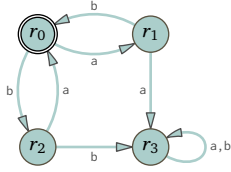
	a	b	a	b
q_3, q_5	q_0, q_2	q_4, q_4	$\mathcal{E}_{1,0}, \mathcal{E}_{1,0}$	$\mathcal{E}_{1,3}, \mathcal{E}_{1,3}$
q_0, q_2	q_1, q_1	q_3, q_5	$\mathcal{E}_{1,1}, \mathcal{E}_{1,1}$	$\mathcal{E}_{1,2}, \mathcal{E}_{1,2}$

0	1	2	3	4	5
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓

There is no splitting so the minimal machine has four states.

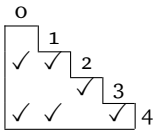
$$r_0 = \mathcal{E}_{2,0} = \{q_0, q_2\} \quad r_1 = \mathcal{E}_{2,1} = \{q_1\} \quad r_2 = \mathcal{E}_{2,2} = \{q_3, q_5\} \quad r_3 = \mathcal{E}_{2,3} = \{q_4\}$$

The transitions from the input machine give this minimized one.



V IV.6.17 The answer is “yes.” The initial state of the minimal machine, r_0 , is the one that contains q_0 . In the minimized machine a state is accepting if and only if it contains any final states in the original machine. So if q_0 is accepting then r_0 is also accepting.

IV.6.18 By inspection all of the states are reachable. So start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

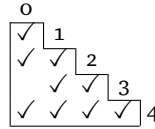


There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_1, q_3\} \quad \mathcal{E}_{0,1} = \{q_2, q_4\}$$

Next we see if any of these classes split.

	0	1	0	1
q_0, q_1	q_1, q_2	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_0, q_3	q_1, q_2	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_1, q_3	q_2, q_2	q_4, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_2, q_4	q_1, q_4	q_4, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$

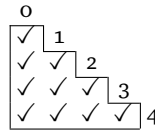


So, q_1 and q_3 are not 1-distinguishable but all the others are 1-distinguishable. These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0\} \quad \mathcal{E}_{1,1} = \{q_1, q_3\} \quad \mathcal{E}_{1,2} = \{q_2\} \quad \mathcal{E}_{1,3} = \{q_4\}$$

Step 2 is to decide if any of the $\sim_1$ classes split.

	0	1	0	1
q_1, q_3	q_1, q_2	q_3, q_4	$\mathcal{E}_{1,1}, \mathcal{E}_{1,2}$	$\mathcal{E}_{1,1}, \mathcal{E}_{1,3}$

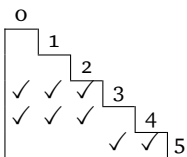


So, q_1 and q_3 are 2-distinguishable. These are the $\sim_2$ classes.

$$\mathcal{E}_{1,0} = \{q_0\} \quad \mathcal{E}_{1,1} = \{q_1\} \quad \mathcal{E}_{1,2} = \{q_2\} \quad \mathcal{E}_{1,3} = \{q_3\} \quad \mathcal{E}_{1,4} = \{q_4\}$$

The starting machine is minimal.

IV.6.19 By eye, all of the states are reachable. So start with the triangular table, checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.



There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_1, q_2, q_5\} \quad \mathcal{E}_{0,1} = \{q_3, q_4\}$$

Next we see if we can 1-distinguish any of these states that cannot be 0-distinguished.

	a	b	a	b	
q_0, q_1	q_1, q_1	q_2, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	
q_0, q_2	q_1, q_2	q_2, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	
q_0, q_5	q_1, q_5	q_2, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	
q_1, q_2	q_1, q_2	q_3, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	
q_1, q_5	q_1, q_5	q_3, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	
q_2, q_5	q_2, q_5	q_4, q_5	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	
q_3, q_4	q_3, q_4	q_5, q_5	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	

These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0, q_5\} \quad \mathcal{E}_{1,1} = \{q_1, q_2\} \quad \mathcal{E}_{1,2} = \{q_3, q_4\}$$

Iterate. See if any of these classes split.

	a	b	a	b	
q_0, q_5	q_1, q_5	q_2, q_5	$\mathcal{E}_{1,1}, \mathcal{E}_{1,0}$	$\mathcal{E}_{1,1}, \mathcal{E}_{1,0}$	
q_1, q_2	q_1, q_2	q_3, q_4	$\mathcal{E}_{1,1}, \mathcal{E}_{1,1}$	$\mathcal{E}_{1,2}, \mathcal{E}_{1,2}$	
q_3, q_4	q_3, q_4	q_5, q_5	$\mathcal{E}_{1,2}, \mathcal{E}_{1,2}$	$\mathcal{E}_{1,0}, \mathcal{E}_{1,0}$	

The class $\mathcal{E}_{1,0}$ splits in two. The pairs that are not 2-distinguishable are: q_1, q_2 , and q_3, q_4 . These are the $\sim_2$ classes.

$$\mathcal{E}_{2,0} = \{q_0\} \quad \mathcal{E}_{2,1} = \{q_1, q_2\} \quad \mathcal{E}_{2,2} = \{q_3, q_4\} \quad \mathcal{E}_{2,3} = \{q_5\}$$

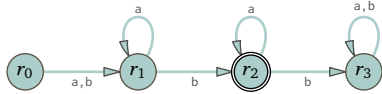
Again.

	a	b	a	b	
q_1, q_2	q_1, q_2	q_3, q_4	$\mathcal{E}_{2,1}, \mathcal{E}_{2,1}$	$\mathcal{E}_{2,2}, \mathcal{E}_{2,2}$	
q_3, q_4	q_3, q_4	q_5, q_5	$\mathcal{E}_{2,2}, \mathcal{E}_{2,2}$	$\mathcal{E}_{2,3}, \mathcal{E}_{2,3}$	

There is no splitting. These are the $\sim$ classes.

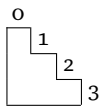
$$r_0 = \mathcal{E}_{3,0} = \{q_0\} \quad r_1 = \mathcal{E}_{3,1} = \{q_1, q_2\} \quad r_2 = \mathcal{E}_{3,2} = \{q_3, q_4\} \quad r_3 = \mathcal{E}_{3,3} = \{q_5\}$$

Here is the minimized machine.



IV.6.20 We will go through the algorithm but before we do, observe that the answer must be that a machine with a minimal number of states and that accepts no string has one state, namely the initial state, and all arrows are loops.

By inspection all of the states are reachable. So start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

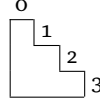


There is one $\sim_0$ -equivalence class.

$$\mathcal{E}_{0,0} = \{q_0, q_1, q_2, q_3\}$$

Next we see if any of these classes split.

	0	1	0	1
q_0, q_1	q_1, q_1	q_2, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_2	q_1, q_0	q_2, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_3	q_1, q_0	q_2, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_2	q_1, q_0	q_3, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_3	q_1, q_0	q_3, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_2, q_3	q_0, q_0	q_2, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$



Of course, if there is only one class then it cannot split, so there is only one $\sim_1$ class.

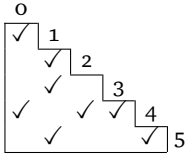
$$\mathcal{E}_{0,0} = \{q_0, q_1, q_2, q_3\}$$

This is the resulting minimized machine, as predicted.



If a machine has that all states are accepting then a similar thing happens. It recognizes the language Σ^* and the minimal such machine has one state, which is both initial and accepting.

IV.6.21 By inspection all of the states are reachable. So start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

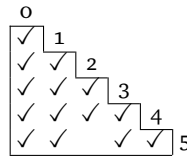


There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_2, q_3, q_5\} \quad \mathcal{E}_{0,1} = \{q_1, q_4\}$$

Next we see if any of these classes split.

	a	b	a	b
q_0, q_2	q_1, q_1	q_0, q_4	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_0, q_3	q_1, q_5	q_0, q_2	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_5	q_1, q_4	q_0, q_1	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_2, q_3	q_1, q_5	q_4, q_2	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,0}$
q_2, q_5	q_1, q_4	q_4, q_1	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$
q_3, q_5	q_5, q_4	q_2, q_1	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$
q_1, q_4	q_3, q_3	q_0, q_4	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$

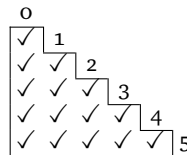


So, q_2 and q_5 are not 1-distinguishable but all the other pairs are 1-distinguishable. These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0\} \quad \mathcal{E}_{1,1} = \{q_1\} \quad \mathcal{E}_{1,2} = \{q_2, q_5\} \quad \mathcal{E}_{1,3} = \{q_3\} \quad \mathcal{E}_{1,4} = \{q_4\}$$

Finally we see if the one remaining multi-state classes splits.

	a	b	a	b
q_2, q_5	q_1, q_4	q_4, q_1	$\mathcal{E}_{1,1}, \mathcal{E}_{1,4}$	$\mathcal{E}_{1,4}, \mathcal{E}_{1,1}$



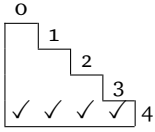
Thus all of the states are not 2-distinguishable. These are the $\sim_2$ classes.

$$\mathcal{E}_{2,0} = \{q_0\} \quad \mathcal{E}_{2,1} = \{q_1\} \quad \mathcal{E}_{2,2} = \{q_2\} \quad \mathcal{E}_{2,3} = \{q_3\} \quad \mathcal{E}_{2,4} = \{q_4\} \quad \mathcal{E}_{2,5} = \{q_5\}$$

The original machine is minimal.

IV.6.22 You will get a machine that recognizes the same language. The reachable states of this machine will form a minimal set. But there will still be unreachable states (although perhaps fewer of them.)

IV.6.23 By inspection all of the states are reachable. So start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.

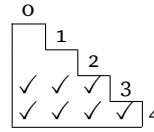


There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{0, 1, 2, 3\} \quad \mathcal{E}_{0,1} = \{4\}$$

Next we see if any of these classes split.

	a	b	a	b
q_0, q_1	q_1, q_2	q_0, q_1	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_2	q_1, q_3	q_0, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_0, q_3	q_1, q_4	q_0, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_2	q_2, q_3	q_1, q_2	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_1, q_3	q_2, q_4	q_1, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$
q_2, q_3	q_3, q_4	q_2, q_3	$\mathcal{E}_{0,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$

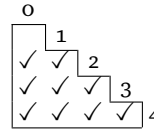


So, q_3 is 1-distinguishable from all the other states, but no other pairs are 1-distinguishable. These are the $\sim_1$ classes.

$$\mathcal{E}_{1,0} = \{q_0, q_1, q_2\} \quad \mathcal{E}_{1,1} = \{q_3\} \quad \mathcal{E}_{1,2} = \{q_4\}$$

Next time through.

	a	b	a	b
q_0, q_1	q_1, q_2	q_0, q_1	$\mathcal{E}_{1,0}, \mathcal{E}_{0,0}$	$\mathcal{E}_{1,0}, \mathcal{E}_{0,0}$
q_0, q_2	q_1, q_3	q_0, q_2	$\mathcal{E}_{1,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{1,0}, \mathcal{E}_{0,0}$
q_1, q_2	q_2, q_3	q_1, q_2	$\mathcal{E}_{1,0}, \mathcal{E}_{0,1}$	$\mathcal{E}_{1,0}, \mathcal{E}_{0,0}$

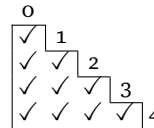


So, q_2 is 2-distinguishable from q_0 and q_1 , giving these $\sim_2$ classes.

$$\mathcal{E}_{2,0} = \{q_0, q_1\} \quad \mathcal{E}_{2,1} = \{q_2\} \quad \mathcal{E}_{2,2} = \{q_3\} \quad \mathcal{E}_{2,3} = \{q_4\}$$

Once more through.

	a	b	a	b
q_0, q_1	q_1, q_2	q_0, q_1	$\mathcal{E}_{2,0}, \mathcal{E}_{2,1}$	$\mathcal{E}_{2,0}, \mathcal{E}_{2,0}$



So, q_1 is 3-distinguishable from q_0 and these are the giving these $\sim_3$ classes.

$$\mathcal{E}_{3,0} = \{q_0\} \quad \mathcal{E}_{3,1} = \{q_1\} \quad \mathcal{E}_{3,2} = \{q_2\} \quad \mathcal{E}_{3,3} = \{q_3\} \quad \mathcal{E}_{3,4} = \{q_4\}$$

No additional splitting happens on the next iteration; indeed, there is nothing left to split. The process stops. The minimized machine has five states; it is the machine we started with.

IV.6.24

- (A) We must show that it is reflexive, symmetric, and transitive. Fix some n . Obviously any state is n -indistinguishable from itself, for any n . Symmetry is also clear. So suppose $q_0 \sim_n q_1$ and $q_1 \sim_n q_2$. Let σ be a string of length less than or equal to n . If σ takes the machine from q_0 to an accepting state then because of $q_0 \sim_n q_1$ it also takes the machine from q_1 to an accepting state. Because of $q_1 \sim_n q_2$ it also takes the machine from q_2 to an accepting state. The same holds for non-accepting states.
- (B) It is an equivalence because it is $\sim_k$ for some k .

IV.6.25

- (A) By inspection all of the states are reachable. So start by checkmarking the i, j entries where one of q_i and q_j is accepting while the other is not.



There are two $\sim_0$ -equivalence classes.

$$\mathcal{E}_{0,0} = \{q_0, q_2\} \quad \mathcal{E}_{0,1} = \{q_1\}$$

Next we see if any of these classes split.

	a	b	a	b	
q_0, q_2	q_1, q_1	q_2, q_2	$\mathcal{E}_{0,1}, \mathcal{E}_{0,1}$	$\mathcal{E}_{0,0}, \mathcal{E}_{0,0}$	

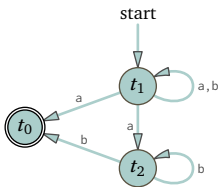
So the class $\mathcal{E}_{0,0}$ does not split. These are the $\sim_1$ classes.

$$r_0 = \mathcal{E}_{0,0} = \{q_0, q_2\} \quad r_1 = \mathcal{E}_{0,1} = \{q_1\}$$

Here is the minimized machine.



- (B) Here is the machine with the arrows reversed, the nodes renamed, and the start node made final, and the final one made a start.

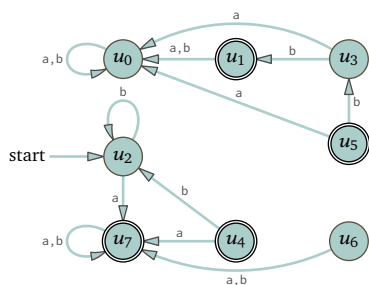


It is nondeterministic; for instance, two edges labeled a leave node t_1 .

- (c) This is the table to convert that nondeterministic Finite State machine to a deterministic one.

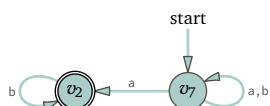
Node name	Set of nodes	a	b
u_0	$\emptyset$	u_0	u_0
+ u_1	$\{t_0\}$	u_0	u_0
u_2	$\{t_1\}$	u_7	u_2
u_3	$\{t_2\}$	u_0	u_1
+ u_4	$\{t_0, t_1\}$	u_7	u_2
+ u_5	$\{t_0, t_2\}$	u_0	u_3
u_6	$\{t_1, t_2\}$	u_7	u_7
+ u_7	$\{t_0, t_1, t_2\}$	u_7	u_7

Here is the arrow diagram.



The only reachable states are u_2 and u_7 .

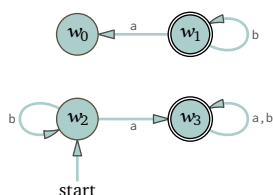
(D) Here is the picture.



(E) Convert that nondeterministic Finite State machine to a deterministic one.

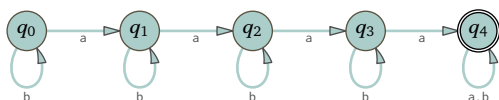
Node name	Set of nodes	a	b
w_0	$\emptyset$	w_0	w_0
+ w_1	$\{v_2\}$	w_0	w_1
w_2	$\{v_7\}$	w_3	w_2
+ w_3	$\{v_2, v_7\}$	w_3	w_3

Here is the arrow diagram.



Once we omit unreachable states, this machine is the same (except for state renaming) as the one in the first item.

IV.6.26 The machine of ?? 6.23



has five states and is minimal for the language described by $b^*ab^*ab^*ab^*a(a|b)^*$. Using induction to generalize this observation is straightforward.

IV.7.8 The machine with these two instructions just keeps popping $\perp$ off the stack and then pushing it back on again.

Inst	Input	Output
0	$q_0, \varepsilon, \perp$	$q_1, \perp$
1	$q_1, \varepsilon, \perp$	$q_0, \perp$

This machine has the stack grow without bound.

Inst	Input	Output
0	$q_0, \varepsilon, \perp$	$q_1, 'g\emptyset \perp'$
1	$q_1, \varepsilon, g\emptyset$	$q_0, 'g\emptyset g\emptyset'$
2	$q_0, \varepsilon, g\emptyset$	$q_1, 'g\emptyset g\emptyset'$

IV.7.9 Each configuration is a triple of present state, remaining tape tokens, and stack.

(A) We have this.

$$\langle q_0, [][]B, \perp \rangle \vdash \langle q_0, [][]B, g\emptyset \perp \rangle \vdash \langle q_0, []B, \perp \rangle \vdash \langle q_0,]B, g\emptyset \perp \rangle \vdash \langle q_0, B, \perp \rangle \vdash \langle q_1, ', \perp \rangle$$

The last configuration has an empty tape and the state q_1 , so the machine accepts the initial string.

(B) This computation is quite different than the prior one.

$$\langle q_0, [][]B, \perp \rangle \vdash \langle q_0, []B, ' \rangle$$

At this point the machine has an empty stack. It can't start the next step because each step starts by popping the top character off the stack, but there is no such character. The computation stops, with the input rejected.

IV.7.10

(A) When this machine consumes a it pushes two $g\emptyset$'s onto the stack. When it sees the middle marker c it switches to popping. The only accepting state is q_2 .

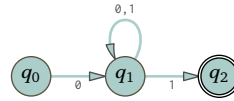
Inst	Input	Output
0	$q_0, a, \perp$	$q_0, 'g\emptyset g\emptyset \perp'$
1	$q_0, c, \perp$	$q_1, '\perp'$
2	$q_0, a, g\emptyset$	$q_0, 'g\emptyset g\emptyset g\emptyset'$
3	$q_0, c, g\emptyset$	$q_1, 'g\emptyset'$
4	$q_1, b, g\emptyset$	$q_1, ''$
5	$q_1, B, \perp$	$q_2, \perp$

(B) This adaptation of the prior item's machine just throws away the first a. The only accepting state is q_3 .

Inst	Input	Output
0	$q_0, a, \perp$	$q_1, '\perp'$
1	$q_1, a, \perp$	$q_1, 'g\emptyset g\emptyset \perp'$
2	$q_1, c, \perp$	$q_2, '\perp'$
3	$q_1, a, g\emptyset$	$q_1, 'g\emptyset g\emptyset g\emptyset'$
4	$q_1, c, g\emptyset$	$q_2, 'g\emptyset'$
5	$q_2, b, g\emptyset$	$q_2, ''$
6	$q_2, B, \perp$	$q_3, \perp$

IV.7.11 This is a regular language so we don't need the stack. This Pushdown machine implements the shown nondeterministic Finite State machine; its accepting state is q_2 .

Inst	Input	Output
0	$q_0, \emptyset, \perp$	$q_1, '\perp'$
1	$q_1, \emptyset, \perp$	$q_1, '\perp'$
2	$q_1, 1, \perp$	$q_1, '\perp'$
3	$q_1, 1, \perp$	$q_2, '\perp'$



IV.7.12 First this machine reads a's and pushes $g\emptyset$'s. When it guesses that the time is right, it transitions to reading a's and popping $g\emptyset$'s. The only accepting state is q_2 .

Inst	Input	Output
0	$q_0, B, \perp$	$q_2, '\perp'$
1	$q_0, a, \perp$	$q_0, 'g\emptyset \perp'$
2	$q_0, a, g\emptyset$	$q_0, 'g\emptyset g\emptyset'$
3	$q_0, \varepsilon, g\emptyset$	$q_1, 'g\emptyset \perp'$
2	$q_1, a, g\emptyset$	$q_1, ''$
3	$q_1, B, \perp$	$q_2, '\perp'$

IV.7.13 This machine pushes a $g\emptyset$ onto the stack for every initial a on the tape. Then when it sees a b it switches to popping two $g\emptyset$'s off for every input. Its accepting state is q_4 .

Inst	Input	Output
2	$q_0, B, \perp$	$q_4, \text{'\perp'}$
0	$q_0, a, \perp$	$q_1, \text{'g0 \perp'}$
1	$q_1, a, g0$	$q_1, \text{'g0 g0'}$
3	$q_1, b, g0$	$q_3, \text{'\text{'}}$
4	$q_2, b, g0$	$q_3, \text{'\text{'}}$
5	$q_3, b, g0$	$q_2, \text{'\text{'}}$
6	$q_2, B, \perp$	$q_4, \text{'\perp'}$

IV.7.14

(A)

Input:	0	1	1	0
				$q_2, \perp$
				$\perp$
$q_2, \perp$		$q_1, g1g0\perp$	$\vdash$	$q_1, g0\perp$
$\perp$				$\vdash$
$q_1, \perp$	$q_1, g0\perp$	$\perp$		$q_1, g1g1g0\perp$
$\perp$				$\perp$
$q_0, \perp$	$\vdash$	$q_0, g0\perp$	$\vdash$	$q_0, g1g1g0\perp$
		$\vdash$	$q_0, g1g0\perp$	$\vdash$
				$q_0, g0g1g1g0\perp$
				$\perp$
				$q_0, g0g1g1g0\perp$
Step:	0	1	2	3
				4

(B)

Input:	0	1	0
$q_2, \perp$			
$\perp$			
$q_1, \perp$	$q_1, g0\perp$	$q_1, g1g0\perp$	$q_1, g0g1g0\perp$
$\perp$	$\perp$	$\perp$	$\perp$
$q_0, \perp$	$\vdash$ $q_0, g0\perp$	$\vdash$ $q_0, g1g0\perp$	$\vdash$ $q_0, g0g1g0\perp$
Step:	0	1	2

IV.7.15 $S \rightarrow oSo \mid 1S1 \mid \varepsilon$ IV.7.16 The language is $\mathcal{L}_{\text{ELP}} = \{\sigma\sigma^R \mid \sigma \in \mathbb{B}^*\}$.

For contradiction, assume that $\mathcal{L}_{\text{ELP}}$ is regular. Then the Pumping Lemma says that it has a pumping length, which we can denote as p . Consider the string $\sigma = 0^p 110^p$.

It is an element of $\mathcal{L}$ of length greater than or equal to p . Therefore it decomposes into three substrings $\sigma = \alpha\beta\gamma$ that satisfy the conditions. Condition (1) is that the length of the prefix $\alpha\beta$ is less than or equal to p . Because the first p -many characters of σ are all 0's, this implies that both α and β contain only 0's. Condition (2) is that β is not empty so it consists of at least one 0.

Consider $\alpha\gamma$. Compared with $\sigma = \alpha\beta\gamma$, this string is missing the β , which means that before $\alpha\gamma$'s substring 11 it has fewer 0's than does σ . But it has the same number of 0's after the 11 substring, and so it is not a member of $\mathcal{L}_{\text{ELP}}$. This contradicts condition (3) of the lemma.

IV.7.17

(A) A Pushdown machine is a finite set of five tuples. Each of the five components comes from a countable set.

So in total the number of Pushdown machines is countable.

(B) Every machine accepts a language. So the number of languages is at most as large as the number of machines.

(C) There are uncountably many languages.

IV.7.18 We can simulate a Pushdown machine with a Turing machine.

IV.A.20

(A) $.^*abc.^*$

- (B) `.*abc+.*`
- (C) `.*abc2.*`
- (D) `.*abc2,5.*`
- (E) `.*abc2,.*`
- (F) `.*a(b|c).*` or `.*a[bc].*`

IV.A.21

- (A) `.*abe.*` Note that in practice you often can just write `abe`; see your language's documentation.
- (B) `[a-zA-Z0-9]*`
- (C) `[^]`

IV.A.22

- (A) `(a|e|i|o|u).*bc.*`
- (B) `(a|e|i|o|u)(bc|.*bc).*`
- (C) `.*a.*e.*i.*o.*u.*`
- (D) `(.*a.*e.*i.*o.*u.*)|(.*a.*e.*i.*u.*o.*)|...|(.*u.*o.*i.*e.*a.*)` There are $5! = 120$ clauses.
You might use a program to write this regex.

IV.A.23 `.*([.*{|{.*[).*`**IV.A.24** `\d{10}`**IV.A.25**

- (A) `(\d{3}|\d{3}\d{3}) \d{3}-\d{4}`
- (B) We must factor out the space as well as the area code: `(\d{3} |\d{3}\d{3})?\d{3}-\d{4}`.

IV.A.27 `\d{2}:\d{2}:\d{2}(\d+)|\d{2}:(\d{2})?`**IV.A.28** One is `[-\+]?(\d)+\.\?|\d)*\.\d+`**IV.A.29**

- (A) `4(\d15|\d12)`
- (B) `(5[1-5]\d2|222[1-9]|22[3-9]\d|2[3-6]\d2|27[01]\d|2720)\d12`
- (C) `3[47][0-9]13`

IV.A.30

- (A) We can do the six as alternatives.

`[A-Z]\d2 \d[A-Z]2|[A-Z]\d \d[A-Z]2|[A-Z]\d[A-Z] \d[A-Z]2|[A-Z]2\d2 \d[A-Z]2|[A-Z]2\d \d[A-Z]2|[A-Z]2\d[A-Z] \d[A-Z]2`

- (B) We can factor out the ending of `[A-Z]2`.

`[A-Z]\d2|[A-Z]\d|[A-Z]\d[A-Z]|[A-Z]2\d2|[A-Z]2\d|[A-Z]2\d[A-Z]) \d[A-Z]2`

V IV.A.31 A suitable regex is `textasciicircum g.n.{3}i.$` and a match is 'gynaecia', meaning the aggregate of carpels or pistils in a flower.

IV.A.32 Factoring out the whitespace would mean that times without `am` or `pm` must have a trailing whitespace.

IV.A.34 This tries to match a character after the end of the string: `$`. (In practice, in a regular expression multiline mode the dollar sign matches a newline character and so this can match. But that lies outside our scope.)

IV.A.35 `^M0,4(CM|CD|D?C0,3)(XC|XL|L?X0,3)(IX|IV|V?I0,3)$`

IV.A.36 Assume it is a regular language and let its pumping length be p . Consider $\sigma = a^p b a^p b$. It is an element of $\mathcal{L}$ whose length is greater than p so By the Pumping Lemma it decomposes as $\sigma = \alpha \beta \gamma$ subject to the three conditions. By the first condition the length of $\alpha \beta$ is less than p , and so these two substrings consist solely of a's. The second condition is that β is not the empty string. The third condition says that the strings $\alpha \gamma, \alpha \beta^2 \gamma, \dots$ are all members of $\mathcal{L}$. Note that the first is not a square because in omitting β it omits at least one a from the prefix but not from the suffix. That's a contradiction.

IV.A.37

- (A) We take the alphabet to be $\Sigma = \mathbb{B}$. Assume that it is regular and let its pumping length be p . Consider $\sigma = 0^p 1 0^p$, which is a member of the language. By the Pumping Lemma it therefore decomposes into

$\sigma = \alpha\beta\gamma$ subject to the three conditions. By the first and second conditions $\alpha\beta$ consists solely of $\emptyset$'s, and is therefore part of the prefix before the 1, and β is nonempty. By the third condition the string $\alpha\gamma$ is also a member of $\mathcal{L}$. But this is a contradiction because the omission of β means that $\alpha\gamma$ has fewer $\emptyset$'s before the 1 than after.

(B) $(a^*)b \setminus 1$

IV.A.38

(A) $.^*r$

(B) $.^*i.^*t.^*$

(C) $.^*foo.^*$

IV.A

(A) HELP

(B)

	(A B C)\1		(AB OE SK)	
.^*M?O.^*	B	O		
(AN FE BE)	B	E		

IV.B.14 We will show there are infinitely many $\sim_{\mathcal{L}}$ equivalence classes by showing that there are infinitely many non-equivalent strings. We claim that no two unequal strings of the form a^n are equivalent. As an example consider $\sigma_2 = aa$ and $\sigma_3 = aaa$. Take $\tau = baa$. Because $\sigma_2 \hat{\ } \tau \in \mathcal{L}$ and $\sigma_3 \hat{\ } \tau \notin \mathcal{L}$, the two are not equivalent. Clearly this works for any two such strings, so there are infinitely many equivalence classes.

Chapter V: Computational Complexity

V 1.29 True. Remember that Big $\mathcal{O}$ means something like “grows at a rate less than or equal to.” Growing at a rate less than or equal to n^2 implies growing at a rate less than or equal to n^3 .

V.1.30 First, the assertion, “I have an algorithm with running time $\mathcal{O}(x^2)$ ” does not mean the time is quadratic, since if $f(x) = x$ then f is $\mathcal{O}(n^2)$.

Also, even if the running time is quadratic, it may involve constants and lower order terms. Thus, $1000 \cdot x^2$ is $\mathcal{O}(n^2)$, as is $x^2 + 3x + 42$. All of the things that impact performance on a particular platform are not part of the algorithm analysis, which only applies to a computing model.

(Another issue is that the runtime behavior is quadratic only for sufficiently large inputs. So unless you know that 5 is greater than the N of the definition, there is no reason to think quadratic at all.)

V 1.31 No doubt they mean that for their problem they have an algorithm that is $\mathcal{O}(n^3)$.

V 1.32

- (A) Because 5 in binary is 101, it takes three bits.
- (B) The number 50 in binary is 11 0010, so it takes six bits.
- (C) In binary the number 500 is 1 1111 0100, so it takes nine bits.
- (D) The binary equivalent of 5 000 is 1 0011 1000 1000, which is thirteen bits.

This table verifies these using the formula $\text{bits}(n) = 1 + \lfloor \lg(n) \rfloor$.

integer n	5	50	500	5 000
binary	101	110 010	1 1111 0100	1 0011 1000 1000
$\lg(n)$	2.322	5.644	8.966	12.288
$1 + \lg(n)$	3	6	9	13

V.1.33 The second is true, the first is not.

Recall the intuition that ‘ f is $\mathcal{O}(g)$ ’ means something like “ f ’s growth rate is less than or equal to g ’s,” while ‘ f is $\Theta(g)$ ’ means something like “ f ’s growth rate is roughly the same as g ’s.” With this in mind, the second one makes sense since if f and g grow at roughly the same rate then that certainly implies f ’s rate is less than or equal to g ’s. More precisely, by definition, if f is $\Theta(g)$ then both f is $\mathcal{O}(g)$ and g is $\mathcal{O}(f)$.

Again with the intuition in mind, assuming that f ’s growth is less than or equal to g ’s does not give that the two rates are equal. That is, there are functions f, g so that f is $\mathcal{O}(g)$ but f is not $\Theta(g)$. One such pair is $f(n) = n$ and $g(n) = n^2$.

V.1.34 We use the principles for simplifying Big $\mathcal{O}$ expressions from page 266.

- (A) $\mathcal{O}(n^2)$
- (B) $\mathcal{O}(2^n)$
- (C) $\mathcal{O}(n^4)$
- (D) $\mathcal{O}(\lg n)$

V.1.35

- (A) For sufficiently large input arguments n , this function is $f(n) = 0$. So it is $\mathcal{O}(1)$.
- (B) For sufficiently large arguments n , this function is $f(n) = n^2$. So it is $\mathcal{O}(n^2)$.
- (C) For sufficiently large n , this function is $f(n) = \lg(n)$. So it is $\mathcal{O}(\lg(n))$.

V.1.36 Because we will apply L’Hôpital’s Rule, to signal that these are functions on $\mathbb{R}$ we convert from n ’s to x ’s.

- (A) The limit of the ratios is this.

$$\lim_{x \rightarrow \infty} \frac{3x^3 + 2x + 4}{\ln(x) + 6}$$

This is an “ ∞/∞ ” limit. Apply L'Hôpital's Rule.

$$\lim_{x \rightarrow \infty} \frac{3x^3 + 2x + 4}{\ln(x) + 6} = \lim_{x \rightarrow \infty} \frac{9x^2 + 2}{1/x} = \lim_{x \rightarrow \infty} \frac{x \cdot (9x^2 + 2)}{1} = \infty$$

So g is $\mathcal{O}(f)$ but f is not $\mathcal{O}(g)$. In short, f 's growth rate is strictly bigger than g 's.

(B) To apply the theorem, consider the limit of the ratio. Use L'Hôpital's Rule.

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x + 4}{x + 5x^3} = \lim_{x \rightarrow \infty} \frac{6x + 2}{1 + 15x^2} = \lim_{x \rightarrow \infty} \frac{6}{30x} = 0$$

Conclude that f is $\mathcal{O}(g)$ but it is not the case that g is $\mathcal{O}(f)$. Briefly, f 's growth rate is strictly less than g 's.

(C) This is the limit of the ratios.

$$\lim_{x \rightarrow \infty} \frac{(1/2)x^3 + 12x^2}{x^2 \ln(x)}$$

We could start by simplifying the fraction by cancelling x^2 from the numerator and the denominator. However, we instead choose to apply L'Hôpital's Rule. Note that we need to use the Product rule on the denominator.

$$\lim_{x \rightarrow \infty} \frac{(1/2)x^3 + 12x^2}{x^2 \ln(x)} = \lim_{x \rightarrow \infty} \frac{(3/2)x^2 + 24x}{2x \ln(x) + x} = \lim_{x \rightarrow \infty} \frac{3x + 24}{2 \ln(x) + 3} = \lim_{x \rightarrow \infty} \frac{3}{2(1/x)} = \lim_{x \rightarrow \infty} \frac{3x}{2} = \infty$$

So g is $\mathcal{O}(f)$ but f is not $\mathcal{O}(g)$.

(D) L'Hôpital's Rule gives this.

$$\lim_{x \rightarrow \infty} \frac{\lg(x)}{\ln(x)} = \lim_{x \rightarrow \infty} \frac{(1/x) \cdot (1/\ln(2))}{(1/x)} = \lim_{x \rightarrow \infty} \frac{1}{\ln(2)} \cdot \frac{(1/x)}{(1/x)} = \lim_{x \rightarrow \infty} \frac{1}{\ln(2)}$$

So both g is $\mathcal{O}(f)$ and f is $\mathcal{O}(g)$, that is, g is $\Theta(f)$. In short, they grow at equivalent rates.

(E) We have this.

$$\lim_{x \rightarrow \infty} \frac{x^2 + \lg(x)}{x^4 - x^3} = \lim_{x \rightarrow \infty} \frac{2x + (1/x)}{4x^3 - 3x^2} = \lim_{x \rightarrow \infty} \frac{2x^2 + 1}{4x^4 - 3x^3} = \lim_{x \rightarrow \infty} \frac{4x}{16x^3 - 9x^2} = \lim_{x \rightarrow \infty} \frac{4}{48x^2 - 18x} = 0$$

So f is $\mathcal{O}(g)$ but g is not $\mathcal{O}(f)$.

(F) We consider the ratio of the functions.

$$\lim_{x \rightarrow \infty} \frac{55}{x^2 + x}$$

This is not an “ ∞/∞ ” limit, so L'Hôpital's Rule doesn't apply. But it doesn't matter because it is easier than that. The denominator grows at a faster rate than the numerator because the numerator doesn't grow at all. That is, f is $\mathcal{O}(g)$ but g is not $\mathcal{O}(f)$.

V.1.37 We follow the principles for simplifying Big $\mathcal{O}$ expressions.

- (A) Both of these are $\mathcal{O}(n^2)$. So both f is $\mathcal{O}(g)$ and also g is $\mathcal{O}(f)$, that is, f is $\Theta(g)$.
- (B) Here, g is a logarithmic function and f is a cubic, so g is $\mathcal{O}(f)$ but it is not the case that f is $\mathcal{O}(g)$.
- (C) The function f is quadratic, while g is a square root, so g is $\mathcal{O}(f)$ but f is not $\mathcal{O}(g)$.
- (D) The function f is $\mathcal{O}(n^{1.2})$ while the function g is $\mathcal{O}(n^{\sqrt{2}})$, which is approximately $\mathcal{O}(n^{1.414})$. Consequently, f is $\mathcal{O}(g)$ but g is not $\mathcal{O}(f)$.
- (E) The exponential function $2^{(1/6)n}$ grows faster than the polynomial function n^6 . So f is $\mathcal{O}(g)$ but it is not the case the g is $\mathcal{O}(f)$.
- (F) As is shown in ?? 1.26, the function 2^n is $\mathcal{O}(3^n)$, while 3^n is not $\mathcal{O}(2^n)$. Thus, g is $\mathcal{O}(f)$ but f is not $\mathcal{O}(g)$.
- (G) Recall the logarithm rule that $\lg(3n) = \lg(3) + \lg(n)$. From this it follows that the two have equivalent growth rates, so f is $\Theta(g)$.

V.1.38 It is (D). This shows that $\mathcal{O}(f_2) < \mathcal{O}(f_3)$

$$\lim_{x \rightarrow \infty} \frac{x^{\log x}}{x\sqrt{x}} = \lim_{x \rightarrow \infty} x^{\log x - \sqrt{x}} = 0$$

as $\lim_{x \rightarrow \infty} (\log x - \sqrt{x}) = -\infty$. For $\mathcal{O}(f_3) < \mathcal{O}(f_1)$ we take the limit of the logarithm of the numerator and denominator. Because the logarithm is an increasing function, this proves the result.

$$\lim_{x \rightarrow \infty} \frac{\log(x\sqrt{x})}{\log(10^x)} = \lim_{x \rightarrow \infty} \frac{\sqrt{x} \log(x)}{x \log(10)} = \lim_{x \rightarrow \infty} \frac{1}{\log(10)} \cdot \frac{\log(x)}{\sqrt{x}}$$

By L'Hôpital's Rule that equals this.

$$= \lim_{x \rightarrow \infty} \frac{1/x}{(1/2)(1/\sqrt{x})} = 2 \cdot \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} = 2 \cdot \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$$

V.1.39 Observe that $\cos(t)$ varies between -1 and 1 . Observe also that by the rules of logarithms $\lg(5^n) = n \cdot \lg(5)$. Thus, the ones with a growth rate that is less than or equal to n^2 's are the first, second, fourth, and fifth. The third one has a growth rate that is $\Theta(n^3)$.

V.1.40

- (A) True. By the simplifying principles from page 266, its growth rate is dominated by the quadratic term, so $5n^2 + 2n$ is $\mathcal{O}(n^2)$, which on the order of growth hierarchy lies below n^3 .
- (B) True. As discussed in the section, all logarithmic functions grow at the same rate. So $\ln n$ is $\Theta(\lg n)$, and hence $\ln n$ is $\mathcal{O}(\lg n)$.
- (C) True. Where $f(n) = n^3 + n^2 + n$, by the simplifying principles from page 266, f is $\mathcal{O}(n^3)$.
- (D) True. For any polynomial function p , we have that p is $\mathcal{O}(2^n)$, by Lemma 1.24.

V.1.41 For each we use the simplifying principles.

- (A) $k = 4$
- (B) $k = 4$
- (C) $k = 3$ (The cosine function is bounded between -1 and 1 .)
- (D) $k = 12$
- (E) This is $\mathcal{O}(n^{7/2})$ but since the question specifies that $k \in \mathbb{N}$, the answer is $k = 4$.

V.1.42 Verify this with $N = 1$ and $C = 1$.

V.1.43

		$n = 1$	$n = 10$	$n = 50$	$n = 100$	$n = 200$
(A)	$\lg n$	–	1.05×10^{-17}	1.79×10^{-17}	2.11×10^{-17}	2.42×10^{-17}
	n	3.17×10^{-18}	3.17×10^{-17}	1.58×10^{-16}	6.34×10^{-16}	6.34×10^{-16}
	$n \lg n$	–	1.05×10^{-16}	8.94×10^{-16}	2.11×10^{-15}	4.85×10^{-15}
	n^2	3.17×10^{-18}	3.17×10^{-16}	7.92×10^{-15}	3.17×10^{-14}	1.27×10^{-13}
	n^3	3.17×10^{-18}	3.17×10^{-15}	3.96×10^{-13}	3.17×10^{-12}	2.54×10^{-11}
	2^n	6.34×10^{-18}	3.24×10^{-15}	3.57×10^{-3}	4.02×10^{12}	5.09×10^{42}
		$n = 1$	$n = 10$	$n = 50$	$n = 100$	
(B)	$\lg n$	–	1.05×10^{-17}	1.79×10^{-17}	2.11×10^{-17}	
	n	3.17×10^{-18}	3.17×10^{-17}	1.58×10^{-16}	6.34×10^{-16}	
	$n \lg n$	–	1.05×10^{-16}	8.94×10^{-16}	2.11×10^{-15}	
	n^2	3.17×10^{-18}	3.17×10^{-16}	7.92×10^{-15}	3.17×10^{-14}	
	n^3	3.17×10^{-18}	3.17×10^{-15}	3.96×10^{-13}	3.17×10^{-12}	
	2^n	6.34×10^{-18}	3.24×10^{-15}	3.57×10^{-3}	4.02×10^{12}	
	3^n	9.51×10^{-18}	1.87×10^{-13}	2.76×10^6	1.63×10^{30}	

V.1.44 The number of seconds in a year is $60 \cdot 60 \cdot 24 \cdot 365.25 = 31\,557\,600$. At 10 000 ticks per second, that is about 3.16×10^{11} ticks in a year (as also described in ?? 1.27).

- (A) With $\lg(n) = 3.16 \times 10^{11}$, we have $n = 2^{3.16 \times 10^{11}} = 4.53 \times 10^{94\,997\,841\,911}$.

(B) $(3.16 \times 10^{11})^2 = 9.96 \times 10^{22}$

(C) 3.16×10^{11}

(D) Because $(3.16 \times 10^{11})^{1/2} \approx 561\,761.52$, the answer is 561 761.

(E) Because $(3.16 \times 10^{11})^{1/3} \approx 6808.24$, the answer is 6 808.

(F) We have $\lg(3.16 \times 10^{11}) \approx 38.20$ so the answer is 38.

V **V.1.45** Solving $100\,000 \cdot n^2 = n^3$ gives $n = 100\,000$. So the first number n where $f(n) = 100 \cdot n^2$ is less than $g(n) = n^3$ is $n = 100\,001$.

V **V.1.46** Linear. Finite State machines consume one character at each step. So they run in time equal to the length of the input. In the notation of Definition 1.28, $t_M(\sigma) = |\sigma|$.

V **V.1.47** Let g be the function $g(x) = 1$.

(A) Directly apply the definition of Big $\mathcal{O}$, Definition 1.6, by taking $N = 1$ and $C = 8$. If $x > N = 1$ then $C \cdot g(x) = 8 \cdot 1 \geq f(x) = 7$.

(B) Recall that $\sin(x)$ is bounded between -1 and 1 . Take $N = 1$ and $C = 9$. If $x > N$ then $C \cdot g(x) = 9 \geq 7 + \sin(x) = f(x)$.

(C) Fix $N = 1$ and $C = 8$. If $x \geq N = 1$ then $C \cdot g(x) = 2$ is greater than $f(x) = 7 + (1/x)$.

(D) We first show that if it is bounded then it is $\mathcal{O}(1)$. Suppose that f is bounded by $K \in \mathbb{R}^+$. Take $N = 1$ and $C = K$. If $x \geq N = 1$ then $C \cdot g(x) = K \geq f(x)$, so f is $\mathcal{O}(g)$.

For the converse, suppose that f is $\mathcal{O}(g)$. Then there exists constants $N, C \in \mathbb{R}^+$ so that $x \geq N$ implies that $C \cdot g(x) = C \geq f(x)$. Let $K = \max(C, f(0), \dots, f(N-1))$. If $x \in \mathbb{R}$ then $f(x) \leq K$, so f is bounded by K .

V **V.1.48** A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is $\mathcal{O}(1)$ if it is bounded by a constant. So $g(x) \leq x^{\mathcal{O}(1)}$ says that the function g has at most a polynomial growth rate.

V **V.1.49** The first half, that f is $\mathcal{O}(g)$, is straightforward. Fix $N = 1$ and $C = 2$. If $x \geq N = 1$ then $C \cdot g(x) = 2x^2 \geq 2x = f(x)$ (because $2x^2 - 2x = 0$ gives $2 \cdot (x(x-1)) = 0$ and so the largest x for which they are equal is $x = 1$).

For the second half, that g is not $\mathcal{O}(f)$, we will show that for every C there is no N that will work. So fix a C , aiming to prove that for any N it is not the case that $x \geq N$ implies that $C \cdot f(x) \geq g(x)$. For any N , consider $x = \max(N, C+1)$, which is greater than or equal to N . Because $x > C$, we have that $g(x) = x^2 > C \cdot x = C \cdot f(x)$.

V **V.1.50** Let $N = 4$ and $C = 1$. If $n \geq N = 4$ then $C \cdot n! = n! = 4! \cdot 5 \cdot 6 \cdots (n-1) \cdot n \geq 2^4 \cdot 2^{n-4} = 2^n$ (note that $4! = 24 > 2^4 = 16$).

V **V.1.51**

(A) This limit equals 0

$$\lim_{x \rightarrow \infty} \frac{(\log_b(x))^2}{x^d} = \lim_{x \rightarrow \infty} \frac{2 \log_b(x) \cdot \frac{1}{x \ln(b)}}{dx^{d-1}} = \frac{2}{d \ln(b)} \cdot \lim_{x \rightarrow \infty} \frac{\log_b(x)}{x^d}$$

by the calculation in Example 1.22.

(B) This limit equals 0

$$\lim_{x \rightarrow \infty} \frac{(\log_b(x))^3}{x^d} = \lim_{x \rightarrow \infty} \frac{3(\log_b(x))^2 \cdot \frac{1}{x \ln(b)}}{dx^{d-1}} = \frac{3}{d \ln(b)} \cdot \lim_{x \rightarrow \infty} \frac{(\log_b(x))^2}{x^d}$$

by the calculation in the prior item.

(c) The prior item suggests an argument by induction. So assume that the statement holds for the powers $n = 1$, $n = 2, \dots, n = k$ and consider the $n = k+1$ case.

$$\lim_{x \rightarrow \infty} \frac{(\log_b(x))^{k+1}}{x^d} = \lim_{x \rightarrow \infty} \frac{(k+1)(\log_b(x))^k \cdot \frac{1}{x \ln(b)}}{dx^{d-1}} = \frac{k+1}{d \ln(b)} \cdot \lim_{x \rightarrow \infty} \frac{(\log_b(x))^k}{x^d}$$

Finish by applying the $n = k$ inductive hypothesis.

V **V.1.52** Fix $N = 1$ and find $C \in \mathbb{R}$ large enough that $C \cdot g(1) \geq f(1)$. Because g is increasing, if $x > N = 1$ then $C \cdot g(x) \geq C \cdot g(1) \geq f(1) = f(x)$.

V **V.1.53**

(A) The function $f_0(x) = 0$ is clearly computable and has output values that grow at a rate that is $\mathcal{O}(1)$. Similarly, the output values of $f_1(x) = x$ grow at a rate that is $\mathcal{O}(1)$, and $f_2(x) = x^2$ is $\mathcal{O}(x^2)$, etc.

(B) The function

$$K(x) = \begin{cases} 0 & \text{if } \phi_x(x) \uparrow \\ 1 & \text{if } \phi_x(x) \downarrow \end{cases}$$

is clearly $\mathcal{O}(1)$.

(c) Let $M(n)$ be the maximum value of $\phi_i(j)$ where $0 \leq i, j \leq n$ and $\phi_i(j) \downarrow$ (if there is no such maximum because none of the $\phi_i(j)$ converge then define $M(n) = 0$). This is not a computable function, but it is a function. Then the function $\hat{M}(n) = M(n) + 1$ has the property that for any computable function ϕ_e , for all $n > e$ if $\phi_e(n) \downarrow$ then $\hat{M}(n) > \phi_e(n)$.

V.1.54 The first equality below comes from $x = 2^{\lg x}$.

$$x^{\lg x} = (2^{\lg x})^{\lg x} = 2^{(\lg x)^2} \in \mathcal{O}(2^{\text{poly}(\lg x)})$$

V.1.55 First we argue that it grows faster than any polynomial. Fix a power $k \in \mathbb{N}$. There is an x_k where if $x \geq x_k$ then $\lg(x) > k + 1$, and thus $\mathcal{O}(x^k) \subsetneq \mathcal{O}(x^{k+1}) \subseteq \mathcal{O}(f)$.

Next we argue that it grows slower than b^n for $b > 1$; actually, we will show it for $b = 2$ but the argument works for any $b > 1$. We use L'Hôpital's Rule. The first equality comes from $x = 2^{\lg x}$.

$$\lim_{x \rightarrow \infty} \frac{x^{\lg x}}{2^x} = \lim_{x \rightarrow \infty} \frac{(2^{\lg x})^{\lg x}}{2^x} = \lim_{x \rightarrow \infty} \frac{2^{(\lg x)^2}}{2^x} = \lim_{x \rightarrow \infty} 2^{(\lg x)^2 - x} = 0$$

The final equality comes from $\lim_{x \rightarrow \infty} (\lg x)^2 - x = -\infty$.

V **V.1.56** Use Theorem 1.17.

$$\lim_{x \rightarrow \infty} \frac{2^x}{3^x} = \lim_{x \rightarrow \infty} \left(\frac{2}{3} \right)^x = 0$$

V **V.1.57**

(A) With an eye toward Theorem 1.17, consider the ratio.

$$\frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n}{n} \leq \frac{1}{n} \cdot 1 \cdots 1$$

Clearly the limit of the ratio $n!/n^n$ is 0.

(B) No, don't overlook the e^{-n} .

V **V.1.58**

(A) Apply L'Hôpital's Rule twice.

$$\lim_{x \rightarrow \infty} \frac{x^2 + 5x + 1}{x^2} = \lim_{x \rightarrow \infty} \frac{2x + 5}{2x} = \lim_{x \rightarrow \infty} \frac{2}{2} = 1$$

(B) Apply L'Hôpital's Rule, remembering to use the Chain Rule for the numerator.

$$\lim_{x \rightarrow \infty} \frac{\lg(x+1)}{\lg(x)} = \lim_{x \rightarrow \infty} \frac{(1/x \ln(2)) \cdot 1}{1/x \ln(2)} = 1$$

V **V.1.59** No. If $\mathcal{O}(f)$ is the set of polynomials then f must be a polynomial. However, then f has some degree, and no polynomial of higher degree is a member of $\mathcal{O}(f)$.

V **V.1.60** Recall the logarithm rules that $\log_b(x^a) = a \cdot \log_b(x)$ and $b^{\log_b(x)} = x$.

(A) L'Hôpital's Rule gives

$$\lim_{x \rightarrow \infty} \frac{\lg(x)}{(\lg(x))^2} = \lim_{x \rightarrow \infty} \frac{1/x \ln(2)}{2 \lg(x) \cdot 1/x \ln(2)} = \lim_{x \rightarrow \infty} \frac{1}{2 \lg(x)} = 0$$

and so Theorem 1.17 gives the required conclusion.

(B) By Theorem 1.17, it suffices to show that this is 0.

$$\lim_{x \rightarrow \infty} \frac{x^k}{x^{\lg(x)}} = \lim_{x \rightarrow \infty} x^{k - \lg(x)}$$

Rewrite.

$$x^{k - \lg(x)} = 2^{\lg(x^{k - \lg(x)})} = 2^{(k - \lg(x)) \cdot \lg(x)} = 2^{k \lg(x) - (\lg(x))^2}$$

Apply the prior item to conclude that the limit of the exponent is $-\infty$, and so the limit of the entire expression is 0.

(C) Taking $\lg$ of $x^{\lg(x)}$ gives $\lg(x^{\lg(x)}) = \lg(x) \cdot \lg(x)$. Taking $\lg$ of the other expression gives $\lg(2^{(\lg(x))^2}) = (\lg(x))^2 \cdot \lg(2)$. Since the logarithm and exponential are inverse functions, that is $(\lg(x))^2$.

(D) Rewrite $x^{\lg(x)}$ as $2^{(\lg(x))^2}$. Theorem 1.17

$$\lim_{x \rightarrow \infty} \frac{2^{(\lg(x))^2}}{2^x} = \lim_{x \rightarrow \infty} 2^{(\lg(x))^2 - x} = 0$$

gives the desired conclusion.

V V.1.61 We will use the notation from the lemma statement.

(A) The assumption is that f is $\mathcal{O}(g)$. So there are constants $N, C \in \mathbb{R}^+$ such that $x \geq N$ implies that $C \cdot g(x) \geq f(x)$. For the function $a \cdot f$, use the constants $\hat{N} = N$ and $\hat{C} = a \cdot C$. Then $x \geq \hat{C}$ implies that $\hat{C} \cdot g(x) \geq f(x)$.

(B) The function f_0 is $\mathcal{O}(g_0)$ and the function f_1 is $\mathcal{O}(g_1)$. So there are constants $N_0, C_0, N_1, C_1 \in \mathbb{R}^+$ such that $x \geq N_0$ implies that $C_0 g_0(x) \geq f_0(x)$, and $x \geq N_1$ implies that $C_1 g_1(x) \geq f_1(x)$. Take $\hat{N} = \max(N_0, N_1)$ and $\hat{C} = \max(C_0, C_1)$. Then $x \geq \hat{N}$ implies that $\hat{C} \cdot (g_0(x) + g_1(x)) = \hat{C} g_0(x) + \hat{C} g_1(x) \geq f_0(x) + f_1(x)$.

(C) As in the prior item, there are constants $N_0, C_0, N_1, C_1 \in \mathbb{R}^+$ such that $x \geq N_0$ implies that $C_0 g_0(x) \geq f_0(x)$, and $x \geq N_1$ implies that $C_1 g_1(x) \geq f_1(x)$. Take $\hat{N} = \max(N_0, N_1)$ and $\hat{C} = C_0 \cdot C_1$. Then $x \geq \hat{N}$ implies that $\hat{C} \cdot (g_0(x) \cdot g_1(x)) = C_0 g_0(x) \cdot C_1 g_1(x) \geq f_0(x) \cdot f_1(x)$.

V V.1.62

(A) We must show that f is in $\mathcal{O}(f)$. Take $N = 1$ and $C = 1$.

(B) Suppose that f_0 is $\mathcal{O}(f_1)$ and that f_1 is $\mathcal{O}(f_2)$. Then there exists $N_0, N_1, C_0, C_1 \in \mathbb{R}^+$ such that if $x \geq N_0$ then $C_0 \cdot f_1(x) \geq f_0(x)$, and if $x \geq N_1$ then $C_1 \cdot f_2(x) \geq f_1(x)$. Take $\hat{N} = \max(N_0, N_1)$ and $\hat{C} = C_0 \cdot C_1$. Then $x \geq \hat{N}$ implies that $\hat{C} \cdot f_2(x) \geq C_0 \cdot f_1(x) \geq f_0(x)$.

V V.1.63

(A) If $\lim_{x \rightarrow \infty} f(x)/g(x)$ exists and equals 0 then for any $\varepsilon > 0$ there is an $N \in \mathbb{R}$ such that $x \geq N$ implies that $f(x)/g(x) < \varepsilon$. That gives $f(x) < \varepsilon \cdot g(x)$. Fix $\varepsilon = 1/2$ and $C = 2$. Take inverses to get that $x \geq N$ implies $C \cdot g(x) = 2g(x) \geq f(x)$.

(B) To show that f is $\mathcal{O}(g)$, take $N = 1$ and $C = 4$.

On the other hand, the limit doesn't exist, as the ratio oscillates between 1 and 2.

V V.1.64 First use L'Hôpital's Rule to get that if $g(x) = a_m x^m + \dots + a_0$ then g is $\Theta(x^m)$. Then Example 1.22 shows that any logarithmic function is $\mathcal{O}(g)$.

For the other half, fix a base $b > 1$ and consider the exponential function $h(x) = b^x$. L'Hôpital's Rule gives this

$$\lim_{x \rightarrow \infty} \frac{x^m}{b^x} = \frac{m}{\ln(b)} \cdot \lim_{x \rightarrow \infty} \frac{x^{m-1}}{b^x} = \frac{m(m-1)}{(\ln(b))^2} \cdot \lim_{x \rightarrow \infty} \frac{x^{m-2}}{b^x} = \dots = \frac{m!}{(\ln(b))^m} \cdot \lim_{x \rightarrow \infty} \frac{1}{b^x} = 0$$

and Theorem 1.17 gives the desired conclusion.

- V V.2.41 There are twenty five: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, and 97.

V.2.42

- (A) Yes, it is prime.
 (B) No, it is not prime. Besides the trivial ones, the divisors of 6 165 are 3, 5, 9, 15, 45, 137, 411, 685, 1 233, and 2 055.
 (C) Yes.
 (D) No. Besides itself and 1, this number is also divisible by 7 and 601.
 (E) No. Besides the trivial divisors of itself and 1, this number also has the divisors 3, 11, 33, 233, 699, and 2 563.

V V.2.43

- (A) One is 3. The entire list is 3, 9, 3469, 10407, and 31221.
 (B) One is 2. The entire list is 2, 4, 8, 6553, 13106, 26212, and 52424.
 (C) One is 5. The entire list is 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 25, 30, 32, 40, 48, 50, 60, 64, 75, 80, 96, 100, 120, 128, 150, 160, 192, 200, 240, 300, 320, 384, 400, 480, 600, 640, 800, 960, 1200, 1600, 1920, 2400, 3200, 4800, and 9600.
 (D) One is 61. The entire list is 61, 71, 4331.
 (E) The only one is 877. It is prime.

V.2.44 The AKS algorithm that solves Primality answers ‘yes’ or ‘no’ without finding divisors.

V.2.45

- (A) This formula is satisfiable.

P	Q	R	$P \wedge Q$	$\neg Q$	$\neg Q \wedge R$	$(P \wedge Q) \vee (\neg Q \wedge R)$
F	F	F	F	T	F	F
F	F	T	F	T	T	T
F	T	F	F	F	F	F
F	T	T	F	F	F	F
T	F	F	F	T	F	F
T	F	T	F	T	T	T
T	T	F	T	F	F	T
T	T	T	T	F	F	T

- (B) This formula is not satisfiable.

P	Q	R	$P \rightarrow Q$	$P \wedge Q$	$\neg P$	$(P \wedge Q) \vee \neg P$	$\neg(P \wedge Q) \vee \neg P$	$(P \rightarrow Q) \wedge \neg((P \wedge Q) \vee \neg P)$
F	F	F	T	F	T	T	F	F
F	F	T	T	F	T	T	F	F
F	T	F	T	F	T	T	F	F
F	T	T	T	F	T	T	F	F
T	F	F	F	F	F	F	T	F
T	F	T	F	F	F	F	T	F
T	T	F	T	T	F	T	F	F
T	T	T	T	T	F	T	F	F

V.2.46

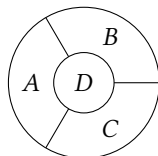
- (A) The three-atom clause is $P \vee \neg Q \vee R$.
 (B) The clause for the F - T - T line is $P \vee \neg Q \vee \neg R$. For the T - F - T line, use $\neg P \vee Q \vee \neg R$. For the T - T - T line, use $\neg P \vee \neg Q \vee \neg R$.
 (C) Here is the entire expression in conjunctive normal form.

$$(P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (\neg P \vee Q \vee \neg R) \wedge (\neg P \vee \neg Q \vee \neg R)$$

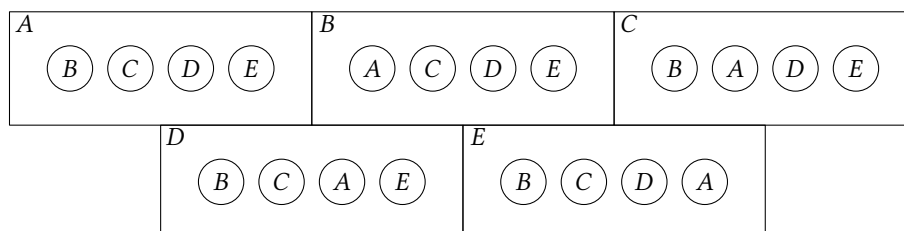
Verifying the truth table is routine.

- V V.2.47 For the octahedron, the path $\langle v_0, v_1, v_5, v_4, v_3, v_2, v_0 \rangle$ will do. For the icosahedron, one path is $\langle v_0, v_{10}, v_{11}, v_8, v_9, v_6, v_7, v_2, v_3, v_4, v_5, v_1, v_0 \rangle$.

V.2.48 This is the natural one.

**V.2.49**

(A) This map shows five countries. They are not contiguous; for instance, country A has five separate components.

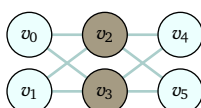


Because of the portion in the upper left, country A must be a different color than any of the others. The other portions similarly give that all the other countries are different colors than each other. In total there are five countries, each a different color than all the others.

(B) Make a circle and split it into five wedges.

V.2.50 The one on the right does not partition the vertices into disjoint sets.

V.2.51 This is a cut set with eight elements.



That is as large as possible because it is all of the edges.

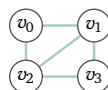
V.2.52

- (A) AT&T connects to Citigroup, which connects to Ford Motor.
- (B) Haliburton connects to Citigroup, which connects to Ford Motor.
- (C) There is no connection between Caterpillar and Ford Motor.
- (D) Also no connection.

V.2.53 If a graph has n vertices $v_0, \dots, v_{n-1}$ then a Hamiltonian path has n edges. Traversing the path has an one and only one edge leaving each vertex.

V.2.54

(A) This graph has a Hamiltonian circuit, around the outside.



To include the vertices v_0 and v_3 , any circuit must include those four outside edges. In every Hamiltonian circuit the number of edges equals the number of vertices and hence no Hamiltonian circuit can contain the edge v_1v_2 , because that would make five edges.

(B) Assume that all the edge weights are positive, and then to the edge e assign the weight $-(\text{number of vertices})(\text{max weight}) - 1$, (Or, we could assign the weight 1 and raise the other edge weights). Every Hamiltonian path has the same number of edges, specifically the number of edges equals the number of vertices, So a path through e has a negative total weight. This is clearly the only way to get a negative total weight.

V.2.55 (A) 2 (B) 2 (C) 0 (D) 4 (I was only in the deleted scenes. But it is all just fun.)

V.2.56 Figure 92 on page 173 shows the result of writing a small script.

V.2.57 A 269-269 tie is easy to concoct. For example, the electoral votes of CA, TX, FL, NY, IL, PA, OH, GA, MI, NC, and VA add to 269.

V.2.58 We can do these by eye.

<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	Weight	Value	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	Weight	Value
N	N	N	N	N	0	0	Y	N	N	N	N	21	50
N	N	N	N	Y	19	40	Y	N	N	N	Y	40	90
N	N	N	Y	N	42	44	Y	N	N	Y	N	63	94
N	N	N	Y	Y	61	84	Y	N	N	Y	Y	82	134
N	N	Y	N	N	49	34	Y	N	Y	N	N	70	84
N	N	Y	N	Y	68	74	Y	N	Y	N	Y	89	124
N	N	Y	Y	N	91	78	Y	N	Y	Y	N	112	128
N	N	Y	Y	Y	110	118	Y	N	Y	Y	Y	131	168
N	Y	N	N	N	33	48	Y	Y	N	N	N	54	98
N	Y	N	N	Y	52	88	Y	Y	N	N	Y	73	138
N	Y	N	Y	N	75	92	Y	Y	N	Y	N	96	142
N	Y	N	Y	Y	94	132	Y	Y	N	Y	Y	115	182
N	Y	Y	N	N	82	82	Y	Y	Y	N	N	103	132
N	Y	Y	N	Y	101	122	Y	Y	Y	N	Y	122	172
N	Y	Y	Y	N	124	126	Y	Y	Y	Y	N	145	176
N	Y	Y	Y	Y	143	166	Y	Y	Y	Y	Y	164	216

FIGURE 92, FOR QUESTION V.2.56: Output of a brute force script to show that there is no solution of an instance of the Knapsack problem.

(A) The shortest is q_2 to q_1 to q_7 , which has length 14.

(B) The shortest is q_0 to q_3 to q_4 to q_8 , of length 23.

(C) There is no path between these two vertices.

V.2.59 Figure 93 on page 174 is a check of every possibility.

V.2.60 A 3-clique is a triangle. A 2-clique is an edge.

V.2.61 A k -clique has $k(k-1)/2$ edges, one for each combination of two vertices.

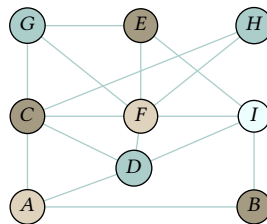
V.2.62

(A) Suppose that a three-coloring suffices. Then the three vertices A , C , and D form a triangle, so they must be different colors. Since CDF is also a triangle, we conclude that A and F are the same colors. So far we have: $C_0 = \{A, F\}$, $C_1 = \{C\}$, $C_2 = \{D\}$.

Next, consider triangle DFI . It's presence gives that C and I are the same colors. So we have $C_0 = \{A, F\}$, $C_1 = \{C, I\}$, $C_2 = \{D\}$. Then triangle EFI leads to $C_0 = \{A, F\}$, $C_1 = \{C, I\}$, $C_2 = \{D, E\}$.

Finally, triangle CFG gives that D and G are the same color, giving $C_0 = \{A, F\}$, $C_1 = \{C, I\}$, $C_2 = \{D, E, G\}$. But this contradicts the presence of triangle GEF .

(B) This is a four-coloring.



V.2.63 It does not imply that. For an example, let the Boolean expression be $P \vee Q$.

<i>P</i>	<i>Q</i>	$P \vee Q$	$\neg(P \vee Q)$
<i>F</i>	<i>F</i>	<i>F</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>T</i>	<i>T</i>	<i>F</i>

It is satisfiable by taking both variables to be T . Its negation is also satisfiable, by taking both variables to be F

21	33	49	42	19	Total	21	33	49	42	19	Total
N	N	N	N	N	0	Y	N	N	N	N	21
N	N	N	N	Y	19	Y	N	N	N	Y	40
N	N	N	Y	N	42	Y	N	N	Y	N	63
N	N	N	Y	Y	61	Y	N	N	Y	Y	82
N	N	Y	N	N	49	Y	N	Y	N	N	70
N	N	Y	N	Y	68	Y	N	Y	N	Y	89
N	N	Y	Y	N	91	Y	N	Y	Y	N	112
N	N	Y	Y	Y	110	Y	N	Y	Y	Y	131
N	Y	N	N	N	33	Y	Y	N	N	N	54
N	Y	N	N	Y	52	Y	Y	N	N	Y	73
N	Y	N	Y	N	75	Y	Y	N	Y	N	96
N	Y	N	Y	Y	94	Y	Y	N	Y	Y	115
N	Y	Y	N	N	82	Y	Y	Y	N	N	103
N	Y	Y	N	Y	101	Y	Y	Y	N	Y	122
N	Y	Y	Y	N	124	Y	Y	Y	Y	N	145
N	Y	Y	Y	Y	143	Y	Y	Y	Y	Y	164

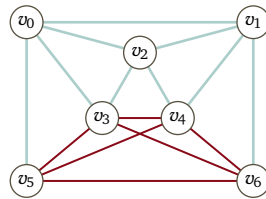
FIGURE 93, FOR QUESTION V.2.59: All possibilities in the Subset Sum instance.

Combination of five vertices	A missing edge	Combination of five vertices	A missing edge
$v_0 v_1 v_2 v_3 v_4$	$v_1 v_3$	$v_0 v_2 v_3 v_4 v_6$	$v_0 v_4$
$v_0 v_1 v_2 v_3 v_5$	$v_2 v_5$	$v_0 v_2 v_3 v_5 v_6$	$v_0 v_6$
$v_0 v_1 v_2 v_3 v_6$	$v_2 v_6$	$v_0 v_2 v_4 v_5 v_6$	$v_0 v_4$
$v_0 v_1 v_2 v_4 v_5$	$v_0 v_4$	$v_0 v_3 v_4 v_5 v_6$	$v_0 v_4$
$v_0 v_1 v_2 v_4 v_6$	$v_0 v_6$	$v_1 v_2 v_3 v_4 v_5$	$v_1 v_3$
$v_0 v_1 v_2 v_5 v_6$	$v_0 v_6$	$v_1 v_2 v_3 v_4 v_6$	$v_1 v_3$
$v_0 v_1 v_3 v_4 v_5$	$v_0 v_4$	$v_1 v_2 v_3 v_5 v_6$	$v_1 v_3$
$v_0 v_1 v_3 v_4 v_6$	$v_0 v_6$	$v_1 v_2 v_4 v_5 v_6$	$v_1 v_5$
$v_0 v_1 v_3 v_5 v_6$	$v_0 v_6$	$v_1 v_3 v_4 v_5 v_6$	$v_1 v_3$
$v_0 v_1 v_4 v_5 v_6$	$v_0 v_6$	$v_2 v_3 v_4 v_5 v_6$	$v_2 v_5$
$v_0 v_2 v_3 v_4 v_5$	$v_0 v_4$		

FIGURE 94, FOR QUESTION V.2.66: A check of all possible 5-cliques.

V.2.64 A finite set of propositional logic statements $\{S_0, \dots, S_{k-1}\}$ is satisfiable if and only if the single statement $S_0 \wedge S_1 \dots \wedge S_{k-1}$ is satisfiable.

V.2.66 It has a 4-clique, and thus it necessarily has a 3-clique.



But it has no 5-clique, as checked by exhaustion in Figure 94 on page 174.

V.2.67

- (A) A vertex cover with $k = 2$ elements is $S = \{q_1, q_3\}$. An independent set with $\hat{k} = 4$ elements is $\hat{S} = \{q_0, q_2, q_4, q_5\}$.
- (B) A vertex cover with $k = 3$ elements is $S = \{q_1, q_2, q_4\}$. An independent set with $\hat{k} = 3$ elements is $\hat{S} = \{q_0, q_3, q_5\}$.
- (C) A vertex cover with $k = 4$ elements is $S = \{q_2, q_5, q_8, q_9\}$. An independent set with $\hat{k} = 6$ elements is

Year	Winner	Other	Tallest?	Year	Winner	Other	Tallest?
1900	W McKinley	W J Bryan	N	1964	L B Johnson	B Goldwater	Y
1904	T Roosevelt	A B Parker	Y	1968	R Nixon	H Humphrey	Y
1908	W H Taft	W J Bryan	Y	1972	R Nixon	G McGovern	N
1912	W Wilson	-2 others-	N	1976	J Carter	G Ford	N
1916	W Wilson	C E Hughes	Y	1980	R Reagan	J Carter	Y
1920	W G Harding	J M Cox	Y	1984	R Reagan	W Mondale	Y
1924	C Coolidge	J W Davis	N	1990	G H W Bush	M Dukakis	Y
1928	H Hoover	A Smith	Y	1992	W Clinton	G H W Bush	-same-
1932	F D Roosevelt	H Hoover	Y	1996	W Clinton	R Dole	Y
1936	F D Roosevelt	A Landon	Y	2000	G W Bush	A Gore	N
1940	F D Roosevelt	W Willkie	N	2004	G W Bush	J Kerry	N
1944	F D Roosevelt	T Dewey	Y	2008	B Obama	J McCain	Y
1948	H S Truman	T Dewey	Y	2012	B Obama	M Romney	N
1952	D Eisenhower	A Stevenson II	Y	2016	D Trump	H Clinton	Y
1956	D Eisenhower	A Stevenson II	Y	2020	J Biden	D Trump	N
1960	J F Kennedy	R Nixon	Y				

FIGURE 95, FOR QUESTION V.3.10: Height comparison of US Presidential winner and main opponent, since 1900.

$$\hat{S} = \{q_0, q_1, q_3, q_4, q_6, q_7\}.$$

- (D) For any edge, if it has at least one endpoint in S then it has at most one endpoint in the complement $\hat{S} = \mathcal{N} - S$. Conversely, If an edge has at most one endpoint in $\hat{S}$ then it has at least one endpoint in the complement $S = \mathcal{N} - \hat{S}$.

V.2.68 The Three-dimensional Matching problem requires that the three sets, the instructors, the courses, and the time slots, all have the same size. In this instance, all three have five elements.

A suitable matching is $\hat{M} = \{\langle A, 2, \epsilon \rangle, \langle B, 0, \alpha \rangle, \langle C, 3, \gamma \rangle, \langle D, 1, \beta \rangle, \langle E, 4, \delta \rangle\}$.

V.2.69

- (A) The set $M = X \times Y \times Z$ contains these twenty seven triples $\langle x, y, z \rangle$.

$z = a$			$z = d$			$z = e$		
$\langle a, b, a \rangle$	$\langle a, c, a \rangle$	$\langle a, d, a \rangle$	$\langle a, b, d \rangle$	$\langle a, c, d \rangle$	$\langle a, d, d \rangle$	$\langle a, b, e \rangle$	$\langle a, c, e \rangle$	$\langle a, d, e \rangle$
$\langle b, b, a \rangle$	$\langle b, c, a \rangle$	$\langle b, d, a \rangle$	$\langle b, b, d \rangle$	$\langle b, c, d \rangle$	$\langle b, d, d \rangle$	$\langle b, b, e \rangle$	$\langle b, c, e \rangle$	$\langle b, d, e \rangle$
$\langle c, b, a \rangle$	$\langle c, c, a \rangle$	$\langle c, d, a \rangle$	$\langle c, b, d \rangle$	$\langle c, c, d \rangle$	$\langle c, d, d \rangle$	$\langle c, b, e \rangle$	$\langle c, c, e \rangle$	$\langle c, d, e \rangle$

- (B) There are lots of correct answers. One is $\hat{M} = \{\langle a, b, a \rangle, \langle b, c, d \rangle, \langle c, d, e \rangle\}$.

V.3.10 Note: the meanings give here are common usage. But because these terms are not formally defined, different authors may use them in different ways.

- (A) A heuristic is a shortcut, an approach to a problem that is faster than a full solution, or at least more reliable than simple guessing.

One example is approaching the Travelling Salesman problem by using a greedy algorithm, that is, always going next to the nearest city. Another example is that antivirus programs use heuristics to try and determine if an executable is malicious, such as comparing the first few bytes to a list of known-bad ones. A third example is based on the data in Figure 95 on page 175. In the contests for US President since 1900, the taller candidate has won 20 out of 30 times, for a sample proportion of 0.67. The chance of such a result by happenstance is less than 0.02. You could decide to bet in the next election on the taller candidate. This isn't logical, but it is a method and it is used in practice.

That is, a heuristic is a method that, while it may be used in practice, is not guaranteed to be optimal, perfect, logical, or even rational. An algorithm for a problem is guaranteed to solve the problem.

- (B) Pseudocode is a description of an algorithm. It gives a sketch or outline, usually at a quite high level. It is often laid out as though it were compilable code but usually it does not conform to any specific language's syntax; for instance it typically does not include details such as declaration of variables.

This is pseudocode for the recursive depth-first traversal of a binary tree.

```

process_subtree(v)
  process_subtree(v's left child)
  process_vertex(v)
  process_subtree(v's right child)

```

Another example describes the most simple algorithm for checking whether a number is prime, by dividing it by each natural smaller than it but bigger than one.

```

is_prime(n)
  for 2 <= i < n
    if i divides n
      return False
  return True

```

We have seen many other examples in this book, including the descriptions in the proof of Chapter Two's Lemma 7.3.

- (c) Of course, in the first section of the first chapter we saw a number of Turing machines, such as one that computes the predecessor function, another that computes the sum of two inputs, and a third that never halts.

A Turing machine is more like a program than an algorithm, in that it is an implementation of an algorithm. For instance, you may have the Bubble Sort algorithm in mind and then you implement it on a Turing machine.

- (d) A flowchart is graphical pseudocode. They describe an algorithm, often at a high level (perhaps even higher than used for pseudocode). We have seen a number of flowcharts in this book, including in this chapter. They differ from an algorithm in the same way that pseudocode does.
- (e) Source code is an implementation of an algorithm (sometimes a number of algorithms are bundled together to make one source). It is targeted at a specific compiler or interpreter, and perhaps at a specific hardware and software platform.

For example, there is source code that compiles to make a simple “Hello World!” program in C, source code in Racket that is interpreted to determine if an input natural number is prime, and source code that gives this book on compilation with $\text{\LaTeX}$ and related programs.

- (f) An executable is a binary file that can be run on an operating system. It is usually the result of compiling source code, and so is an implementation of an algorithm. Three examples are the executable that results from compiling (and linking) the “Hello World!” program in C, the executable for your browser, and the executable that performs $\text{\LaTeX}$.
- (g) A process is an instance of a program that has been loaded into a machine's memory for execution. As such, it is an implementation of an algorithm (or perhaps a number of algorithms). Three examples are the result of calling the executable of the “Hello World!” program, of calling the executable for your browser, and the of invoking the executable for $\text{\LaTeX}$.

Some processes, such as an operating system shell or a web server, are designed to run forever, while typically in computational complexity discussions the term ‘algorithm’ is used for something that is guaranteed to halt.

V.3.11 A solution is an algorithm that acts as the characteristic function of the set.

V.3.12 A decision problem is any one that ends in a ‘yes’ or ‘no’ answer. A language decision problem is a specialization of that, to ‘yes’ or ‘no’ answers about membership in the given language.

V.3.13

- (A) The multiplication is easy: $(1/0.01)^2 \cdot 21 \cdot 30 = 6.30 \times 10^6$.
- (B) The answer is $\lceil \lg(1\,000\,000) \rceil$. If all you have is a pocket calculator with a natural log function button, then you can get $6 \ln(10)/\ln(2) \approx 19.93$, so the answer is 20.
- (c) Multiply $(6.30 \times 10^6) \cdot 20 = 1.26 \times 10^8$. That's less than 16 megabytes, about eight books from Project Gutenberg.

V V.3.14 One is the smell of an baby's head. They have a great smell. But we don't compute it.

V V.3.15 Yes, two programs that implement the same algorithm must compute the same function. However, the converse does not hold; two programs computing the same function need not implement the same algorithm. For instance, two programs may both sort input strings but one uses Bubble Sort while the other uses Quick

Sort.

V.3.17 These algorithms mark that input is accepted or not by printing 1 or 0.

- (A) Because the number 4 is a square that is one greater than a prime, three members are $\langle 0, 4 \rangle$, $\langle 1, 3 \rangle$, and $\langle 2, 2 \rangle$. For an algorithm computing the characteristic function, first input the sequence and unpack it to get $n, m \in \mathbb{N}$. Then add the two numbers. If the sum is both square and one greater than a prime then print 1, and otherwise print 0.
- (B) Three string members of $\mathcal{L}$ are $\sigma_0 = 200$, $\sigma_1 = 100$, and $\sigma_2 = 300$. The algorithm is: if the input is a single 0 or ends in two 0's, then print 1. Otherwise print 0.
- (C) The language $\mathcal{L}_2$ contains 11, 110, and 1. A natural algorithm is to scan the input string, keeping a running count of the number of 0's and 1's. At the end, if there are more 1's then print 1, otherwise print 0.
- (D) Recall that strings equal to their reverse are called palindromes. Three palindromes over $\mathbb{B}^*$ are 101, 11, and 1. To compute the characteristic function, read and save the input string. Then, compare the characters starting from the back with those starting from the front. If each matches (or the string is empty) then print 1. Otherwise print 0.

V.3.18

- (A) The empty language is $\mathcal{L} = \emptyset = \{ \}$. Its characteristic function is $\mathbb{1}_{\mathcal{L}}(n) = 0$. The algorithm to solve this language decision problem is to, for all input, return 0.
- (B) This language contains two strings, $\mathbb{B} = \{ \emptyset, 1 \}$ (technically, these are characters, but we don't distinguish between characters and length-1 strings). The characteristic function is this.

$$\mathbb{1}_{\mathbb{B}}(\sigma) = \begin{cases} 1 & \text{if } \sigma = \emptyset \text{ or } \sigma = 1 \\ 0 & \text{otherwise} \end{cases}$$

The algorithm that implements that function is easy.

- (C) The language $\mathbb{B}^*$ contains all bitstrings. To solve the decision problem for this language, the algorithm returns 1 on all inputs.

V.3.19

- (A) For the algorithm, fix a Finite State machine $\mathcal{M}$ that recognizes the language described by a^*ba^* . Now, read the input σ and simulate running $\mathcal{M}$ on that input. This will terminate in a finite number of steps. If $\mathcal{M}$ ends in an accepting state then print 1 (or yes, or some other value signaling acceptance), otherwise print 0.
- (B) The grammar defines the language $\mathcal{L} = \{ a^n b^m \mid n, m \in \mathbb{N} \}$. The algorithm for that language is simple. Input the string and if all of the b's come after all of the a's then output 1 (or 'yes', or whatever indicates acceptance). Otherwise, output 0.

V.3.20 First observe that there is an easy algorithm that can, given a machine $\mathcal{M}$ and a state q in that machine, find the set of states reachable from q . The algorithm proceeds in steps. At step 1, initialize by taking R_1 to be the set of states that are reachable from q in one transition. Step $i + 1$ begins with the set of states that are reachable from q with i -many transitions or fewer. In that step, for each state in R_i , find if any states reachable from it in one transition are not in R_i . Make the set R_{i+1} by adding them all to R_i . If there are no such new states then the algorithm stops. Obviously it must stop at some point because the number of states in the machine is finite.

- (A) The language is empty if and only if from the start state, no accepting state is reachable.
- (B) The language is infinite if and only if there is a state q that can be reached from the initial state such that q can be reached from itself by a nonempty string, and some accepting state can be reached from q .
- (C) Given $\mathcal{M}$, obtain a new machine $\hat{\mathcal{M}}$ that recognizes the complement of $\mathcal{L}(\mathcal{M})$ by copying $\mathcal{M}$, except setting each accepting state of $\mathcal{M}$ to be non-accepting in $\hat{\mathcal{M}}$, and setting each non-accepting state of $\mathcal{M}$ to be accepting in $\hat{\mathcal{M}}$. Now, $\mathcal{L}(\mathcal{M})$ is empty if and only if $\mathcal{L}(\hat{\mathcal{M}})$ is empty.

V.3.21

- (A) See if the input string starts with a 1 bit.
- (B) See if the input string has length less than or equal to ten. If the length equals ten then there are some cases to do but that doesn't change that the algorithm is $\mathcal{O}(1)$.

V.3.22 This is a search problem, since we need only produce a single bridge.

V.3.23 As remarked in the section, a decision problem involves finding a function — a Boolean function. So in that sense, multiple answers are possible.

- (A) This is a yes-or-no question, so it is a decision problem.
- (B) Decision problem
- (C) Function problem.
- (D) Decision problem.
- (E) Search problem.
- (F) Decision problem.
- (G) Language decision problem, for $\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents a number that in decimal has only odd digits}\}$.

V.3.24

- (A) Decision problem
- (B) Search problem
- (C) Function problem
- (D) Function problem (although it is a search problem in that any satisfying assignment at all will do for output). The ‘F’ in F-SAT stands for ‘functional’.
- (E) Decision problem

V.3.25

- (A) A language suited to Turing machines is $\mathcal{L}_0 = \{1^n \mid n \in \mathbb{N} \text{ is a square}\}$. A language suited to more everyday machines is $\mathcal{L}_1 = \{d \in \{0, \dots, 9\}^* \mid d \text{ represents a perfect square in base 10}\}$.
- (B) One language is the set of triples $\langle x, y, z \rangle \in \{0, \dots, 9\}^*$ such that x, y, z represent integers in base 10 where the squares of the first two add to the square of the third.
- (C) A language is $\{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents a graph } \mathcal{G} \text{ with an even number of edges}\}$.
- (D) The language $\{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents a pair } \langle \mathcal{G}, p \rangle \text{ where } p \text{ is a path with a repeated vertex}\}$ will do.

V.3.26

- (A) $\{n \in \mathbb{N} \mid \text{the factors of } n \text{ sum to more than } 2n\}$.
- (B) $\{\langle \mathcal{P}, i \rangle \mid \text{machine } \mathcal{P} \text{ on input } i \text{ halts in less than ten steps}\}$
- (C) $\{E \mid \text{the logic expression } E \text{ can be satisfied by three distinct assignments}\}$
- (D) $\{\langle \mathcal{G}, B \rangle \mid \text{for all } v_0, v_1 \in \mathcal{G} \text{ there is a path of cost less than } B\}$

V.3.27

- (A) The Graph Colorability problem is stated as a decision problem. Imagine that we had an algorithm to recognize membership in this language.

$$\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents } \langle \mathcal{G}, k \rangle, \text{ where } \mathcal{G} \text{ is } k\text{-colorable}\}$$

Then, given an graph and an integer k , we could use that algorithm to decide whether the graph is k -colorable. Convert the graph and integer to a string representing the pair, and use the algorithm to decide if that pair is in the language.

- (B) Consider this.

$$\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents a graph with a path that traverses each edge exactly once}\}$$

The Euler Circuit problem is given as a function problem. If we had an algorithm to recognize membership in the language then, given an graph, we could use that algorithm to decide whether the graph has a circuit: we just convert the graph to its representation and apply the algorithm. If we know there is no Euler circuit then we are done. If we know there is an Euler circuit then a search will find it.

- (C) The Shortest Path problem is given as a function problem. If we had an algorithm to recognize membership in the language

$$\mathcal{L} = \{\sigma \in \mathbb{B}^* \mid \sigma \text{ represents } \langle \mathcal{G}, v_0, v_1, p \rangle, \text{ where } p \text{ is a minimal-length path between the vertices}\}$$

then, given an graph and two vertices, we could find the minimal path by using that algorithm to test all possible paths from one vertex to the other for minimality by seeing if the representation of the quad is in the language.

V.3.28 It is a matter of deciding membership in this language. $\{\langle \mathcal{P}_e, e \rangle \mid e \in \mathbb{N}\}$.

V.3.29 Take $B \in \mathbb{N}$ as a bound and consider this family.

$$\mathcal{L}_B = \{\langle \mathcal{G}, v_0, v_1 \rangle \mid \text{There is a path from } v_0 \text{ to } v_1 \text{ of distance less than or equal to } B\}$$

Given a graph and two vertices we can solve the **Shortest Path** problem for it by searching through $\mathcal{L}_1, \mathcal{L}_2, \dots$ until we find the smallest bound B for which our triple is a member of the language $\mathcal{L}_B$.

V.3.30

(A) Take $B \in \mathbb{N}$ to be a bound and suppose that we can solve this language decision problem, where $\mathcal{B}$ is a game board.

$$\mathcal{L}_B = \{\mathcal{B} \mid \mathcal{B} \text{ can be solved in fewer than } B \text{ slide moves}\}$$

Given a board, to find the minimum number of moves, iterate through $B = 0, B = 1$, etc., looking to see if the given board is a member of $\mathcal{L}_B$. Stop at the first B where it is in the language. That is the minimum.

(B) Take $B \in \mathbb{N}$ as a bound. Suppose that we have an algorithm to solve this language decision problem, where $\mathcal{R}$ is a Rubik's cube arrangement.

$$\mathcal{L}_B = \{\mathcal{R} \mid \mathcal{R} \text{ can be solved in fewer than } B \text{ cube moves}\}$$

Given a cube, to find its minimum number of moves, iterate through $B = 0, B = 1$, etc., until a B appears so that the given cube is in $\mathcal{L}_B$. That is the minimum number of moves.

(C) Take $B \in \mathbb{N}$. Suppose that we have an algorithm to solve this language decision problem, where S is a specification a car's assembly jobs.

$$\mathcal{L}_B = \{S \mid S \text{ can be accomplished in less time than } B\}$$

Given a specification, find the minimum time by iterating through $B = 0, B = 1$, etc., until a B appears so that the give specification is in $\mathcal{L}_B$.

V.3.31 A function version would input a graph and output a circuit (if there is no circuit then it would output something like the string None).

An optimization version would return a circuit that is best by some criterion. For instance, in a weighted graph it would return the circuit of least weight; this is the **Traveling Salesman** problem.

V.3.32

- (A) Search problem
- (B) Decision problem
- (C) Function problem
- (D) Optimization problem

A language suitable for an associated language problem is $\{M \mid M \text{ has a } 3 \times 3 \text{ invertible submatrix}\}$.

V.3.33

- (A) Given a triple $\langle n_0, n_1, p \rangle \in \mathbb{N}^3$, decide if $p = n_0 \cdot n_1$.
- (B) Given a triple $\langle \mathcal{G}, v_0, v_1 \rangle$, where $\mathcal{G}$ is a weighted graph and $v_0 \neq v_1$ are vertices in that graph, decide if v_1 is a vertex that is as near as possible to v_0 ,

V.3.34

- (A) Use the language $\{\langle s_0, \dots, s_{n-1} \rangle \in \mathbb{Q}^n \mid a_{i,0}s_0 + \dots + a_{i,n-1}s_{n-1} \leq b_i \text{ for all } i\}$.
- (B) It is a search problem as it is stated. We need only find one feasible sequence, if there is one.
- (C) The function could produce one feasible sequence, if one exists.
- (D) To make it an optimization function, the natural thing is to optimize a linear function $P(x_0, \dots, x_{n-1}) = c_0x_0 + \dots + c_{n-1}x_{n-1}$, for some $c_0, \dots, c_{n-1} \in \mathbb{Q}$.

V.3.35 A set with one vertex is independent, as is the empty set. So every graph has a subset of its vertices that is independent. Because of that, these answers at least state a version for some number of vertices $k \in \mathbb{N}$.

- (A) Given a graph, decide if it has an independent set with k vertices.
- (B) Use the language $\{\mathcal{G} \mid \mathcal{G} \text{ has } k \text{ vertices that do not share an edge}\}$.

- (c) Given a graph G , find an independent set with k -many vertices. (There may be more than one, or none, but we only look for one.)
- (d) Given a graph, return an independent set of size k or the string None.
- (e) Given a graph, return an independent set that is maximal, in that it is as large as possible for the graph.

V.3.36 Every list of numbers has a maximum, but finding it appears to mean at least looking through the list.

V.3.38 Imagine that we have a routine `shortest_path` that takes in a weighted graph and two vertices, and returns the shortest path between them. That also solves the decision problem of whether there is any path at all.

If the graph is not weighted, convert it to a weighted graph by taking each edge to be of weight 1.

V.3.39 By taking the numbers in the set $S = \{s_0, \dots, s_{n-1}\}$ one at a time: $S_0 = \{s_0\}$, then $S_1 = \{s_0, s_1\}$, $\dots$, you can figure out from the decision problem solver what is the largest number s_j in the sum. Subtract that from the goal and then iterate using $S - \{s_j\}$.

V.3.40 Check all pairs of vertices.

- V V.4.12** True. Such problems have solution algorithms that are basically a fixed number of if ... then ... branches, so there is a maximum number of steps that any input can trigger. Thus, the solution algorithm is $\mathcal{O}(1)$.
- V V.4.13** Algorithms are not in P ; rather, problems are in P . Specifically, a problem is in P if from among all the algorithms that solves it, at least one runs in polynomial time, is $\mathcal{O}(n^c)$ for some constant c . So, better is to say something like, "I've got a problem whose solution algorithm takes polytime." (It is absolutely right that we are not too careful to distinguish between problems, meaning language decision problems, and languages. But problems and algorithms are very different.)
- V V.4.14** An order of growth is a set of functions. A complexity class is a set of problems, language decision problems.
- V V.4.15** Your friend is observing that the circuit representation may have length exponential in the circuit depth, since an r -deep circuit may have 2^r many vertices. But the definition of polynomial time is that the number of steps taken is a polynomial function of the size of the input representation. If the representation is very large then that only means that the polynomial receives a large input. That is, we are not comparing the time taken to the depth of the circuit, but rather to the size of the representation.
- V V.4.16** The student's machine accepts all strings. The definition requires a machine that determines whether the input string is in the language.
- V.4.17** True.
- V V.4.18** True, although not by the same machine. Let $\mathcal{L}$ be computed by $\mathcal{P}$, so that the machine has this input-output behavior.

$$\phi_{\mathcal{P}}(\sigma) = \mathbb{1}_{\mathcal{L}}(\sigma) = \begin{cases} 1 & \text{-- if } \sigma \in \mathcal{L} \\ 0 & \text{-- otherwise} \end{cases}$$

Modify the machine by interchanging the output 0's and 1's, to get a new machine $\hat{\mathcal{P}}$ with the complementary behavior.

$$\phi_{\hat{\mathcal{P}}}(\sigma) = \mathbb{1}_{\mathcal{L}^c}(\sigma) = \begin{cases} 0 & \text{-- if } \sigma \in \mathcal{L} \\ 1 & \text{-- otherwise} \end{cases}$$

- V V.4.19** We must produce an algorithm that runs in polytime. Given the input σ , find its length. If the length is not divisible by three then return 0. Otherwise, find the substring τ that is the first third of σ , and check whether $\tau \cap \tau \cap \tau = \sigma$. If so, return 1, and if not, return 0.
- V V.4.20** The algorithm is straightforward. Given σ , for $i \in \{0, 1, \dots, |\sigma| - 1\}$, read the characters $\sigma[i]$ and compare them to the corresponding character read from the back, $\sigma[(|\sigma| - 1) - i]$. (You can stop once you get through the halfway point, $i = \lfloor |\sigma|/2 \rfloor$.) If each comparison gives equality then return 1, else return 0. This algorithm is polynomial since it is basically a loop.
- V.4.21** In each case we must produce an algorithm that runs in polynomial time.

- (A) Given σ , decide if its length is divisible by three. If not, return $\emptyset$. If it is divisible by three, set τ to be the substring that is the first third of σ . Then just check, by moving through the string once, whether $\sigma = \tau^3$.
- (B) Stated as a language decision problem, we have $\{i \in \mathbb{N} \mid \mathcal{P}_i \text{ halts on the empty string in 10 steps}\}$. There is some overhead in going from the input i to the Turing machine $\mathcal{P}_i$, simulating it with a universal Turing machine, etc. But putting aside that overhead, then no matter what is the input i , we then just run for ten steps. So it is an element of P .

V V.4.22

- (A) Because the size of the clique is fixed. In the Clique problem, the size of the clique is a parameter, so that we are given a pair $\langle G, B \rangle$ and asked if the graph has a B -sized clique.
- (B) Given G , enumerate all triples $\langle v_0, v_1, v_2 \rangle \in \mathcal{N}^3$ and check whether all three of v_0v_1 , v_0v_2 , and v_1v_2 are edges, members of $\mathcal{E}$. This takes polynomial time. Specifically, it is $\mathcal{O}(|\mathcal{N}|^3 \cdot |\mathcal{E}|)$. Because $|\mathcal{E}| \leq |\mathcal{N}|^2$, the algorithm is $\mathcal{O}(|\mathcal{N}|^5)$.

V V.4.23

- (A) To determine whether the string is in alphabetical order, a single pass through it suffices. We can assume that each comparison takes constant time. So, overall this one-pass algorithm takes polytime.
- (B) The elementary school algorithm of adding in columns from the least significant digit (or bit) and then off to the left shows that the runtime of addition is proportional to the number of digits in the larger of the two input numbers (or the number of bits, which is $\lg(10)$ times the number of digits). We can do addition of individual digits (or bits) with a lookup table, so that step happens in constant time. Therefore, overall, addition takes time proportional to the length of the input representation, which is polytime.
- (C) The naive algorithm for matrix multiplication gives a triply nested loop, and so is $\mathcal{O}(n^3)$ where n is the largest of the number of rows or columns in either of A or B .
- (D) As mentioned in Problem 2.40, determining whether a number is composite or prime is known to be possible in polytime.
- (E) We can give an algorithm that runs in time polynomial in the length of the graph description. Fix the set $S_0 = \{v_0\}$. Next, follow each edge leading out of v_0 , to get the set of vertices that are connected to v_0 in at most one step, $S_1 = S_0 \cup \{v \mid \langle v_0, v \rangle \in \mathcal{E}\}$. After that, for all the vertices in S_1 , follow all the edges to get the set $S_2 = S_1 \cup \{v \mid \langle \hat{v}, v \rangle \in \mathcal{E} \text{ where } \hat{v} \in S_1\}$ of all vertices that are connected to v in at most two steps. This process must stop at some point, with $S_{i-1} = S_i$, because the graph is finite. Then, either $v_1 \in S_i$ or not, which determines whether there is a path from v_0 to v_1 . The number of steps in this algorithm, i , is bounded by the number of vertices in the graph, so the algorithm is polynomial in the size of the graph representation.

V V.4.24

- (A) By Dijkstra's algorithm, the Shortest Path problem can be done in polynomial time. So the problem is in P .
- (B) Knapsack is believed to be quite hard. No one knows of a polynomial algorithm.
- (C) Euler Path is easy; we just need to check whether each node has even degree. There are algorithms that are linear.
- (D) No one knows of a polynomial algorithm for Hamiltonian Circuit.

V.4.25 Yes. The algorithm that takes in a single input and returns 0 computes the characteristic function of the empty language, and clearly runs in polytime.

V V.4.26

To show it is in P we must produce a solution algorithm for it that runs in polytime. For that, start by picking a vertex, v . Let the set S_0 equal $\{v\}$. Follow each edge leading out of v to get all of the vertices that are connected to v in at most one step, and call that set $S_1 \supseteq S_0$. Iterate: for all the vertices in S_1 , follow all the edges to get the set S_2 of all of the vertices that are connected to v in at most two steps. Clearly this process must stop at some point, with $S_{i-1} = S_i$, because the graph is finite.

This algorithm takes a number of steps, i , that is bounded by the number of vertices in the graph, and so the algorithm is polynomial in the size of the graph.

V V.4.27

- (A) Our definition of complexity class is completely general—it is a collection of languages. So under the definition that we gave, the collection of regular languages is a complexity class.
As to the additional condition, a language is regular if it is accepted by a Finite State machine. So the

regular languages are computed by a device. Further, a Finite State machine meets the constraint that it runs in a number of steps that is equal to the length of its input, σ .

- (B) The question notes that we can define regular languages using a type of Turing machine. What remains is to show that each regular language is computable in polytime. As in the prior item, the fact that such a machine consumes its input and never writes to the tape means that it runs in linear time, $\mathcal{O}(|\sigma|)$.

V V.4.28

- (A) It is a complexity class because it is a collection of languages. In addition, each such language $\mathcal{L}$ satisfies that there is a computable function ϕ_e with $\mathcal{L} = \{k \in \mathbb{N} \mid k = \phi_e(i) \text{ for some input } i \in \mathbb{N}\}$.
 (B) The resource specification is that there is no finite time bound.

V.4.29 There are countably many Turing machines to use as deciders, so P is countable.

V.4.30 No. Take $\mathcal{L}_0 = \emptyset$ and $\mathcal{L}_1 = \mathcal{P}(\mathbb{B}^*)$. Because P is countable, as there are only countably many Turing machines to use as deciders, and the collection of all languages $\mathcal{P}(\mathbb{B}^*)$ is uncountable, there are uncountably many languages with $\mathcal{L}_0 \subseteq \mathcal{L} \subseteq \mathcal{L}_1$ that are not in P .

- V V.4.31** No. Certainly we can state the Halting problem as a language decision problem, using the language $\mathcal{L} = \{n \in \mathbb{N} \mid \phi_n(n) \downarrow\}$. But to be a member of P , there must be a Turing machine that solves the problem (and that runs in polytime). No Turing machine solves the Halting problem, so it is not in P .

- V V.4.32** A language is a member of P if there is an algorithm for it that runs in polytime. That is, there is a Turing machine that computes the characteristic function of the language in polytime. Clearly, existence of such a machine is equivalent to the existence of a machine with a designated acceptance state (and that runs in polytime). So the set of languages in P is a subset of the set of all decidable languages.

- V V.4.33** Here is the desired input-output behavior for the circuit.

b_2	b_1	b_0	$b_2 + b_1 + b_0$	$b_2 + b_1 + b_0 \pmod{2}$
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	2	1
1	0	0	1	0
1	0	1	2	1
1	1	0	2	1
1	1	1	3	0

This propositional logic formula gives the behavior in that table.

$$(\neg b_2 \wedge \neg b_1 \wedge \neg b_0) \vee (\neg b_2 \wedge b_1 \wedge b_0) \vee (b_2 \wedge \neg b_1 \wedge b_0) \vee (b_2 \wedge b_1 \wedge \neg b_0)$$

Figure 96 on page 183 shows the circuit, using the standard notation of $\sqcap$ for ‘and’ gates, $\supset$ for ‘or’ gates, and $\triangleright$ for ‘not’ gates. But this circuit does not perfectly match the problem since some gates do not have two inputs and one output. Figure 97 on page 183 modifies the prior diagram to draw the circuit as an acyclic directed graph, by folding the ‘not’ gates into the boolean binary functions. These are the Boolean functions used there.

P	Q	$\wedge$	P	Q	$\vee$	P	Q	f_0	P	Q	f_1	P	Q	f_2
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	1	0	0	1	1	0	1	0	0	1	0	0	1	1
1	0	0	1	0	1	1	0	0	1	0	1	1	0	0
1	1	1	1	1	1	1	1	0	1	1	0	1	1	0

So, f_0 is equivalent to $\neg P \wedge \neg Q$, and f_1 is equivalent to $P \wedge \neg Q$, and f_2 is equivalent to $\neg P \wedge Q$.

- V V.4.34** A complexity class is just a set, a collection of languages. So the union of two is also a complexity class. The same holds for the intersection, and for the complement (where the complement is taken inside the universe $\mathcal{P}(\mathbb{B}^*)$).
- V V.4.35** Let two languages $\mathcal{L}_0, \mathcal{L}_1 \in \mathcal{P}(\mathbb{B}^*)$ be elements of P . Then there are computable functions to determine membership in the two with runtimes $\mathcal{O}(x^{n_0})$ and $\mathcal{O}(x^{n_1})$. The algorithm to compute membership in the

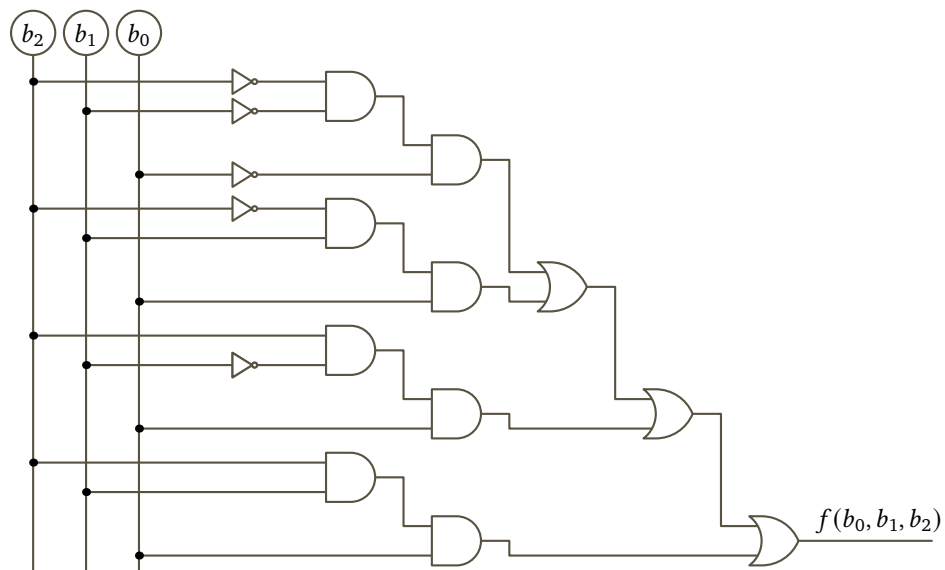


FIGURE 96, FOR QUESTION V.4.33: Initial circuit for the Circuit Evaluation problem.

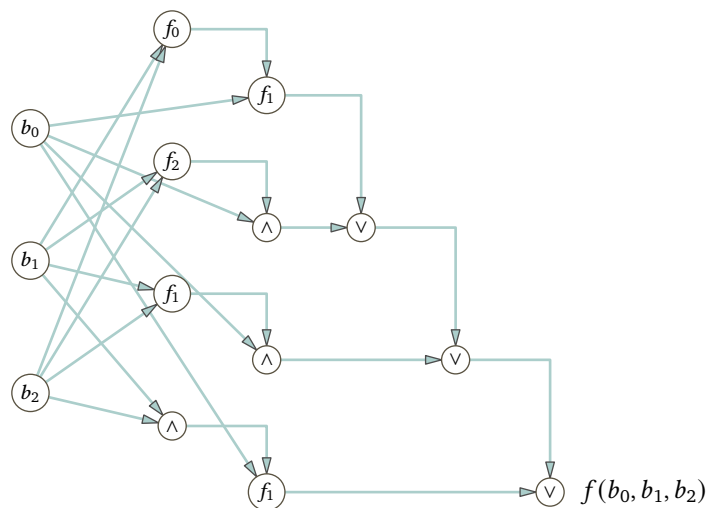


FIGURE 97, FOR QUESTION V.4.33: Directed acyclic graph for the Circuit Evaluation problem.

union gets as input a string $\sigma \in \mathbb{B}^*$. It first checks whether the string is a member of the first language, and then checks whether it is a member of the second. Together, that takes a running time of $\mathcal{O}(x^{\max(n_0, n_1)})$, which means that the algorithm runs in polytime.

The argument for the union of any finite number of languages proceeds by induction and is straightforward.

- V **V.4.36** Assume that $\mathcal{L} \in P$. Then there is a deterministic Turing machine $\mathcal{P}$ that runs in polynomial time, and that accepts an input string $\sigma \in \mathbb{B}^*$ if and only if $\sigma \in \mathcal{L}$. Restated, there is a computable function $f: \mathbb{B}^* \rightarrow \mathbb{B}^*$ such that

$$f(\sigma) = \begin{cases} 0 & \text{if } \sigma \notin \mathcal{L} \\ 1 & \text{if } \sigma \in \mathcal{L} \end{cases}$$

and such that computing f is $\mathcal{O}(p)$ for some polynomial p . (Specifically, what is $\mathcal{O}(p)$ is the function $t(n)$, defined as the maximum runtime of $f(\tau)$ over all $\tau \in \mathbb{B}^*$ with $n = |\tau|$.)

Then clearly this function is also computable, and also in polytime: $\bar{f}(\sigma) = 1 - f(\sigma)$. Consequently, $\mathcal{L}^c \in P$.

- V **V.4.37** Fix a language $\mathcal{L} \in P$. Then there is a Turing machine that computes the characteristic function $f: \mathbb{B}^* \rightarrow \mathbb{B}^*$ of $\mathcal{L}$, with runtime $\mathcal{O}(x^n)$ for some power $n \in \mathbb{N}$. To compute the characteristic function of the reversal, $\mathcal{L}^R = \{\tau^R \mid \tau \in \mathcal{L}\}$, the algorithm is given an input string σ . It constructs the reversal, σ^R , and decides whether $\sigma^R \in \mathcal{L}$. The runtime is: the algorithm can construct the reversal from the input in one pass through the input, and deciding whether that reversal string is in $\mathcal{L}$ is $\mathcal{O}(x^n)$, to in total the runtime is $\mathcal{O}(x^{n+1})$.
- V **V.4.38** Consider two languages, $\mathcal{L}_0, \mathcal{L}_1 \in \mathcal{P}(\mathbb{B}^*)$, that are members of P . Then there are powers $n_0, n_1 \in \mathbb{N}$ so that $\mathcal{L}_0$ is accepted by a Turing machine with runtime $\mathcal{O}(x^{n_0})$, and $\mathcal{L}_1$ is accepted by a Turing machine with runtime $\mathcal{O}(x^{n_1})$.

Here is the algorithm to decide if a given input string $\sigma \in \mathbb{B}^*$ is a member of $\mathcal{L}_0 \cap \mathcal{L}_1$. We will pass through the string, first decomposing it as $\sigma = \varepsilon \sigma$, then as $\sigma = \sigma[0] \cap \sigma[1:]$, etc. At each stage of this pass we have $\sigma = \alpha \cap \beta$ and we check whether $\alpha \in \mathcal{L}_0$ and $\beta \in \mathcal{L}_1$.

Checking whether $\alpha \in \mathcal{L}_0$ takes time $\mathcal{O}(|\alpha|^{n_0})$, and checking whether $\beta \in \mathcal{L}_1$ takes time $\mathcal{O}(|\beta|^{n_1})$. It simplifies things to overestimate and use $|\sigma|$, rather than worry about the length of the substrings. Hence each step of the pass can be done in $\mathcal{O}(|\sigma|^{\max(n_0, n_1)})$ time. The pass itself takes $|\sigma| + 1$ many steps, so overall the algorithm runs in polytime, $\mathcal{O}(|\sigma|^{1+\max(n_0, n_1)})$.

- V.4.39 We are given a language $\mathcal{L} \in \mathcal{P}(\mathbb{B}^*)$ such that there is a Turing machine, $\mathcal{P}_{\mathcal{L}}$, that decides the language in polytime, in time that is some power, $n \in \mathbb{N}$, of the length of the input string.

We will follow the hint that $\sigma \in \mathcal{L}^*$ if $\sigma = \varepsilon$, or $\sigma \in \mathcal{L}$, or $\sigma = \alpha \cap \beta$ for some $\alpha, \beta \in \mathcal{L}^*$. Rather than present a flowchart of pseudocode, we will work through the example of $\sigma = \langle s_0, s_1, s_2, s_3 \rangle$. This table lists the substrings $\sigma[i:j]$.

	$j = 0$	$j = 1$	$j = 2$	$j = 3$
$i = 0$	$\langle s_0 \rangle$	$\langle s_0 s_1 \rangle$	$\langle s_0 s_1 s_2 \rangle$	$\langle s_0 s_1 s_2 s_3 \rangle = \sigma$
$i = 1$		$\langle s_1 \rangle$	$\langle s_1 s_2 \rangle$	$\langle s_1 s_2 s_3 \rangle$
$i = 2$			$\langle s_2 \rangle$	$\langle s_2 s_3 \rangle$
$i = 3$				$\langle s_3 \rangle$

We start on the main diagonal, which holds all of the length one substrings. We can determine whether each of these is a member of $\mathcal{L}$ by using $\mathcal{P}_{\mathcal{L}}$. If it is a member, put a checkmark next to the substring.

Next we move to the diagonal above, with the length two substrings. We can determine whether each is a member of $\mathcal{L}$ by using $\mathcal{P}_{\mathcal{L}}$, or, taking the hint, by knowing whether both length one substrings are members of $\mathcal{L}^*$ (which we can do quickly using the checkmarks). Again, when a substring in this diagonal is a member then checkmark it.

Now iterate. On the diagonal containing the length three substrings, we can determine whether each is a member of $\mathcal{L}^*$ by first asking $\mathcal{P}_{\mathcal{L}}$ if it is a member of $\mathcal{L}$. If not, we then determine whether it is the

concatenation of two substrings. We've already checked or not checked all the members of both of the earlier diagonals, so determining that is very fast.

In this way, we proceed through diagonals until we get to the table's upper right. It holds the given string, σ , so we can determine whether $\sigma \in \mathcal{L}^*$ in the same way.

Now for the runtime analysis. Let $m = |\sigma|$. This is a triply-nested loop. The outer loop moves through the diagonals, from the length one substrings to the length m substring σ . Inside that is a loop that moves through a particular diagonal, starting at $i = 0$, and covering at most m substrings. Finally, inside of that is a loop that runs the machine $\mathcal{P}_{\mathcal{L}}$ and then possibly loops through the substring pairs. The runtime on this inner loop is $\mathcal{O}(m^n) + \mathcal{O}(m)$, and so in total we have $\mathcal{O}(m) \cdot \mathcal{O}(m) \cdot (\mathcal{O}(m^n) + \mathcal{O}(m))$, which is polynomial in $m = |\sigma|$.

- V **V.4.40** Suppose for contradiction that we could solve this problem, that there is a Turing machine that, given $\mathcal{P}$ as input, decides whether it runs in polytime on the empty input. We will show how to, from $\mathcal{P}$, construct a new machine $\mathcal{N}_{\mathcal{P}}$ that runs in polynomial time if and only if $\mathcal{P}$ does not halt. That is a contradiction, because then being able to test machines for running in polytime allows us to solve the Halting problem.

The machine $\mathcal{N}_{\mathcal{P}}$ takes as input bitstrings. For input σ , it uses a universal Turing machine to run $\mathcal{P}$ on the empty tape. If $\mathcal{P}$ does not halt within $|\sigma|$ steps, then $\mathcal{N}_{\mathcal{P}}$ halts. If $\mathcal{P}$ does halt within that time then $\mathcal{N}_{\mathcal{P}}$ will run for $2^{|\sigma|}$ many steps. Clearly, if $\mathcal{P}$ halts then $\mathcal{N}_{\mathcal{P}}$ runs in exponential time, and if $\mathcal{P}$ does not halt then $\mathcal{N}_{\mathcal{P}}$ runs in polynomial time, as required.

V.4.41

- (A) Recall that a function is a kind of relation. Assume that there are computable functions $f_0, f_1: \mathbb{N} \rightarrow \mathbb{N}$ with associated Turing machines $\mathcal{P}_0$ and $\mathcal{P}_1$ that, given $a \in \mathbb{N}$, can find $f_0(a)$ or $f_1(a)$ in times $\mathcal{O}(x^{n_0})$ and $\mathcal{O}(x^{n_1})$, for some powers $n_0, n_1 \in \mathbb{N}$.

For addition the algorithm can, given $a \in \mathbb{N}$, compute $f_0(a)$ and $f_1(x)$ in time $\mathcal{O}(x^{\max(n_0, n_1)})$. The grade school addition algorithm brings the time up to $\mathcal{O}(x^{1+\max(n_0, n_1)})$.

The scalar multiplication algorithm can, given $a \in \mathbb{N}$, compute $f_0(a)$ in time $\mathcal{O}(x^{n_0})$. The naive multiplication algorithm brings the time up to $\mathcal{O}(x^{2+n_0})$.

To subtract we can use a branch that tests if the second number is greater than the first, and if so returns 0. In any event, clearly polytime.

Multiplication is similar, and also in polytime.

For composition, if f_0 is $\mathcal{O}(x^{n_0})$ and f_1 is $\mathcal{O}(x^{n_1})$ then the composition is $\mathcal{O}(x^{n_0 n_1})$, also polytime.

- (B) Here is the naive algorithm. The solution algorithm gets a string σ . it can determine in polytime that $\sigma = \text{str}(\langle a, y \rangle)$ for some $a, y \in \mathbb{N}$ (if σ does not have that form then the algorithm rejects it). Now, the algorithm uses $\mathcal{P}_0$ to determine whether $f_0(a) = 0, f_0(a) = 1, \dots, f_0(a) = y$. If none of those relations are true then the algorithm rejects σ . Otherwise, let $k \in \mathbb{N}$ be such that $f(a) = k$. The algorithm uses TM_1 to determine whether $f_1(a) = y - k$. If so, the algorithm accepts σ , otherwise it rejects.

The reason it is pseudopolynomial and not properly polynomial lies in the testing of the $y + 1$ many pairs. Inside of the input σ , the number y is represented in binary. So taking $y + 1$ steps is taking time that is exponential in the length of the input.

- V.4.42 Because $\mathcal{L}_0 \in P$, there is a Turing machine with runtime $\mathcal{O}(x^n)$ for some power $n \in \mathbb{N}$, and whose input/output behavior is the characteristic function of $\mathcal{L}_0$. Because $\mathcal{L}_1 \leq_p \mathcal{L}_0$, there is a Turing machine whose runtime is $\mathcal{O}(x^m)$ for some $m \in \mathbb{N}$, whose input/output behavior is the function $f: \mathbb{N} \rightarrow \mathbb{N}$, and such that $\sigma \in \mathcal{L}_1$ if and only if $f(\sigma) \in \mathcal{L}_0$.

So, to decide questions about whether $\sigma \in \mathcal{L}_1$, first translate them, at a cost of $\mathcal{O}(|\sigma|^m)$, to questions about whether $f(\sigma) \in \mathcal{L}_0$. Those questions we can settle in $\mathcal{O}(|\sigma|^n)$. The result, $\mathcal{O}(|\sigma|^{\max(n, m)})$, is polytime.

- V **V.5.13** That lemma requires that the verifier have the property that if $\sigma \in \mathcal{L}$ then there is a witness leading to acceptance, while if $\sigma \notin \mathcal{L}$ then there exists no witness that would cause the verifier to accept.

For instance, consider the problem $\mathcal{L} = \{n \in \mathbb{N} \mid n \text{ is even}\}$. Your study partner's proposed verifier inputs a pair $\langle \sigma, \omega \rangle$, where σ is a number and ω is a bit, and then accepts the pair if $\omega = 1$. Absolutely, if you give your verifier the pair $\langle 101, 0 \rangle$ then it will not accept (here, 101 represents the number 5, and is not an element of $\mathcal{L}$). However, give it $\langle 101, 1 \rangle$ and your verifier will accept. So, for $\sigma = 101$ there exists an ω leading to acceptance. Consequently, your verifier is wrong.

- V.5.14 It is none. The negation is that every branch fails to accept. That does not mean that every branch

rejects; some of them could be computations that never halt.

V.5.15

(A) This formula is satisfiable.

P	Q	R	$P \wedge Q$	$\neg Q$	$\neg Q \wedge R$	$(P \wedge Q) \vee (\neg Q \wedge R)$
F	F	F	F	T	F	F
F	F	T	F	T	T	T
F	T	F	F	F	F	F
F	T	T	F	F	F	F
T	F	F	F	T	F	F
T	F	T	F	T	T	T
T	T	F	T	F	F	T
T	T	T	T	F	F	T

(B) This formula is not satisfiable.

P	Q	R	$P \rightarrow Q$	$P \wedge Q$	$\neg P$	$(P \wedge Q) \vee \neg P$	$\neg((P \wedge Q) \vee \neg P)$	$(P \rightarrow Q) \wedge \neg((P \wedge Q) \vee \neg P)$
F	F	F	T	F	T	T	F	F
F	F	T	T	F	T	T	F	F
F	T	F	T	F	T	T	F	F
F	T	T	T	F	T	T	F	F
T	F	F	F	F	F	F	T	F
T	F	T	F	F	F	F	T	F
T	T	F	T	T	F	T	F	F
T	T	T	T	T	F	T	F	F

V.5.16 True, by Lemma 5.6.

V.5.17 The machine accepts σ because there is at least one accepting branch.

V.5.18

- (A) Backtracking will solve the problem. But it is a deterministic algorithm. (Technically, a deterministic algorithm is a special case of a nondeterministic one, but when you ask about it, your professor says, “The point of this exercise is to demonstrate understanding of nondeterminism, not to quibble like a sea lawyer.”)
- (B) Your algorithm chooses, but it does so at random. If there are paths through the maze, but the random choices happens not to fall on such a path, then your algorithm will not detect them. A nondeterministic algorithm is required to output 1 if there is a path (and not to output 1 if there is not).
- (C) We will give two answers, one using the unbounded parallelism imagery and one using choosing.

The unbounded parallelism algorithm has a computational history that is a tree. Whenever the algorithm finds itself at a place in the maze where it can do more than one thing, it forks child processes to handle each possibility. So it may have lots of branches, and lots of them may go to dead ends, but if there is a way through the maze then this algorithm will find it. So it reports 1 if there is a path, and never reports 1 when there is not.

The choosing algorithm is to choose. That is, the machine guesses at the correct path (and then verifies it against the input maze). By definition, if there is a way for this guess to be right then the algorithm is right. (Note again that this algorithm is not guessing at random. It is guessing correctly.)

V.5.19 The algorithm is given an unordered array $\langle a_0, a_1, \dots, a_{n-1} \rangle$. It returns Yes if $k = a_i$ for some index i .

In the unbounded parallelism model, the computation tree of the algorithm is a single node with n children, each of which is a process. Child i checks whether $a_i = k$, and if so causes the algorithm to output Yes.

The guessing model is much the same, except that the machine does not compute the tree. It guesses at a child process, and if the array entry equals k then it causes the algorithm to output Yes. This model, by definition, is the same as the other because it says that an algorithm is right if there is a way for the guess to correctly result in Yes, which will happen if and only if there is an array entry that equals k .

V.5.20

- (A) A deterministic algorithm is to try every permutation, one after the other, until the program recognizes the result as correctly decoding.

- (b) The unbounded parallelism model is to do the same, except that it tries them simultaneously. That is, the computation tree has a single root node with one child process for each of the $26!$ -many permutations. That child takes its single permutation, substitutes it, and then checks it. This machine has $403\,291\,461\,126\,605\,635\,584\,000\,000 - 1$ child processes that fail the check. But one of the substitutions is right, so overall you get the correctly decoded string.
- (c) The guessing model guesses. It guesses a permutation (or is given it in a hint from the demon), and then it works like any of the prior item's processes: it makes the substitution and then applies the program to check it. This is correct because under this model of nondeterminism it just needs to be possible to have a guess that checks out.

V **V.5.21** First the unbounded parallelism description, which works by brute force. Where the number of vertices in the graph is k , the number of colorings is bounded, by 4^k . The computation tree has a single root node with that many branches. Each branch tries its coloring and then checks to see whether it is correct, that is, whether all connected vertices are different colors. If there is a correct coloring then this algorithm will find it and output Yes.

In the second model, the machine does not spawn 4^k -many branches. The demon gives the machine a branch and the machine just verifies that all connected vertices are different colors. If a correct coloring exists for this graph then there exists a hint that results in the machine verifying a correct coloring and outputting Yes, so this satisfies the definition of a guessing algorithm working.

V **V.5.22** We recast this as a sequence of language decision problems for a family of languages parametrized by the bound $B \in \mathbb{N}$.

$$\mathcal{L}_B = \{ \langle x_0, x_1, \dots, x_n \rangle \in \mathbb{Z}^{n+1} \mid \text{the } x_j \text{ satisfy all the constraints and } f(x_0, x_1, \dots, x_n) \leq B \}$$

An unboundedly parallel algorithm tries all possible assignments, in some way guaranteed to try every assignment eventually (as with Cantor's enumeration). If there is a satisfying assignment then this algorithm will find it and accept that branch. If there is no satisfying assignment then the computation history tree is infinite, but with no accepting branch.

A guessing algorithm guesses an assignment $\langle x_0, \dots, x_n \rangle$ and then verifies that the assignment has $f(x_1, \dots, x_n) \leq B$. If there is a satisfying assignment then there is a way to guess it so, by definition, this machine recognizes the language $\mathcal{L}_B$.

V **V.5.23** The language is $\mathcal{L} = \{ n \in \mathbb{N} \mid n \text{'s prime factorization has exactly two nontrivial primes} \}$.

One unbounded parallelism algorithm tries each conceivable prime factorization, simultaneously. For instance, given the input 8, it could check the products $2^0 3^0 5^0 7^0$, $2^1 3^0 5^0 7^0$, $\dots$, $2^7 3^7 5^7 7^7$, to see whether any of them equals 8 and nontrivially involves exactly two primes. Multiplying is fast and checking whether a number is composite or prime can also be done in polytime. Thus each branch check happens in polytime.

A guessing algorithm is that the machine inputs the number n and guesses a two-prime factorization (or is given one by a demon). For instance, given 45, the machine might guess $3^2 5^1$. (It could also guess $3^1 5^2$, but that won't verify.) Again, the verification is polytime because multiplication is polytime, as is checking whether the two numbers are composite or prime. By definition, the machine recognizes the language $\mathcal{L}$ if, given $n \in \mathbb{N}$, when $n \in \mathcal{L}$ then there exists a guess that verifies, and when $n \notin \mathcal{L}$ then there is no way for a guess to verify. So this algorithm recognizes $\mathcal{L}$.

V **V.5.24**

(A) This is the language.

$$\mathcal{L} = \{ M \in X \times Y \times Z \mid M \text{ has an } n\text{-element subset } \hat{M} \text{ where no triples agree on any coordinate} \}$$

A nondeterministic algorithm, framed in terms of guessing, is: given M , the machine guesses a size n subset $\hat{M}$ meeting the conditions. Verifying the conditions clearly takes only polytime. If there is such an $\hat{M}$ then there is a way for the machine to guess it, and it will verify. If there is no such $\hat{M}$ then there is no way for the machine to guess one that verifies. So by the definition of this model of nondeterminism, the machine recognizes the language in polytime.

(B) Here is the language.

$$\mathcal{L} = \{ A \mid A \text{ is a multiset of natural numbers with an } \hat{A} \subseteq A \text{ so that } \sum_{a \in \hat{A}} a = \sum_{a \in A - \hat{A}} a \}$$

A nondeterministic algorithm stated in guessing terms is that, when asked to determine if the given A is a member of $\mathcal{L}$, the machine guesses at a $\hat{A}$. (That is, $\hat{A}$ is the witness.) Verifying that $\sum_{a \in \hat{A}} a = \sum_{a \notin \hat{A}} a$ is clearly polytime. If there is such a partition then for this machine there exists a guess that would pass verification. If there is no such partition then no guess will verify. So by the definition, this machine recognizes the language $\mathcal{L}$.

- V **V.5.25** For unbounded parallelism, we can try every possible B -coloring of the graph's nodes. There are a large but finite number of those (where the graph has k vertices, there are B^k -many colorings). So the computation tree has a single node with that many children. Each coloring is easy to check for validity, since we just have to check every pair of vertices to see if they are the same color. If any of the child nodes is OK, then there is a B -coloring.

For guessing, the machine nondeterministically selects one coloring (that is, it guesses one or is given one by a demon). It then checks, by iterating through every vertex pair to see if they are the same color. If there is a way to correctly nondeterministically guess then by definition this algorithm succeeds.

- V **V.5.26**

(A) This is the language $\mathcal{L}$.

$$\{\sigma \mid \sigma \text{ represents a Propositional Logic statement where at least two lines of the truth tables end in } T\}$$

These are suitable fill-ins: (1) a Propositional Logic statement, (2) a pointer to two lines of the truth table, (3) for those two truth table lines, the statement returns T , (4) two truth table lines that cause the expression σ to evaluate to T , and (5) pair of truth table lines.

(B) Here is the problem cast as the decision problem for a language.

$$\mathcal{L} = \{\langle S, T \rangle \mid \text{a subset of } A \subseteq S \text{ has } \sum_{a \in A} a = T\}$$

These are fill-ins: (1) a set and number pair $\langle S, T \rangle$, (2) a set of numbers, (3) that is a subset of the given S and that its elements sum to T , (4) a subset of S whose elements sum to T , and (5) subset of S .

- V **V.5.27** Lemma 5.9 requires that we produce a deterministic Turing machine verifier, $\mathcal{V}$. It must input pairs of the form $\langle \sigma, \omega \rangle$, where $\sigma = \langle s_0, \dots, s_5, T \rangle$. It must have the property that if $\sigma \in \mathbf{CD}$ then there is a ω such that $\mathcal{V}$ accepts the input, while if $\sigma \notin \mathbf{Countdown}$ then there is no such witness ω . And it must run in time polynomial in $|\sigma|$.

The witness ω is an arithmetic expression (or, more precisely, a string representation of an arithmetic expression) that evaluates to the target T and that involves a subset of the six numbers $s_0, \dots, s_5$, each used at most once. The verifier checks that the expression does indeed evaluate to the target, and does indeed use each of the six numbers at most once. Clearly those checks can be done in polytime.

If $\sigma \in \mathbf{CD}$ then by definition there is an arithmetic expression, and so a witness ω exists that will cause $\mathcal{V}$ to accept the input. If $\sigma \notin \mathbf{CD}$ then there is no such arithmetic expression, and therefore no witness ω will cause $\mathcal{V}$ to accept.

- V **V.5.28** We recast the **Traveling Salesman** problem from an optimization problem by using bounds $B \in \mathbb{N}$ to get this parametrized set of bitstrings.

$$\mathcal{TS}_B = \{G \in \mathbb{B}^* \mid G \text{ is a weighted graph with a circuit of length less than } B\}$$

To show that this problem is in NP , Lemma 5.9 requires that we produce a deterministic Turing machine verifier, $\mathcal{V}$. It must input pairs of the form $\langle \sigma, \omega \rangle$, where σ is a graph, $\mathcal{G}$ (technically, a bitstring representation of a graph). The verifier must have the property that if $\sigma \in \mathcal{TS}_B$ then there is a ω such that $\mathcal{V}$ accepts the input, while if $\sigma \notin \mathcal{TS}_B$ then there is no such ω . This verifier must run in time polynomial in $|\sigma|$.

For a witness we can use a circuit $\omega = \langle v_{i_0}, v_{i_1}, \dots, v_{i_k} \rangle$ that is of total cost less than the bound. The verifier gets an input pair $\langle \sigma, \omega \rangle$ where σ is a weighted graph, and checks that ω is indeed a Hamiltonian circuit of the graph, and that it costs less than the bound. A verifier can do those checks in time polynomial in $|\sigma|$. If the graph σ is indeed an element of $\mathcal{TS}_B$ then there is such an ω , so there is a pair $\langle \sigma, \omega \rangle$ that the verifier can accept. If σ is not an element of $\mathcal{TS}_B$ then there is no such ω , and so the verifier will never accept such a pair.

V **V.5.29** The Independent Sets problem is the decision problem for this language.

$$\mathcal{L} = \{ \langle \mathcal{G}, n \rangle \mid \text{the graph } \mathcal{G} \text{ has at least } n \text{ independent vertices} \}$$

To show that this language decision problem is in *NP*, Lemma 5.9 requires that we produce a deterministic Turing machine verifier, $\mathcal{V}$. This verifier must input pairs of the form $\langle \sigma, \omega \rangle$, where σ is a graph-natural number pair, $\langle \mathcal{G}, n \rangle$. The verifier must satisfy that if $\sigma \in \mathcal{L}$ then there is a ω such that $\mathcal{V}$ accepts the input, while if $\sigma \notin \mathcal{L}$ then there is no such ω . This verifier must run in time polynomial in $|\sigma|$.

For a witness, take a size- n set of independent vertices from the graph, $\omega = \langle v_0, \dots, v_{n-1} \rangle$. The verifier gets an input pair $\langle \sigma, \omega \rangle$ where σ is a graph, and checks that the vertices are in the graph and that no two of them are connected. Those checks take time polynomial in $|\sigma|$ (obviously, this depends on the graph being represented with reasonable efficiency, but we always make that assumption). If the pair σ is indeed an element of $\mathcal{L}$ then there is such a witness ω , so there exists a pair $\langle \sigma, \omega \rangle$ that the verifier can accept. If σ is not an element of $\mathcal{L}$ then there is no such ω , and so there does not exist a pair that the verifier will accept.

V **V.5.30** The Knapsack problem is the decision problem for this language, $\mathcal{L}$.

$$\{ \langle (w_0, v_0), \dots, (w_{n-1}, v_{n-1}), B, T \rangle \mid \text{there is } I = \{i_0, \dots, i_k\} \subseteq \{0, \dots, n-1\} \text{ so } \sum_{i \in I} w_i \leq B \text{ and } \sum_{i \in I} v_i \geq T \}$$

To show that this language decision problem is in *NP*, Lemma 5.9 requires that we produce a deterministic Turing machine verifier, $\mathcal{V}$. The verifier input pairs $\langle \sigma, \omega \rangle$, where $\sigma = \langle (w_0, v_0), \dots, (w_{n-1}, v_{n-1}), B, T \rangle$. The verifier must have the property that if $\sigma \in \mathcal{L}$ then there is a ω such that $\mathcal{V}$ accepts the input, while if $\sigma \notin \mathcal{L}$ then there is no such ω . This verifier must also run in time polynomial in $|\sigma|$.

For a witness, take a set $\omega = \{i_0, \dots, i_k\} \subseteq \{0, \dots, n-1\}$. The verifier gets an input pair $\langle \sigma, \omega \rangle$ and checks that the sum of the weights of the indicated pairs, $\sum_{i \in \omega} w_i$, is less than or equal to the bound B , and also that the sum of the values of pairs, $\sum_{i \in \omega} v_i$, is greater than or equal to the target T . Those checks clearly take time polynomial in $|\sigma|$.

If σ is indeed an element of $\mathcal{L}$ then there is such a selection of indices, so there is such a witness ω , so there exists a pair $\langle \sigma, \omega \rangle$ that the verifier can accept. If σ is not an element of $\mathcal{L}$ then there is no such ω , and so there does not exist a pair that the verifier will accept.

V **V.5.31** By Lemma 5.6 it suffices to show that the language decision problem is in *P*. But that's clear: given a triple $\langle a, b, c \rangle$, checking whether the first two sum to the third is clearly a polytime task.

V **V.5.32** We start by expressing this as a language decision problem.

$$\mathcal{L} = \{ \langle \mathcal{G}, B \rangle \mid \mathcal{G} \text{ has a simple path of length at least } B \}$$

To show that this problem is in *NP*, Lemma 5.9 wants a deterministic Turing machine verifier, $\mathcal{V}$. The verifier input pairs $\langle \sigma, \omega \rangle$, where $\sigma = \langle \mathcal{G}, B \rangle$. The verifier must satisfy that if $\sigma \in \mathcal{L}$ then there is a ω such that $\mathcal{V}$ accepts the input pair, while if $\sigma \notin \mathcal{L}$ then there is no such ω . Also, the verifier must run in time polynomial in $|\sigma|$.

For a witness, take a path $\omega = \langle v_0, \dots, v_{n-1} \rangle$. The verifier gets an input pair $\langle \sigma, \omega \rangle$ and checks that the path is in $\mathcal{G}$, is simple (that no two vertices are equal), and has length at least B . Those checks clearly can be done in time polynomial in $|\sigma|$.

If σ is indeed an element of $\mathcal{L}$ then there is such a path ω , so there exists a pair $\langle \sigma, \omega \rangle$ that the verifier accepts. If σ is not an element of $\mathcal{L}$ then there is no such ω , and so there does not exist a pair that the verifier accepts.

V **V.5.33**

(A) Here is the recasting, as the decision problem for this language.

$$\mathcal{L}_0 = \{ \langle a, b \rangle \in \mathbb{N}^2 \mid \text{there exists } x \in \mathbb{N} \text{ with } ax + 1 = b \}$$

For Lemma 5.9, we will produce a deterministic Turing machine verifier, $\mathcal{V}$. It inputs pairs $\langle \sigma, \omega \rangle$, where σ is the number pair $\langle a, b \rangle$. The verifier must satisfy that if $\sigma \in \mathcal{L}_0$ then there is a witness ω such that $\mathcal{V}$ accepts the input pair, while if $\sigma \notin \mathcal{L}_0$ then there is no such ω . The verifier must run in time polynomial in $|\sigma|$.

For a witness, take a number $\omega = x$. The verifier gets an input pair $\langle \sigma, \omega \rangle$ and checks that $ax + 1 = b$. That can be done in time polynomial in $|\sigma|$.

If σ is indeed an element of $\mathcal{L}_0$ then there is such a number $\omega = x$, so there exists a pair $\langle \sigma, \omega \rangle$ that the verifier can accept. If σ is not an element of $\mathcal{L}_0$ then there is no such ω , and so there does not exist a pair that the verifier accepts.

(B) This is the associated language.

$$\mathcal{L}_1 = \{ M \mid \text{there is a set } P \subset \mathbb{Z} \text{ of numbers such that } M \text{ is the multiset of pairwise distances} \}$$

Consider a deterministic Turing machine verifier, $\mathcal{V}$, that inputs pairs $\langle \sigma, \omega \rangle$, where σ is a multiset M . For a witness, take a set of positions $\omega = P$. The verifier gets an input pair $\langle \sigma, \omega \rangle$ and checks that the pairwise distances among elements of P equals M . That can be done in time polynomial in $|\sigma|$.

If $\sigma \in \mathcal{L}_1$ then there is such a set of positions $\omega = P$, so there exists a pair $\langle \sigma, \omega \rangle$ that the verifier accepts. If $\sigma \notin \mathcal{L}_1$ then there is no such set ω , and so there does not exist a pair that the verifier accepts.

V.5.34 Countable. A verifier is a deterministic Turing machine, and there are countably many such machines.

V.5.35 We use Lemma 5.9. This is a suitable language (where the road map is $\mathcal{G} = \langle \mathcal{N}, \mathcal{E} \rangle$).

$$\mathcal{L} = \{ N \subseteq \mathcal{N} \mid N \text{ is a set of vertices where there are two suitable cycles} \}$$

To use Lemma 5.9, we must produce a verifier and suitable witnesses. The verifier inputs $\langle \sigma, \omega \rangle$, where $\sigma = N$ (technically, σ is a string representation of a set $N \subseteq \mathcal{N}$). A witness is the two cycles, $\omega = \langle c_0, c_1 \rangle$. The verifier checks that every vertex $v \in V$ is on at least one of the two cycles, and that each of the cycles is of length at most B . Clearly that check can be done in time polynomial in $|\sigma|$.

If $\sigma \in \mathcal{L}$ then there is a pair of cycles, so there exists a suitable witness ω such that the verifier accepts the pair $\langle \sigma, \omega \rangle$. If $\sigma \notin \mathcal{L}$ then there are not two such cycles. So there is no such witness ω that checks out, and so the verifier will not accept for any such ω .

V.5.36

(A) The natural language is $\mathcal{L} = \{ \langle \mathcal{G}_0, \mathcal{G}_1 \rangle \mid \text{they are isomorphic} \}$.

(B) We use Lemma 5.9, so we must produce a verifier and suitable witnesses. The verifier inputs pairs $\langle \sigma, \omega \rangle$ where σ is a pair $\langle \mathcal{G}_0, \mathcal{G}_1 \rangle$. The witness is a function, a set of ordered pairs $\langle v, f(v) \rangle$. The verifier checks that the function satisfies the conditions, that it is one-to-one and onto, and that $\{v, \hat{v}\}$ is an edge of $\mathcal{G}_0$ if and only if $\{f(v), f(\hat{v})\}$ is an edge of $\mathcal{G}_1$. That check takes time polynomial in $|\sigma|$, because it depends only on the size of the two graphs.

If $\sigma \in \mathcal{L}$ then there is such a function, so there exists a hint ω that will allow the verifier to accept the pair $\langle \sigma, \omega \rangle$. If the two graphs are not isomorphic then no witness ω will verify.

V.5.37 Suppose that a nondeterministic Turing machine $\mathcal{P}$ recognizes a language $\mathcal{L}$. We will describe a deterministic machine $\mathcal{Q}$ that does the same. Given an input σ , the nondeterministic machine $\mathcal{P}$ generates a computation tree and $\mathcal{Q}$ can do a breadth-first traversal of that tree. If the tree triggered by σ has an accepting branch then $\mathcal{Q}$ will eventually reach it, and will accept that input. If the tree has no accepting branch then $\mathcal{Q}$ will never find one, and so will not accept σ .

V.5.38 A configuration of a nondeterministic Turing machine $\mathcal{P}$ is a set of four-tuples, $\langle q, s, \tau_L, \tau_R \rangle$, where $q \in Q$, $s \in \Sigma$, and where τ_L and τ_R are strings of elements from the tape alphabet, $\tau_L, \tau_R \in \Sigma^*$. These signify the current state, the character under the read/write head, and the tape contents to the left and right of the head. We write $\mathcal{C}(t)$ for the machine's configuration after the t -th transition, and say that this is the configuration at step t . The initial configuration has the form $\mathcal{C}(0) = \{ \langle q_0, s, \alpha, \varepsilon \rangle \}$. We say that $\alpha \wedge \langle s \rangle$ is the machine's input.

We next describe how the machine $\mathcal{P}$ transitions from being in configuration $\mathcal{C}(t)$ to being in the next configuration, $\mathcal{C}(t+1)$. Start with an empty set and then get $\mathcal{C}(t+1)$ by adding instructions to it as follows.

For each $\langle q, s, \tau_L, \tau_R \rangle \in \mathcal{C}(t)$, compute the set $\Delta(q, s)$. That set may be empty, but if not then for each instruction $I = q_p T_p T_n q_n$ in that set, there are three possibilities. (1) If $T_n \in \Sigma$ then write that character to the tape, so that $\langle q_n, T_n, \tau_L, \tau_R \rangle \in \mathcal{C}(t+1)$. We say that $I \vdash \langle q_n, T_n, \tau_L, \tau_R \rangle$. (2) If $T_n = \mathbb{L}$ then the machine moves the tape head to the left, that is, $\langle q_n, \hat{s}, \hat{\tau}_L, \hat{\tau}_R \rangle \in \mathcal{C}(t+1)$, where $\hat{s}$ is the rightmost character of the string τ_L (if $\tau_L = \varepsilon$ then $\hat{s}$ is the blank character), where $\hat{\tau}_L$ is τ_L with its rightmost character omitted (if $\tau_L = \varepsilon$ then $\hat{\tau}_L = \varepsilon$

also), and where $\hat{\tau}_R$ is the concatenation of $\langle s \rangle$ and τ_R . We say that $I \vdash \langle q_n, \hat{s}, \hat{\tau}_L, \hat{\tau}_R \rangle$. (3) If $T_n = R$ then the machine moves the tape head to the right. This is like (2) so we omit the details.

A computation tree branch is a sequence of configurations $I_0 \vdash C(1) \vdash \dots \vdash C(h)$, starting with the initial configuration $C(0)$, and such that there is no instruction I where $I_k \vdash I$. If the instruction $I_k = \langle \hat{q}, \hat{s}, \beta, \gamma \rangle$ is such that its state is accepting, $\hat{q} \in A$, then we say the machine $\mathcal{P}$ accepts the input.

V.5.39

(A) The class NP is the set of languages decidable by a nondeterministic machine in polytime. Every problem decidable by a nondeterministic Turing machine is decidable by a deterministic Turing machine. The Halting problem is not decidable by a deterministic Turing machine, so it is not in NP .

(B) There is no polynomial bound on the number of steps, and consequently on the length of ω .

V.6.11 A fast algorithm for $\mathcal{L}_0$ would give a fast one for $\mathcal{L}_1$.

V.6.12 When $\mathcal{L}_1 \leq_p \mathcal{L}_0$ then $\mathcal{L}_1$ reduces to $\mathcal{L}_0$.

V.6.13 The same is true about NP . Both are immediate from Lemma 6.10.

V.6.14 Lemma 6.10 does not say that every $\mathcal{L} \in P$ is $\leq_p$ to every other language. It says that it is related in that way to every other language in the collection Rec . Each language in that collection has a Turing machine decider and so there are only countably many such languages.

V.6.15

(A) This has nothing to do with the ‘polytime’. If they were reducible then there would be a function f such that $n \in \mathbb{N}$ if and only if $f(n) \in \emptyset$. But the former is always true while the latter is always false.

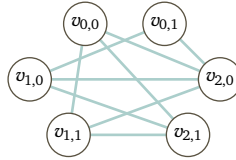
(B) Suppose that there is a computable function f such that $x \in A$ if and only if $f(x) \in B$, and suppose that B is computably enumerable. Then computably enumerate the set A by dovetailing an enumeration of B with an enumeration of the range of f : $f(0), f(1), \dots$; when an element $f(i) = b_j$ appears on both lists then enumerate i into A .

As to $K^c \not\leq_p K$, the set K is computably enumerable but K^c is not.

V.6.16 Assume $\mathcal{L} \leq_p \mathcal{L}^c$. Then there is a polytime f so that $\sigma \in \mathcal{L}$ if and only if $f(\sigma) \in \mathcal{L}^c$. That’s equivalent to $\sigma \notin \mathcal{L}$ if and only if $f(\sigma) \notin \mathcal{L}^c$, which holds when $\sigma \in \mathcal{L}^c$ if and only if $f(\sigma) \in \mathcal{L}$.

V.6.17 There is indeed a subset that adds to T , namely $A = \{23, 31, 72\}$.

V.6.18 It has three clauses, $x_0 \vee x_1$, $\neg x_0 \vee \neg x_1$, and $x_0 \vee \neg x_1$, each with two literals. This is the graph.



V.6.19 For each vertex there is a clause.

$$a_{0,0} \vee b_{0,0} \vee c_{0,0} \quad a_{0,1} \vee b_{0,1} \vee c_{0,1} \quad a_{1,0} \vee b_{1,0} \vee c_{1,0} \quad a_{1,1} \vee b_{1,1} \vee c_{1,1} \quad a_{2,0} \vee b_{2,0} \vee c_{2,0} \quad a_{2,1} \vee b_{2,1} \vee c_{2,1}$$

For each edge there are three clauses. For instance, for the edge between $v_{0,0}$ and $v_{1,1}$ the three clauses are here.

$$(\neg a_{0,0} \vee \neg a_{1,1}) \quad (\neg b_{0,0} \vee \neg b_{1,1}) \quad (\neg c_{0,0} \vee \neg c_{1,1})$$

Construct the entire expression by taking the conjunction of the clauses.

V.6.20

(A) This is wrong. For instance, if $A = \emptyset$ then $\emptyset \leq_p B$ for any language B but while there is a decider for $\emptyset$ that runs in time $\mathcal{O}(1)$, there may be no polynomial time decider for B (for instance, if $B = K$ then there is no decider at all). The correct statement is the opposite: a polytime decider for B can be used to decide the set A in polytime.

(B) This is also wrong. As in the prior item, if A is the empty set, which is decidable, and $B = K$, then $A \leq_p B$ but B is not decidable in polytime. The reason that $A \leq_p B$ is that A can be decided in polytime without any reference to B at all; in fact, deciding A is $\mathcal{O}(1)$. Instead, the other way around is right: if B is polytime decidable then A is polytime decidable also.

(c) This is true.

V.6.21

(A) Three are: $\beta_2 = 1001001$, $\beta_1 = 1100100$, and $\beta_0 = 0110010$.

(B) It is a cyclic shift, starting from index $k = 6$.

(C) It is the decision problem for this language.

$$\mathcal{L}_0 = \{ \langle \sigma, \tau \rangle \in \Sigma^2 \mid \tau \text{ is a substring of } \sigma \}$$

(D) It is the decision problem for this.

$$\mathcal{L}_1 = \{ \langle \alpha, \beta \rangle \in \Sigma^2 \mid \beta \text{ is a cyclic shift of } \alpha \}$$

(E) Following the hint, the reduction function f inputs $\langle \alpha, \beta \rangle$ and if the two are the same length then it outputs $\langle \alpha\alpha, \beta \rangle$ (else it outputs some pair sure to fail, such as $\langle \varepsilon, \emptyset \rangle$). Clearly f is computable in polytime, and $\langle \alpha, \beta \rangle \in \mathcal{L}_1$ if and only if $f(\langle \alpha, \beta \rangle) \in \mathcal{L}_0$. Therefore $\mathcal{L}_1 \leq_p \mathcal{L}_0$.

V.6.22

(A) The Independent Set problem is the decision problem for this language.

$$\mathcal{L}_0 = \{ \langle \mathcal{G}, B \rangle \mid \text{there is } I \subseteq \mathcal{N}(\mathcal{G}) \text{ with } |I| \leq B \text{ such that every edge contains a vertex from } I \}$$

Vertex Cover is the decision problem for this language.

$$\mathcal{L}_1 = \{ \langle \mathcal{H}, D \rangle \mid \text{there is } C \subseteq \mathcal{N}(\mathcal{H}) \text{ with } |C| \leq D \text{ such that every edge contains a vertex from } C \}$$

(B) A vertex cover with $k = 4$ elements is $S = \{q_2, q_5, q_8, q_9\}$.

(C) An independent set with $\hat{k} = 6$ elements is $\hat{S} = \{q_0, q_1, q_3, q_4, q_6, q_7\}$.

(D) For any edge, if it has at least one endpoint in S then it has at most one endpoint in the complement $\hat{S} = \mathcal{N} - S$. Conversely, if an edge has at most one endpoint in $\hat{S}$ then it has at least one endpoint in the complement $S = \mathcal{N} - \hat{S}$.

(E) We will do Independent Set $\leq_p$ Vertex Cover; the other reduction is similar. The definition requires that we produce a polytime computable function f . Given an instance of Independent Set $\sigma = \langle \mathcal{G}, \hat{k} \rangle$, then we convert that to an instance of Vertex Cover $f(\sigma) = \langle \mathcal{G}, |\mathcal{G}| - \hat{k} \rangle$.

V.6.23

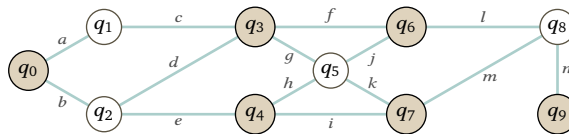
(A) Vertex Cover is the decision problem for this language.

$$\mathcal{L}_0 = \{ \langle \mathcal{G}, B \rangle \mid \text{there is } C \subseteq \mathcal{N}(\mathcal{G}) \text{ with } |C| \leq B \text{ such that every edge contains a vertex from } C \}$$

Set Cover is the decision problem for this language.

$$\mathcal{L}_1 = \{ \langle S, S_0, \dots, S_{k-1}, D \rangle \mid \text{each } S_j \text{ is a subset of } S, \text{ and } k \leq D, \text{ and } S_0 \cup \dots \cup S_{k-1} = S \}$$

(B) One vertex cover is $C = \{q_0, q_3, q_4, q_6, q_7, q_9\}$.



(c) The set of edges is $S = \{a, b, \dots, n\}$. The subsets are these.

$$\begin{aligned} S_0 &= \{a, b\}, S_1 = \{a, c\}, S_2 = \{b, d, e\}, S_3 = \{c, d, f, g\}, S_4 = \{e, h, i\}, \\ S_5 &= \{g, h, j, k\}, S_6 = \{f, j, l\}, S_7 = \{i, k, m\}, S_8 = \{l, m, n\}, S_9 = \{n\} \end{aligned}$$

The union of the subsets associated with vertices in the vertex cover

$$S_{q_0} \cup S_{q_3} \cup S_{q_4} \cup S_{q_6} \cup S_{q_7} \cup S_{q_9} = \{a, b\} \cup \{c, d, f, g\} \cup \{e, h, i\} \cup \{f, j, l\} \cup \{i, k, m\} \cup \{n\}$$

equals S .

- (D) We must produce a reduction function, f . It inputs an instance $\langle \mathcal{G}, B \rangle$ of the **Vertex Cover** problem. The computation is: construct the set S of all edges, and for each vertex v , construct $S_v \subseteq S$ consisting of the edges incident on v . Then the output of the function is an instance of the **Set Cover** problem, $f(\langle \mathcal{G}, B \rangle) = \langle S, S_{v_0}, \dots, B \rangle$. Clearly we can compute this function in polytime, and clearly also $\langle \mathcal{G}, B \rangle \in \mathcal{L}_0$ if and only if $\langle S, S_{v_0}, \dots, B \rangle \in \mathcal{L}_1$.

V.6.24

- (A) The **Hamiltonian Circuit** problem is the decision problem for this language.

$$\mathcal{L}_0 = \{ \langle \mathcal{H} \mid \text{the graph contains a circuit that includes each vertex exactly once} \}$$

The **Traveling Salesman** problem is this decision problem.

$$\mathcal{L}_1 = \{ \langle \mathcal{G}, B \rangle \mid \text{some circuit of } \mathcal{G} \text{ that includes each vertex has total cost less than or equal to } B \}$$

- (B) The reduction function inputs an instance of **Hamiltonian Circuit**, a graph $\mathcal{H} = \langle \mathcal{N}, \mathcal{E} \rangle$ whose edges are unweighted. It returns the instance of **Traveling Salesman** that uses the vertex set $\mathcal{N}$ as cities and that takes the distances between the cities to be: $d(v_i, v_j) = 1$ if $v_i v_j$ is an edge of $\mathcal{G}$ and $d(v_i, v_j) = 2$ if $v_i v_j \notin \mathcal{E}$, and such that the bound B is the number of vertices $|\mathcal{N}|$.

The bound means that there will be a **Traveling Salesman** solution if and only if there is a **Hamiltonian Circuit** solution, namely the **Traveling Salesman** solution uses the edges of the Hamiltonian circuit.

Clearly $\mathcal{H} \in \mathcal{L}_0$ if and only if $f(\mathcal{H}) \in \mathcal{L}_1$. What remains is to show that f runs in polytime. The number of edges is less than twice the number of vertices, so that polytime in the input graph size is the same as polytime in the number of vertices. To output the **Traveling Salesman** instance, the algorithm examines all the pairs of vertices, which is a nested loop and so takes time that is quadratic in the number of vertices. So f runs in polytime.

V.6.25

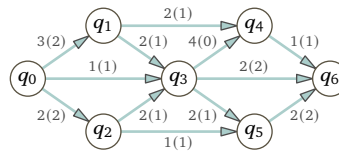
- (A) **Max-Flow** has this language.

$$\mathcal{L}_0 = \{ \langle F, \mathcal{G}, B \rangle \mid \text{the flow } F \text{ meets the constraints of } \mathcal{G} \text{ and the flow into the sink is at least } B \}$$

This is the language for the **Linear Programming** optimization problem.

$$\mathcal{L}_1 = \{ \langle \vec{x}, C, Z, B \rangle \mid \text{the point } \vec{x} \text{ meets the constraints } C \text{ and makes } Z(\vec{x}) \geq B \}$$

- (B) There are algorithms to solve these problems, but for small ones we can do it by eye. The maximum flow is 5, with each edge's numbers given in parentheses.



- (C) The variables are $x_{0,1}, \dots, x_{5,6}$. These are the capacity constraints.

$$\begin{aligned} 0 \leq x_{0,1} \leq 3, \quad 0 \leq x_{0,2} \leq 2, \quad 0 \leq x_{0,3} \leq 1, \quad 0 \leq x_{1,3} \leq 2, \quad 0 \leq x_{1,4} \leq 2, \\ 0 \leq x_{2,3} \leq 2, \quad 0 \leq x_{2,5} \leq 1, \quad 0 \leq x_{3,4} \leq 4, \quad 0 \leq x_{3,5} \leq 2, \quad 0 \leq x_{3,6} \leq 2, \\ 0 \leq x_{4,6} \leq 1, \quad \text{and} \quad 0 \leq x_{5,6} \leq 2 \end{aligned}$$

These are the flow constraints.

$$\begin{aligned} x_{0,1} = x_{1,3} + x_{1,4}, \quad x_{0,2} = x_{2,3} + x_{2,5}, \quad x_{0,3} + x_{1,3} + x_{2,3} = x_{3,4} + x_{3,5} + x_{3,6}, \\ x_{1,4} + x_{3,4} = x_{4,6}, \quad \text{and} \quad x_{2,5} + x_{3,5} = x_{5,6} \end{aligned}$$

We want to maximize $Z = x_{3,6} + x_{4,6} + x_{5,6}$. Observe that all of these equations and inequalities are linear.

- (c) The reduction function inputs propositional logic expressions in Conjunctive Normal form $\sigma = \sigma_0 \wedge \sigma_1 \wedge \dots$, where the clauses are disjunctions of three or fewer literals. It must output propositional logic expressions again in Conjunctive Normal form but where the clauses are disjunctions of exactly three literals. The prior item shows how to do that, so that the output and input expressions are equivalent. Obviously the function can run in polytime.

V.6.28

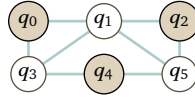
- (A) An instance of **3-SAT** is a propositional logic expression, in Conjunctive Normal form and such that the clauses have at most three literals, that is satisfiable. An instance of **SAT** is a propositional logic expression in Conjunctive Normal form that is satisfiable. So take the reduction function f to be the identity $f(\sigma) = \sigma$. Obviously f runs in polytime.
- (B) We can talk our way through it but the most straightforward thing is to exhibit a truth table.

P	Q	$P \rightarrow Q$	$\neg P$	$\neg P \vee Q$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	T	F	T

- (c) That $A \rightarrow (P \vee Q)$ is equivalent to $(P \vee Q \vee \neg A)$ is an instance of the prior item. For the other direction, by the prior item $(P \vee Q) \rightarrow A$ is equivalent $\neg(P \vee Q) \vee A$. Both are clauses with three literals.
- (D)
- (E) The reduction function inputs Conjunctive Normal form expressions $\sigma = \sigma_0 \wedge \sigma_1 \dots$. It must output Conjunctive Normal form expressions where the clauses have three or fewer literals. The prior items show how to do that, and clearly those transformations can be done in polytime.

V.6.29

- (A) The natural independent set is this.

**V.6.30**

- (A) These constraints

$$0 \leq z_0 \leq 1, \quad 0 \leq z_1 \leq 1, \quad 0 \leq z_2 \leq 1$$

ensure, in an Integer Programming context, that each variable is either 0 or 1. As to the expression, use $z_0 + (1 - z_1) + (1 - z_2) > 0$.

- (B) We must produce a reduction function f . It is given a propositional logic expression. For each variable P_i in that expression, create a variable z_i for the Integer Linear Programming problem. To force each z_i to be either 0 or 1, include the constraint $0 \leq z_i \leq 1$. For every clause in the propositional logic expression, add a constraint to the Integer Linear Programming instance as in the prior item.

$$P_i \vee \neg P_j \vee \neg P_k \quad \text{is associated with} \quad z_i + (1 - z_j) + (1 - z_k) > 0$$

Clearly this function can be done in polytime.

V.6.31

- (A) This is just the definition of 'or'.
- (B) Suppose first that all three polynomials in $S_{E_0} = \{x_0(1 - x_0), x_1(1 - x_1), x_0(1 - x_1)\}$ have a value of 0. The first polynomial $x_0(1 - x_0)$ evaluates to 0 if and only if x_0 is either 0 or 1. The second polynomial gives the same for x_1 . The third evaluates to 0 if and only if either $0 = x_0$ or $0 = 1 - x_1$.
- (c) For $E_1 = (P_0 \vee \neg P_1 \vee \neg P_2) \wedge (P_1 \vee P_2 \vee \neg P_3)$, the analogous set is this.

$$S_{E_1} = \{x_0(1 - x_0), x_1(1 - x_1), x_2(1 - x_2), x_3(1 - x_3), x_0(1 - x_1)(1 - x_2), x_1x_2(1 - x_3)\}$$

(D) Take the sum of squares.

$$p(x_0, x_1, x_2, x_3) = [x_0(1-x_0)]^2 + [x_1(1-x_1)]^2 + [x_2(1-x_2)]^2 + [x_3(1-x_3)]^2 + [x_0(1-x_1)(1-x_2)]^2 + [x_1x_2(1-x_3)]^2$$

(E) We must produce the reduction function f , which inputs an instance E of 3-Satisfiability, a Boolean expression in which all clauses have three or fewer literals, and outputs an instance p of D. We must verify that E is satisfiable if and only of p has zero for some integer n -tuple. (And we must show it runs in polytime, although that will be clear.)

The function f constructs the polynomial p following the pattern outlined in the prior items. For each variable P_i used in the input expression E , put $x_i(1-x_i)$ into the set S_E . In addition, for each clause $L_{i_0} \vee L_{i_1} \vee L_{i_2}$, where L_i is a literal, add an at most three-factor polynomial: it has the factor x_i if $L_i = P_i$, and for a negation $L_i = \neg P_i$ has the factor $(1-x_i)$. Finally, the polynomial p is the sum of the squares of the polynomials from S_E .

For one direction of the verification, suppose that E is satisfiable. For all atoms P_i that are assigned T , take $x_i = 0$, while if P_i is assigned F then take $x_i = 1$. Notice that as a result each polynomial member of S_E has the value of 0. Consequently, p has a value of 0, and so this set of x_i 's is a zero.

Now for the other direction. Suppose that we have a tuple of integers $\vec{c}$ that make the polynomial p have value 0. Because p is a sum of squares, each term must have a value of 0. Notice that each term of the form $x_i(1-x_i)$ is *zero*, and therefore $x_i = 0$ or $x_i = 1$. Notice also that each term that matches the form of a clause must have a value of 0, and therefore there is a matching assignment to the atoms P_i that makes the clause true (namely, $x_i = 0$ is associated with $P_i = T$, and $x_i = 1$ is associated with $P_i = F$).

V.6.32

(A) A symmetric instance is a special case of an asymmetric instance.

(B) This is the matrix. It is clearly symmetric.

$$\begin{array}{c} h_0 \quad h_1 \quad h_2 \quad \hat{h}_0 \quad \hat{h}_1 \quad \hat{h}_2 \\ \begin{pmatrix} - & & & & 3 & 5 \\ & - & & 1 & & 6 \\ & & - & 2 & 4 & \\ \hat{h}_0 & & 1 & 2 & - & \\ \hat{h}_1 & 3 & & 4 & & - \\ \hat{h}_2 & 5 & 6 & & & - \end{pmatrix} \end{array}$$

(c) The adjusted matrix is also symmetric.

$$\begin{array}{c} h_0 \quad h_1 \quad h_2 \quad \hat{h}_0 \quad \hat{h}_1 \quad \hat{h}_2 \\ \begin{pmatrix} - & & & -19 & 3 & 5 \\ & - & & 1 & -19 & 6 \\ & & - & 2 & 4 & -19 \\ \hat{h}_0 & -19 & 1 & 2 & - & \\ \hat{h}_1 & 3 & -19 & 4 & & - \\ \hat{h}_2 & 5 & 6 & -19 & & - \end{pmatrix} \end{array}$$

(D) The solution for $\mathcal{G}$ is $g_0 \rightarrow g_i \rightarrow g_2 \rightarrow g_0$, of total length 10. For $\mathcal{H}$ the solution is $h_0 \rightarrow \hat{h}_0 \rightarrow h_1 \rightarrow \hat{h}_1 \rightarrow h_2 \rightarrow \hat{h}_2 \rightarrow \hat{h}_2 \rightarrow h_0$. The large negative weights ensure that those edges must be part of a circuit. (Any circuit must have a number of edges equal to the number of vertices, and within that limit substituting larger numbers for the -19 's must increase the total.) The solution thus alternates between edges of the form $h_i \hat{h}_i$ and those of the form $\hat{h}_i h_j$, where $g_i g_j$ are in $\mathcal{G}$'s optimal circuit. These reductions can clearly be done in polytime.

V.6.33

(A) A little time playing with the table numbers

$C(t_i, w_j)$	w_0	w_1	w_2	w_3
t_0	13	4	7	6
t_1	1	11	5	4
t_2	6	7	2	8
t_3	1	3	5	9

shows that the optimum solution is to pair t_0, w_1 , along with t_1, w_3 , and t_2, w_2 , and finally t_3, w_0 . The total cost is 11.

- (B) As in the hint, the reduction function f inputs a triple $\langle W, T, C \rangle$ of the set of workers, the same-sized set of tasks, and the table of costs. It outputs a graph $f(\langle W, T, C \rangle) = \mathcal{G}$ that is bipartite, meaning that there are no inter- w edges or inter- t edges. For each w and t there is a directed edge in each direction.

For each i and j give the edge from w_i to t_j the weight $C(w_i, t_j)$. Give the edge from t_j to w_i the weight 0. (If you want all weights to be positive then these can be $C(w_i, t_j) + 1$ and 1.) This function f is clearly computable in polytime.

The function g inputs an assignment of workers to tasks, $w_0 \mapsto t_{i_0}, w_1 \mapsto t_{i_1} \dots$ and outputs an associated circuit through $\mathcal{G}$. That circuit starts at the vertex w_0 , then travels to vertex t_{i_0} , then to w_1 and t_{i_1} , etc, until returning back to w_0 . Clearly this map g is polytime computable and also clearly its input is optimal if and only if its output is also optimal.

V.6.34

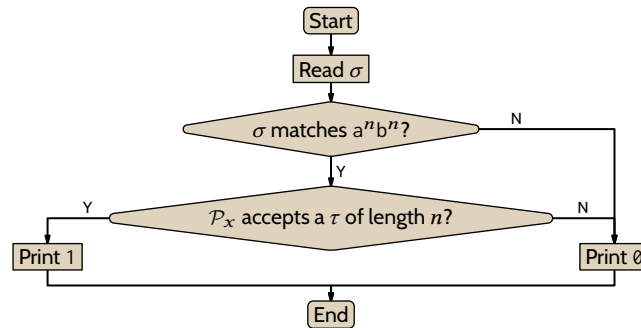
- (A) The tweaks are small, but here is the entire argument. Let $\mathcal{L}$ be an infinite subset of $\{a^n b^n \mid n \in \mathbb{N}\}$. For contradiction, assume that it is regular. Then the Pumping Lemma says that $\mathcal{L}$ has a pumping length, p .

Consider the list of strings $a^p b^p, a^{p+1} b^{p+1}, \dots$. Because $\mathcal{L}$ is infinite, at least one member of that list is a member of $\mathcal{L}$; let it be σ . It is an element of $\mathcal{L}$ and its length is greater than or equal to p so the Pumping Lemma applies. So σ decomposes into substrings $\sigma = \alpha \beta \gamma$ satisfying the three conditions. Condition (1) is that the length of the prefix $\alpha \beta$ is less than or equal to p . This implies that both α and β are composed exclusively of a's. Condition (2) is that β is not the empty string, so it contains at least one a.

Condition (3) is that all of the strings $\alpha \gamma, \alpha \beta^2 \gamma, \alpha \beta^3 \gamma, \dots$ are members of $\mathcal{L}$. To get the desired contradiction, consider $\alpha \beta^2 \gamma$. Compared with $\sigma = \alpha \beta \gamma$, this string has an extra β , which adds at least one a without adding any balancing b's. In short, $\alpha \beta^2 \gamma$ has more a's than b's and is therefore not a member of $\mathcal{L}$. But the Pumping Lemma says that it must be a member of $\mathcal{L}$, and therefore the assumption that $\mathcal{L}$ is regular is incorrect.

- (B) The flowchart makes clear that this behavior can be programmed on an everyday computer. Thus Church's Thesis says there is a Turing machine with this behavior. Let that machine have index e .

Applying the s - m - n lemma gives a family of machines, of which this is the x -th.



- (C) The reduction function f must have domain $\mathbb{N}$ and codomain $\mathbb{N}$. It must have the property that $i \in F$ if and only if $x \in R$. And it must run in polynomial time.

The function f is $i \mapsto s(e, i)$. Suppose that $i \in F$. Note first that the machine $\mathcal{P}_i$ is a language decider, and so halts on all inputs τ . Because the language of $\mathcal{P}_i$ is finite, the language of $\mathcal{P}_{s(e,i)}$ is finite also. Any finite language is regular, so the language of $\mathcal{P}_{s(e,i)}$ is regular.

Now suppose that $i \notin F$. If the machine $\mathcal{P}_{s(e,i)}$ has even one input σ where the second test box fails to halt on some τ then this machine is not a language decider, and so $s(e, i) \notin R$. If, on the other hand, for all input σ , the second test box halts on all relevant τ 's then because $i \notin F$, the language decided by $\mathcal{P}_{s(e,i)}$ is

One more thing wrong is that the statement seems to say that the problem is in that class because there is not algorithm yet known. It is not a question of whether we know of such an algorithm, it is a question of whether one exists.

- V **V.7.13** This algorithm is indeed exponential, not polynomial. But how to prove that this is the best possible algorithm? No one knows.
- V **V.7.14** In saying that the problem has a polytime algorithm, your friend is saying that there is a deterministic Turing machine that solves this problem and that runs in polytime. But a deterministic Turing machine is a special case of a nondeterministic one. That is, $P \subseteq NP$, so this problem is in NP .
- V **V.7.15** Computational classes such as NP do not contain algorithms, they contain languages. (We don't distinguish too sharply between problems and languages, and so may say that a problem is in a complexity class, but we do distinguish between languages and algorithms to decide those languages.)

V **V.7.16**

- (A) The statement “ NP is a subset of NP complete” is false. Rather, NP complete is a subset of NP . The other part, that NP complete is a subset of NP hard is true. It is the intersection of NP hard and NP .
- (B) This is a false statement, because nondeterministic machines are not a specialization. Rather, nondeterministic machines are a generalization of deterministic machines, that is, any deterministic machine is a nondeterministic one.

However, the clause about subset is trickier. We know that $P \subseteq NP$, which is the opposite direction than the one stated. Most experts judge that NP is not a subset of P , but of course we have no proof.

V **V.7.17**

- (A) True. Because $\text{Primality} \in NP$, it can be solved on a nondeterministic Turing machine. We can simulate such a machine with a deterministic one, and so we can solve the problem on a “regular” computer. (Of course, the solution algorithm may be quite slow.)
- (B) False; we cannot infer this from the given facts. It may be that there is an efficient algorithm, but we don't know. (We don't know from the statement as given, and we also don't know in general.)
- (C) False. The $\text{Traveling Salesman}$ problem is NP complete. Thus, from $\text{Prime Factorization} \in NP$, we can deduce that $\text{Prime Factorization} \leq_p \text{Traveling Salesman}$, and from an efficient algorithm for $\text{Traveling Salesman}$ we would get an efficient algorithm for $\text{Prime Factorization}$. But this is the reverse of what the question asks. The question says that $\text{Prime Factorization}$ is not known to be NP complete, so we don't know the other direction.

- V.7.18** (A) No, from $\mathcal{L}_1 \leq_p \mathcal{L}_0$ and $\mathcal{L}_0$ is NP complete, you cannot conclude that $\mathcal{L}_1$ is NP complete. An example is where $\mathcal{L}_0$ is the Satisfiability problem, and $\mathcal{L}_1$ is the problem of determining whether an input number is even.
- (B) False. It may be that $\mathcal{L}_0$ is not a member of NP . (C) False. This is a reprise of the first item: from $\mathcal{L}_1 \leq_p \mathcal{L}_0$ and $\mathcal{L}_0$ is NP complete, you cannot conclude that $\mathcal{L}_1$ is NP complete, even knowing that $\mathcal{L}_1 \in NP$. An example is where $\mathcal{L}_0$ is the Satisfiability problem, and $\mathcal{L}_1$ is the problem of determining whether an input number is even.
- (D) True; this is Lemma 7.4. If $\mathcal{L}_1$ is NP complete then any $\mathcal{L} \in NP$ is reducible to $\mathcal{L}_1$, so $\mathcal{L} \leq_p \mathcal{L}_1$, and transitivity of $\leq_p$ gives that $\mathcal{L} \leq_p \mathcal{L}_0$. That, along with the statement that $\mathcal{L}_0 \in NP$, gives that $\mathcal{L}_0$ is NP complete.
- (E) False. A problem is NP complete if it is in NP and every problem $\mathcal{L}$ in NP reduces to it. So two NP complete problems reduce to each other. For example, any of the problems on the Basic List are reducible to any of the others, or to Satisfiability .
- (F) False. An example is that $\mathcal{L}_1$ is the problem of determining whether a number is even, and $\mathcal{L}_0$ is a problem whose fastest solution algorithm is exponential (such as chess).
- (G) True. This is Lemma 6.10, that P is closed downward.

V.7.19

- (A) Clearly $\mathcal{L} = \{n \mid n \text{ is even}\}$ is in P . That implies it is in NP . Showing that $\mathcal{L}$ is NP complete would show that $P = NP$.
- (B) This is like the prior item, except that it is not as trivially a member of P . To see that it is a member of P , just note that there is an algorithm to check all 4-tuples with a nested loop.

- V.7.20** Fix a language $\mathcal{L} \in P$ that is nontrivial, so $\mathcal{L} \neq \emptyset$ and $\mathcal{L} \neq \mathbb{N}$. Because $P \subseteq NP$ we have $\mathcal{L} \in NP$. What remains is to show that $\mathcal{L}$ is NP hard, that every language $\hat{\mathcal{L}} \in NP$ satisfies that $\hat{\mathcal{L}} \leq_p \mathcal{L}$. But if $P = NP$ then $\hat{\mathcal{L}} \in P$, and Lemma 6.10 says that every nontrivial computable language is P hard, so $\hat{\mathcal{L}} \leq_p \mathcal{L}$.

- V.7.21** (A) False; for example, the brute force algorithm solves all instances. (B) True, where we use Cobham's

Thesis to take ‘quickly’ to mean polytime. (c) False. If *Traveling Salesman* $\in P$ then every *NP* problem is in *P*, because every *NP* problem reduces to *Traveling Salesman*. That contradicts the $P \neq NP$ assumption. (d) False. This would imply that *Traveling Salesman* is in *P*, which the prior item says is impossible.

- V **V.7.22** The obvious thing to do is to prove that 3-Satisfiability $\leq_p$ 4-Satisfiability. Briefly, given a propositional logic expression in which the clauses have three literals, such as $P_i \vee P_j \vee \neg P_k$, convert it into an expression in which the clauses have four literals by repeating the final literal, as in $P_i \vee P_j \vee \neg P_k \vee \neg P_k$.

V **V.7.23**

- (A) In short, the witness ω is the path.

The longer version is that, to apply Lemma 5.9, we will produce a deterministic Turing machine verifier, $\mathcal{V}$. It inputs pairs $\langle \mathcal{G}, \omega \rangle$, and must satisfy that if $\mathcal{G}$ has a Hamiltonian path then there is a witness ω such that $\mathcal{V}$ accepts the input pair, while if $\mathcal{G}$ does not have a Hamiltonian path then no witness will result in the pair being accepted. The verifier must run in polytime.

The verifier here interprets the witness ω to be a Hamiltonian path through the graph, so it checks that it is a legal path in the given graph and that every vertex is in the path. If there is such a path then there is a witness that will check out. If there is no such path then no witness will suffice. The verifier clearly runs in polytime (the size of a graph is polynomial in the number of vertices since the number of edges is $\mathcal{O}(|V|)$).

- (B) A Hamiltonian path is $\langle v_0, v_7, v_1, v_2, v_3, v_6, v_5, v_4, v_8 \rangle$. The two v_0 and v_8 must be the start and end of the path because they have degree 1.
 (C) We must produce a reduction function f . It must input a graph $\mathcal{G}$, outputs a graph $\mathcal{H}$, and have the property that $\mathcal{G}$ has a Hamiltonian circuit if and only if $\mathcal{H}$ has a Hamiltonian path.

Given $\mathcal{G}$, we form the graph $\mathcal{H}$ by adding three vertices. First, fix a vertex $v \in \mathcal{G}$. Add a new vertex $\hat{v} \in \mathcal{H}$ that has the same connections as v . That is, for every edge $v_i v$ or vv_j , add $\hat{v} v_i$ or $\hat{v} v_j$. Also add two vertices of degree one: w connects only to v , and $\hat{w}$ connects only to $\hat{v}$.

We must show that there is a Hamiltonian path in $\mathcal{H}$ if and only if there is a Hamiltonian circuit in $\mathcal{G}$. If there is a Hamiltonian circuit in $\mathcal{G}$ then the Hamiltonian path in $\mathcal{H}$ is derived by splitting the circuit at v , and hooking it to a path in $\mathcal{H}$ that starts at w , goes to v , then through the circuit, out to $\hat{v}$, and then to $\hat{w}$.

For the other direction suppose that there is a Hamiltonian path in $\mathcal{H}$. Then, as in the prior item, its start and end must be w and $\hat{w}$. Further, it must then proceed to v and $\hat{v}$. Because $\hat{v}$ is a copy of v , a Hamiltonian path in $\mathcal{H}$ that starts off at these vertices corresponds to a Hamiltonian circuit in $\mathcal{G}$.

- (D) Because the Hamiltonian Circuit problem is on the list of Basic *NP* Complete Problems, the fact that it reduces to Hamiltonian Path, and the latter is a member of *NP*, means that Hamiltonian Path is *NP* complete.

V.7.24

- (A) There is a Hamiltonian path: $\langle q_3, q_6, q_4, q_7, q_8, q_5, q_0, q_1, q_2 \rangle$.
 (B) Take $B \in \mathbb{N}$ to be the parameter and consider this family of languages.

$$\mathcal{L}_B = \{ \langle \mathcal{G}, B \rangle \mid \mathcal{G} \text{ has a simple path of length at least } B \}$$

For the reduction, the witness ω is the path. That is, consider a Turing machine verifier $\mathcal{V}$ that inputs $\langle \mathcal{G}, B, \omega \rangle$. It accepts the triple if and only if ω represents a path a simple path through the graph, of length at least B . Clearly that can be done in polytime. If $\langle \mathcal{G}, B \rangle \in \mathcal{L}_B$ then there is such a witness ω , so $\mathcal{V}$ accepts the input. If $\langle \mathcal{G}, B \rangle \notin \mathcal{L}_B$ then there is no such witness, and so the verifier $\mathcal{V}$ does not accept the triple.

- (c) Given an instance $\mathcal{G}$ of the Hamiltonian Path problem the reduction function converts it into the pair $\langle \mathcal{G}, |\mathcal{G}| - 1 \rangle$. Obviously that computation can be done in polytime. Further, where a graph has n -many vertices, a simple path containing $|G| - 1$ edges must be Hamiltonian.
 (D) The prior exercise shows that the Hamiltonian Path problem is *NP* complete, and that, along with the other parts of this exercise, gives that Longest Path is *NP* complete.

V.7.25

- (A) The sum of elements in $\hat{S} = \{6, 7, 19\}$ equals the sum of elements in $S - \hat{S} = \{3, 4, 12, 13\}$.
 (B) The sum of the elements of $\hat{T} = \{4, 6, 7, 13\}$ is $B = 30$.
 (C) If we can partition a set $S \subset \mathbb{N}$ into $\hat{S} \subset S$ and $S - \hat{S}$ with equal sums, then the total of the elements of S is twice the sum of the elements of $\hat{S}$, and so is even.

(D) This is the Subset Sum problem.

$$\mathcal{L}_0 = \{ \langle T, B \rangle \mid \text{a subset } \hat{T} \subseteq T \text{ has } \sum_{t \in \hat{T}} t = B \}$$

This is the Partition problem.

$$\mathcal{L}_1 = \{ \langle S \rangle \mid \text{there is } \hat{S} \subset S \text{ with } \sum_{s \in \hat{S}} s = \sum_{s \in S - \hat{S}} s \}$$

(E) We must produce a polytime computable function that takes in $\sigma = \langle S \rangle$ and outputs $f(\sigma) = \langle T, B \rangle$, such that $\sigma \in \mathcal{L}_0$ if and only if $f(\sigma) \in \mathcal{L}_1$.

Given $\sigma = \langle S \rangle$, consider the sum of its elements $x = \sum_{s \in S} s$. Where

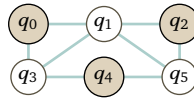
$$f(\sigma) = \begin{cases} \langle S, x/2 \rangle & \text{-- if } x \text{ is even} \\ \langle S, x + 1 \rangle & \text{-- otherwise} \end{cases}$$

we have that $\sigma \in \mathcal{L}_1$ if and only if $f(\sigma) \in \mathcal{L}_0$, as required (and f is clearly computable in polytime).

(F) **Partition** is one of the *NP* complete sets listed in the section. So **Subset Sum** is *NP* hard. But **Subset Sum** is clearly *NP*; as a witness ω we can use the subset.

V.7.26

(A) The natural independent set is this.



(B) This is the language.

$$\mathcal{L}_0 = \{ \langle \mathcal{G}, B \rangle \mid \mathcal{G} \text{ has at least } B\text{-many disconnected vertices} \}$$

(C) One way to satisfy it is by taking $P_0 = T$, $P_1 = P_2 = P_3 = F$.

(D) This is the language.

$$\mathcal{L}_1 = \{ E \mid E \text{ is a satisfiable expression with at most 3 literals per clause} \}$$

(E) The graph associated with the expression is this.



(F) We must produce a reduction function, f . It inputs an instance of the 3-Satisfiability problem, E . It constructs a graph $\mathcal{G}$ as in the prior item. That is, for the j -th clause it adds three vertices, labeling with $v_{j,i}$ if the literal is P_i , and with $\bar{v}_{j,i}$ if the literal is $\neg P_i$. It connects these three vertices completely, making a triangle. Further, the reduction function connects any two vertices labeled with the same index i and where one has an overline and the other does not, that is, it connects all pairs of $v_{j_0,i}$ and $\bar{v}_{j_1,i}$. Clearly this construction can happen in polytime.

The expression E is satisfiable if and only if in each clause at least one literal evaluates to T . Let S be a minimal such set of literals. Consider the associated vertices in the graph $\mathcal{G}$. In each triangle there must be at least one vertex associated with a literal from S because in each clause there must be a literal from S that evaluates to T . Further, for each literal in S , the negation evaluates to F , and so each associated vertex is not connected to an associated vertex in another triangle. Thus this forms an independent set of vertices.

The converse works the same way.

V.7.27 Move the head left, write a 1, and halt. This is a constant time algorithm.

V.7.28 No. If we knew of a problem in $NP - P$ then we'd know that $P \neq NP$. We don't know that. (Experts suspect that some problems are in this area, such as the **Vertex-to-Vertex Path** problem, but this is guesswork.)

V.7.29 Let $\mathcal{L}$ be any *NP* complete language of bitstrings. Take $\mathcal{L}_2 = \{\emptyset \frown \sigma \mid \sigma \in \mathcal{L}\}$, and $\mathcal{L}_1 = \{\emptyset \frown \sigma \mid \sigma \in \Sigma^*\}$, and $\mathcal{L}_0 = \mathcal{L}_1 \cup \{1 \frown \sigma \mid \sigma \in \mathcal{L}\}$. (The language $\mathcal{L}_1$ is an element of *P* because testing membership is as simple as testing whether the first character is 0.)

V.7.30 Assume that $\mathcal{L}_0$ is *NP* complete. Assume also that $\mathcal{L}_0 \leq_p \mathcal{L}_1$ and $\mathcal{L}_1 \in \text{NP}$. To verify that $\mathcal{L}_1$ is *NP* complete we must show that it is *NP* hard, that for every $\mathcal{L} \in \text{NP}$ we have $\mathcal{L} \leq_p \mathcal{L}_1$.

The assumption that $\mathcal{L}_0$ is *NP* complete gives $\mathcal{L} \leq_p \mathcal{L}_0$. Thus $\mathcal{L} \leq_p \mathcal{L}_0 \leq_p \mathcal{L}_1$ and transitivity of polytime reduction from Lemma 6.10 gives that $\mathcal{L} \leq_p \mathcal{L}_1$.

V.7.31

- (A) Fix $\mathcal{L}, \hat{\mathcal{L}} \in \text{NP}$. There are associated polytime verifiers $\mathcal{V}$ and $\hat{\mathcal{V}}$. A verifier for the union can just run those two on the input, and accept if either accepts.
- (B) Fix $\mathcal{L}, \hat{\mathcal{L}} \in \text{NP}$. There are associated polytime verifiers $\mathcal{V}$ and $\hat{\mathcal{V}}$. A verifier for the concatenation inputs a string σ , and loops over splitting it into substrings $\sigma = \sigma[0 : i] \frown \sigma[i :]$ for $i \in [0 \dots |\sigma|]$, testing whether both verifiers accept. Each verifier is polytime, and running the loop still keeps the overall algorithm inside of polytime.
- (C) The class *P* is closed under complement. Consequently, if *NP* is not closed under complement then $P \neq \text{NP}$. (But it is possible that *NP* is closed under complement but is still different than *P*.)

V.7.32 Countable. It is a subset of *NP*, which is countable because a verifier is a deterministic Turing machine, and there are countably many such machines.

V.7.33

- (A) It is not a member of *NP* because it is not decidable.
- (B) Since $\mathcal{L} \in \text{NP}$ there is an associated nondeterministic machine $\mathcal{P}$ that decides membership. Consider the Turing machine $\hat{\mathcal{P}}_{\mathcal{L}}$ that, given an input τ , constructs the computation tree of $\mathcal{P}_{\mathcal{L}}$ on input τ and halts if and only if $\mathcal{P}_{\mathcal{L}}$ accepts τ .

The reduction $\mathcal{L} \leq_p \text{HP}$ will be done by a function $f_{\mathcal{L}}$. This function will, given σ , return the pair $\hat{\mathcal{P}}, \sigma$ where $\hat{\mathcal{P}}$. (Note that this function is polynomial since, while running $\hat{\mathcal{P}}_{\mathcal{L}}$ may not be polynomial, constructing it from $\mathcal{P}$ is polynomial.)

V.8.14

- (A) The naive algorithm is to try all subsets. Where the set has n elements, there are 2^n subsets.
- (B) The naive algorithm is to check each assignment of the k colors to the n vertices. There are k^n many assignments.

V.8.15 First, $n! \leq n^n = (2^{\lg n})^n = 2^{n \lg n} \leq 2^{n^2}$. With that, the naive algorithm is to compute all circuits, and find the one of minimum cost. Where there are n cities, there are $n!$ many circuits.

V.8.16

- (A) Googling gives 10^{86} .
- (B) $\lg(10^{86}) = 86 \cdot \lg(10) \approx 86 \cdot 3.32 = 285.52$
- (C) The average chess game is about 40 moves (see (Stack Exchange author babou and various others 2013)). The longest game in the cited database is 277 moves, so 286 is a long chess game, but not completely out of range.

V.8.17

V.A.11 They are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, and 97.

V.A.12

- (A) $n = pq = 134$, $(p-1) \cdot (q-1) = 120$
- (B) Use $e = 7$ (the three 2, 3, and 5 are not relatively prime).
- (C) The multiplicative inverse of $e = 7$ modulo $n = 134$ is $d = 103$.
- (D) The encryption is $m^e \bmod n = 9^7 \bmod 143 = 48$. The decryption is $48^{103} \bmod 143 = 9$.

V.A.13

- (A) With two factors $n = k_0 \cdot k_1$ if one is greater than $\sqrt{n}$ then the other must be less.
- (B) If n is a perfect square $n = k \cdot k$ then its first nontrivial factor is $k = \sqrt{n}$.
- (C) The square root of 10^{12} is $10^6 = 1\,000\,000$, a million.

(D) The input n has size about $\lg(n)$ bits. Thus $\sqrt{n} \approx \sqrt{2^{\text{size of } n}} \approx (1/2) \cdot \text{size of } n$.

V.B.1 The main point relevant to this section is that for an input as small as size 100 we have very capable solvers, which will return the answer quite quickly. The fact that a problem is *NP* hard does not mean that most of its instances are slow to solve.

V.B.2 $n!$

V.B.3 Here is the exhaustive search.

<i>City sequence</i>	<i>First edge</i>	<i>Second</i>	<i>Third</i>	<i>Return</i>	<i>Total</i>
$v_0-v_1-v_2-v_3$	3	1	4	7	15
$v_0-v_1-v_3-v_2$	3	5	4	2	14
$v_0-v_2-v_1-v_3$	2	1	5	7	15
$v_0-v_2-v_3-v_1$	2	4	5	3	14
$v_0-v_3-v_1-v_2$	7	5	1	2	15
$v_0-v_3-v_2-v_1$	7	4	1	3	15
$v_1-v_0-v_2-v_3$	3	2	4	5	14
$v_1-v_0-v_3-v_2$	3	7	4	1	15
$v_1-v_2-v_0-v_3$	1	2	7	5	15
$v_1-v_2-v_3-v_0$	1	4	7	3	15
$v_1-v_3-v_0-v_2$	5	7	2	1	15
$v_1-v_3-v_2-v_0$	5	4	2	3	14
$v_2-v_0-v_1-v_3$	2	3	5	4	14
$v_2-v_0-v_3-v_1$	2	7	5	1	15
$v_2-v_1-v_0-v_3$	1	3	7	4	15
$v_2-v_1-v_3-v_0$	1	5	7	2	15
$v_2-v_3-v_0-v_1$	4	7	3	1	15
$v_2-v_3-v_1-v_0$	4	5	3	2	14
$v_3-v_0-v_1-v_2$	7	3	1	4	15
$v_3-v_0-v_2-v_1$	7	2	1	5	15
$v_3-v_1-v_0-v_2$	5	3	2	4	14
$v_3-v_1-v_2-v_0$	5	1	2	7	15
$v_3-v_2-v_0-v_1$	4	2	3	5	14
$v_3-v_2-v_1-v_0$	4	1	3	7	15

Basically, you can go around the outside for 14 or go through the middle for 15.

Appendices

A.1

- (A) $\sigma \wedge \tau = 10110 \wedge 110111 = 10110110111$.
- (B) $\sigma \wedge \tau \wedge \sigma = 10110 \wedge 110111 \wedge 10110 = 101101101110110$
- (C) $\sigma^R = 10110^R = 01101$
- (D) $\sigma^3 = 10110 \wedge 10110 \wedge 10110 = 101101011010110$
- (E) $\emptyset^3 \wedge \sigma = 000 \wedge 10110 = 00010110$

A.2

- (A) abbca
- (B) ababbcabca
- (C) baacb
- (D) ababab

A.3 There are $2^4 = 16$ bit strings of length 4 and half of them start with 0, so there are 8 strings in the language.

A.4

- (A) Yes.
- (B) Yes.
- (C) No.
- (D) Yes.

A.5 Show that the length of a concatenation is the sum of the two lengths: $|\sigma \wedge \tau| = |\langle s_0 \dots s_{i-1}, t_0, \dots t_{j-1} \rangle| = i + j = |\sigma| + |\tau|$. Proving this by induction is routine.

A.7 One example using $\mathbb{B}$ is $\sigma = \langle 0 \rangle$ and $\tau = \langle 1 \rangle$. Then $\sigma \wedge \tau = \langle 0, 1 \rangle$ while $\tau \wedge \sigma = \langle 1, 0 \rangle$.

V

B.1

- (A) For one-to-one, suppose that $f(x_0) = f(x_1)$ for $x_0, x_1 \in \mathbb{R}$. Then $3x_0 + 1 = 3x_1 + 1$. Subtract the 1's and divide by 3 to conclude that $x_0 = x_1$.
For onto, fix a member of the codomain $c \in \mathbb{R}$. Observe that $d = (c - 1)/3$ is a member of the domain and that $f(d) = 3 \cdot ((c - 1)/3) + 1 = (c - 1) + 1 = c$. Thus f is onto.
- (B) The function g is not one-to-one because $g(2) = g(-2)$. It is not onto because no element of the domain $\mathbb{R}$ is mapped by g to the element 0 of the codomain.

V

B.2

- (A) It is a left inverse because $g \circ f$ has the action $(x, y) \mapsto (x, y, 0) \mapsto (x, y)$ for all x and y . It is not a right inverse because there is an input on which the composition $f \circ g$ is not the identity, namely the action is $(1, 2, 3) \mapsto (1, 2) \mapsto (1, 2, 0)$.
- (B) Observe that $f(2) = f(-2) = 4$. Any left inverse would have to map 4 to 2 and also to map 4 to -2. That would be not well-defined, so no function is the left inverse.
- (C) One right inverse is $g_0: C \rightarrow D$ given by $10 \mapsto 0, 11 \mapsto 1$. It is a right inverse because $f \circ g_0(10) = f(g_0(10)) = f(0) = 10$ and $f \circ g_0(11) = f(g_0(11)) = f(1) = 11$. A second right inverse is $g_1: C \rightarrow D$ given by $10 \mapsto 2, 11 \mapsto 3$.

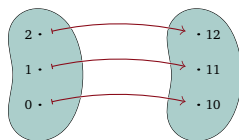
V

B.3

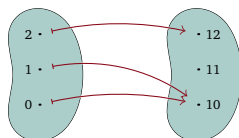
- (A) First, the domain of each equals the codomain of the other so both compositions $f \circ g$ and $g \circ f$ are defined. Next, $g \circ f(a) = g(f(a)) = g(a + 3) = (a + 3) - 3 = a$ is the identity map. Similarly, $f \circ g(a) = (a - 3) + 3$ is also the identity map. Thus they are inverse to each other.
- (B) The inverse of h is itself. The domain and codomain are equal, as required. In the case that n is odd, $h \circ h(n) = h(n - 1) = (n - 1) + 1 = n$ (the second equality holds because $n - 1$ is even in this case). Similarly if n is even then $h \circ h(n) = (n + 1) - 1 = n$.

- (c) The inverse of s is the map $r: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ given by $r(x) = \sqrt{x}$ (that is, the positive root). The domain of each is the codomain of the other, and each of $s \circ r$ and $r \circ s$ is the identity function.

V B.4 This is a bean diagram of the function f



and this is a diagram of g .

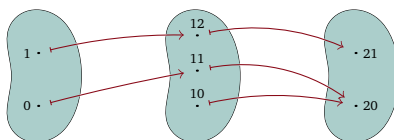


- (A) By inspection f is both one-to-one and onto.
 (B) Reverse the association made by f so that $f^{-1}(10) = 0$, $f^{-1}(11) = 1$, and $f^{-1}(12) = 2$.
 (C) The map g is not one-to-one since $g(0) = g(1)$. (In addition, g is not onto since no input is sent to $11 \in C$.)
 (D) If a h were the inverse of g then we would have both $h(10) = 0$ and $h(10) = 1$, which would violate the definition of a function.

V B.5

- (A) Let $f: D \rightarrow C$ and $g: C \rightarrow B$ be one-to-one. Suppose that $g \circ f(x_0) = g \circ f(x_1)$ for some $x_0, x_1 \in D$. Then $g(f(x_0)) = g(f(x_1))$. The function g is one-to-one and so, since the two values $g(f(x_0))$ and $g(f(x_1))$ are equal, the arguments $f(x_0)$ and $f(x_1)$ are also equal. The function f is one-to-one and so $x_0 = x_1$. Because $g \circ f(x_0) = g \circ f(x_1)$ implies that $x_0 = x_1$, the composition is one-to-one.
 (B) Let $f: D \rightarrow C$ and $g: C \rightarrow B$ be onto. If B is the empty set then by the definition of function, C is empty also and, again by the definition of function, in turn $D = \emptyset$, and so the composition is onto, vacuously. Otherwise, fix an element of the codomain of the composition, $b \in B$. The function g is onto so there is a $c \in C$ such that $g(c) = b$. The function f is onto so there is a $d \in D$ such that $f(d) = c$. Then $g \circ f(d) = b$, and so the composition is onto.
 (C) Let $f: D \rightarrow C$ and $g: C \rightarrow B$ be such that the composition $g \circ f$ is one-to-one. Suppose that f is not one-to-one. Then there are $d_0, d_1 \in D$ with $d_0 \neq d_1$ such that $f(d_0) = f(d_1)$. But this implies that $g(f(d_0)) = g(f(d_1))$, contradicting that the composition is one-to-one.
 (D) Let $f: D \rightarrow C$ and $g: C \rightarrow B$ be such that the composition $g \circ f$ is onto. Suppose for a contradiction that g is not onto. Then for some $b \in B$ there is no $c \in C$ with $g(c) = b$. But this means that there is no $d \in D$ such that $b = g(f(d))$, contradicting that the composition is onto. So g must be onto.
 (E) The answer to both questions is “no.”

Let $D = \{0, 1\}$, $C = \{10, 11, 12\}$, and $B = \{20, 21\}$. Let $f: D \rightarrow C$ and $g: C \rightarrow B$ be as shown below. Then $g \circ f$ is onto because $g \circ f(0) = 20$ and $g \circ f(1) = 21$. But f is not onto because no element of its domain D is mapped to the element 10 of its codomain C .



The same function is an example of a composition $g \circ f$ that is one-to-one but where the function performed second, g , is not one-to-one. In particular, $g \circ f$ is one-to-one because only one element of its domain D is mapped to its codomain element 20. Similarly only one element of its domain is mapped to 21.

V B.6

- (A) We first prove that if the function $f: D \rightarrow C$ has an inverse $f^{-1}: C \rightarrow D$ then it must be a correspondence. To verify that f is one-to-one, suppose that $d_0, d_1 \in D$ are such that $f(d_0) = f(d_1) = c$ for some $c \in C$.

Consider $f^{-1}(c)$. Because $f^{-1} \circ f$ is the identity map on D we have both that $f^{-1}(c) = f^{-1}(f(d_0)) = d_0$ and that $f^{-1}(c) = f^{-1}(f(d_1)) = d_1$. Thus $d_0 = d_1$ and so f is one-to-one.

Next we verify that f is onto. Suppose otherwise, so that there is a $c \in C$ that is not associated with any $d \in D$. But that gives the contradiction that the map $f \circ f^{-1}$ cannot give $c = f \circ f^{-1}(c) = f(f^{-1}(c))$ because $c \notin \text{ran}(f)$. Therefore f must be onto.

To finish, we prove that if a map $f: D \rightarrow C$ is a correspondence then it has an inverse. We will consider an inverse association $g: D \rightarrow C$ and show that it is well-defined. For each $c \in C$, let $g(c)$ be the element $d \in D$ such that $f(d) = c$. Note that there is such an element because f is onto, and also note that there this element is unique because f is one-to-one. Therefore, g is well-defined, that is, it is a function. That the compositions $g \circ f$ and $f \circ g$ are identity maps is clear.

- (B) Suppose that $f: D \rightarrow C$ has two inverses $g_0, g_1: C \rightarrow D$. We will show that the two same associate the same elements of the domain and codomain, that is, we will show that they have the same graph and are thus the same function.

Let c be any element of the domain C of the two g_0 and g_1 . The prior item of this exercise shows that f is onto so there exists a $d \in D$ such that $f(d) = c$. The prior item also shows that f is one-to-one, so there is only one d with that property: $d = g_0(c)$ and $d = g_1(c)$. Therefore $g_0(c) = g_1(c)$.

- (C) Observe that f^{-1} is a function that has an inverse, namely f . This is simply because in the equations $g \circ f^{-1} = \text{id}$ and $f^{-1} \circ g = \text{id}$ we can take g to be f . Then by the first item of this exercise, f^{-1} is a correspondence.
- (D) Here is the verification that composition in one order gives the identity function.

$$(f^{-1} \circ g^{-1}) \circ (g \circ f)(x) = (f^{-1} \circ g^{-1})(g(f(x))) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x$$

Verification of the other order is similar.

B.7

- (A) We will do induction on the number of elements in the range. Before the proof we give an illustration. Fix $D = \{0, 1, 2, 3\}$ and $C = \{10, 11, 12\}$. Consider the function $f: D \rightarrow C$ given here, which is such that $\text{ran}(f) = C$.

$$0 \mapsto 10 \quad 1 \mapsto 10 \quad 2 \mapsto 11 \quad 3 \mapsto 12$$

To do the induction, we pick an element of the range and omit it; suppose that we omit 10. We then want to consider the restriction map that is onto $C - \{10\}$. For this map to be well-defined, we must omit from its domain all of the elements that map to 10, the set $f^{-1}(10) = \{0, 1\}$. We then have the function $\hat{f}$ with domain $\hat{D} = \{2, 3\}$ and codomain $\hat{C} = \{11, 12\}$ given here.

$$2 \mapsto 11 \quad 3 \mapsto 12$$

Note that the range of this restriction map $\hat{f}$ is $\hat{C}$. The inductive hypothesis therefore gives that $\hat{f}$'s domain, $\hat{D}$, has at least as many elements as its range, $\hat{C}$. To finish the argument, we will add back the omitted elements to get f , adding back two elements from $\hat{f}$'s domain $\hat{D}$ and one from its range C , and conclude that its domain has at least as many elements as its range.

With that example in mind we can go to the induction proof. The base step is to consider a function f whose range is empty. The domain must also be empty since the definition of function requires that every element of the domain is mapped somewhere. Hence $|\text{ran}(f)| \leq |D|$ because both of them are zero.

For the inductive step assume that the statement holds for all functions between finite sets where the range has 0 elements, or 1 element, ... or k elements. Consider $f: D \rightarrow C$ with $C = \text{ran}(f)$ and $|C| = k + 1$.

Fix some $c_0 \in C$ (the range is not empty because $k + 1 > 0$). Let $\hat{C} = C - \{c_0\}$. Because c_0 is in the range of f , the set $f^{-1}(c_0) = \{d \in D \mid f(d) = c_0\}$ has at least one element. Restated, where $\hat{D} = D - f^{-1}(c_0)$ we have $|\hat{D}| < |D|$, with a strict inequality. Now the key point: the map $\hat{f}: \hat{D} \rightarrow \hat{C}$ is onto, as any element $c \in \text{ran}(f)$ is the image of some $d \in D$ and so any such element with $c \neq c_0$ must be the image of some $d \in \hat{D}$. Therefore the inductive hypothesis applies, giving that $|\hat{C}| \leq |\hat{D}|$. Finish by adding one, $|\text{ran}(f)| = |\hat{C}| + 1 \leq |\hat{D}| + 1 \leq |D|$.

Alternate proof. We can instead do induction on the number of elements in the domain D . As above, we first illustrate the argument. Fix $D = \{0, 1, 2, 3\}$ and codomain $C = \{10, 11, 12, 13\}$. To do induction, we will omit an element of the domain, in this example 0, giving $\hat{D} = D - \{0\}$. There are two cases. The first is that the omitted element 0 maps to the same member of the codomain as at least one other, not omitted, element of the domain. This happens with this function f .

$$0 \mapsto 10 \quad 1 \mapsto 10 \quad 2 \mapsto 11 \quad 3 \mapsto 12$$

Consider the restriction, $\hat{f}: \hat{D} \rightarrow C$ given by $\hat{f}(d) = f(d)$. The inductive hypothesis applies, giving $|\text{ran}(\hat{f})| \leq |\hat{D}|$. In this first case, 0 is not the only element of D associated with 10, so while $|\hat{D}| < |D|$ we have $|\text{ran}(f)| = |\text{ran}(\hat{f})|$ and the desired conclusion is immediate, $|\text{ran}(f)| = |\text{ran}(\hat{f})| \leq |\hat{D}| < |D|$.

The other case is that the omitted domain element 0 does not map to the same member of the codomain as any other domain element. That happens for this function f .

$$0 \mapsto 10 \quad 1 \mapsto 11 \quad 2 \mapsto 12 \quad 3 \mapsto 13$$

To do the induction, we omit 0 from the domain, giving $\hat{D} = D - \{0\}$. Consider the restriction map $\hat{f}: \hat{D} \rightarrow C$ defined by $\hat{f}(d) = f(d)$. The inductive hypothesis applies, so $|\text{ran}(\hat{f})| \leq |\hat{D}|$. Then adding 1 to both sides gives $|\text{ran}(f)| = |\text{ran}(\hat{f})| + 1 \leq |\hat{D}| + 1 = |D|$.

We are ready for the proof. The base step is to consider a function f whose domain D has no elements at all. The only function with an empty domain is the empty function, whose range is empty, and so $0 = |\text{ran}(f)| \leq |D| = 0$.

For the inductive step, assume that the statement is true for functions between finite sets when the domain of that function has 0 elements, or 1 element, $\dots$ or k elements. Consider $f: D \rightarrow C$ where $|D| = k + 1$.

Fix some $d_0 \in D$ (there must be such an element since the domain is not empty, as $k + 1 > 0$). Let $\hat{D} = D - \{d_0\}$ and let $\hat{f}: \hat{D} \rightarrow C$ be the restriction of f to $\hat{D}$, defined by $\hat{f}(d) = f(d)$. The inductive hypothesis gives $|\text{ran}(\hat{f})| \leq |\hat{D}|$.

Consider the codomain element $c_0 = f(d_0)$. There are two cases: either (1) c_0 is also the image of another domain element, so that there is a $d \in \hat{D}$ with $d \neq d_0$ and $f(d) = c_0$, or else (2) it is not the image of another such element. In the first case, $\text{ran}(f) = \text{ran}(\hat{f})$ and we have $|\text{ran}(f)| = |\text{ran}(\hat{f})| \leq |\hat{D}| < |D|$. In the second case, $\text{ran}(f) = \text{ran}(\hat{f}) + 1$ and we have $|\text{ran}(f)| = |\text{ran}(\hat{f})| + 1 \leq |\hat{D}| + 1 = |D|$.

(B) We will do induction on the number of elements in the range. The base step is that the range is empty. The only function with an empty range is the empty function, which has an empty domain, giving $0 = |\text{ran}(f)| = |D| = 0$.

For the inductive step, assume that the statement is true for any one-to-one function between finite sets with whose range has 0 elements, or 1 element, $\dots$ or k elements. Consider a one-to-one $f: D \rightarrow C$ with $|\text{ran}(f)| = k + 1$.

Fix $c_0 \in \text{ran}(f)$ (such an element exists because $k + 1 > 0$). Let $\hat{C} = C - \{c_0\}$. Because c_0 is an element of the range of f , there is a domain element that maps to it. Because f is one-to-one, there is only one such element. Call it d_0 and let $\hat{D} = D - \{d_0\}$.

The restriction function $\hat{f}: \hat{D} \rightarrow \hat{C}$ is defined by $\hat{f}(d) = f(d)$. This map is also one-to-one: thinking of f as a set of ordered pairs, the assumption that it is one-to-one means that no two pairs have the same second element, and the restriction $\hat{f}$ is a subset of f so it also has no two pairs with the same second element. By the inductive hypothesis, $|\text{ran}(\hat{f})| = |\hat{D}|$. Adding 1 gives $|\text{ran}(f)| = |\text{ran}(\hat{f})| + 1 = |\hat{D}| + 1 = |D|$.

Alternate proof. We can instead do induction on the number of elements in the domain D . The base step is that the domain is empty, $|D| = 0$. The only function with an empty domain is the empty function and we have $0 = |\text{ran}(f)| = |D| = 0$.

For the inductive step, assume that the statement is true for any one-to-one function between finite sets with whose domain has 0 elements, $\dots$ k elements. Consider a one-to-one function $f: D \rightarrow C$ whose domain has $|D| = k + 1$.

Fix some $d_0 \in D$ (the set D is not empty because $k + 1 > 0$) and let $\hat{D} = D - \{d_0\}$. Consider $c_0 = f(d_0)$. Because f is one-to-one, c_0 is not the image of any element of the domain other than d_0 . So the restriction map $\hat{f}: \hat{D} \rightarrow C$ has one less element in its range, $\text{ran}(f) = \text{ran}(\hat{f}) + 1$.

The key point is that because f is one-to-one, $\hat{f}$ is also one-to-one: thinking of f as a set of ordered pairs, one-to-one-ness means that no two pairs have the same second element, and $\hat{f}$ is a subset of f so it has the same property. Therefore the inductive hypothesis applies, giving $|\text{ran}(\hat{f})| = |\hat{D}|$. Add 1 to get the desired conclusion, that $|\text{ran}(f)| = |\text{ran}(\hat{f})| + 1 = |\hat{D}| + 1 = |D|$.

C.1

(A) This is a table for $(P \wedge Q) \wedge R$.

P	Q	R	$P \wedge Q$	$(P \wedge Q) \wedge R$
F	F	F	F	F
F	F	T	F	F
F	T	F	F	F
F	T	T	F	F
T	F	F	F	F
T	F	T	F	F
T	T	F	T	F
T	T	T	T	T

(B) And this is a table for $P \wedge (Q \wedge R)$. It has the same final column as the one from the prior item.

P	Q	R	$Q \wedge R$	$P \wedge (Q \wedge R)$
F	F	F	F	F
F	F	T	F	F
F	T	F	F	F
F	T	T	T	F
T	F	F	F	F
T	F	T	F	F
T	T	F	F	F
T	T	T	T	T

(C) This is a table for $P \wedge (Q \vee R)$.

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$
F	F	F	F	F
F	F	T	T	F
F	T	F	T	F
F	T	T	T	F
T	F	F	F	F
T	F	T	T	T
T	T	F	T	T
T	T	T	T	T

(D) And this is a table for $(P \wedge Q) \vee (P \wedge R)$. It has the same final column as the prior item.

P	Q	R	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \vee (P \wedge R)$
F	F	F	F	F	F
F	F	T	F	F	F
F	T	F	F	F	F
F	T	T	F	F	F
T	F	F	F	F	F
T	F	T	F	T	T
T	T	F	T	F	T
T	T	T	T	T	T

C.2

(A) This is a table for $\neg(P \vee Q)$.

P	Q	$P \vee Q$	$\neg(P \vee Q)$
F	F	F	T
F	T	T	F
T	F	T	F
T	T	T	F

(B) The same for $\neg P \wedge \neg Q$.

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	F	F	F

It has the same final column as the prior item.

(c) This is a table for $\neg(P \wedge Q)$.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$
F	F	F	T
F	T	F	T
T	F	F	T
T	T	T	F

(D) This is the one for $\neg P \vee \neg Q$.

P	Q	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	T
T	T	F	F	F

The final column is the same as the final column of the prior item.

C.3

(A) The second, third and sixth lines show T . The associated clauses are $\neg P \wedge \neg Q \wedge R$, $\neg P \wedge Q \wedge \neg R$, and $P \wedge \neg Q \wedge R$. The resulting CNF expression is this.

$$(\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge Q \wedge \neg R) \vee (P \wedge \neg Q \wedge R)$$

(B) The T lines give four clauses.

$$(\neg P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge R)$$

C.4

(A) The F lines are first, fourth, fifth, seventh, and eighth. So we have this.

$$E_0 \equiv \neg \left((\neg P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R) \vee (P \wedge Q \wedge \neg R) \vee (P \wedge Q \wedge R) \right)$$

Apply DeMorgan's law to distribute the negation across the $\vee$'s.

$$\neg(\neg P \wedge \neg Q \wedge \neg R) \wedge \neg(\neg P \wedge Q \wedge R) \wedge \neg(P \wedge \neg Q \wedge \neg R) \wedge \neg(P \wedge Q \wedge \neg R) \wedge \neg(P \wedge Q \wedge R)$$

Now distribute the $\neg$'s across the $\wedge$'s.

$$(P \vee Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee \neg Q \vee R) \wedge (\neg P \vee \neg Q \vee \neg R)$$

(B) Find the F lines.

$$E_1 \equiv \neg \left((\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R) \vee (P \wedge \neg Q \wedge R) \right)$$

Distribute the negation $\neg$ across the $\vee$'s.

$$\neg(\neg P \wedge \neg Q \wedge R) \wedge \neg(\neg P \wedge Q \wedge R) \wedge \neg(P \wedge \neg Q \wedge \neg R) \wedge \neg(P \wedge \neg Q \wedge R)$$

Distribute the four $\neg$'s across the $\wedge$'s.

$$(P \vee Q \vee \neg R) \wedge (P \vee \neg Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$$

C.5 Here are the sixteen.

$\begin{array}{c c c} P & Q & O_0 \\ \hline F & F & F \\ F & T & F \\ T & F & F \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_1 \\ \hline F & F & F \\ F & T & F \\ T & F & F \\ T & T & T \end{array}$	$\begin{array}{c c c} P & Q & O_2 \\ \hline F & F & F \\ F & T & F \\ T & F & T \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_3 \\ \hline F & F & F \\ F & T & F \\ T & F & T \\ T & T & T \end{array}$
$\begin{array}{c c c} P & Q & O_4 \\ \hline F & F & F \\ F & T & T \\ T & F & F \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_5 \\ \hline F & F & F \\ F & T & T \\ T & F & F \\ T & T & T \end{array}$	$\begin{array}{c c c} P & Q & O_6 \\ \hline F & F & F \\ F & T & T \\ T & F & T \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_7 \\ \hline F & F & F \\ F & T & T \\ T & F & T \\ T & T & T \end{array}$
$\begin{array}{c c c} P & Q & O_8 \\ \hline F & F & T \\ F & T & F \\ T & F & F \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_9 \\ \hline F & F & T \\ F & T & F \\ T & F & F \\ T & T & T \end{array}$	$\begin{array}{c c c} P & Q & O_{10} \\ \hline F & F & T \\ F & T & F \\ T & F & T \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_{11} \\ \hline F & F & T \\ F & T & F \\ T & F & T \\ T & T & T \end{array}$
$\begin{array}{c c c} P & Q & O_{12} \\ \hline F & F & T \\ F & T & T \\ T & F & F \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_{13} \\ \hline F & F & T \\ F & T & T \\ T & F & F \\ T & T & T \end{array}$	$\begin{array}{c c c} P & Q & O_{14} \\ \hline F & F & T \\ F & T & T \\ T & F & T \\ T & T & F \end{array}$	$\begin{array}{c c c} P & Q & O_{15} \\ \hline F & F & T \\ F & T & T \\ T & F & T \\ T & T & T \end{array}$

The operators are: O_0 is F (sometimes called falsum and represented $\perp$), O_1 is $P \wedge Q$, O_2 is $\neg(P \rightarrow Q)$, O_3 is P , O_4 is $\neg(Q \rightarrow P)$, O_5 is Q , O_6 is $\neg(P \leftrightarrow Q)$, O_7 is $P \vee Q$, O_8 is $\neg(P \vee Q)$, O_9 is $P \leftrightarrow Q$, O_{10} is $\neg Q$, O_{11} is $Q \rightarrow P$, O_{12} is $\neg P$, O_{13} is $P \rightarrow Q$, O_{14} is $\neg(P \wedge Q)$, and O_{15} is T (sometimes represented $\top$).